Manufacturing

Running the ADM in a manufacturer — a step-by-step workbook for the repository you have been handed

The companion volume. Not something to read — something to work through. Every phase of the ADM as numbered steps: what you produce, which sheet of the repository it goes in, which columns, who signs it, how you know it is done, and the way it usually goes wrong. Written against a real populated metamodel and a real comparison tool, not against the standard in the abstract.

Use it with the learner guide. South African lens. October 2026.

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Thotnr Consulting

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CHAPTER 1

How to use this workbook

What you were handed, how a step is laid out, and what is actually inside the 105 sheets before you trust a single one of them.

This volume runs the TOGAF Architecture Development Method against one populated repository, one phase at a time. The repository is `Manufacturing_TOGAF_Metamodel.xlsx` — 105 sheets built for **Kestrel Industrial Group**, an invented manufacturer whose data is real inside these pages and nowhere else. Every instruction in the eleven phase chapters points at a sheet in that file by name and at columns in that sheet by header.

Read this item once, end to end, before you do anything. Then stop reading the volume and start working it: open at the phase you are on, do the step, close it.

What you were handed

File	What it is	What this volume does with it
`Manufacturing_TOGAF_Metamodel.xlsx`	105 sheets. A populated TOGAF content metamodel for Kestrel Industrial Group	It is the target of almost every step. You correct it, extend it and sign it
`Manufacturing_Architecture_Comparison_Tool_TEMPLATE.xlsx`	A baseline-against-target diff tool, empty	Phase E and Phase F load it. Phase G reads its output at the gate
`Manufacturing_ArchiMate4_Views.pptx`	Worked sample views	The modelling appendix works from it. No phase step depends on it
Volume 1, *Manufacturing — a starter pack for enterprise architects*	234 pages of industry teaching	Every step with teaching behind it names the chapter and the section to read

Volume 1 taught you the industry. This volume assumes you know it. Where a step needs a fact about how plants actually run — what a changeover costs, why two plants report different availability,

which tree the equipment hangs off — it sends you back with a **Look it up.** line rather than re-teaching it in half a paragraph.

Volume 1 has its own invented group, **Karoo Industrial Group**, with its own plants and its own systems. Kestrel and Karoo are both inventions and their data must never be mixed. When this volume points back at volume 1 it is pointing at teaching, not at numbers you can carry across.

How a step is laid out

Every numbered step is a single block with eight labelled parts, always present, always in this order. The labels are run-in bold at the start of their paragraph, so a step reads as one thing rather than as a form.

Label	What it answers	A good one looks like
Produce.	What artefact exists at the end that did not exist at the start	One sentence, one noun. "A measure definition register with fourteen columns and sixteen rows"
You need.	Inputs: prior steps, people by role, volume 1 sections	Named steps by number and people by the role they hold, not by department
Where it lands.	The sheet, the columns, and what goes in the non-obvious ones	A bordered sheet block. Real sheet, real headers, spelled as the file spells them
How.	The actions, in order, six or fewer	Each item starts with a verb you can do in a room
Signed by.	Who is accountable for the content being right	A row of `057 Governance Body`, or a named role from `011 Role`
Done when.	A condition someone else can test without asking you	A count, a comparison or a signature. Never "when the team is comfortable"
Shows up in.	Which `RPT` sheet moves, and what the number should become	"`RPT05 Data Ownership` shows fourteen named individuals, not nine job titles"
Goes wrong when.	The failure mode that actually happens, concretely	The specific shortcut, and what it costs three months later

Three disciplines make the template work.

Do one step at a time, to the end. A step that is half done is worse than a step not started, because the repository now holds rows nobody has signed and a later step will trust them. If you run out of time, finish the step you are in and leave the next one untouched.

If you cannot fill in `Signed by`, stop. Not "proceed and find a signatory later". The absence of a signatory is the finding. Record it, escalate it, and go to the next step. Several steps in this volume exist only because the repository holds twenty-four principles owned by a committee and twelve applications owned by departments, and nobody noticed for long enough that it became the shape of the file.

Treat `Goes wrong when` as a pre-mortem, not as a postscript. Read it before **How.**

> **WHEN A STEP NAMES SOMETHING THE FILE DOES NOT HAVE**
>
> Several steps point at a sheet or a column that is not in the repository. That is not an error in the step. It is the finding, and the step will say so on its face and give you the columns of the thing you are to create. There are four of these in the first three chapters alone: a production line sheet, a stakeholder sheet, a measure definition register with usable columns, and change control. Build them. Nothing downstream works without them.

The repository's shape, stated plainly

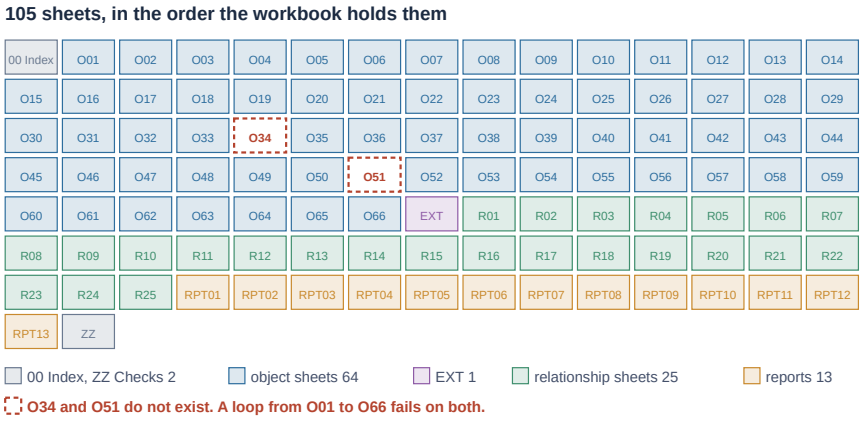


Figure 1.1 The repository, all 105 sheets. Two object numbers are absent, so an index loop is not a safe way to walk the file.

The file holds **105 sheets**: `00 Index`, **64 object sheets**, `EXT Industry Extension`, **25 relationship sheets** `R01`–`R25`, **13 report sheets** `RPT01`–`RPT13`, and `ZZ Checks`. `00 Index` counts the extension as the sixty-fifth object type, which is how the file arrives at *65 object types* in its own header line [1].

The two holes

The object sheets run `O01` to `O66`, but **`O34` and `O51` do not exist**. There is no `O34`-anything and no `O51`-anything; the numbering steps over them. Sixty-six numbers, sixty-four sheets.

This matters the moment you automate anything. A loop over `range(1, 67)` formatting `O%02d` and asking the workbook for a sheet by prefix fails twice, and in most spreadsheet libraries it fails with a key error that names the missing sheet and tells you nothing about why. Enumerate the sheet names the file actually has and filter them; never generate them.

Where the data starts

O14 Value Stream, as every sheet in the file is laid out

1	**Value Stream**			sheet title
2	Kestrel Industrial Group · Business domain · 5 records			subtitle, with the row count
3				blank, always
4	**Ref**	**Value stream**	**Stages**	headers, always row 4
5	VS01	Concept to Product	4 stages	first data row, always row 5
6	VS02	Plan to Produce	5 stages	
7	VS03	Source to Pay	6 stages	

Column A is Ref on every object sheet. O46 has two columns headed Ref.
Read by column index, not by header name, or the second one wins.

Figure 1.2 Every sheet has the same preamble. Headers on row 4, data from row 5, and column A headed `Ref` on every object sheet.

Every sheet in the file — object, relationship and report alike — uses the same four-row preamble. Row 1 is the sheet title. Row 2 is a subtitle carrying the organisation name, the domain and the record count. Row 3 is blank. **Row 4 holds the column headers. Row 5 is the first data row.**

The first column of every object sheet is headed `Ref`. The relationship sheets do not carry a `Ref` column at all, with one exception: `R06 Application interfaces with` has one, holding `INT-001` to `INT-030`. That exception is the hinge of the record-count problem later in this chapter.

The tab names are cut at thirty-one characters

Excel will not hold a sheet name longer than thirty-one characters, and twenty-three of the names in this file are at exactly that limit, truncated mid-word. `O26 Logical Technology Componen` has lost its final *t*. `O27 Physical Technology Compone` has lost *nt*. Among the relationship sheets only `R01`, `R11`, `R12`, `R21` and `R22` survived intact.

One of them is worse than truncated. **`R04 Business Service supported` ends in a space.** It is thirty-one characters of which the last is a space, and nothing in any spreadsheet interface will show you that. A lookup on the name you can see fails, the failure looks like a missing sheet, and you will spend twenty minutes on it. Strip and compare, or match on the `R04` prefix.

`O46` has two columns headed `Ref`

`O46 Technology Building Block` has five columns and two of them are headed `Ref`. Column A holds `TBB-01` to `TBB-07`. Column C holds `T01` to `T07`, a key space that appears nowhere else in the file and resolves to nothing. When you must distinguish them in writing, call the second one `Ref (T)`.

The practical consequence is a silent one. Any reader that builds a dictionary per row from the header line loses column A entirely, because the second `Ref` overwrites the first. The sheet then appears to have references `T01` – `T07` and no building-block identifiers at all. Read that sheet by column index.

THREE READING TRAPS, ALL VERIFIED AGAINST THE FILE

1. **Do not generate sheet names.** `034` and `051` are absent. List what is there.
2. **Do not match a sheet name exactly without stripping.** `R04 Business Service supported` ends in a space and twenty-three names are truncated at thirty-one characters.
3. **Do not key a row by its header on `046`.** Two columns are headed `Ref` and the second wins.

The reference prefixes, derived from the file

There is no convention. There are sixty-five conventions, and the only way to know a sheet's is to open it. This table is derived from the file, row by row [3]. The irregular ones are in bold.

Sheet	Prefix	First – last	Rows	Note
`001 Driver`	`DRV-`	`DRV-01` – `DRV-12`	12	
`002 Goal`	`GOL-`	`GOL-01` – `GOL-06`	6	
`003 Objective`	`OBJ-`	`OBJ-01` – `OBJ-14`	14	
`004 Measure`	`MSR-`	`MSR-01` – `MSR-16`	16	
`005 Principle`	`PRN-`	`PRN-01` – `PRN-24`	24	
`006 Requirement`	**`REQ-`**	`REQ-001` – `REQ-031`	31	**Three digits**, not two
`007 Constraint`	`CON-`	`CON-01` – `CON-08`	8	
`008 Assumption`	`ASM-`	`ASM-01` – `ASM-06`	6	
`009 Organization Unit`	`ORG-`	`ORG-01` – `ORG-12`	12	
`010 Actor`	`ACT-`	`ACT-01` – `ACT-18`	18	
`011 Role`	`ROL-`	`ROL-01` – `ROL-18`	18	
`012 Location`	`LOC-`	`LOC-01` – `LOC-08`	8	
`013 Business Capability`	**`C`**	`C01` – `C10.03.02`	100	**Dotted, and no letter prefix at all**
`014 Value Stream`	**`VS`**	`VS01` – `VS05`	5	**No hyphen**
`015 Business Service`	`BSV-`	`BSV-01` – `BSV-10`	10	
`016 Business Process`	`BPR-`	`BPR-01` – `BPR-20`	20	
`017 Business Function`	`BFN-`	`BFN-01` – `BFN-10`	10	
`018 Product`	`PRD-`	`PRD-01` – `PRD-04`	4	
`019 Contract`	`CTR-`	`CTR-01` – `CTR-10`	10	

O20 Data Entity	**E**	E01 – E14	14	**No hyphen**
O21 Logical Data Component	LDC-	LDC-01 – LDC-12	12	
O22 Physical Data Component	PDC-	PDC-01 – PDC-12	12	
O23 Application Component	**A**	A01 – A12	12	**No hyphen**
O24 Information System Service	ISS-	ISS-01 – ISS-12	12	
O25 Technology Service	TSV-	TSV-01 – TSV-12	12	
O26 Logical Technology Componen	LTC-	LTC-01 – LTC-07	7	
O27 Physical Technology Compone	PTC-	PTC-01 – PTC-12	12	
O28 Platform Service	PSV-	PSV-01 – PSV-08	8	
O29 Gap	GAP-	GAP-01 – GAP-02	2	
O30 Work Package	WP-	WP-01 – WP-12	12	
O31 Deliverable	DLV-	DLV-01 – DLV-20	20	
O32 Standard	STD-	STD-01 – STD-16	16	
O33 Architecture Contract	ACO-	ACO-01 – ACO-08	8	
O35 Business Building Block	BBB-	BBB-01 – BBB-10	10	
O36 Data Entity Relationship	REL-	REL-01 – REL-11	11	
O37 Data Attribute	**ATR-**	ATR-001 – ATR-060	60	**Three digits**, not two
O38 Data Set	DST-	DST-01 – DST-06	6	
O39 Data Migration	MIG-	MIG-01 – MIG-06	6	
O40 Source To Target Mapping	MAP-	MAP-01 – MAP-12	12	

`O41 Security Control`	`SEC-`	`SEC-01` – `SEC-08`	8	
`O42 Metadata Element`	`MTD-`	`MTD-01` – `MTD-14`	14	
`O43 Master Data Rule`	`MDR-`	`MDR-01` – `MDR-10`	10	
`O44 Data Building Block`	`DBB-`	`DBB-01` – `DBB-10`	10	
`O45 Application Building Block`	`ABB-`	`ABB-01` – `ABB-10`	10	
`O46 Technology Building Block`	**`TBB-`**	`TBB-01` – `TBB-07`	7	**Two columns headed `Ref`.** The second holds `T01` – `T07`
`O47 Reference Model Entry`	`RME-`	`RME-01` – `RME-11`	11	
`O48 Environment`	`ENV-`	`ENV-01` – `ENV-04`	4	
`O49 Resilience Tier`	`RES-`	`RES-01` – `RES-12`	12	
`O50 Network Link`	`NWL-`	`NWL-01` – `NWL-10`	10	
`O52 Implementation Factor`	`IF-`	`IF-01` – `IF-08`	8	
`O53 Transition Architecture`	**`TA-`**	`TA-1` – `TA-3`	3	**Unpadded** — `TA-1`, not `TA-01`
`O54 Architecture Alternative`	**`ALT-A`**	`ALT-A` – `ALT-E`	5	**Lettered**, not numbered
`O55 Building Block`	**`BBB-`**	`BBB-01` – `TBB-07`	37	**No space of its own.** Reuses `BBB-` 10, `DBB-` 10, `ABB-` 10, `TBB-` 7
`O56 Project`	`PRJ-`	`PRJ-01` – `PRJ-12`	12	
`O57 Governance Body`	`GB-`	`GB-01` – `GB-05`	5	
`O58 Governance Gate`	**`GT-`**	`GT-0` – `GT-3`	4	**Starts at zero**, and the gate names are G0–G3
`O59 Compliance Assessment`	—	—	0	Empty. Header row only
`O60 Governance Decision`	`DEC-`	`DEC-01` – `DEC-07`	7	
`O61 Governance Change Request`	**`CR-G`**	`CR-G01` – `CR-G05`	5	**Two letters and a letter-number** — `CR-G01`

`O62 Delivery Risk`	`RSK-`	`RSK-01` – `RSK-08`	8	
`O63 Control Objective`	`CO-`	`CO-01` – `CO-06`	6	
`O64 Security Policy`	`POL-`	`POL-01` – `POL-04`	4	
`O65 Enforcement Point`	`EP-`	`EP-01` – `EP-05`	5	
`O66 Architecture Increment`	**`INC-`**	`INC-1` – `INC-6`	6	**Unpadded** — `INC-1`, not `INC-01`
`EXT Industry Extension`	`X`	`X01` – `X06`	6	Type register, **no instances**
`R06 Application interfaces with`	**`INT-`**	`INT-001` – `INT-030`	30	**The only reference space on a relationship sheet**

Nine distinct irregularities, and every one of them will bite a join:

- **`O06 Requirement` uses three digits** (`REQ-001`) where almost every other sheet uses two. So does **`O37 Data Attribute`** (`ATR-001`). A sort that treats the reference as text puts `REQ-031` before `REQ-004` on sheets that use two digits and gets it right on these two.
- **`O13 Business Capability` has no letter prefix and uses dots.** `C01`, then `C01.01`, then `C01.01.01`. Three levels in one column, with the level also carried in a `Level` column and the parent in a `Parent` column. A string sort gives you `C1`-nothing, because there is no `C1`.
- **`O14`, `O20` and `O23` drop the hyphen.** `VS01`, `E01`, `A01`. A regular expression tuned to `[A-Z]{2,4}-\d+` misses all three of the sheets you will join against most often.
- **`O53` and `O66` do not pad.** `TA-1`, `TA-2`, `TA-3`; `INC-1` to `INC-6`. Both would sort wrongly past nine, and neither gets past nine, which is how it survived.
- **`O54 Architecture Alternative` is lettered.** `ALT-A` to `ALT-E`.
- **`O58 Governance Gate` starts at zero.** `GT-0` to `GT-3`, carrying gates named G0 to G3. The reference and the gate name are one apart in form and identical in meaning, which reads as an off-by-one error every time.
- **`O61` is `CR-G01`.** A prefix, a letter and a number.
- **`O55 Building Block` has no reference space of its own.** Its thirty-seven rows reuse the references of `O35`, `O44`, `O45` and `O46` — ten `BBB-`, ten `DBB-`, ten `ABB-`, seven `TBB-`. A reference is therefore not unique across the file. `BBB-01` is a row of `O35` and a row of `O55`, and they are the same building block seen twice.
- **`R06` carries references on a relationship sheet.** `INT-001` to `INT-030`, the only such space in the file.

When you add rows to this repository — and almost every step in this volume adds rows — match the sheet's own convention, however poor it is. A single consistent sheet is worth more than a sheet half-corrected.

From ADM phase to chapter, and from chapter back to volume 1

ADM phase	This volume	Deliverables in O31	The sheets it mostly writes
Preliminary	Chapter 2	—	`O04`, `O05`, `O12`, `O31`, `O57`, `O58`, `EXT`
A, Architecture Vision	Chapter 3	`DLV-01`, `DLV-02`, `DLV-03`	`O01`–`O03`, `O09`–`O12`, `O14`, `O18`
B, Business Architecture	Chapter 4	`DLV-04`, `DLV-05`, `DLV-06`	`O10`, `O11`, `O13`, `O15`–`O17`, `O35`, `O63`
C, Data Architecture	Chapter 5	`DLV-07`, `DLV-08`	`O20`–`O22`, `O36`–`O40`, `O42`, `O43`
C, Application Architecture	Chapter 6	`DLV-09`, `DLV-10`	`O23`, `O24`, `O45`, `R05`, `R06`
D, Technology Architecture	Chapter 7	`DLV-11`, `DLV-12`	`O25`–`O28`, `O32`, `O46`, `O48`–`O50`
E, Opportunities and Solutions	Chapter 8	`DLV-13`, `DLV-14`, `DLV-15`	`O29`, `O30`, `O52`–`O55`
F, Migration Planning	Chapter 9	`DLV-16`, `DLV-17`	`O30`, `O53`, `O56`, `O62`, `O66`
G, Implementation Governance	Chapter 10	`DLV-18`, `DLV-19`	`O33`, `O57`–`O60`, plus the four sheets you add
H, Architecture Change Management	Chapter 11	—	`O60`, `O61`, plus the four sheets you add
Requirements Management	Chapter 12	`DLV-20`	`O06`–`O08`, `O63`, `R08`, `R09`

The phase names and their order are The Open Group's [4]; what each phase writes into is this repository's.

`O31 Deliverable` holds twenty rows with a `Phase` column reading `A` through `G` and `RM`. Fifteen of them are `Approved`; the five that are `Draft` are the Phase F and G ones. Treat the `Approved` status with suspicion — `DLV-03 Stakeholder Map` is marked `Approved` and there is no stakeholder sheet anywhere in the repository for it to have been built from.

This volume	Read first, in volume 1
Chapter 2, Preliminary	7 The operating model and the planning cadence · 19 Your first ninety days
Chapter 3, Phase A	2 The archetypes · 5 Who does what · 19 Your first ninety days
Chapter 4, Phase B	5 Who does what · 7 The operating model · 8 The process model · 16 Regulation
Chapter 5, Phase C data	9 The conceptual data model · 10 The information model
Chapter 6, Phase C application	11 The conceptual application model · 12 The application landscape
Chapter 7, Phase D	13 The plant floor · 14 The reference architecture · 15 Technology providers
Chapters 8 and 9, Phases E and F	6 How manufacturers make money · 18 Karoo Industrial Group, end to end
Chapters 10 and 11, Phases G and H	16 Regulation, risk and what the inspector asks for
Chapter 12, Requirements Management	3 The vocabulary · 16 Regulation · 20 Appendix A, The cross-maps

`EXT Industry Extension`, and the trap inside it

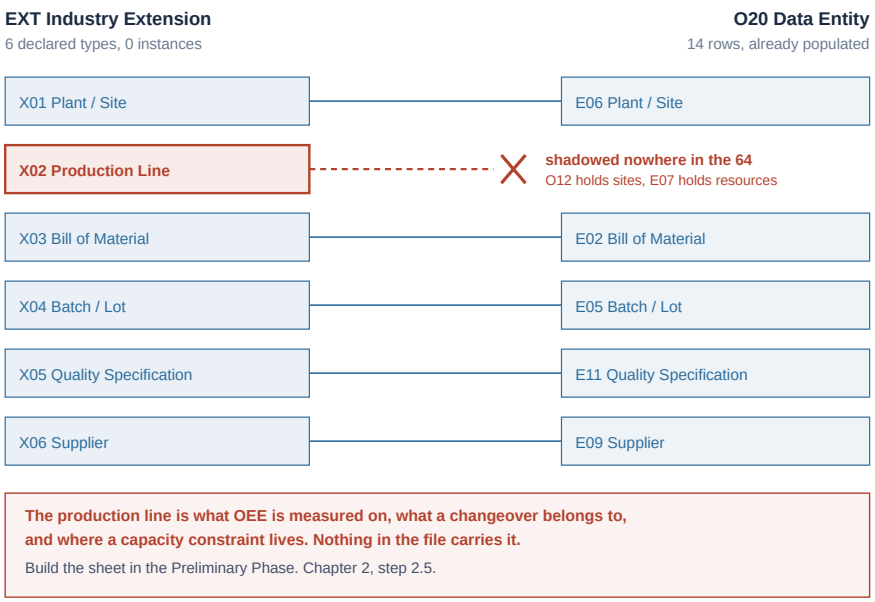


Figure 1.3 `EXT Industry Extension` **against** `O20 Data Entity`**.** Five of the six declared types are already in the core under another reference. `X02 Production Line` is in neither, and it is the one a manufacturer cannot do without.

`EXT Industry Extension` is three columns wide and six rows deep. It declares six object types the core metamodel does not carry: `X01 Plant / Site`, `X02 Production Line`, `X03 Bill of Material`, `X04 Batch / Lot`, `X05 Quality Specification` and `X06 Supplier`. The third column says, for each, why the core is not enough.

It holds no instances. It is a register of types. There is no `X01` sheet, no `X02` sheet and no rows anywhere carrying an `X` reference. A step that says "use the industry extension" is telling you to build something.

Five of the six already exist in the core under another name. `O20 Data Entity` carries `E06 Plant / Site`, `E02 Bill of Material`, `E05 Batch / Lot`, `E11 Quality Specification` and `E09 Supplier`. So for five of the six, the extension is a duplicate declaration and you can resolve it by deciding which sheet is authoritative and deleting the other claim.

`X02 Production Line` is shadowed nowhere. It is not in `O20`. `O12 Location` holds eight rows of which three are plants, and a plant is not a line. `E07 Work Centre` holds resources, and a work centre is not a line either — a line is an ordered set of work centres with a shared rate. Nothing in the sixty-four sheets carries it.

For a manufacturer that is the most consequential absence in the file, because the production line is the object three of the most important facts in the business hang off:

- **It is what effectiveness is measured on.** An availability, performance and quality figure belongs to a line running a product family over a shift. Computed over a plant it is an average of incomparable things.
- **It is what a changeover belongs to.** A changeover is a transition between two products on one line. There is no other object it can attach to.
- **It is where a capacity constraint lives.** The bottleneck is a line, or a work centre on a line. A plant does not have a rate; a line does.

Look it up. Volume 1 chapter 7, *Sites, plants, lines, cells and stations*, for what a line is and is not, and *Why OEE is not comparable, stated hard* for what goes wrong when the denominator moves.

Chapter 2, step 2.5 builds the sheet and sets the pattern the other five follow.

BEFORE YOU ACCEPT THE REPOSITORY AS A BASELINE

Which three sheets will your first automated read of this file touch, and have you checked each one for a duplicate header and a truncated name?

Name the object in your plant that OEE is measured on. Which sheet of this repository holds it today?

If a reference in this file is not unique across sheets, what does that do to the joins your reports depend on?

`ZZ Checks`, counted against the data

`ZZ Checks` holds fifteen checks and a closing note, each check carrying a result and a threshold, under the claim *Computed over this workbook* [2]. Most are correct. Four are not, and one is true only

because of how the data was made. Count them yourself; it takes twenty minutes and it is the best twenty minutes you will spend on this file.

Check	Stated	Counted	Verdict
Object types populated	65	64 of 65	**Fails.** `059 Compliance Assessment` is empty
Records	843	813	**Fails by 30**
Relationship types	25	25	Holds
Relationship instances	430	430	Holds
Reports	13	13	Holds
Interfaces unmanaged	22	22 of 30	Holds
Interfaces with no logging	21	21 of 30	Holds
Direct database reads	4	4	Holds
Platform components past support	4	2	**Fails.** Typed, not computed
Security controls absent	1	1	Holds. `SEC-02 Design review`
Control objectives without evidence	5	5 of 6	Holds
Data entities without lineage	6	6 of 14	Holds
Vacant architecture roles	0	0 of 18	True by construction
Work packages past end date	4	4 labels, 0 real	**Fails.** Typed, not computed
Records validated against a real organisation	0	0	Holds, and says so

Records: 843 claimed, 813 counted

Add the row counts of the sixty-four object sheets and `EXT Industry Extension` and you get **813**. The header of `00 Index` and line 6 of `ZZ Checks` both say **843**. The difference is **exactly 30**, and 30 is the row count of `R06 Application interfaces with`.

So the total reconciles on one condition: that an interface counts as a record. Which is an admission worth keeping. An interface in this file has a reference space of its own, eight attributes, a managed flag and a logging flag — everything an object has except a sheet. It is an object that was modelled as a relationship, and the record count is where that shows.

Object types populated: 64 of 65, not 65

The threshold column reads *Every type carries records*. `059 Compliance Assessment` holds a header row and nothing else. `R24 Compliance Assessment tests` is header-only in consequence, since it has nothing to resolve. `029 Gap` holds two rows, which is not empty but is not a gap analysis either.

Platform components past support: 2, not 4

`O27 Physical Technology Compone` has a `Position` column and four rows read `Past support`: `PTC-06 Shop-floor Windows terminals`, `PTC-07 Plant network switches`, `PTC-08 Core database cluster` and `PTC-09 Enterprise storage array`. Their `Support end` dates are **2027-07, 2026-05, 2027-04 and 2026-05**. Two of those four are in the future. `PTC-06` and `PTC-08` are labelled unsupported and are not.

`Position` is a typed string, not a formula over `Support end`. `RPT10 Technology Currency` and line 15 of `RPT01 Dashboard` reprint the same four, so the error propagates into two reports.

Work packages past end date: four labels, none real

`O30 Work Package` carries a `Delivery` column and four rows read `Past end date`: `WP-02`, `WP-03`, `WP-09` and `WP-12`. Their `End` dates are **2030-10, 2031-06, 2029-07 and 2030-01**. Every `End` on the sheet falls between 2029-07 and 2031-10. Nothing on this sheet can be past its end date. The count of four is a correct count of a wrong label.

Vacant architecture roles: zero, by construction

All eighteen rows of `R21 Role performed by Actor` read `Filled`, and `O10 Actor` holds exactly the same eighteen names as `O11 Role`, one to one. The check cannot return anything but zero. That is not a healthy estate; it is a sheet that was generated from another sheet.

> **THE HABIT THIS CHAPTER IS TRYING TO INSTALL**
>
> A repository tells you two things: what it holds, and what it believes about what it holds. They are different, and the distance between them is the first finding of every engagement. Count before you cite. Every number in this volume was counted.

The change control the file does not have

There is no change control anywhere in the 105 sheets. No version column, no baseline, no change log, no change request against an architecture object, no effective date on any row. `O61 Governance Change Request` holds five rows, but they are governance dispensations with an `Expires` date, not changes to the architecture. `O60 Governance Decision` records seven decisions and has no date column at all.

So nothing in this repository can answer the three questions a governance body asks: what changed, who approved it, and against which baseline. This volume adds four sheets and five columns.

These four sheets do not take `O` numbers, and they do not go at the end. **They sit in front of `00 Index`, as the first four tabs of the file**, in the order below. Anyone scripting the workbook needs to know that, because a reader that assumes `00 Index` is sheet one will now be off by four.

Sheet to create	Tab	Columns, in order	Refs run
`01 Version Control`	First	`Version`, `Date`, `Author`, `Approved by`, `Scope of change`, `Change refs`, `Baseline ref`	`1.0` is the file as received
`02 Change Log`	Second	`Log ref`, `Date`, `Sheet`, `Record ref`, `Field`, `Old value`, `New value`, `Change Ref`, `Changed by`, `Reason`	`LOG-0001` onward
`03 Change Requests`	Third	`Change ref`, `Raised`, `Raised by`, `Against`, `Type`, `Description`, `Classification`, `Impact`, `Decision body`, `Decided`, `Status`, `Expires`, `Decision ref`	`CR-0001` onward
`04 Baselines`	Fourth	`Baseline ref`, `Date`, `Label`, `Scope`, `Record count`, `Approved by`, `Decision ref`, `Supersedes`	`BL-00` is the received state

Three things about them are easy to get wrong. **`02 Change Log`'s grain is one row per field**, not per record and not per change request: correcting `Disposition`, `Health` and `Criticality` on `A04 SCADA / Historian` is three rows, and the `Old value` is copied before the edit, which is the only reason the sheet is worth keeping. **`03 Change Requests` runs `CR-0001`, not `CR-G`** — that space belongs to `O61 Governance Change Request` and reusing it collides on the first row. And **there is no separate dispensation sheet**: a dispensation is a `03 Change Requests` row with `Type` = `Dispensation` and a mandatory `Expires` date, alongside `Correction`, `Scope`, `Standard` and `Compliance`.

And five columns on **every** object sheet, appended to the right of the existing ones so no existing column index moves:

Column	What goes in it	Why
`Version`	The `01 Version Control` version in which this row last changed	So a row can be cited
`Record Status`	One of `Proposed`, `Approved`, `Superseded`, `Retired`	So a draft row cannot be read as agreed
`Changed By`	A person's name	Not a department, not a committee
`Change Ref`	The `CR-` row in `03 Change Requests` that authorised it	So the row traces to a decision
`Approved By`	The `GB-` body from `O57` that approved it	So a gate can see who signed

Append them by walking the sheet names the file actually has. A loop over `O01` to `O66` misses `EXT`, fails outright at `O34` and `O51`, and leaves the four new sheets out altogether.

Add them in the Preliminary Phase — chapter 2, step 2.13 — not when you need them. Chapter 11 builds the same four sheets out in full, with the trigger list that governs `Against` and `Type`; chapter 10 is where the gates start reading `04 Baselines`. A change log started in month four records nothing about months one to three, and months one to three are when the repository changes most.

Printing this volume, and the transcription loop

The forms in this volume are built to be printed at A4 and written on with a pen, in a room, with the people who know the answer. They carry at most six columns because more will not fit the text column, and eight to twelve blank rows because that is a page. Where a sheet has more than six columns the form is split into an **A** form that identifies and a **B** form that assesses, and each says which columns it covers and which you type straight into Excel.

Every form carries four things on its face: a form number, the sheet it feeds, how many rows a first pass usually needs, and who signs it. A form that feeds a sheet the repository does not have says so on its face and gives the columns of the sheet you are to create.

The loop is three steps and it has to close the same day.

1. **Paper in the room.** You are standing next to a line, or sitting across from a planner. You are writing. Nobody is looking at a laptop. The form's columns are the sheet's columns, so you are not translating anything later.
2. **Spreadsheet the same afternoon.** Transcribe into the repository while you still remember which answer was confident and which was a guess. Mark the guesses. A row you cannot source is a row you delete, not a row you keep and hope about.
3. **Repository record with a change reference.** Every transcribed row gets its `Version`, `Record Status`, `Changed By`, `Change Ref` and `Approved By` filled in, and one line per field in `02 Change Log`. If those four sheets do not exist yet, you are not past chapter 2, step 2.13, and you should be.

Leave it a week and the loop does not close. You will have paper nobody can read, rows nobody can source, and a repository that is worse than the one you were handed because it now looks maintained.

Form 1-1 — Repository count sheet

Feeds nothing. This is the day-one count that tells you whether to believe `ZZ Checks`. A first pass needs all fifteen lines. Signed by the **Chief Enterprise Architect** before any other step starts. Do it by reading the file, not by reading `00 Index`.

Check	ZZ Checks says	You counted	Agrees?	How you counted it
Records	843	EXAMPLE 813 — delete this row	No	Summed rows 5 down, 64 object sheets plus EXT

THE REVIEWER'S ASK, BEFORE YOU START CHAPTER 2

Which of the fifteen `ZZ Checks` lines did your own count disagree with, and by how much?

Who in this organisation will be shown the record-count discrepancy, and what do you want them to do about it?

The repository has no change control. What is the date you will have `02 Change Log` running by, and what happens to every row you add before then?

THE CHAPTER IN TEN LINES

- 105 sheets: 00 Index, 64 object sheets, EXT, 25 relationship sheets, 13 reports and ZZ Checks.
- O34 and O51 do not exist. Enumerate the sheet names the file has; never generate them.
- Headers are on row 4 and data starts on row 5, on every sheet without exception.
- R04 Business Service supported ends in a space, and 23 tab names are cut at 31 characters.
- O46 Technology Building Block has two columns headed Ref. Read that sheet by column index.
- There is no single reference convention. There are 65, and the irregular ones break joins.
- O55 Building Block reuses the references of O35, O44, O45 and O46, so a ref is not unique.
- EXT declares six industry types and holds no instances. Five are shadowed in O20 Data Entity.
- X02 Production Line is shadowed nowhere, and the line is what effectiveness is measured on.
- ZZ Checks claims 843 records against a counted 813. The difference is exactly R06's 30 rows.
- Platform components past support is 2, not 4, and work packages past end date is 0, not 4.
- The file has no change control at all. This volume adds four sheets and five columns.
- Print the forms, fill them in the room, transcribe the same afternoon, log every change.

SOURCES FOR THIS CHAPTER

- **1.** Manufacturing_TOGAF_Metamodel.xlsx, sheet 00 Index — header line, sheet inventory and record counts
- **2.** Manufacturing_TOGAF_Metamodel.xlsx, sheet ZZ Checks — fifteen checks with results and thresholds, plus a closing note
- **3.** Manufacturing_TOGAF_Metamodel.xlsx, sheets O01–O66 and EXT Industry Extension — row counts and reference ranges counted directly, 16 October 2026
- **4.** The Open Group, the TOGAF Standard and the Architecture Development Method https://www.opengroup.org/togaf

CHAPTER 2

Standing the practice up

Thirteen steps to make the repository governable, the measures definable and the principles testable, before Phase A writes a single row.

The Preliminary Phase is where you decide what the architecture capability is, who it answers to, and what it is allowed to say. In this repository it is also where you repair the foundations, because several of the sheets the later phases depend on were generated rather than assessed.

Thirteen steps. Work them in order. Steps 2.1 to 2.3 can be done in the first fortnight; 2.4 to 2.7 need the plant; 2.8 to 2.13 need a decision from somebody senior. Kestrel Industrial Group is an invention and its data is real only inside this volume.

Scope, count and measure

Step 2.1 — Fix the boundary of the enterprise and the plants inside it

Produce. A one-page scope statement naming every plant in scope, every plant explicitly out of scope, the legal entities involved, and the period the architecture addresses.

You need. The group structure, the statutory accounts, a site list from operations, and thirty minutes with the Chief Operating Officer. Volume 1 chapter 2 for the archetype each plant runs.

Where it lands.

> **`012 Location` — 8 rows, 4 columns**
>
> `Ref` · `Location` · `Type` · `Scale`. Three rows are `Production`: `LOC-02 Northern Plant`, `LOC-03 Coastal Plant`, `LOC-04 Components Plant`. `Scale` is a text string — "198 people" — so it cannot be summed.
>
> Add six columns to the right: `Province`, `Legal entity`, `Ownership` (owned, leased, tolled), `Line count`, `Shift pattern`, `Archetype` (discrete, batch, process, mixed). Mark each existing row in or out of scope in a seventh column, `In scope`.

How.

1. List every site the group operates, including the ones nobody calls a plant: a packing hall, a satellite workshop, a contract manufacturer running your tooling.
2. Set `Archetype` per plant, not per group. A group with a discrete plant and a process plant has two data models and no amount of scoping will merge them.
3. Fill `Line count` from a walk, not from a document. This is the number the rest of the volume turns on.

4. Mark the three `Technology` and `Logistics` rows out of scope for business architecture and in scope for technology architecture.
5. Have the scope statement initialled by the people whose plants are listed.

Signed by. `GB-04 Executive Committee`, on the recommendation of the Chief Enterprise Architect (`ROL-08`).

Done when. Every row of `O12 Location` carries an archetype, a line count and an in-scope flag, and no plant discovered later is missing from the list.

Shows up in. Nowhere, and that is the finding. No report in the thirteen reads `O12 Location`. `RPT10 Technology Currency` carries a `Location` column, but it comes from `O27`, not from here.

Goes wrong when. You take the eight rows as the site list. `LOC-08 Cloud Region` carries a `Scale` of **261 people**, which is more than any plant in the file. A cloud region is not a place with people in it, and a sheet that says it is will be read as authoritative by the fourth person down the chain.

Look it up. Volume 1 chapter 7, *Sites, plants, lines, cells and stations*, and chapter 2, *What to do with this in week one*.

Step 2.2 — Count the repository instead of believing it

Produce. A completed Form 1-1, signed, showing which of the fifteen `ZZ Checks` lines your own count agrees with.

You need. The file, a spreadsheet, two hours. No interviews.

Where it lands.

> **`ZZ Checks` — 15 checks and a closing note, 3 columns**
>
> `Check` · `Result` · `Threshold`, headers on row 4, checks on rows 5 to 19, and a note on row 21. Nothing in this sheet is a formula. Every result is a typed value. Add a fourth column, `Counted`, and a fifth, `Counted on`, and never delete the original `Result` — the distance between the two columns is the finding.

How.

1. Sum the data rows of the sixty-four object sheets plus `EXT Industry Extension`. You will get **813**. The sheet says **843**.
2. Subtract. The difference is **30**, which is the row count of `R06 Application interfaces with` — the only relationship sheet carrying a `Ref` column.
3. Open `O59 Compliance Assessment`. It is header-only, so *object types populated* is 64 of 65, not 65, and `R24 Compliance Assessment tests` has nothing to resolve.

4. Check `O27`'s four `Past support` rows against their `Support end` dates. Two of the four are future-dated. The `Position` column is typed, not computed.
5. Check `O30`'s four `Past end date` rows the same way. Every `End` on the sheet falls between 2029-07 and 2031-10, so none of them can be past anything.
6. Write the five discrepancies into your scope statement as assumptions you have tested.

Signed by. Chief Enterprise Architect (`ROL-08`). This one does not need a committee.

Done when. Form 1-1 has all fifteen checks filled, each with the method you used, and the four failing lines are named in writing to whoever handed you the file.

Shows up in. `RPT01 Dashboard`, which reprints the same figures with the same errors — line 15 repeats *platform components past vendor support 4*, and line 21 repeats *work packages past end date 4*.

Goes wrong when. You quote 843 records in a steering pack in week two. Somebody counts it in month five, and from then on every number you produce is checked before it is read.

Look it up. Volume 1 chapter 19, *What to count before you model anything*.

AFTER YOU HAVE COUNTED THE FILE YOURSELF

Which number in `ZZ Checks` were you most tempted to take on trust, and why?

The record count reconciles only if an interface is a record. Do you accept that an interface is an object in this repository? What follows if it is?

Step 2.3 — Stand up the measure definition register before anything else

Produce. A measure definition register: one row per measure, with the words, the numerator, the denominator, the exclusions and the source, agreed across all three plants.

You need. Step 2.1. The plant managers, the controller, and whoever produces the monthly operations pack at each site. Half a day per plant, and it will take two rounds.

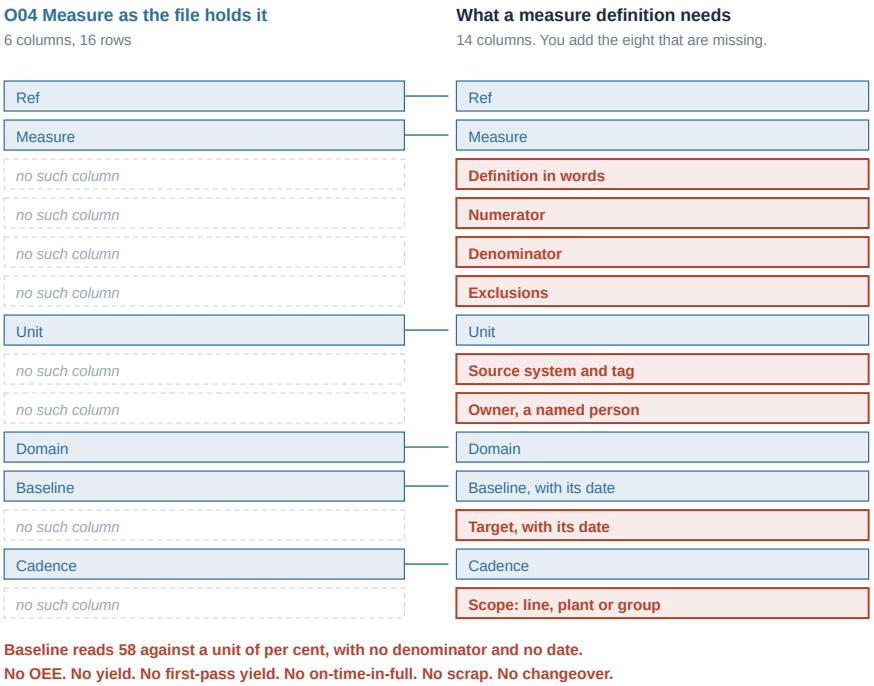


Figure 2.1 `O04 Measure` against what a measure definition needs. Six columns present, eight absent, and nothing on the sheet that an operations manager would recognise as theirs.

Where it lands.

> **`O04 Measure` — 16 rows, 6 columns**
>
> `Ref` · `Measure` · `Unit` · `Domain` · `Baseline` · `Cadence`. There is no `Definition`, no `Numerator`, no `Denominator`, no `Exclusions`, no `Source`, no `Owner`, no `Target` and no date anywhere. `MSR-01 Obligations with evidence` carries a `Unit` of `%` and a `Baseline` of `58`, which is a number with no denominator and no date attached to it.
>
> Add eight columns: `Definition in words`, `Numerator`, `Denominator`, `Exclusions`, `Source system and tag`, `Owner`, `Target`, `Scope` (line, plant or group). Date the baseline in its own column, `Baseline taken`.

How.

1. Read the sixteen existing rows aloud to an operations audience. `MSR-05 Single customer view coverage` and `MSR-06 Cross-sell ratio` are not manufacturing measures and should be challenged on the spot.
2. Add the measures a plant actually runs on and the file does not carry: effectiveness, yield, first-pass yield, scrap, on-time-in-full, changeover time, schedule adherence.
3. For each, write the denominator first. The denominator is where plants disagree, every time.
4. Name the source system and the tag or table the number comes from, not the report it appears on.
5. Give every row one named owner who is a person.
6. Take the baseline on a stated date, from a stated run, and write both down.

Signed by. `GB-03 Data Governance Council` for the definitions, and each plant manager for the baselines taken at their site.

Done when. The same measure computed at all three plants uses the same denominator, and you can show the arithmetic from the tag to the published number for at least one measure end to end.

Shows up in. Nowhere yet. No report reads `004 Measure`. That is itself the argument for doing this now: the reporting layer in this repository has no definitions behind it.

Goes wrong when. You let the register wait for Phase B. By then three plants have published three effectiveness figures to the executive, the figures differ by twenty points, and the conversation is about whose plant is worse rather than about what the number means.

Look it up. Volume 1 chapter 7, *So the first deliverable is a measure definition register*, and chapter 10, *The measure definition register*.

BEFORE YOU WRITE A SINGLE MEASURE DEFINITION

For the effectiveness measure at your worst-performing plant, what exactly is in the denominator? Write it as a number of hours and say what was excluded.

Which of the sixteen rows in `004 Measure` would an operations manager recognise as theirs?

Who signs a measure definition, and what happens when a plant disagrees with it after signing?

Form 2-1A — Measure definition, identify

Feeds `004 Measure` plus the eight columns you add in step 2.3. One row per measure. A first pass needs twelve to twenty rows. Signed by the **Data Governance Council** (`GB-03`). Form 2-1B covers `Exclusions`, `Owner`, `Target`, `Baseline taken` and `Scope`; type those straight into Excel.

Ref	Measure	Definition in words	Numerator	Denominator	Source system and tag
MSR-17	EXAMPLE — delete this row				

The plant, and the objects the file does not have

Step 2.4 — Settle the plant hierarchy against the ERP hierarchy

Produce. A signed mapping between the control-side equipment hierarchy and the hierarchy the ERP holds, with the levels that do not join named as such.

You need. Step 2.1. The process engineer or control engineer at one plant, the ERP functional lead, and a printout of the plant's equipment list. Two hours, then a walk.

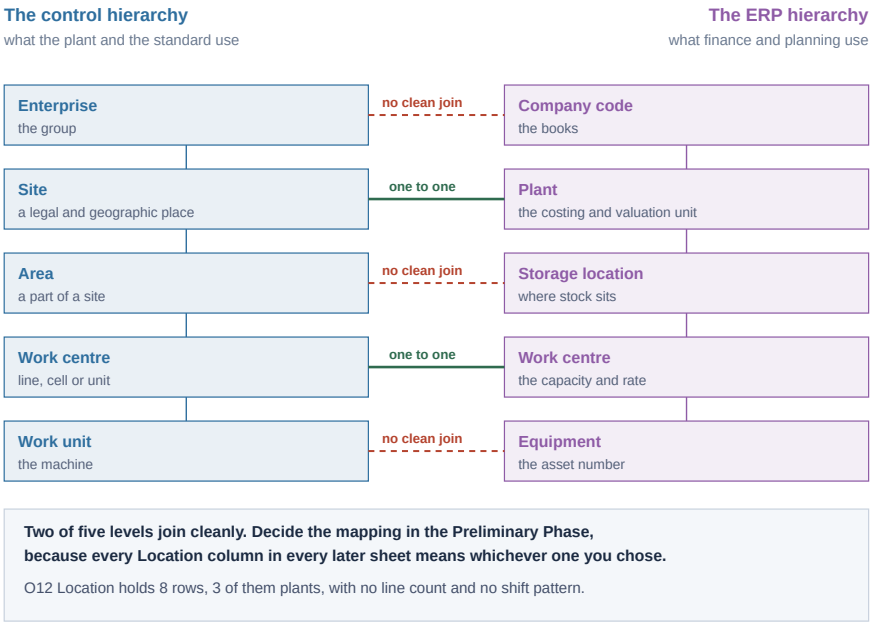


Figure 2.2 The reconciliation to settle in the Preliminary Phase. Two levels join cleanly, three do not, and every later `Location` column means whichever side you chose.

Where it lands.

> **`O12 Location` and the new `X02 Production Line` sheet**
>
> Decide which hierarchy level each `Location` row represents, and record the answer in a new column `Hierarchy level`. Every later sheet with a location column — `O23` through `R20`, and `O27`'s `Location` — then means that level and nothing else.

How.

1. Draw both hierarchies on one page: the control hierarchy down the left, the ERP hierarchy down the right.
2. Join the levels that genuinely correspond. In most plants two of five join cleanly and the rest do not.
3. Where they do not join, say which one the repository will use and write the other into a cross-reference column rather than forcing a merge.
4. Test the decision against one real question: *where does this piece of equipment sit?* If two people give different answers, the mapping is not settled.
5. Record the decision in `O60 Governance Decision` as a new row, and note that `O60` has no date column, so put the date in the decision text.

Signed by. `GB-01 Architecture Board`, with the plant's engineering manager present.

Done when. One sentence defines what `Location` means on every sheet, and a named person at each plant can place any asset in the hierarchy without asking.

Shows up in. `RPT10 Technology Currency`, whose `Location` column currently places `PTC-03 SCADA control servers` at `Primary Data Centre` and `PTC-07 Plant network switches` at `Central Distribution Centre`. Control servers sit at the plant they control. After this step those two rows should move.

Goes wrong when. You treat this as a Phase D question. It is not. It determines what the location column of every sheet written in Phases A to C means, and rewriting those columns later means re-signing every one of them.

Look it up. Volume 1 chapter 9, *The plant hierarchy, and the reconciliation nobody budgets for*, and chapter 20, *Map 2 — The ISA-95 equipment hierarchy to the ERP plant hierarchy*.

Step 2.5 — Build the `X02 Production Line` sheet

Produce. A new sheet, `X02 Production Line`, with one row per production line across every plant in scope, and the precedent that the other five extension types will follow.

You need. Steps 2.1 and 2.4. A walk of each plant with the plant manager or production manager. Half a day per plant.

Where it lands.

> **`X02 Production Line` — the sheet you are creating**
>
> `EXT Industry Extension` declares `X02 Production Line` and holds no instances. Nothing in the sixty-four object sheets carries a line. Build the sheet with the file's own conventions — title row 1, subtitle row 2, blank row 3, headers row 4, data from row 5, `Ref` in column A.
>
> Columns: `Ref` (`LIN-01` onward) · `Production line` · `Plant` (a `LOC-` reference) · `Product family` · `Work centres` (count) · `Design rate` · `Rate unit` · `Shift pattern` · `Changeover class` · `Constraint` (yes or no) · `Effectiveness measured` (yes or no) · `Owner` (a person).

How.

1. Walk each plant end to end and write one row per line. A line is an ordered set of work centres with a shared rate, not a room and not a cost centre.
2. Where one physical line runs two product families on different setups, record two rows and say so in `Changeover class`.
3. Set `Design rate` and `Rate unit` together — tonnes per hour and units per hour are not the same column.
4. Mark the constraint. Every plant has one, the plant knows which it is, and nobody has written it down.
5. Link the sheet to `O12 Location` by the `LOC-` reference, not by plant name.
6. Repeat the pattern for `X01`, `X03`, `X04`, `X05` and `X06` only where the core sheet does not already carry them. Five of the six are shadowed in `O20 Data Entity`; this one is not.

Signed by. The plant manager for the rows at their site; Chief Enterprise Architect for the sheet.

Done when. The line count in `O12 Location` from step 2.1 equals the row count of `X02 Production Line`, plant by plant.

Shows up in. Nothing today, because no report knows the sheet exists. Phase B adds it to the capability map and Phase E to the gap analysis.

Goes wrong when. You map lines onto `E07 Work Centre` because it is the nearest thing. A work centre is a resource with a capacity; a line is a sequence with a rate and a changeover. Collapsing them loses the changeover, and the changeover is where most of the lost capacity in a discrete plant goes.

Look it up. Volume 1 chapter 7, *The four capacities*, and chapter 3, *OEE, and the three traps inside it*.

Form 2-2 — Production line register

This form feeds a sheet the repository does not have. `EXT Industry Extension` declares `X02 Production Line` and holds no instances; you are creating the sheet in step 2.5. A first pass needs one row per line, typically four to twelve per plant. Signed by the **plant manager**. Columns not on this form — `Design rate`, `Rate unit`, `Changeover class`, `Constraint`, `Effectiveness measured`, `Owner` — go straight into the sheet.

Ref	Production line	Plant	Product family	Work centres	Shift pattern
LIN-01	EXAMPLE — delete this row	LOC-02			

STANDING IN THE PLANT, BEFORE YOU LEAVE

How many production lines does this plant run, and who gave you that number?

Which line is the constraint today, and was it the constraint last quarter?

If you asked the planner and the plant manager to each name the lines, would the two lists match?

The principles, and whether a gate can test them

Step 2.6 — Give `005 Principle` the columns a gate can test

Produce. `005 Principle` extended with the five columns that make a principle testable, and every row filled.

You need. Step 2.2. The Architecture Board's existing terms of reference, and a working session with the four domain architects.

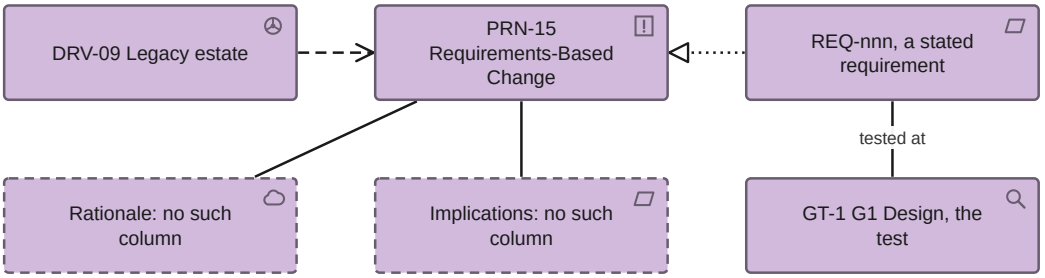


Figure 2.3 Why a principle in `005` cannot be tested at a gate. The driver and the requirement are in the file. The rationale and the implications have no column to live in, and the implications are what a reviewer actually holds a design against.

Where it lands.

> **`005 Principle` — 24 rows, 6 columns**
>
> `Ref` · `Principle` · `Category` · `Statement` · `Owner` · `Status`. `Category` holds `Business`, `Application`, `Data`, `Technology` and `Security`.
>
> There is **no `Tier`**, **no `Rationale`**, **no `Implications`**, **no `Approved` date and no `Review date`**. A principle with a statement and nothing else cannot be tested, because the test of compliance is the implications, not the slogan. Add all five columns.

How.

1. Add `Tier` and set it: a tier-1 principle is one you will stop a project over; a tier-2 principle is one you will argue about at design.
2. Write `Rationale` in one sentence per row, answering *why would anybody agree to this?*
3. Write `Implications` as the thing a project must now do differently. If you cannot write one, the principle is a slogan and should be deleted rather than kept.
4. Date `Approved` and `Review date`. A principle with no review date is never revisited.
5. Test three live decisions against the extended sheet. If none of the twenty-four changes the answer, the set is decorative.

Signed by. `GB-01 Architecture Board`, and the decision recorded in `060 Governance Decision`.

Done when. Every one of the twenty-four rows carries an implication a reviewer could hold a design against, and at least three rows have been used to change an actual decision.

Shows up in. Nowhere. No report reads `O05 Principle`. `RPT12 Standards Conformance` reports against `O32 Standard`, which is a different thing — a standard is a product or protocol choice, a principle is a rule about choices.

Goes wrong when. You keep twenty-four principles because deleting one feels like a loss. Six principles a board will enforce beat twenty-four it will not. `PRN-10 Data is Shared` and `PRN-15 Requirements-Based Change` are the two the rest of this volume leans on; make sure those two at least survive with implications attached.

Look it up. Volume 1 chapter 14, *The constructive answer: build the house reference model*, and *Then declare it*.

Form 2-3 — Principle test card

Feeds `005 Principle` plus the five columns you add in step 2.6. One card per principle, worked in the board session. A first pass needs all twenty-four, though most boards cut the set to eight. Signed by the **Architecture Board** (`GB-01`). `Approved` and `Review date` go straight into the sheet.

Ref	Principle	Tier	Rationale	Implications	Owner (a person)
PRN-13	EXAMPLE — delete this row	1			

Step 2.7 — Give every principle a person, and re-assess the Draft rows

Produce. `005 Principle` with twenty-four named individual owners, and a status on each row that came from an assessment rather than from a pattern.

You need. Step 2.6. The Architecture Board, in session, with the sheet on screen.

Where it lands.

> **`005 Principle` — the `Owner` and `Status` columns**
>
> **All twenty-four rows carry `Owner` = `Architecture Board`.** That is a committee. The sheet's own `PRN-13 Data Ownership is Individual` states: *"An owner is a person, not a committee."* The sheet violates its own principle on every row, including that row.
>
> `Status` holds `Approved` on twenty rows and `Draft` on four: `PRN-05`, `PRN-10`, `PRN-15`, `PRN-20`. Those are rows 5, 10, 15 and 20 — **every fifth row**. That is a generation pattern, not an assessment.

How.

1. Replace every `Owner` with the name of one person. The Architecture Board can still approve; it cannot own.
2. Show the board the every-fifth-row pattern on screen. It is the fastest way to establish that the repository was generated and must be re-assessed rather than inherited.
3. Re-assess all twenty-four statuses from scratch. Do not assume the twenty `Approved` rows were approved.
4. Pay attention to the two `Draft` rows the rest of this volume depends on: `PRN-10 Data is Shared` ("Data is shared through published interfaces, not copies") and `PRN-15 Requirements-Based Change` ("Change is driven by stated requirements"). Both are draft. Both need to be approved before Phase C and Phase E can cite them.
5. Record the re-assessment as a single decision in `O60 Governance Decision`.

Signed by. `GB-01 Architecture Board`, chaired by the Chief Enterprise Architect (`ROL-08`).

Done when. No `Owner` cell in `O05 Principle` names a body, and the `Draft` rows no longer fall on a regular interval.

Shows up in. Nowhere directly, but `RPT05 Data Ownership` carries the same disease on a different sheet: nine of fourteen `O20 Data Entity` owners are job titles that resolve to nothing in `O11 Role`. Fixing the principle without fixing the practice changes nothing.

Goes wrong when. You fix the owner column and leave the statuses. Six months later somebody asks which principles are approved, discovers four drafts at a regular interval, and concludes the whole repository is synthetic. They will be right about those four cells.

Look it up. Volume 1 chapter 5, *Why you need the cast list before you need the model*.

AFTER THE BOARD HAS SEEN THE PRINCIPLE SHEET

Which three of the twenty-four principles would your board actually stop a project over?

Who is the named person behind `PRN-13 Data Ownership is Individual`, now that it cannot be the Architecture Board?

What changed in the board's view of the repository when they saw the every-fifth-row pattern?

Governance, deliverables and the tooling

Step 2.8 — Stand up the governance bodies and put a plant voice in them

Produce. `057 Governance Body` extended with a chair, a quorum and an escalation path, and at least one body with operations representation.

You need. Steps 2.6 and 2.7. The existing terms of reference for each body.

Where it lands.

> **`057 Governance Body` — 5 rows, 5 columns**
>
> `Ref` · `Governance body` · `Cadence` · `Membership` · `Mandate`. `GB-01 Architecture Board`, `GB-02 Investment Committee`, `GB-03 Data Governance Council`, `GB-04 Executive Committee`, `GB-05 Architecture Function`.
>
> Add `Chair`, `Quorum`, `Escalates to` and `Decision rights`. Note that `GB-01`'s membership names "CIO and CISO delegates" and there is no CISO anywhere in `010 Actor` — the nearest row is `ACT-17 Security Architect`.

How.

1. Name the chair of each body as a person, not as a role title that resolves to nothing.
2. Set a quorum. A board that can sit with two people will sit with two people.
3. Read the five membership lists and find the plant. There is no plant manager, no operations engineer and no quality manager in any of the five bodies.
4. Add operations representation to `GB-01 Architecture Board` — a rotating plant seat is enough, and it changes what gets approved.
5. Reconcile `GB-03 Data Governance Council`'s membership — "business data owners, stewards" — against `020 Data Entity`, whose owners do not exist as people.

Signed by. `GB-04 Executive Committee`.

Done when. Each of the five bodies has a named chair, a quorum, an escalation route, and at least one body includes somebody who works at a plant.

Shows up in. Nowhere. No report reads `057`. The visible effect appears in Phase G, when `059 Compliance Assessment` starts to fill and the verdicts carry a body's name.

Goes wrong when. You accept a governance structure with no operations voice because it is faster. Every gate then tests corporate concerns against plant designs, and the plants learn to route around the gate.

Look it up. Volume 1 chapter 5, *Above the plant*, and chapter 7, *Centralised, or plant-autonomous*.

Step 2.9 — Give the four gates a holder and an entry condition

Produce. `058 Governance Gate` extended so a gate can be scheduled, held and evidenced.

You need. Step 2.8. The delivery method the organisation actually uses, whatever it is called.

Where it lands.

> **`058 Governance Gate` — 4 rows, 4 columns**
>
> `Ref` · `Gate` · `When` · `What it tests`. `GT-0 G0 Charter`, `GT-1 G1 Design`, `GT-2 G2 Build complete`, `GT-3 G3 Benefit`. The references are `GT-0` to `GT-3` and the gate names are G0 to G3, one apart in form.
>
> There is **no holder column**. `057`'s `Mandate` text says `GB-01` holds G1 and G2 and `GB-02` holds G0 and G3, but that is prose on another sheet, not a key. Add `Held by`, `Entry criteria`, `Evidence required` and `Outcome recorded in`.

How.

1. Add `Held by` and populate it from `057`'s mandate text, then delete the claim from the prose so there is one source.
2. Write `Entry criteria` as a list a delivery lead can check before booking the gate.
3. Set `Evidence required` to the specific artefact — a filled sheet, a signed form, a test result — not "evidence of compliance".
4. Point `Outcome recorded in` at `059 Compliance Assessment`, which is empty and will stay empty until this is done.
5. Walk one in-flight project through all four gates on paper to find out which criteria are unmeetable.

Signed by. `GB-01 Architecture Board` for G1 and G2, `GB-02 Investment Committee` for G0 and G3.

Done when. A delivery lead can read `058` alone and know who to book, what to bring and what gets written down.

Shows up in. `RPT11 Security Position`, indirectly: `SEC-02 Design review` is the one control reading `Absent`, and its `Enforced at` value is `Design gate`. The gate exists on paper and the control does not exist at all.

Goes wrong when. You design four beautiful gates and hold none of them, because no sheet says who books them. `059 Compliance Assessment` is the evidence: zero rows, in a repository with twelve work packages and twelve projects.

Look it up. Volume 1 chapter 14, *Then declare it*.

Step 2.10 — Date and own the deliverables register

Produce. `031 Deliverable` with an owner, a due date and a location for each of the twenty rows, and the false `Approved` statuses corrected.

You need. Step 2.2, the phase map from chapter 1, and the project list in `056 Project`.

Where it lands.

> **`031 Deliverable` — 20 rows, 4 columns**
>
> `Ref` · `Deliverable` · `Phase` · `Status`. `Phase` holds `A` to `G` and `RM`. Fifteen rows are `Approved` and five are `Draft`.
>
> Add `Owner`, `Due`, `Approved on`, `Where it lives` and `Supersedes`. Then test the fifteen `Approved` rows: `DLV-03 Stakeholder Map` is `Approved` and **there is no stakeholder sheet anywhere in the sixty-four** for it to have been built from.

How.

1. For each `Approved` row, ask for the artefact. If nobody can produce it, change the status to `Not started` and date the change.
2. Set `Owner` to a person for all twenty.
3. Set `Due` from the phase plan, not from the project plan — a deliverable is due before the project that consumes it starts.
4. Put `Where it lives` on every row. A deliverable nobody can find has not been delivered.
5. Re-check `DLV-19 Compliance Assessment` against `O59`, which is empty.

Signed by. Chief Enterprise Architect (`ROL-08`), tabled at `GB-01 Architecture Board`.

Done when. No row reads `Approved` without a named approver, a date and a location.

Shows up in. Nowhere. No report reads `031`. The register is for you and for the board, and the absence of a report over it is one of the reasons it drifted.

Goes wrong when. You inherit fifteen approvals. Phase A then starts on the assumption that the vision, the statement of work and the stakeholder map already exist, and the first two weeks are spent looking for documents that were never written.

Look it up. Volume 1 chapter 19, *The ninety-day shape*.

Step 2.11 — Choose the tool, and declare what you have extended

Produce. A tooling decision and a written metamodel extension statement listing every sheet, column and reference space you have added to the delivered file.

You need. Steps 2.3, 2.5 and 2.6, so you know what you have already added.

Where it lands.

> **`EXT Industry Extension` — 6 rows, 3 columns**
>
> `Ref` · `Industry extension type` · `What it holds and why the core metamodel is not enough`. Six declared types, no instances.
>
> Add a fourth column, `Realised by`, and fill it: `X01` by `O20.E06`, `X03` by `O20.E02`, `X04` by `O20.E05`, `X05` by `O20.E11`, `X06` by `O20.E09`, and `X02` by the sheet you built in step 2.5. A declared type with nothing behind it is a note, not a metamodel.

How.

1. Decide whether the spreadsheet stays the repository or becomes an export from a tool. Both are defensible; undecided is not.
2. If a tool is coming, list the sheets whose structure will not survive import: `O13` with its three levels in one column, `O55` with its borrowed references, `O46` with two columns headed `Ref`.
3. Write the extension statement: every added sheet, every added column, and the reference space each uses.
4. Check any vendor claim about notation support against a dated, product-specific source rather than a marketing page.
5. Put the extension statement where a new joiner finds it in their first hour.

Signed by. `GB-01 Architecture Board` for the extension, `GB-02 Investment Committee` for the tool spend.

Done when. Someone who has never seen the file can read the extension statement and know which sheets came with it and which you added.

Shows up in. `RPT13 Orphan Analysis`, which today finds two orphans — `A11 Track & Trace` and `A12 Plant Analytics`, both with no business service mapped. Added sheets with nothing pointing at them will appear there too, which is the point of running it.

Goes wrong when. You extend quietly. Four months later nobody can tell your columns from the delivered ones, and the first person to re-import the original file silently deletes your work.

Look it up. Volume 1 chapter 15, *Enterprise architecture tooling, and the honest treatment of ArchiMate 4*, and chapter 17, *The ground rules, stated once*.

Step 2.12 — Write down the reference conventions, including the bad ones

Produce. A one-page reference convention note covering all sixty-five reference spaces, the nine irregular ones, and the rule for anything you add.

You need. The prefix table in chapter 1, and an hour.

Where it lands.

> **`Every object sheet, column A`**
>
> The file has no single convention. `O06` and `O37` use three digits; `O13` uses dots and no letter prefix; `O14`, `O20` and `O23` drop the hyphen; `O53` and `O66` do not pad; `O54` is lettered; `O58` starts at zero; `O61` is `CR-G01`; `O55` borrows four other sheets' spaces; and `R06 Application interfaces with` carries `INT-001` to `INT-030` on a relationship sheet.

How.

1. Record the convention each sheet actually uses. Do not normalise them — every existing join depends on the current strings.
2. State the rule for new rows: match the sheet you are adding to, exactly.
3. State the rule for new sheets: three letters, a hyphen, two digits, zero-padded.
4. Flag `O55 Building Block` in the note: its references are not unique across the file, so a lookup on a bare reference can return two rows.
5. Circulate the note to anyone who will touch the file, including the delivery teams.

Signed by. Chief Enterprise Architect (`ROL-08`).

Done when. A new joiner can add a row to any sheet without asking what the reference should look like.

Shows up in. `RPT06 Requirements Traceability`, which joins `O06`'s three-digit `REQ-` space to `O35`'s two-digit `BBB-` space across `R09`. Any join between two differently-shaped spaces is where a convention note pays for itself.

Goes wrong when. Somebody tidies. A well-meaning analyst renumbers `TA-1` to `TA-01` and breaks `R15 Transition contains Work Pa`, which carries the old strings.

Look it up. Volume 1 chapter 3, *How to use a list like this*.

Step 2.13 — Put change control in place before you change anything

Produce. Four new sheets and five new columns on every object sheet, in place and in use, dated from today.

You need. Steps 2.11 and 2.12. Thirty minutes to build, and the discipline to use it.

Where it lands.

> **Four new sheets in front of `00 Index`, and five columns on all 64**
>
> There is no change control anywhere in the 105 sheets: no version, no baseline, no change log, no effective date. `O61 Governance Change Request` holds five dispensations with an `Expires` date, not changes to the architecture. `O60 Governance Decision` has no date column at all.
>
> Create `01 Version Control`, `02 Change Log`, `03 Change Requests` and `04 Baselines` with the columns set out in chapter 1. **They are the first four tabs of the workbook, ahead of `00 Index`**, and they are not numbered as object sheets because they are not object types.
>
> Append `Version`, `Record Status`, `Changed By`, `Change Ref` and `Approved By` to the right of every object sheet, so no existing column index moves. `Record Status` takes exactly one of `Proposed`, `Approved`, `Superseded` and `Retired`.

How.

1. Build the four sheets in front of `00 Index`, with the file's own preamble convention: title, subtitle, blank, headers on row 4.
2. Append the five columns to all sixty-four object sheets plus `EXT` and the sheets you added, walking the sheet names the file has rather than looping `O01` to `O66`.
3. Back-fill `Record Status` = `Proposed` on all 813 rows and cut `BL-00` in `04 Baselines` today, with `Record count` = 813 — the counted figure from step 2.2, never the stated 843. Publish version `1.0` in `01 Version Control` as the file received, and `1.1` as the back-fill.
4. Log every change from this point, one row per field in `02 Change Log`, with `Old value` copied before the edit. A change not in `02 Change Log` did not happen.
5. Move the five `O61` rows into `03 Change Requests` as rows with `Type` = `Dispensation` and their `Expires` dates carried across, and leave `O61` as it was found, with a note.

Signed by. `GB-01 Architecture Board`, as the decision that makes the repository governable.

Done when. Any row in the file answers three questions without a conversation: what changed, who approved it, and against which baseline.

Shows up in. Nothing yet. From Phase G onward, every gate reads `04 Baselines`, and every decision in Phase H writes a `CR-` row to `03 Change Requests`. Chapter 11 builds all four out in full, with the trigger list that governs what may appear in `Against` and `Type`.

Goes wrong when. You defer it to Phase G, which is the natural place for it and the wrong one. Phases A to F are when the repository changes most, and a change log that starts in month six has nothing to say about the months that mattered.

Look it up. Volume 1 chapter 9, *Revision, and engineering change as a data event* — the discipline is the same one an engineering change order imposes on a parts list.

THE REVIEWER'S ASK, AT THE END OF THE PRELIMINARY PHASE

Name the three decisions taken in this phase that every later phase depends on. Who signed each one?

Of the thirteen steps, which did you shorten, and what will that cost in Phase C?

The measure definition register, the production line sheet and change control did not exist in the file you were handed. Which of the three is furthest behind, and what is the date it will be complete?

THE CHAPTER IN TEN LINES

- The Preliminary Phase is thirteen steps here, because the foundations in the file were generated, not assessed.
- Fix the plant list first. O12 Location has 8 rows, 3 plants, no line count and no shift pattern.
- Count the file on day one. 813 rows against a claimed 843, and the difference is R06's 30 interfaces.
- Stand up the measure definition register before anything else. O04 Measure has no definition, numerator, denominator, source or owner.
- O04 carries single customer view coverage and cross-sell ratio, and no OEE, yield, scrap or on-time-in-full.
- Settle the plant hierarchy against the ERP hierarchy now, because it defines what every later location column means.
- Build the X02 Production Line sheet. EXT declares the type, holds no instances, and nothing else in the file carries a line.
- O05 Principle has no Tier, no Rationale, no Implications, no Approved date and no Review date, so no gate can test it.
- All 24 principles are owned by the Architecture Board, against PRN-13, which says an owner is a person, not a committee.
- Draft status falls on PRN-05, PRN-10, PRN-15 and PRN-20 — every fifth row. That is a pattern, not an assessment.
- No governance body in O57 has a plant seat, and O58 has four gates with no holder column.
- Fifteen deliverables in O31 read Approved, including a stakeholder map built from a sheet that does not exist.
- Put change control in place before Phase A, not in Phase G. Four sheets in front of 00 Index, five columns on every object sheet, baselined today.

SOURCES FOR THIS CHAPTER

- **1.** Manufacturing_TOGAF_Metamodel.xlsx, sheet O05 Principle — 24 rows, Owner and Status columns counted directly
- **2.** Manufacturing_TOGAF_Metamodel.xlsx, sheet O04 Measure — 16 rows, 6 columns
- **3.** Manufacturing_TOGAF_Metamodel.xlsx, sheets O12 Location, O57 Governance Body, O58 Governance Gate and O31 Deliverable
- **4.** Manufacturing_TOGAF_Metamodel.xlsx, sheet EXT Industry Extension and sheet ZZ Checks
- **5.** The Open Group, the TOGAF Standard and the Architecture Development Method https://www.opengroup.org/togaf

CHAPTER 3

Architecture Vision

Twelve steps to a mandate, a stakeholder map that does not exist yet, a cast list with the plant in it, and a target the sponsor will pay for.

Phase A sets the mandate, the stakeholders, the scope and a target worth the money. In this repository it is also where you find out that the organisation modelled in the file has no factory in it, and that is the finding Phase A exists to surface.

Twelve steps. Steps 3.1 to 3.4 are about people and can be run in a fortnight. Steps 3.5 to 3.10 are the vision itself. Steps 3.11 and 3.12 are the paperwork that makes it binding. Kestrel Industrial Group is an invention; so is volume 1's Karoo Industrial Group, and the two sets of data must never be mixed.

The mandate and the people

Step 3.1 — Secure the mandate and name a sponsor who exists

Produce. A request for architecture work, one page, naming the sponsor by name, the question the architecture must answer, and the date an answer is needed.

You need. Chapter 2 complete, particularly steps 2.1, 2.8 and 2.13. Thirty minutes with the executive who is paying.

Where it lands.

> **`002 Goal` and `001 Driver`**
>
> `002 Goal` — `Ref` · `Goal` · `Description` · `Owner` · `Status`. Six rows, all `Active`. `001 Driver` — `Ref` · `Driver` · `Type` · `Description` · `Origin` · `Owner` · `Pressure`. Twelve rows: six `Regulatory` owned by `Chief Compliance Officer`, six `Business` owned by `Group Executive`.
>
> **`Group Executive` is not a row in `010 Actor`.** Neither is `Chief Commercial Officer`, who owns `GOL-03`. Seven of the eighteen motivation rows name an owner who does not exist as a person anywhere in the file.

How.

1. Ask the sponsor for the question in one sentence. If it takes three, the mandate is not agreed.
2. Replace `Group Executive` on `DRV-07` to `DRV-12` with the name of the executive who will actually answer for each.
3. Resolve `GOL-03`'s owner, or accept that it is a goal no named person holds.

4. Record the date an answer is needed. Every scope decision later in this chapter is made against that date.
5. Table the request at `GB-02 Investment Committee`, which holds the G0 gate.

Signed by. The sponsor, personally, and `GB-02 Investment Committee`.

Done when. Every `Owner` cell in `O01` and `O02` names a person who appears in `O10 Actor`, and the sponsor has initialled the one-sentence question.

Shows up in. `RPT06 Requirements Traceability`, which carries a `Driver` column against all thirty-one requirements. A driver nobody owns produces requirements nobody defends.

Goes wrong when. You accept a sponsor who is a committee. The repository already does this with `O05 Principle`, where all twenty-four rows are owned by `Architecture Board`, and chapter 2 step 2.7 exists to undo it. Do not repeat the pattern in Phase A.

Look it up. Volume 1 chapter 19, *Week one: see the plant*.

Step 3.2 — Build the stakeholder sheet, because there is not one

Produce. A new sheet, `O67 Stakeholder`, with one row per stakeholder, their interest, their influence and the view that will answer their question.

You need. Step 3.1. The organisation chart, the plant visits booked, and an hour per stakeholder.

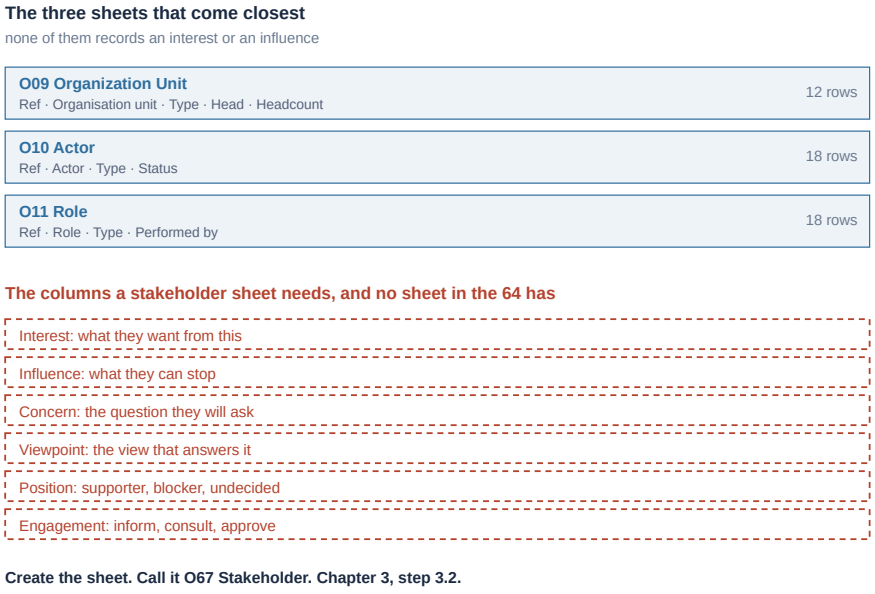


Figure 3.1 The stakeholder gap. `O09`, `O10` and `O11` are the nearest the repository comes, and none of them records what Phase A needs.

Where it lands.

> **`067 Stakeholder` — the sheet you are creating**
>
> **There is no stakeholder sheet anywhere in the sixty-four.** The three sheets that come nearest hold none of what Phase A needs: `009 Organization Unit` (`Ref` · `Organisation unit` · `Type` · `Head` · `Headcount`), `010 Actor` (`Ref` · `Actor` · `Type` · `Status`) and `011 Role` (`Ref` · `Role` · `Type` · `Performed by`). None records an interest, an influence or a concern.
>
> Build it with the file's conventions: headers on row 4, data from row 5, `Ref` in column A as `STK-01` onward. Columns: `Ref` · `Stakeholder` · `Role ref` (a `ROL-` value) · `Organisation unit` (an `ORG-` value) · `Location` (a `LOC-` value) · `Interest` · `Influence` (high, medium, low) · `Concern` · `Viewpoint` · `Position` (supporter, blocker, undecided) · `Engagement` (inform, consult, approve) · `Last engaged`.

How.

1. List everyone who can stop the work, then everyone who can help it, then everyone who will be affected. The three lists are different lengths and the first is the shortest.
2. Write `Concern` as the question they will ask in the room, in their words.
3. Set `Viewpoint` to the specific view that answers that concern, not to a layer name.
4. Mark `Influence` honestly. A plant manager who can refuse access has high influence whatever the chart says.
5. Note `DLV-03 Stakeholder Map` in `031 Deliverable` reads `Approved`. Change it to reflect the fact that it is being written now.

Signed by. Chief Enterprise Architect (`ROL-08`), tabled at `GB-01 Architecture Board`.

Done when. Every row has a concern written in the stakeholder's own words and a named view that answers it, and nobody who can stop the work is missing.

Shows up in. Nowhere. No report reads any of `009`, `010` or `011` either, so the whole stakeholder layer of this repository is invisible to the reporting pack.

Goes wrong when. You build the map from the organisation chart. The chart holds twelve corporate units and no plants, so a map built from it will have nobody from the factory on it, and the first design review will be the first time anybody from operations sees the work.

Look it up. Volume 1 chapter 5, *Who to see first*.

Form 3-1 — Stakeholder register

This form feeds a sheet the repository does not have. You are creating `067 Stakeholder` in step 3.2. A first pass needs fifteen to twenty-five rows. Signed by the **Chief Enterprise Architect**. `Viewpoint`, `Position`, `Engagement`, `Location` and `Last engaged` go straight into the sheet.

Ref	Stakeholder	Role ref	Org unit	Interest	Influence
STK-01	EXAMPLE — delete this row	ROL-14	ORG-07		High

BEFORE THE FIRST PLANT VISIT

Who can stop this work, and have you met them?

What is the question the plant manager will ask in the first thirty seconds, and which view answers it?

Step 3.3 — Put the plant on the role sheets

Produce. `010 Actor` and `011 Role` extended with every role that actually runs the plant, each one named, counted and located.

You need. Step 3.2 and a plant walk on shift, including a night shift. The plant manager's permission and a safety induction. Two days per plant.

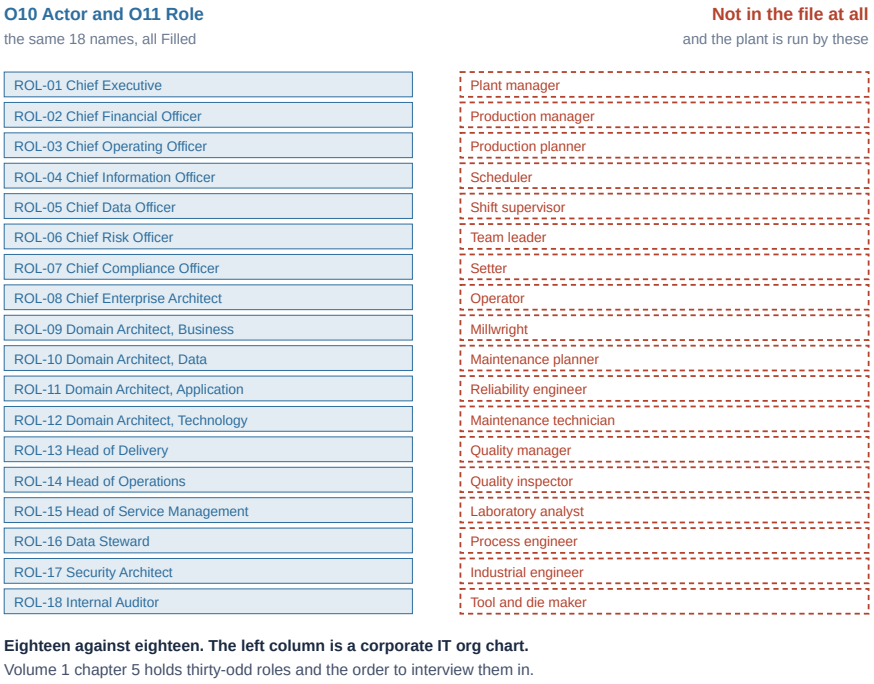


Figure 3.2 Eighteen roles in the file, and the plant in none of them. The right-hand column is not exhaustive; volume 1 chapter 5 carries the full cast.

Where it lands.

> **`O10 Actor and O11 Role — 18 rows each, joined by R21`**
>
> `O10 Actor` and `O11 Role` hold **the same eighteen names**: Chief Executive, CFO, COO, CIO, CDO, CRO, Chief Compliance Officer, Chief Enterprise Architect, four domain architects, Head of Delivery, Head of Operations, Head of Service Management, Data Steward, Security Architect, Internal Auditor. `R21 Role performed by Actor` joins them one to one, `ROL-01` to `ACT-01` and so on down, and every row reads `Filled`. Every row of `O11` reads `Performed by = Person`.
>
> **There is no plant manager, no production planner, no scheduler, no maintenance planner, no reliability engineer, no quality manager, no process engineer, no operator, no setter and no millwright anywhere in the file.** This is a corporate IT org chart wearing a manufacturer's name.
>
> Add rows from `ROL-19` and `ACT-19` onward, and add four columns to `O11 Role`: `Plant` (a `LOC-` value), `Reports to` (a `ROL-` value), `Shift` and `Headcount`.

How.

1. Walk the plant on shift. Write down every role you see performed, including the ones with no job title — the person who does the changeover, the person who signs the batch record.
2. Add them to `O11 Role` first, then `O10 Actor`, then the `R21` join. Keep the one-to-one convention the file uses, or say in writing that you have broken it.

3. Set `Headcount` per role per plant. A plant with four shift teams has four shift supervisors, not one.
4. Fix the `Status` column on `O10 Actor` while you are there. Every row reads `Filled`, which is why `ZZ Checks` reports zero vacant architecture roles. Count the vacancies.
5. Do not delete the eighteen corporate rows. They are real; they are just not the whole cast.

Signed by. The plant manager for the rows at their site; Head of Operations (`ROL-14`) for the set.

Done when. Somebody who has never visited the plant can read `O11 Role` and name who to ask about a changeover, a quality hold and a breakdown.

Shows up in. `RPT05 Data Ownership`, eventually. Nine of fourteen `O20 Data Entity` owners are job titles — `Head of Quality Management`, `Head of Maintenance`, `Head of Warehouse` — that resolve to nothing in `O11 Role`. Add the roles and those owners can finally be made to resolve.

Goes wrong when. You add the roles from the organisation chart without the walk. You get a plant manager and a quality manager and nothing below them, and the roles that actually hold the knowledge — the setter, the maintenance planner, the shift team leader — stay missing.

Look it up. Volume 1 chapter 5, *The plant*, for the full cast, and chapter 19, *The interview order*, for who to see in what sequence.

Form 3-2 — Plant role register

Feeds `010 Actor` and `011 Role`, plus the four columns added in step 3.3. One row per role per plant. A first pass needs twenty-five to forty rows across three plants — there are eighteen rows in the file today and none of them work in a factory. Signed by the **plant manager**. `Reports to`, `Shift` and the `R21` join go straight into the sheets.

Ref	Role	Type	Performed by	Plant	Headcount
ROL-19	EXAMPLE — delete this row	Operations	Person	LOC-02	

AFTER THE FIRST SHIFT ON THE FLOOR

Name five roles you watched being performed that do not appear in `011 Role`.

Which of them holds knowledge that exists nowhere in writing?

`ZZ Checks` reports zero vacant architecture roles. How many vacancies did you actually find?

Step 3.4 — Repair the organisation units and make their heads resolve

Produce. `009 Organization Unit` with the plants in it, numeric headcounts and heads that resolve to people.

You need. Step 3.3. The payroll headcount by site, and the group structure from step 2.1.

Where it lands.

> **`009 Organization Unit` — 12 rows, 5 columns**
>
> `Ref` · `Organisation unit` · `Type` · `Head` · `Headcount`. `Type` holds `Group`, `Corporate`, `Line` and `Assurance`.
>
> **Six of the twelve `Head` values resolve to nothing in `010 Actor`:** `Chief People Officer`, `Chief Commercial Officer`, `Head of Service`, `Head of Supply`, `Head of Internal Audit` and `General Counsel`. **Every `Headcount` cell is text, not a number**, so the column cannot be summed without a cast. And `ORG-07 Operations` carries a headcount of **102** while the three plants in `012 Location` hold **568 people** between them.

How.

1. Convert `Headcount` to numbers. The twelve rows sum to 1,896; check that against payroll.
2. Add a row per plant, or add a `Location` column and attach the existing units to sites. A manufacturer with three plants and no plant in its organisation sheet cannot do business architecture.
3. Resolve the six missing heads: either add them to `010 Actor` or correct the title to one that exists.
4. Reconcile `ORG-07 Operations` against the plant headcounts. One of the two numbers is wrong and it matters which.
5. Re-run `R22 Org Unit owns Capability` after the change. It currently maps `ORG-01` to `C01`, `ORG-02` to `C02` and so on down the row numbers, which hands capability ownership to whichever unit shares a row index.

Signed by. Head of Operations (`ROL-14`) and the Chief Financial Officer for the headcounts.

Done when. Every `Head` resolves to a row of `010 Actor`, every headcount is numeric, and the plants appear.

Shows up in. Nowhere directly, but `R22`'s diagonal is what feeds capability ownership into Phase B. A wrong organisation sheet produces a wrong owner on every capability.

Goes wrong when. You fix the titles and leave the diagonal. `R22` then maps corrected units to capabilities by row number, which is exactly as wrong as before and now looks deliberate.

Look it up. Volume 1 chapter 5, *Where these people live*.

The scope and the vision

Step 3.5 — Set the scope: plants, lines, products and period

Produce. A scope table: which plants, which lines, which products, which systems, and the period, each with an explicit in or out.

You need. Steps 2.1, 2.5 and 3.1. The product list and the revenue split by product.

Where it lands.

> **`O18 Product — 4 rows, 4 columns`**
>
> `Ref` · `Product` · `Type` · `Description`. `PRD-01 Industrial Assemblies` (`Primary`), `PRD-02 Spare Parts` (`Aftermarket`), `PRD-03 Service Contracts` (`Service`), `PRD-04 Custom Engineering` (`Project`).
>
> There is no volume, no revenue, no margin, no plant and no archetype. Add `Revenue share`, `Plants` (`LOC-` values), `Archetype` and `In scope`. Note that `R25 Application supports Produc` maps `PRD-01` to `A01`, `PRD-02` to `A04`, `PRD-03` to `A07` and `PRD-04` to `A10` — every third application, by row position.

How.

1. Put revenue share against each of the four products. The four are plausibly four different businesses with four different order-entry points.
2. Attach each product to the plants that make it, using `LOC-` references.
3. Set the archetype per product, not per group: `PRD-04 Custom Engineering` is engineered to order and `PRD-02 Spare Parts` is made to stock, and no single planning design serves both.
4. Mark the lines in scope using the `X02 Production Line` sheet built in step 2.5.
5. State the period: the baseline date and the target year.

Signed by. `GB-02 Investment Committee`.

Done when. Every plant, line and product carries an in-scope flag, and the sponsor has agreed in writing what is out.

Shows up in. `RPT13 Orphan Analysis`, which finds `A11 Track & Trace` and `A12 Plant Analytics` referenced by no business service. A product-to-application map that is a row-position diagonal is how orphans like that arise.

Goes wrong when. You scope by system rather than by product. The four products run through the same twelve applications in four different ways, and a scope written in application names hides the differences you were hired to find.

Look it up. Volume 1 chapter 2, *The full matrix*, and chapter 4, *The customer writes your architecture*.

Step 3.6 — Test the drivers against the jurisdiction the plant is in

Produce. A corrected driver list: the obligations that bind this manufacturer, with the ones that do not marked as such.

You need. Step 3.1. The compliance officer, the quality manager, and the plant's licences.

Where it lands.

> **`001 Driver` — the six `Regulatory` rows**
>
> `DRV-01 ISO 9001`, `DRV-02 ISO 14001 / 45001`, `DRV-03 Product safety and CE / UKCA marking`, `DRV-04 REACH / RoHS`, `DRV-05 Customs and origin rules`, `DRV-06 Extended producer responsibility`. The same six appear as the six obligations in `063 Control Objective`.
>
> **This is a European obligation set.** CE and UKCA marking have no application to goods placed on the South African market; the domestic regime runs on compulsory specifications and letters of authority. REACH and RoHS bind only on export to the markets that impose them. And nothing in the six covers the instruments that actually inspect a plant here.

How.

1. Mark `DRV-03` and `DRV-04` as export-conditional, not as general obligations. They are real for the products you export and absent for the rest.
2. Add the domestic instruments: occupational health and safety, air quality, water use, legal metrology, carbon tax and the data-protection regime.
3. For each, name the regulator and what that regulator asks to see.
4. Set `Pressure` from the inspection calendar, not from a committee's opinion.
5. Leave the rebuild of `063 Control Objective` to the Requirements Management chapter, but record the finding here so Phase B inherits it.

Signed by. Chief Compliance Officer (`ROL-07`), with the plant's safety officer present.

Done when. Every regulatory driver names a regulator and an evidence type, and the two export-conditional rows say which products and which markets.

Shows up in. `RPT09 Regulatory Coverage`, which today reports five of six obligations with no mechanism or a mechanism in breach. The count will rise before it falls, because you are adding obligations nobody had listed.

Goes wrong when. You inherit the six. A compliance programme then spends two years building evidence for conformity marking that does not apply, while the air emission licence conditions — which an inspector will ask for — are tracked in a spreadsheet on somebody's laptop.

Look it up. Volume 1 chapter 16, *Compulsory specifications, letters of authority, and section 20*, and *Who inspects, and what they ask for*.

Step 3.7 — Put dates on the objectives and attach them to measures

Produce. `O03 Objective` with a start date, an end date and a measure that is actually the measure of that objective.

You need. Step 2.3, the measure definition register. Steps 3.1 and 3.6.

Where it lands.

> **`O03 Objective` — 14 rows, 5 columns**
>
> `Ref` · `Objective` · `Goal` · `Horizon` · `Status`. **`Horizon` is a duration string** — "2 years", "18 months", "3 years" — with no start date, so no objective in this repository can be put on a calendar.
>
> `R10 Objective measured by Measu` maps `OBJ-01` to `MSR-01`, `OBJ-02` to `MSR-02` and so on down the row numbers. Add `Starts`, `Due`, `Owner` and `Target value` to `O03`, and rebuild `R10` by meaning.

How.

1. Convert every `Horizon` to a dated `Starts` and `Due`. The conversion will produce arguments, which is the point.
2. Read the fourteen objectives against the industry. `OBJ-05 Single view of customer across channels` and `OBJ-06 Cross-sell rate improved from baseline` belong to a different sector. Not one of the fourteen mentions effectiveness, yield, scrap or on-time delivery.
3. Add the objectives a manufacturer's executive would recognise, each tied to a measure from the register in step 2.3.
4. Rebuild `R10` row by row from meaning, and record how many of the fourteen the diagonal got right by accident.
5. Give every objective one named owner.

Signed by. The goal owner from `O02 Goal` for each objective under their goal.

Done when. Every objective has a start, a due date, a measure with a definition behind it, and a target value.

Shows up in. `RPT01 Dashboard` carries no objective line, so the effect is indirect. The visible proof is that a steering pack can now show progress against a dated target rather than against a horizon.

Goes wrong when. You leave the horizons. Eighteen months later nobody can say whether a two-year objective is late, because nobody wrote down when it started.

Look it up. Volume 1 chapter 20, *Map 5 — Measure to definition to source*.

The spine, the baseline and the target

Step 3.8 — Take the value streams as the spine, and repair them

Produce. `O14 Value Stream` with named stages, a capability list that matches `R02`, and a stage count that matches `O16 Business Process`.

You need. Steps 3.3 and 3.5. A workshop with operations, planning, quality and commercial in one room. Half a day.

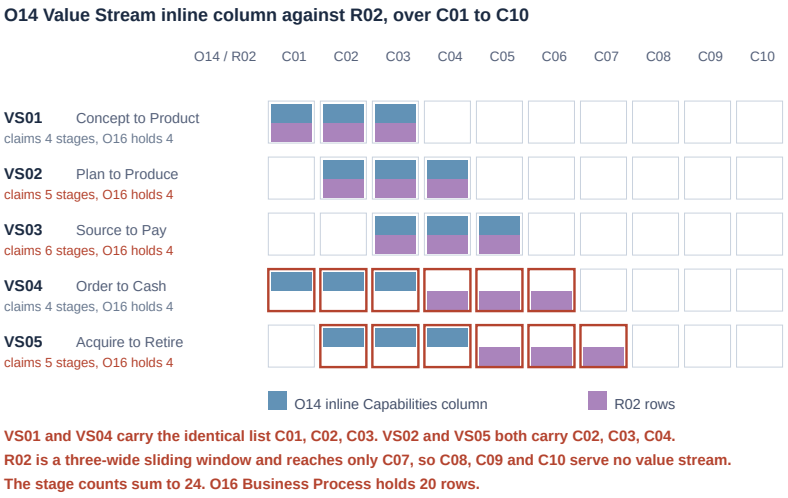


Figure 3.3 `O14 Value Stream` **against** `R02`**.** Two streams carry identical capability lists, the two sources disagree on the last two streams, and the stage counts do not match `O16 Business Process`.

Where it lands.

> `O14 Value Stream` **— 5 rows, 5 columns**
>
> `Ref` · `Value stream` · `Scope` · `Stages` · `Capabilities`. `VS01 Concept to Product`, `VS02 Plan to Produce`, `VS03 Source to Pay`, `VS04 Order to Cash`, `VS05 Acquire to Retire`.
>
> Three defects, all verifiable in a minute. `VS01` **and** `VS04` **carry the identical capability list** `C01, C02, C03`, and `VS02` and `VS05` both carry `C02, C03, C04`. `Stages` **is a count** — "4 stages", "5 stages", "6 stages" — not stage names. **The counts sum to 24** against the twenty rows in `O16 Business Process`, which holds exactly four stages for each of the five streams.
>
> Replace `Stages` with named stages, one row per stage in `O16`.

How.

1. Name the stages. "Stage 1 of plan to produce" is what `O16` holds today for all twenty rows; replace each with what actually happens.
2. Fix the capability lists. Order to cash and concept to product do not use the same three capabilities, and a sheet that says they do was generated.

3. Reconcile `O14` with `R02 Value stream uses Capabilit`, which is a three-wide sliding window: `VS01` gets `C01` – `C03`, `VS02` gets `C02` – `C04`, down to `VS05` getting `C05` – `C07`. The two agree for the first three streams and disagree for the last two.
4. Note that `R02` reaches only `C07`, so `C08 Sales & Order Management`, `C09 Health, Safety & Environment` and `C10 Finance & Cost Control` serve no value stream at all.
5. Make the stage count and the `O16` row count equal, in both directions.

Signed by. Head of Operations (`ROL-14`), with the commercial and quality leads signing their own streams.

Done when. Every stage has a name, every stream's capability list was argued over, and the sum of the stage counts equals the row count of `O16 Business Process`.

Shows up in. `RPT03 Capability Heatmap`, which reports the ten level-1 capabilities. After this step the three capabilities no stream uses should either gain a stream or lose their place.

Goes wrong when. You take the five streams as given because five is a reasonable number. The `Scope` descriptions are good; the capability lists and the stage counts are generated, and everything Phase B builds on top of them inherits the error.

Look it up. Volume 1 chapter 17, *Value stream*, and chapter 8, *The twelve processes a manufacturer runs*.

Form 3-3 — Value stream stage card

Feeds `014 Value Stream` and `016 Business Process`. One card per value stream, one row per stage. A first pass needs four to seven stages per stream. Signed by the **stream owner** named in step 3.8. The `Automation` column of `016` is set in the same session and typed straight into the sheet.

Stream	Stage no	Stage name	Starts when	Ends when	Capability
VS02	1	EXAMPLE — delete this row			C02

AFTER THE VALUE STREAM WORKSHOP

Which two streams were described identically in the file, and did the room agree they are different?

Three level-1 capabilities serve no value stream. Name them, and say whether that is a modelling error or a real finding.

Step 3.9 — Sketch the baseline on one page

Produce. A one-page baseline: the twelve applications, the ten capabilities, the three plants, and the four or five facts that will decide the target.

You need. Steps 3.5 and 3.8. `RPT02 Application Portfolio` and `RPT03 Capability Heatmap` printed.

Where it lands.

> **`O23 Application Component` — 12 rows, 10 columns**
>
> `Ref` · `Application` · `Type` · `Description` · `Disposition` · `Criticality` · `Health` · `In service` · `Users` · `Owner`.
>
> Read the sheet before you draw. `A04 SCADA / Historian` is the **only** `Critical` application and its `Disposition` is **`Replace`**, with nothing named to replace it. `A03 MES` has been in service since 2010, `Health` = `Poor`, `Disposition` = `Retain`. Four of twelve read `Replace`. Every `Owner` is an organisation unit, not a person: `A01 ERP Core` is owned by `Legal`, `A05 APS` by `Risk & Compliance`, `A04` and `A07` by `Internal Audit`.

How.

1. Draw three bands: plant floor, plant systems, corporate systems. Put the twelve applications in the band they actually sit in.
2. Mark the one `Critical` application and the fact that it is slated for replacement with no successor. That single pairing will carry the conversation.
3. Overlay the capability heatmap. `C04 Manufacturing Operations` is rated `Low` criticality in a manufacturer, which is worth saying out loud.
4. Ignore `Users` as a sizing proxy. `A05 APS` records 266 users against the ERP's 196, which no plant resembles.
5. Keep it to one page. If it needs two, you have drawn the target, not the baseline.

Signed by. Chief Enterprise Architect (`ROL-08`). The baseline is a statement of fact and does not need a committee.

Done when. The sponsor can look at the page for ninety seconds and name the problem.

Shows up in. `RPT02 Application Portfolio`, which reprints `O23` column for column, and `RPT03 Capability Heatmap`, which adds a computed `Gap` column over `O13`'s `Current` and `Target`.

Goes wrong when. You draw the landscape the documents describe rather than the one the plant runs. The repository places `PTC-03 SCADA control servers` at `Primary Data Centre`; the real ones are in a cabinet at the plant, and that difference changes the resilience argument entirely.

Look it up. Volume 1 chapter 12, *The shape of a South African estate*.

Step 3.10 — State the target in capability terms, and reject the arithmetic

Produce. A target capability state: current and target maturity per level-1 capability, with a disposition that came from a judgement.

You need. Steps 3.8 and 3.9. The capability owners from `R22`, corrected in step 3.4.

Where it lands.

> **`013 Business Capability — 100 rows, 9 columns`**
>
> `Ref` · `Capability` · `Level` · `Parent` · `Description` · `Criticality` · `Current` · `Target` · `Disposition`. Ten level-1 capabilities, thirty level-2, sixty level-3.
>
> **`Disposition` is pure arithmetic.** It reads `Build` where `Target` minus `Current` is 2 or more and `Sustain` everywhere else, on all one hundred rows without exception. There are only two values on the sheet. **Criticality does not inherit**: `C01 Product Development` is `High` while `C01.01` is `Low` and `C01.01.01` is `Medium`, and `C01.02.01` is `Critical` under a `Low` parent. The ninety sub-rows appear in no report.

How.

1. Work the ten level-1 rows only. The ninety below them are a generated cross-product of a word list and can wait for Phase B.
2. Set `Current` from evidence — what the plant can actually do today — not from an opinion in a workshop.
3. Set `Target` from the objectives dated in step 3.7. A target with no objective behind it is a wish.
4. Replace `Disposition` with a judgement, and use more than two values. A capability can be invested in, sustained, outsourced or deliberately left alone.
5. Check criticality against the archetype from step 3.5. `C04 Manufacturing Operations` rated `Low` in a manufacturer is either a data error or a finding about who filled the sheet in.

Signed by. `GB-01 Architecture Board`, with each capability's owning executive signing their own rows.

Done when. Each of the ten level-1 rows has a disposition somebody argued for, and no disposition can be reproduced by subtracting two columns.

Shows up in. `RPT03 Capability Heatmap`, which carries exactly the ten level-1 rows with a `Gap` column. The heatmap is only as good as the two numbers behind it.

Goes wrong when. You accept the hundred rows because a hundred looks thorough. Ten real capabilities with defended numbers beat a hundred generated ones, and the ninety are not referenced by anything in the file.

Look it up. Volume 1 chapter 17, *Capability map*.

BEFORE YOU WRITE THE VISION DOCUMENT

In one sentence, what will be different in three years, and who said so?

Which capability will you deliberately not improve, and have you told its owner?

The only `Critical` application in the file is marked for replacement with nothing named to replace it. What is your answer when the sponsor asks what replaces it?

Making it binding

Step 3.11 — Write the Architecture Vision and take it through G0

Produce. The Architecture Vision document, and a G0 decision recorded against it.

You need. Steps 3.1 to 3.10. `GB-02 Investment Committee` in session.

Where it lands.

> **`O31 Deliverable`, `O54 Architecture Alternative`, `O58 Governance Gate`, `O60 Governance Decision`**
>
> `DLV-01 Architecture Vision` in `O31` reads `Approved` already; correct it, then re-approve it with a date in the `Approved on` column added in step 2.10.
>
> `O54 Architecture Alternative` holds five: `ALT-A` $24m `Rejected`, `ALT-B Foundations first` $46m `Recommended`, `ALT-C` $52m `Rejected`, `ALT-D` $88m `Rejected`, `ALT-E Compliance led` $31m **`Partly adopted`**. No decision in `O60 Governance Decision` records that partial adoption, and `DEC-01 Adopt the recommended architecture alternative` still reads `Outstanding`.
>
> `O30 Work Package` sums to **$65.0m** against `ALT-B`'s $46m. The whole file is in dollars; read the magnitudes, not the currency.

How.

1. Write the vision against the five alternatives, not in the abstract. The sponsor is choosing between them.

2. Reconcile the money before the meeting. A $19m difference between the recommended alternative and the work packages is the first question a committee will ask.
3. Resolve `ALT-E`'s `Partly adopted` status. Either a decision was taken and not recorded, or the status is wrong.
4. Take it to `GT-0 G0 Charter`, held by `GB-02 Investment Committee`, with the entry criteria written in step 2.9.
5. Record the outcome in `O60 Governance Decision`, and put the date in the decision text because the sheet has no date column.

Signed by. `GB-02 Investment Committee` at G0, countersigned by the sponsor.

Done when. `DEC-01` no longer reads `Outstanding`, and the chosen alternative's cost and the work package total agree or the difference is explained in writing.

Shows up in. `RPT08 Roadmap`, which reprints `O30` with cost, start, end, approval and delivery. The totals there are what the committee will be shown.

Goes wrong when. You present a vision with no price. The committee then approves a direction rather than a spend, and Phase E discovers that nobody agreed to $65m.

Look it up. Volume 1 chapter 18, *The decisions that must be taken before anything can start*.

Step 3.12 — Write the Statement of Architecture Work and sign it

Produce. A signed statement of architecture work: scope, deliverables, constraints, assumptions, resources, dates and the gates it will pass.

You need. Steps 3.5, 3.11 and the deliverables register from step 2.10.

Where it lands.

> **`O31`, `O33 Architecture Contract`, `O07 Constraint`, `O08 Assumption`**
>
> `O33 Architecture Contract` holds eight rows — `Ref` · `Architecture contract` · `Work package` · `Compliance status`. There is no party column, no date and no signatory, so a contract in this file cannot say who agreed to what.
>
> `O07 Constraint` holds eight rows with `Type` values `Financial`, `Resource`, `External`, `Regulatory`, `Operational` and `Organisational`. `O08 Assumption` holds six, two of them `At risk` — `ASM-03 Vendor selections complete before the dependent phase` and `ASM-04 Delivery capacity can be staffed to plan`.

How.

1. Copy the agreed scope from step 3.5 verbatim. Do not paraphrase it; paraphrase is how scope moves.
2. List the deliverables by their `DLV-` reference, with the dates set in step 2.10.

3. Test the eight constraints against the plant. `CON-06 Change freeze windows` and `CON-08 Union and works council consultation` are the two most likely to be underestimated in a manufacturer.
4. Convert the two `At risk` assumptions into either a dependency with an owner or a risk in `O62 Delivery Risk`.
5. Add `Parties`, `Signed on` and `Signed by` to `O33 Architecture Contract`, then sign it.

Signed by. The sponsor and the Chief Enterprise Architect (`ROL-08`), tabled at `GB-01 Architecture Board`.

Done when. `DLV-02 Statement of Architecture Work` carries a real approval date, a named approver and a location, and every assumption has an owner.

Shows up in. `RPT07 Gap Analysis` and `RPT08 Roadmap` in due course, since both read work packages the contract will govern. `O33`'s own `Compliance status` column becomes meaningful only once the gates in step 2.9 are being held.

Goes wrong when. You sign a statement of work with assumptions nobody owns. Both `At risk` assumptions in this file are about things outside the architecture team's control, and an unassigned assumption becomes an excuse rather than a dependency.

Look it up. Volume 1 chapter 19, *The ninety-day checklist*.

THE REVIEWER'S ASK, AT THE END OF PHASE A

There is no stakeholder sheet in the file and no plant role in it either. What did you build, and who signed it?

Which finding from this phase will change what Phase B does first?

If the sponsor withdrew tomorrow, which three artefacts from this phase would still be worth what they cost?

THE CHAPTER IN TEN LINES

- Phase A is where you find out that the organisation in this repository has no factory in it.
- Seven of eighteen motivation rows name an owner, Group Executive or Chief Commercial Officer, who is not a person in O10 Actor.
- There is no stakeholder sheet anywhere in the 64. Build O67 Stakeholder with interest, influence, concern and viewpoint.
- O10 Actor and O11 Role hold the same eighteen names, joined one to one by R21, every row Filled.
- No plant manager, planner, scheduler, maintenance planner, reliability engineer, quality manager, process engineer, operator, setter or millwright appears anywhere.
- Six of twelve heads in O09 resolve to nobody, every headcount is text, and Operations shows 102 people against 568 in the plants.
- The six regulatory drivers are a European set. CE and UKCA marking have no application to goods placed on the South African market.
- O03 Objective carries a Horizon as a duration string, so no objective in the file can be put on a calendar.
- VS01 and VS04 carry the identical capability list C01, C02, C03, and VS02 and VS05 both carry C02, C03, C04.
- O14 Stages is a count, not stage names, and the counts sum to 24 against O16's 20 process rows.
- O13 Disposition is arithmetic: Build where Target minus Current is 2 or more, on all 100 rows, with only two values used.
- The only Critical application is marked Replace with nothing named to replace it, and it is owned by Internal Audit.
- O30 sums to 65.0m against the recommended alternative at 46m, and DEC-01 still reads Outstanding.

SOURCES FOR THIS CHAPTER

- **1.** Manufacturing_TOGAF_Metamodel.xlsx, sheets O09 Organization Unit, O10 Actor, O11 Role and R21 Role performed by Actor — counted directly, 16 October 2026
- **2.** Manufacturing_TOGAF_Metamodel.xlsx, sheets O01 Driver, O02 Goal and O03 Objective
- **3.** Manufacturing_TOGAF_Metamodel.xlsx, sheets O14 Value Stream, O16 Business Process and R02 Value stream uses Capability
- **4.** Manufacturing_TOGAF_Metamodel.xlsx, sheets O13 Business Capability, O23 Application Component, O54 Architecture Alternative and O30 Work Package
- **5.** The Open Group, the TOGAF Standard and the Architecture Development Method https://www.opengroup.org/togaf

CHAPTER 4

Business Architecture

The repository holds an IT organisation chart and a generated capability tree. Phase B is where you put the factory back.

What Phase B has to produce here

Phase B is where this repository stops being plausible. The motivation layer reads like a manufacturer's. The business layer holds a corporate IT organisation chart, a generated capability tree and five value streams that disagree with themselves [1]. Nothing in `009` through `019` would survive ten minutes in front of a plant manager, and the reason is worth stating once: the business layer was built from the architecture function outwards, not from the factory inwards.

So Phase B here is a rebuild. Fourteen steps: three build sheets the file does not have, four undo arithmetic mistaken for assessment. The order matters — the archetype decision in step 4.1 changes what a correct answer looks like in every step after it.

Kestrel Industrial Group, the manufacturer in the repository, is an invention, as is volume 1's Karoo Industrial Group. Their data is never mixed; every sheet reference here is Kestrel.

BEFORE YOU OPEN A BUSINESS-LAYER SHEET, ANSWER THESE

Which plant are you modelling first, and why that one?

Who will be asked to defend the capability map, and to whom?

What is the one sentence that decides whether a row belongs in `013 Business Capability`?

Name three roles performed daily in the plant that appear nowhere in `010 Actor` or `011 Role`.

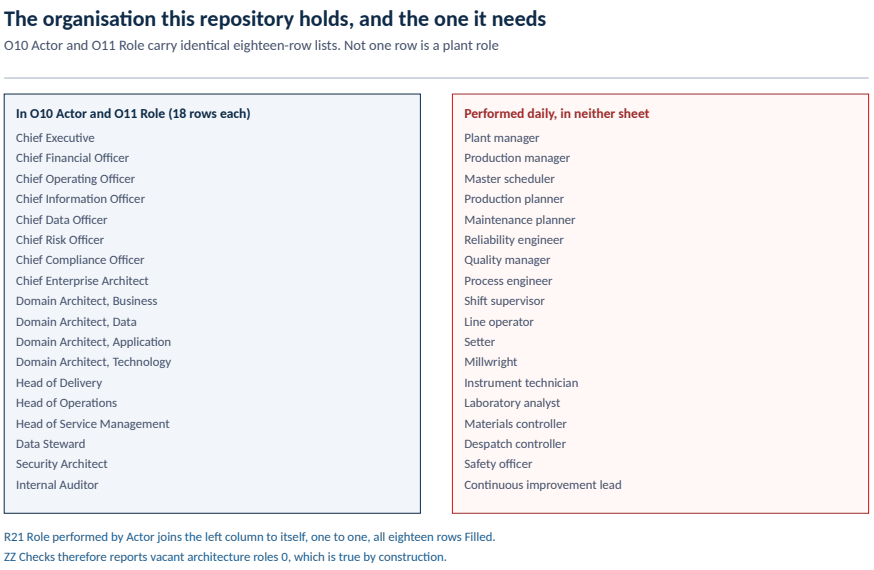


Figure 4.1 The sharpest defect in the file. `O10 Actor` and `O11 Role` hold the same eighteen names, and not one of them works in a factory. Rebuilding the right column is step 4.5; volume 1 chapter 5 holds the full cast list and the interview order.

Start with the archetype, not with a sheet

Step 4.1 — Settle the archetype, order type by order type

Produce. A one-page archetype statement per plant: its position on the discrete / process / batch axis, the order-entry point for each product family it runs, and the consequences you accept from both.

You need. `O18 Product` (four rows), `O12 Location` rows `LOC-02`, `LOC-03` and `LOC-04`, the plant manager and master scheduler of each plant, and two hours on the floor per plant.

Where it lands.

> **`O08 Assumption` · columns `Ref`, `Assumption`, `Status`**
>
> No sheet has a column for an archetype, so the decision lands as an assumption until step 4.2 builds one that can hold it. One row per plant per order type. Write `Assumption` as a testable sentence — "Coastal Plant runs the Spare Parts family make-to-stock on a repetitive line" — not as a label. `Status` takes `Validated` only after the plant manager has read the sentence back to you.

How.

1. List every product family each plant runs. `O18 Product` gives four — `PRD-01` Industrial Assemblies, `PRD-02` Spare Parts, `PRD-03` Service Contracts, `PRD-04` Custom Engineering — and `PRD-03` is not a manufactured thing, so expect to split and extend.
2. For each family, ask where the customer order enters: before design, before the parts list, before the works order, or after the finished goods are in the store.

3. Decide discrete, process or batch from the object model the plant keeps, not the industry label.
4. Write the sentence. Read it back to the plant manager. Change the sentence, not the plant.
5. Record the consequence accepted: a make-to-order family cannot be promised from stock, and an engineer-to-order family cannot be scheduled against a standard route.

Look it up. Volume 1 chapter 2, *The archetype is a property of the order, not only of the plant*, with the matrix in *The full matrix* and the mixed case in *Where the axes meet: the mixed-mode plant*.

Signed by. The plant manager of each plant, countersigned by `ROL-14 Head of Operations`, the only accountable operations role in `O11 Role` today.

Done when. Every row of `O18 Product`, after extension, maps to at least one archetype sentence, and no sentence covers two order-entry points.

Shows up in. Nowhere yet — no report reads `O08`, and say so to the sponsor. After step 4.2 it reaches `RPT03 Capability Heatmap` indirectly, through the target you set for `C02 Demand & Supply Planning`.

Goes wrong when. You take the archetype from the group's self-description. A group calling itself a discrete manufacturer will have a coating line or a paint shop running batch, and the parts list under it does not behave like the rest.

The plant, the line and the shift pattern

Eight locations, three typed `Production`. That is the whole of what this file knows about the factories [4]. There is no column for line count, format, ownership or shift pattern — and those four facts carry most of the capacity, cost and resilience argument you will make later.

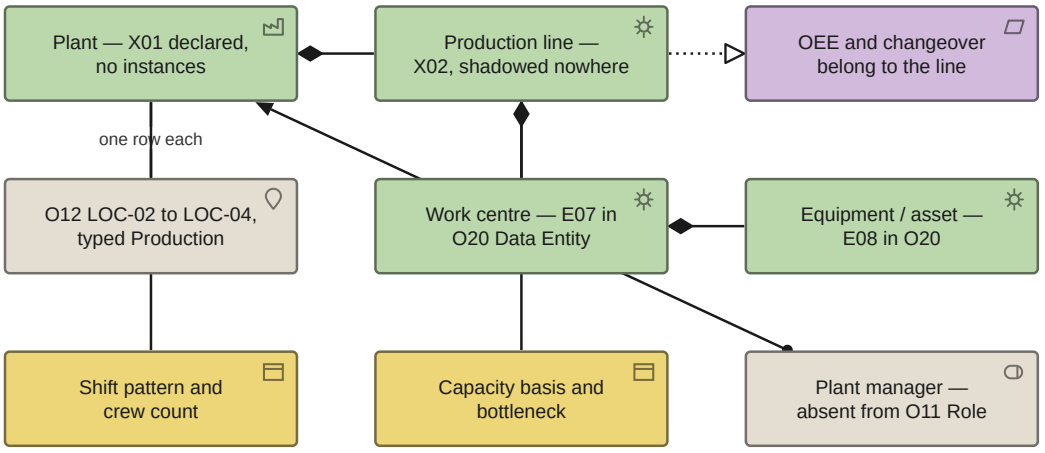


Figure 4.2 What a manufacturer hangs its architecture from, and what the repository can hold. The plant and the line are declared in `EXT Industry Extension` with no instances; the work centre and the asset exist only as data entities in `O20`. Steps 4.2 and 4.3 build the register that carries the rest.

BEFORE YOU WRITE THE PLANT REGISTER, ANSWER THESE

How many lines does each plant run, counted by walking them?

Which line sets the capacity of each plant, and in what unit?

Is any capacity in this group tolled, leased or shared?

Form B-2 — Plant and line register

This form feeds a sheet the repository does not have. `EXT Industry Extension` declares `X01 Plant / Site` and `X02 Production Line` and holds **no instances** of either. Build the sheets on the file's own convention — title row 1, subtitle row 2, headers row 4, data from row 5, first column `Ref` — using the columns below for `X01`. The Preliminary-phase chapter builds `X02`; use the same column set there so the two join on `Plant`.

First pass: one row per plant (three, from `O12 Location`), then one per line — eight to fourteen rows. Signed by the plant manager, countersigned by `ROL-09 Domain Architect, Business`.

Ref	Plant or line	Archetype	Lines	Shift pattern	Ownership
X01-01	EXAMPLE — delete this row	Discrete, MTS	3	Two-shift, 5 days	Owned

Continue on a second copy if the group runs more than three plants.

Step 4.2 — Build the plant register the repository does not have

Produce. A populated `X01 Plant / Site` sheet, one row per plant, carrying the facts `O12 Location` cannot hold.

You need. Form B-2, `O12 Location`, the archetype statements from step 4.1, and the responsible person at each plant for the licence and format fields.

Where it lands.

> **`X01 Plant / Site` · sheet to create · columns `Ref`, `Plant`, `Archetype`, `Format`, `Line count`, `Shift pattern`, `Ownership`, `Capacity basis`, `Bottleneck`**
>
> `Format` takes one of `Continuous`, `Batch`, `Repetitive discrete`, `Job shop`, `Cell`. `Ownership` distinguishes owned, leased and tolled capacity, because tolling changes who owns the material on the line. `Capacity basis` names the unit the plant plans in — tonnes per hour, units per shift, line-hours — and `Bottleneck` the work centre that sets it. Carry `Location` as a foreign key to `O12 Location` so `R20 Application located at Loca` still resolves.

How.

1. Create the sheet on the file's convention. Keep `Ref` as `X01-01` upward so it cannot collide with `LOC-` or `E06`.
2. Transcribe Form B-2: three rows, from `LOC-02`, `LOC-03` and `LOC-04`.
3. Count lines by walking the floor, not by asking for a number. A plant that says "four lines" usually runs three and a rework cell.
4. Record `Capacity basis` in the unit the plant plans in. Two functions giving two units for one line is a measure-definition dispute; log it and move on.
5. Cross-check `E06 Plant / Site` in `O20 Data Entity`: the entity exists and your sheet is its instance register. `O20` says `E06` is mastered in `APS`, which chapter 5 corrects.

Look it up. Volume 1 chapter 7, *Sites, plants, lines, cells and stations*, and chapter 9, *The plant hierarchy, and the reconciliation nobody budgets for*. For the notation, chapter 17, *The plant hierarchy*.

Signed by. Each plant manager for their own row; `ROL-14 Head of Operations` for the set.

Done when. Every row typed `Production` in `O12 Location` has exactly one `X01` row, and every `X01` row has a non-empty `Line count` and `Shift pattern`.

Shows up in. No report, because no report knows the sheet exists. Add that to `O29 Gap` as a reporting gap and expect to extend `RPT13 Orphan Analysis` yourself.

Goes wrong when. You treat the plant register as `O12 Location` with more columns. `O12` holds eight locations including a data centre and a cloud region; only three are plants.

Step 4.3 — Add the columns `O12 Location` is missing, and take the headcount off the cloud

Produce. A corrected `O12 Location` sheet with a defensible `Scale` column and a type that distinguishes a place from a hosting arrangement.

You need. `O12 Location`, `O27 Physical Technology Compone`, `R20 Application located at Loca`, and the headcount by site.

Where it lands.

> **`O12 Location` · columns `Ref`, `Location`, `Type`, `Scale`**
>
> `Scale` reads "92 people" through "261 people" on all eight rows, including `LOC-08 Cloud Region` at **261 people**, which is nobody. Keep the column, but populate it only for rows that are places. For `LOC-07 Primary Data Centre` and `LOC-08 Cloud Region`, set `Type` to `Hosting` and leave `Scale` blank rather than inventing an occupancy. Add `Line count` and `Shift pattern` here only if you decided against building `X01`; two sheets holding one fact is worse than one sheet missing it.

How.

1. Re-type `LOC-07` and `LOC-08`. A cloud region is a contract with a hosting provider, not a site with a turnstile.
2. Correct the remaining headcounts against the HR number, by site, on one dated extract.
3. Check `O27`: `PTC-03 SCADA control servers` sits at `Primary Data Centre` and `PTC-07 Plant network switches` at `Central Distribution Centre`. Both are wrong — control servers sit at the plant they control. Raise it for Phase D.
4. Re-run `R20`: twelve applications placed by row number into eight locations, wrapping after `LOC-08`, so `A09 CAD / CAE` lands at Head Office and `A11 Track & Trace` at Coastal Plant. Every reference resolves and not one placement was decided.

Look it up. Volume 1 chapter 7, *Sites, plants, lines, cells and stations*. For why the control server belongs at the plant, chapter 13, *The air gap, and where the real path is*.

Signed by. `ROL-15 Head of Service Management` for the hosting rows; `ROL-14 Head of Operations` for the plants.

Done when. No row carries a headcount for something that is not a place, and at least one `R20` placement has been checked against where the application actually runs.

Shows up in. `RPT13 Orphan Analysis`, which finds two orphans today and will never flag a wrongly placed application: row-position placement leaves no orphan to detect.

Goes wrong when. You delete `LOC-08` instead of re-typing it. One `R20` row and one `O27` row point at it; deleting converts a wrong number into a broken link.

Step 4.4 — Record the shift pattern and the crew count

Produce. A shift-pattern statement per line, with the crew count the pattern implies and the calendar the plant actually works.

You need. The `X01` rows from step 4.2, the plant HR manager, the shop steward or the relevant bargaining-council agreement, and `O04 Measure` for any availability measure already defined.

Where it lands.

> **`X01 Plant / Site` · columns `Shift pattern`, `Crews`, `Calendar basis`**
>
> `Shift pattern` takes a named pattern, not a number of hours — `Two-shift, 5 days`, `Three-shift, 6 days`, `Continuous 24/7`. `Crews` is the count that pattern requires: a continuous 24/7 pattern needs **four or five crews**, not three, because somebody has to be off. `Calendar basis` records planned non-production time — holidays, the annual shutdown, maintenance windows — the denominator every availability figure is divided by.

How.

1. Ask each plant what pattern each line runs, and separately what pattern it is *supposed* to run. Check the answer against the capacity basis from step 4.2: a tonnes-per-hour basis on a two-shift line and a tonnes-per-year figure in the board pack will not reconcile.
2. Derive the crew count from the pattern and compare it with the establishment. A shortfall shows up as overtime, not as a gap on any sheet.
3. Tell the sponsor what this does to `O13 Business Capability`: its `Current` and `Target` columns are ratings out of five with no stated basis, and a capacity claim without a shift pattern behind it means nothing. Record any pattern change already agreed with labour in `O08`, with its date.

Look it up. Volume 1 chapter 7, *Shifts, crews, and what a pattern costs*, and *The four capacities*. For the denominator argument, chapter 7, *Why OEE is not comparable, stated hard*.

Signed by. The plant HR manager and the plant manager jointly.

Done when. Every line has a named pattern, a crew count and a calendar basis with dates, and the crew count has been checked against the establishment rather than assumed.

Shows up in. `RPT03 Capability Heatmap` becomes defensible, because the rating for `C04 Manufacturing Operations` can be argued from a capacity figure with a basis.

Goes wrong when. You write "24/7" and move on. A 24/7 pattern run on three crews is an industrial-relations problem wearing an architecture label, and it surfaces as unexplained overtime in the business case.



Shift pattern, crews and what the pattern costs

Record the pattern per line in X01. A capacity figure without it means nothing

Pattern	Covered	Crews	Consequence for the architecture
Day only, 5 days	40 h	1	Capacity is one eighth of the week. Overtime is the only flex.
Two-shift, 5 days	80 h	2	Changeover between shifts becomes a handover record.
Three-shift, 5 days	120 h	3	Weekend is the maintenance window. Protect it in the plan.
Three-shift, 6 days	144 h	4	One crew is always off cycle. Training time disappears.
Continuous 24/7	168 h	4 or 5	No natural maintenance window. Shutdowns must be planned years out.

Four crews is the arithmetic of 168 hours against a 40-hour week plus leave, training and absence cover.

Figure 4.3 The pattern decides the crew count, and the crew count decides the cost. Record it per line in the `X01` sheet step 4.2 builds. Volume 1 chapter 7, *Shifts, crews, and what a pattern costs*, does the arithmetic properly.

The people

`O10 Actor` and `O11 Role` hold the **same eighteen names**: Chief Executive, CFO, COO, CIO, CDO, CRO, Chief Compliance Officer, Chief Enterprise Architect, four domain architects, Head of Delivery, Head of Operations, Head of Service Management, Data Steward, Security Architect, Internal Auditor. `R21 Role performed by Actor` joins them one to one, all `Filled`, and every `O11` row reads `Performed by` = `Person`. `ZZ Checks` therefore reports *vacant architecture roles 0*, true by construction [2].

There is no plant manager, production planner, scheduler, maintenance planner, reliability engineer, quality manager, process engineer, operator, setter or millwright anywhere in the file. For a manufacturing repository that is the sharpest defect in it.

Form B-1 — Role register

Feeds `O11 Role` directly, in sheet order. One row per role **actually performed**, plant first and office second. A three-plant group needs thirty to thirty-five rows on a first pass; the eighteen already there stay, so you are adding, not replacing. `Type` takes a value you define and then use consistently — `Production`, `Planning`, `Maintenance`, `Quality`, `Engineering`, `Logistics`, `Executive`, `Architecture`. `Performed by` takes `Person`, `Team` or `Crew`; a shift crew is not a person. Signed by the plant manager for the plant rows and `ROL-08 Chief Enterprise Architect` for the set.

Ref	Role	Type	Performed by
ROL-19	EXAMPLE — delete this row	Production	Crew

Continue on a second copy of this form. Do not fill `O10 Actor` from it — step 4.6 explains why the two sheets must stop holding the same list.

Step 4.5 — Rebuild the role register against the plant

Produce. `O11 Role` extended from eighteen rows to the set of roles the group actually performs, with the plant roles present and typed.

You need. Form B-1, `O11 Role`, `O09 Organization Unit`, and interviews in the order volume 1 sets out. Budget two days per plant.

Where it lands.

> **`O11 Role` · columns `Ref`, `Role`, `Type`, `Performed by`**
>
> Continue the reference space from `ROL-19`. Do not renumber the existing eighteen — `R21` joins on them. `Performed by` matters most: it reads `Person` on all eighteen rows, and should read `Crew` for a shift-based production role and `Team` for one held jointly. The sheet has no `Organisation unit` and no `Headcount` column, so the link to `O09` is carried in `Type` or added as a column; say which in the subtitle row.

How.

1. Walk the floor before you write a row. You cannot name a setter's role from an org chart.
2. Interview in volume 1's order: plant manager, production manager, planner, maintenance planner, quality manager, process engineer, then the systems people.
3. Write one row per role, not one per person. Three shift supervisors on three crews are one role.
4. Type each role, keeping the vocabulary to eight values or fewer, written into the subtitle row, and mark `Performed by` honestly: a crew behaves differently under access control and under an incident investigation.

Look it up. Volume 1 chapter 5, *The plant*, then *Engineering*, *Maintenance*, *Quality* and *Planning and supply chain* for the role lists, and *Who to see first*; then volume 1 chapter 19, *The interview order*, for the same sequence as a ninety-day plan.

Signed by. Each plant manager for their plant's roles; `ROL-08 Chief Enterprise Architect` for the register.

Done when. Every role named in a volume 1 chapter 5 section appears in `O11 Role` or has a written reason for not applying here, and no role on the sheet is unknown to the plant it sits in.

Shows up in. `ZZ Checks` line *vacant architecture roles*, which will stop reading zero as soon as the register contains roles nobody holds. That change is the point.

Goes wrong when. You add the plant roles and leave `R21` at eighteen rows against a forty-row sheet. The next report that joins the two reports the plant as unstaffed.

Step 4.6 — Separate the actor from the role, and name people

Produce. `O10 Actor` holding named individuals, `O11 Role` holding roles, and `R21` carrying a real many-to-many assignment between them.

You need. The role register from step 4.5, the HR establishment list, and `O05 Principle` row `PRN-13 Data Ownership is Individual`.

Where it lands.

> **`O10 Actor` · columns `Ref`, `Actor`, `Type`, `Status`**
>
> `Actor` takes a person's name, or a named position held by one identified person — not a job title reused as a role name. `Status` takes `Filled`, `Vacant`, `Interim` or `Shared`, and for several plant roles the honest answer is `Vacant`. `R21` then stops being a diagonal: one role may be performed by three actors across three crews, and one actor four roles in a small plant.

How.

1. Read the two sheets side by side and confirm that all eighteen names are identical. That is the finding; do not take it on trust from this chapter.
2. Replace the eighteen `O10` rows with named individuals or named positions, keeping `ACT-` references stable where a row genuinely survives.
3. Rebuild `R21` as many-to-many. Expect eighteen rows to become sixty or more.
4. Quote `PRN-13` — *"An owner is a person, not a committee"* — in the covering note, and point out that `O05 Principle` gives all twenty-four of its own rows an `Owner` of `Architecture Board`, a committee, in breach of its own thirteenth principle.

Look it up. Volume 1 chapter 5, *Where these people live*, for the role-versus-holder distinction across sites, and volume 1 chapter 18, *Finding 8 — Biometric clocking, with consent taken in the employment contract*, for what follows when an actor record is a job title.

Signed by. `ROL-08 Chief Enterprise Architect`, with the HR establishment list attached.

Done when. No name appears in both `O10 Actor` and `O11 Role` as the same string, and `R21` has more rows than either sheet.

Shows up in. `RPT01 Dashboard` line *Vacant architecture roles*, which reads 0 today and should not.

Goes wrong when. You put real names into a repository shared more widely than you think. Decide the access position first; if you cannot, use named positions rather than names.

Step 4.7 — Put the plant into `O09 Organization Unit`

Produce. An organisation sheet that contains the factories, with heads that resolve to roles.

You need. `O09 Organization Unit`, the corrected `O11 Role`, and the group's actual reporting lines, which are usually not the ones on the published chart.

Where it lands.

> **`O09 Organization Unit` · columns `Ref`, `Organisation unit`, `Type`, `Head`, `Headcount`**
>
> Twelve rows — Executive, Finance, Risk & Compliance, Human Resources, Technology, Data & Analytics, Operations, Commercial, Customer Service, Supply & Logistics, Internal Audit, Legal — and not one is a plant. `Type` holds `Group`, `Corporate`, `Line` and `Assurance`; add `Plant`. `Head` holds job titles, and **six of the twelve resolve to nothing**: `Chief People Officer`, `Chief Commercial Officer`, `Head of Service`, `Head of Supply`, `Head of Internal Audit` and `General Counsel` appear in neither `O10 Actor` nor `O11 Role`.

How.

1. Add one unit per plant, typed `Plant`, and make every `Head` value a string that exists in `O11 Role`. A missing head means the role register from step 4.5 is incomplete. There is no parent column, so record whether plants report to `ORG-07 Operations` or to the group in `Type`.
2. Check the headcounts. `ORG-02 Finance` carries **273** people against `ORG-07 Operations` at **102**, in a group whose three plants hold 568 people between them on `O12 Location`. One set counts something else; find out which before you quote either.
3. Raise as one finding that `O23 Application Component` uses a third owner vocabulary — unit names such as `Legal` and `Internal Audit` — while `O20 Data Entity` holds job titles such as `Head of Finance`. Three sheets, three vocabularies.

Look it up. Volume 1 chapter 7, *Centralised, or plant-autonomous*, and *Multi-plant, multi-company and the transfer price*, which is the decision behind step 4 above.

Signed by. `ROL-03 Chief Operating Officer`.

Done when. Three plant units exist, every `Head` resolves against `O11 Role`, and the sum of `Headcount` has been reconciled against `O12 Location`'s `Scale` column or the difference explained in writing.

Shows up in. `R22 Org Unit owns Capability`, which step 4.10 rebuilds and which cannot be right until the organisation sheet contains the factories.

Goes wrong when. You add a single unit called "Manufacturing" covering all three plants. It hides the plant-autonomy question that decides whether one capability map or three is correct.

The capability model

`O13 Business Capability` holds 100 rows and is a generated cross-product: ten level-1 capabilities, each with three level-2 children drawn from a word list — Administration, Optimisation, Onboarding, Reporting, Assurance, Planning, Execution, Monitoring, Support, Design — and each of those with two level-3 children suffixed `– Define` and `– Operate`. Descriptions restate the names: *"Product Development Administration within product development."*

`Disposition` is pure arithmetic: `Build` exactly when `Target – Current ≥ 2`, `Sustain` otherwise, on all 100 rows with no exceptions, and those are the only two values on the sheet.

Criticality does not inherit. `C01 Product Development` is `High` while `C01.01` is `Low` and `C01.01.01` is `Medium`; `C01.02.01` is `Critical` under a `Low` parent. **Thirty-four of the ninety child rows are rated more critical than their parent.** And the ninety sub-rows appear in no report: `RPT03 Capability Heatmap` prints the ten level-1 rows with a computed `Gap` column, nothing else [3][8].

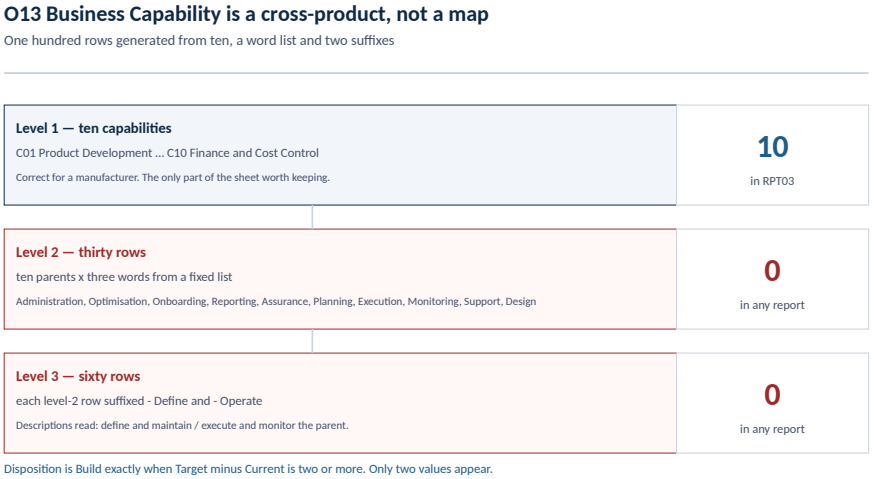


Figure 4.4 One hundred rows, ninety of which no report reads. The generation pattern is visible from the names alone. Step 4.8 rebuilds level 2 against real manufacturing capabilities and deletes level 3 unless a decision needs it.

BEFORE YOU TOUCH `013`, ANSWER THESE

What test makes a row a capability here rather than a process step?

What does `Current` = 3 mean, in a sentence a plant manager would accept?

Why is `C04 Manufacturing Operations` rated `Low` criticality, and who agrees?

Form B-3 — Capability decomposition, level 2

Feeds `013 Business Capability`, covering five of that sheet's nine columns: `Ref`, `Capability`, `Level`, `Parent` and `Criticality`. Type `Description`, `Current`, `Target` and `Disposition` straight into Excel — a description hand-written on a form becomes a restated name, the defect you are fixing.

One row per level-2 capability. Ten level-1 capabilities decompose into forty to sixty real level-2 children, not thirty; `C04 Manufacturing Operations` alone carries six or seven. Work one parent per copy. Signed by the executive who owns that parent, named in `R22` after step 4.10.

Ref	Capability	Level	Parent	Criticality
C04.01	EXAMPLE — delete this row	2	C04	High

Continue on a second copy of this form for each further level-1 parent.

Step 4.8 — Rebuild the level-2 decomposition against real manufacturing capabilities

Produce. A level-2 decomposition under each of the ten level-1 rows that a plant manager recognises, replacing the word-list children.

You need. Form B-3, `013 Business Capability`, `R01 Capability decomposition` (90 rows), the value streams from step 4.11, and two workshops per level-1 parent.

Where it lands.

> **`O13 Business Capability` · columns `Ref`, `Capability`, `Level`, `Parent`, `Description`, `Criticality`, `Current`, `Target`, `Disposition`**
>
> Keep the ten level-1 rows `C01` – `C10`: they are the only part of the sheet worth keeping and read correctly for a manufacturer. Replace the thirty level-2 rows. Keep the dotted reference convention — `C04.01`, no hyphen — because `R01` and `Parent` both join on it. `Description` must say what the capability *does*, in a sentence that could not be generated from the name. Delete the sixty level-3 rows unless a decision needs them; a row in no report informing no decision is cost without benefit.

How.

1. Take one level-1 parent. Ask the owning executive and the plant what distinct things the business does under it, and write the answers as verb phrases. Test each: can it be sourced, bought, outsourced or measured on its own? If not it is a step in a process, not a capability.
2. Write the row. `C04 Manufacturing Operations` decomposes into scheduling, order release, execution and confirmation, changeover, in-process quality, and shop-floor data capture — six children, not three.
3. Rebuild `R01 Capability decomposition` to match. It carries ninety rows mirroring the generated tree exactly.

Look it up. Volume 1 chapter 17, *Capability map*, for the viewpoint and its traps, and volume 1 chapter 11, *From business capability to application capability*, for the mapping this supports in Phase C.

Signed by. The executive owning each level-1 capability, established in step 4.10.

Done when. No level-2 description can be derived from its own name, and every level-2 row has been read back to someone who performs it.

Shows up in. `RPT03 Capability Heatmap` — unchanged in row count, because it reads only level 1. That is itself the finding to report: ninety rows of work that no report consumes.

Goes wrong when. You keep the generated level-2 names because deleting them looks like losing content. Thirty rows that restate their parent's name are worse than none, because they make the sheet look populated.

Step 4.9 — Re-rate criticality, and stop deriving disposition from arithmetic

Produce. A criticality rating that inherits, a maturity rating with a stated basis, and a disposition that records a decision rather than a subtraction.

You need. The rebuilt `O13`, `O04 Measure` for the basis, `RPT03 Capability Heatmap`, and the executive owners.

Where it lands.

> **`O13 Business Capability` · columns `Criticality`, `Current`, `Target`, `Disposition`**
>
> `Criticality` must not be lower on a parent than on any child. Rate the level-1 row from the business consequence of its failure, then let a child inherit at most its parent's rating. **`C04 Manufacturing Operations` is rated `Low` today** — in a manufacturer — with `Current` 4 and `Target` 5. `Disposition` needs more than two values: add `Retire`, `Partner` and `Acquire`, and record who decided and when in two new columns.

How.

1. Re-rate the ten level-1 rows first, from consequence of failure. Ask what stops if the capability stops, and for how long.
2. Check inheritance. Thirty-four child rows are rated above their parent; each is either a mis-rated child or a mis-rated parent.
3. State the maturity basis. `Current` and `Target` are integers 1 to 5 with no scale anywhere in the file. Write the five-point scale into the subtitle row before you change a number.
4. Re-rate `C04`. A plant whose manufacturing operations capability is `Low` criticality and 4 out of 5 mature has either the wrong rating or the wrong plant.
5. Break the arithmetic. Set at least one `Disposition` that `Target − Current` would not produce and record its reason; that row proves the column is now an assessment.

Look it up. Volume 1 chapter 6, *Gross margin is not a property of the company*, for why criticality follows product mix rather than function, and volume 1 chapter 7, *So the first deliverable is a measure definition register*, for the basis problem.

Signed by. `GB-01 Architecture Board`, which holds G1 and G2 and whose mandate in `O57 Governance Body` covers dispensations with an expiry [9].

Done when. No child is rated above its parent, a five-point scale is written into the sheet, and at least one `Disposition` value cannot be produced by subtraction.

Shows up in. `RPT03 Capability Heatmap`, where `Disposition` should stop being predictable from `Gap`. If a reader can still derive one from the other, the step is not done.

Goes wrong when. You re-rate criticality in a workshop with no plant representative. Everything becomes `High`, and a map on which everything is critical prioritises nothing.

Step 4.10 — Reassign capability ownership in `R22`

Produce. A capability-ownership map in which each capability is owned by the organisation unit that performs it, covering every level of the tree.

You need. `R22 Org Unit owns Capability` (10 rows), the corrected `O09` from step 4.7, and the rebuilt `O13`.

Where it lands.

> **`R22 Org Unit owns Capability` · columns `Org unit`, `Unit name`, `Capability`, `Capability name`**
>
> A straight index diagonal: `ORG-01→C01` through `ORG-10→C10`. Ownership falls to whichever unit shares a row number, which is how `ORG-04 Human Resources` comes to own `C04 Manufacturing Operations` and `ORG-03 Risk & Compliance` to own `C03 Procurement & Supplier Management`. The ninety sub-capabilities have **no owner at all**. Rebuild row by row and add rows for level 2.

How.

1. Print the diagonal and read the ten pairs aloud to the Chief Operating Officer. The step usually ends here. Note on the way past that `C09 Health, Safety & Environment` and `C10 Finance & Cost Control` are both `Critical` and both owned by whichever unit shares the number.
2. Assign each level-1 capability to the unit that performs it, then name the accountable executive from `O11 Role`. The sheet has no owner column, so the executive goes in `O09.Head`.
3. Add a row for every level-2 capability: ninety unowned rows become forty to sixty owned ones.

Look it up. Volume 1 chapter 7, *Centralised, or plant-autonomous*, which determines whether a capability is owned once at the group or three times at the plants.

Signed by. `ROL-03 Chief Operating Officer`, countersigned by each owning executive.

Done when. No `R22` row pairs a unit with a capability that shares its row number by accident, and every level-1 and level-2 capability has exactly one owning unit.

Shows up in. `RPT03 Capability Heatmap`, which cannot show ownership at all — a reporting gap worth raising, since the heatmap is what the board reads.

Goes wrong when. You accept "Operations owns everything on the floor". Scheduling is usually owned by planning, in-process quality by quality, and changeover by engineering — and those three boundaries are where the real arguments live.

Value streams, processes and services

BEFORE YOU REDRAW A VALUE STREAM, ANSWER THESE

How many stages does `VS02 Plan to Produce` actually have, counted on the floor?

Which of `C08`, `C09` and `C10` genuinely belongs to no value stream?

What triggers a works order in this plant, and what closes it?

O14 Value Stream against R02, and the stages that do not add up

Two sources for one fact. The inline column repeats itself; R02 slides a three-wide window

Ref	Value stream	O14 Capabilities	R02 window	O14 Stages	Verdict
VS01	Concept to Product	C01, C02, C03	C01, C02, C03	4 stages	agree
VS02	Plan to Produce	C02, C03, C04	C02, C03, C04	5 stages	agree
VS03	Source to Pay	C03, C04, C05	C03, C04, C05	6 stages	agree
VS04	Order to Cash	C01, C02, C03	C04, C05, C06	4 stages	clash
VS05	Acquire to Retire	C02, C03, C04	C05, C06, C07	5 stages	clash

VS01 and VS04 carry an identical inline list. So do VS02 and VS05. R02 reaches only C01 to C07, so C08, C09 and C10 are used by no value stream. The stage counts sum to 24; O16 holds 20 rows.

Figure 4.5 Two sources for one fact, and they disagree on the later streams. Step 4.11 makes `R02` the authority, deletes the inline column, and reconciles the stage count against `O16 Business Process`.

Step 4.11 — Rebuild the value streams and name their stages

Produce. Five value streams whose stages are named, whose capability lists agree with `R02`, and whose stage count matches the process rows in `O16`.

You need. `O14 Value Stream`, `R02 Value stream uses Capabilit` (15 rows), `O16 Business Process` (20 rows) [5][6], and a workshop per stream with the people who run it.

Where it lands.

> **`O14 Value Stream` · columns `Ref`, `Value stream`, `Scope`, `Stages`, `Capabilities`**
>
> `Stages` holds a **count** — "4 stages", "5 stages", "6 stages" — not stage names, and the counts sum to **24** against `O16`'s 20 process rows. Replace the count with an ordered list of stage names. `Capabilities` holds an inline list that contradicts `R02`: `VS01 Concept to Product` and `VS04 Order to Cash` carry the **identical** list `C01, C02, C03`, and `VS02 Plan to Produce` and `VS05 Acquire to Retire` both carry `C02, C03, C04`. Make `R02` the authority and delete the inline column.

How.

1. Put the two capability lists side by side. `R02` is a three-wide sliding window — `VS01` gets `C01, C02, C03`, up to `VS05` with `C05, C06, C07` — so it reaches only `C01` through `C07`. Three level-1 capabilities are used by no value stream.
2. Decide which of `C08 Sales & Order Management`, `C09 Health, Safety & Environment` and `C10 Finance & Cost Control` genuinely sits outside every stream, and add the rest to `R02`.
3. Name the stages. Walk `VS02 Plan to Produce` with the master scheduler and write down what happens between the forecast and the confirmed quantity.
4. Count the stages you named and reconcile against `O16`. If you named six and `O16` holds four, the process sheet is short by two rows, not your workshop by two stages.
5. Check `VS04 Order to Cash` against the archetype from step 4.1. In a make-to-order family the stream begins with a promise, not with a pick.

Look it up. Volume 1 chapter 8, *The twelve processes a manufacturer runs*, for the inventory, and *Plan to produce* for the stage sequence; volume 1 chapter 17, *Value stream*, for the viewpoint.

Signed by. The executive owning the stream's dominant capability, from the rebuilt `R22`.

Done when. `Stages` holds names, the sum of stage counts equals the row count of `O16`, and no two value streams carry identical capability lists.

Shows up in. No report reads `O14` directly, which is worth saying.

Goes wrong when. You name stages from the stream's own title. *Concept to Product — stage 1* through *stage 4* is what `O16` already contains, and it carries no information.

Step 4.12 — Give `O16 Business Process` a level, a trigger, an outcome, an owner and a volume

Produce. A process sheet that can hold a level-1 and a level-2 row at once, and that can carry the BPMN models volume 1 draws.

You need. `O16 Business Process`, the named stages from step 4.11, the process owners, and volume throughput figures from the plant.

Where it lands.

> **`O16 Business Process` · columns `Ref`, `Business process`, `Value stream`, `Description`, `Automation`**
>
> Five columns, and **no `Level`, no `Trigger`, no `Outcome`, no `Owner` and no `Volume`**. Without `Level` the sheet cannot hold a level-1 process and its level-2 children at once, which is what every manufacturing process model needs. Without `Trigger` and `Outcome` a process cannot be tested for completeness. Add all five. The twenty rows are named *"Concept to Product — stage 1"* through *"Acquire to Retire — stage 4"*; rename them from the stages settled in step 4.11.

How.

1. Add the five columns before you touch a row, then set `Level` on the twenty existing rows; most are level 2.
2. Write `Trigger`, `Outcome` and `Volume` for each, with the volume's period and source. A process with no trigger is a function; move it to `O17 Business Function` and say so.
3. Name an `Owner` from `O11 Role`, as a role and not a unit. The process owner is whoever can change the process, which is rarely whoever performs it.
4. Register each BPMN model as a row in `O31 Deliverable` with its phase; there is no artefact store.

Look it up. Volume 1 chapter 8, *The four levels, and who reads each*, then *The conventions this book uses at level 2*, *Testing a level-2 model for completeness* and *The notation exception*; chapter 17 restates the notation decision in *The BPMN exception, stated once*.

Signed by. `ROL-14 Head of Operations` for the operational processes, `ROL-13 Head of Delivery` for the change processes.

Done when. Every row has a level, a trigger, an outcome and an owner, and at least one level-1 process has level-2 children on the same sheet.

Shows up in. `RPT06 Requirements Traceability`, indirectly: a requirement that names a process can now be traced to a row with an owner who can accept it.

Goes wrong when. You model twelve processes in BPMN and leave `O16` holding twenty rows named after stage numbers. The models and the repository then describe different businesses, and the models get thrown away.

Step 4.13 — Re-point the business services and functions at something real

Produce. A business-service catalogue whose exposure and criticality are assessed, and a function sheet that says what each function does.

You need. `O15 Business Service` (10 rows), `O17 Business Function` (10 rows), `R03 Capability realised by Busi`, `R04 Business Service supported ` (note the trailing space in the sheet name), and the rebuilt `O13`.

Where it lands.

> **`O15 Business Service` · columns `Ref`, `Business service`, `Capability`, `Description`, `Exposure`, `Criticality`**
>
> Ten rows, one per level-1 capability, named by appending "Service" to the capability name. Every `Description` reads *"Externally consumable service realising …"* — including the five rows whose `Exposure` is `Internal`, contradicting the description on its own row. `R03` is a diagonal `C01→BSV-01` through `C10→BSV-10`; `R04` continues it into applications, `BSV-01→A01` through `BSV-10→A10`, which is how *Quality Management Service* comes to be supported by whichever application shares the row. `O17 Business Function` is worse: all ten descriptions read *"Organisational function performing this capability."*

How.

1. Delete any business service that exists only because a capability existed.
2. Fix the five rows where `Description` says *externally consumable* and `Exposure` says `Internal` — `BSV-02`, `BSV-03`, `BSV-04`, `BSV-06` and `BSV-09`. One of the two is wrong on each.
3. Rebuild `R04` from the portfolio, not from row position, and write a real description for each `O17 Business Function` row: what it does, where it sits, who staffs it.
4. Re-run `RPT13 Orphan Analysis`, which reports only `A11 Track & Trace` and `A12 Plant Analytics` today, both because no business service maps to them. After step 3 that list changes.

Look it up. Volume 1 chapter 11, *The twelve*, for the application capabilities these services sit over, and *Three things that are not application capabilities*; then chapter 4, *What is actually sold*, and chapter 4, *Product, conversion, or capacity*, for what a manufacturer's external services are.

Signed by. `ROL-08 Chief Enterprise Architect`, with each service's named consumer recorded.

Done when. No `O15` description contradicts its own `Exposure` value, no `O17` description is generic, and `R04` has at least one pair the diagonal would not produce.

Shows up in. `RPT13 Orphan Analysis`. Expect the orphan count to rise before it falls — that is the step working, not failing.

Goes wrong when. You rename the ten services and keep the diagonal. The names improve, the map stays wrong, and the next reader trusts it more than they did before.

Step 4.14 — Fix the product, the contract and the counterparty

Produce. A product sheet that distinguishes what is made from what is sold, an application map that is not a diagonal, and a contract sheet whose counterparty is a party.

You need. `O18 Product` (4 rows), `O19 Contract` (10 rows), `R25 Application supports Produc` (4 rows) [7], `O23 Application Component`, and the commercial and procurement teams.

Where it lands.

> **`O19 Contract` · columns `Ref`, `Contract`, `Counterparty`, `Type`, `Expiry`**
>
> **The `Counterparty` column holds the system name, not the party.** `CTR-01 ERP Core vendor agreement` has `Counterparty` = `ERP Core`; `CTR-04` has `SCADA / Historian`. A software product is not a counterparty — the vendor organisation is. All ten rows are vendor agreements, so the sheet holds **no customer supply agreement, no direct-material contract, no tolling arrangement and no service-level commitment to a customer**, which between them are every contract that binds a plant. There is no `Start`, `Value`, `Notice period` or `Owner` column, and `A11 Track & Trace` and `A12 Plant Analytics` have no contract row at all.

How.

1. Replace every `Counterparty` value with the contracting organisation's legal name, and add `Start`, `Value`, `Notice period` and `Owner`.
2. Add the contracts that bind production: supply and offtake agreements, direct-material contracts, tolling arrangements, maintenance agreements on the bottleneck equipment.
3. Split `O18 Product`. `PRD-03 Service Contracts` is a commercial offer, not a manufactured item, and `PRD-04 Custom Engineering` is an engineer-to-order family.
4. Rebuild `R25`. It maps `PRD-01→A01`, `PRD-02→A04`, `PRD-03→A07`, `PRD-04→A10` — every third application, in order — which puts Spare Parts on the historian and Service Contracts on the quality system.
5. Record ownership of material wherever a tolling arrangement exists. It is an attribute of the inventory and cannot be retrofitted onto historic stock movements.

Look it up. Volume 1 chapter 6, *Tolling, where you never own the material*, and chapter 3, *Ownership: tolling and contract manufacturing*, for the vocabulary, and chapter 4, *Ownership is an attribute of inventory, and this one you cannot fix later*, before step 6.

Signed by. `ROL-02 Chief Financial Officer` for the contract values; the commercial head for the customer agreements.

Done when. No `Counterparty` value matches an application name in `O23`, every `O18` row has an archetype from step 4.1, and `R25` contains at least one pair the diagonal would not produce.

Shows up in. `RPT02 Application Portfolio` and `RPT12 Standards Conformance` become readable against contract expiry once the counterparty is a party. Today neither can be joined to `O19` at all.

Goes wrong when. You fix the ten vendor rows and stop. The missing contracts are the finding; the wrong counterparty is only how you noticed.

Closing Phase B

Phase B here produced sheets that did not exist, corrected two that recorded arithmetic as assessment, and replaced an organisation chart with an organisation. The capability map still has ten level-1 rows,

which is correct — worth saying to a sponsor who expected the number to move.

Two things carry forward. The owner-vocabulary problem — `O09.Head` uses one set of job titles, `O20.Owner` another, `O23.Owner` unit names — is a Phase C and Phase G problem too, and chapter 5 settles it for data. And `O63 Control Objective` is populated for a European manufacturer: its obligations include product safety and CE or UKCA marking, which have no application in South Africa [10], and omit every instrument that actually inspects a plant here. Rebuilding it belongs to Requirements Management; it is Phase B that notices.

THE REVIEWER'S ASK — BEFORE YOU SIGN PHASE B OFF

Which plant has not been visited, and what in this chapter rests on its data?

Name the role that signed the capability map, and the date they read it.

Which `O13` disposition value cannot be produced by subtracting `Current` from `Target`?

How many rows did you delete, and who agreed to the deletion?

What in Phase B will be wrong again in six months, and what would catch it?

THE CHAPTER IN TEN LINES

- O10 Actor and O11 Role hold the same eighteen names, joined one to one by R21, every row Filled.
- There is no plant manager, production planner, scheduler, maintenance planner, quality manager or operator anywhere in the file.
- Six of the twelve Head values in O09 Organization Unit resolve to no role and no actor.
- O12 Location gives LOC-08 Cloud Region a headcount of 261 people, and has no column for lines, format, ownership or shift pattern.
- EXT declares X01 Plant / Site and X02 Production Line and holds no instances of either, so the plant register is a sheet you build.
- A continuous 24/7 pattern needs four or five crews, not three, and no capacity figure means anything without the pattern behind it.
- O13 Business Capability is a generated cross-product: ten level-1 rows, thirty level-2 from a word list, sixty level-3 suffixed Define and Operate.
- Disposition on O13 reads Build exactly when Target minus Current is two or more, on all 100 rows, and only two values appear.
- Thirty-four of the ninety child capability rows are rated more critical than their parent, and C04 Manufacturing Operations is rated Low.
- R22 Org Unit owns Capability is an index diagonal, so Human Resources owns Manufacturing Operations and ninety capabilities have no owner at all.
- O14 Value Stream gives VS01 and VS04 identical capability lists, states Stages as a count summing to 24 against twenty process rows, and disagrees with R02.
- O16 Business Process has no Level, Trigger, Outcome, Owner or Volume column, so it cannot hold a level-1 and a level-2 row at once.
- O19 Contract records the system name as the counterparty on all ten rows, and holds no customer, supply or tolling contract.

SOURCES FOR THIS CHAPTER

- **1.** Manufacturing_TOGAF_Metamodel.xlsx, sheets O09 Organization Unit through O19 Contract, the populated business layer for Kestrel Industrial Group
- **2.** Manufacturing_TOGAF_Metamodel.xlsx, sheets O10 Actor and O11 Role (eighteen rows each, identical names) with R21 Role performed by Actor, and ZZ Checks
- **3.** Manufacturing_TOGAF_Metamodel.xlsx, sheet O13 Business Capability (100 rows: ten level 1, thirty level 2, sixty level 3) with R01 Capability decomposition
- **4.** Manufacturing_TOGAF_Metamodel.xlsx, sheet O12 Location (eight rows; columns Ref, Location, Type, Scale) with R20 Application located at Loca and O27 Physical Technology Compone
- **5.** Manufacturing_TOGAF_Metamodel.xlsx, sheet O14 Value Stream and R02 Value stream uses Capabilit (fifteen instances, a three-wide sliding window)
- **6.** Manufacturing_TOGAF_Metamodel.xlsx, sheet O16 Business Process (twenty rows; columns Ref, Business process, Value stream, Description, Automation)
- **7.** Manufacturing_TOGAF_Metamodel.xlsx, sheets O18 Product and O19 Contract with R25 Application supports Produc
- **8.** Manufacturing_TOGAF_Metamodel.xlsx, sheets RPT01 Dashboard, RPT03 Capability Heatmap and RPT13 Orphan Analysis
- **9.** Manufacturing_TOGAF_Metamodel.xlsx, sheets O57 Governance Body (five bodies) and O58 Governance Gate (GT-0 to GT-3)
- **10.** National Regulator for Compulsory Specifications Act 5 of 2008, consolidated text (SAFLII), section 20 https://www.saflii.org/za/legis/consol_act/nrfcsa5o2008543/

CHAPTER 5

Data Architecture

Two sheets claim the golden source and disagree nine times out of ten. Everything else in this chapter follows from fixing that.

The chapter this repository exists to teach

Two sheets in this file claim the golden source, and they disagree on nine of the ten entities they share. `O20 Data Entity` holds a `Master system` column. `O43 Master Data Rule` holds a `Golden source` column. They name the same ten entities, and they agree on exactly one row — `E01 Product`, where both say `SCADA / Historian`, which is the wrong answer anyway [1][2].

Meanwhile `O06 Requirement` carries eight `Must` requirements, `REQ-007` through `REQ-014`, each reading *"<entity> is mastered in one system and consumed elsewhere through a published interface"*, all eight `Approved` [9]. The repository's own data falsifies its own approved requirements, and no report notices, because `RPT05 Data Ownership` reprints `O20` and never joins to `O43` [11].

That is Phase C for data in one paragraph. Fourteen steps follow; three build sheets the file declares and does not populate. Kestrel Industrial Group is an invention, as is volume 1's Karoo Industrial Group; their data is never mixed.

BEFORE YOU OPEN A DATA SHEET, ANSWER THESE

Which system is the system of record for the item master in this group, and who told you?

Which entity, if it were wrong for a week, would stop a plant?

What does your traceability obligation require you to reconstruct, and from what?

Two sheets claim the golden source. They agree on one row

O20 Data Entity, column Master system, against O43 Master Data Rule, column Golden source

Ref	Data entity	O20 Master system	O43 Golden source	Verdict
E01	Product	SCADA / Historian	SCADA / Historian	agree
E02	Bill of Material	PLM	Track and Trace	clash
E03	Routing / Recipe	Plant Analytics	Supplier Portal	clash
E04	Work Order	APS	Supplier Portal	clash
E05	Batch / Lot	WMS	Supplier Portal	clash
E06	Plant / Site	APS	QMS	clash
E07	Work Centre	QMS	WMS	clash
E08	Equipment / Asset	SCADA / Historian	MES	clash
E09	Supplier	APS	QMS	clash
E10	Purchase Order	ERP Core	QMS	clash
E11	Quality Specification	APS	no row	absent
E12	Non-conformance	ERP Core	no row	absent
E13	Inventory Position	Supplier Portal	no row	absent
E14	Customer Order	Track and Trace	no row	absent

Nine disagreements on the ten entities the two sheets share. The one agreement, E01 Product, agrees on the wrong answer: an append-only historian cannot master an item. Four entities have no rule at all.

Meanwhile REQ-007 to REQ-014 in O06 Requirement all read mastered once, Must, Approved.

Figure 5.1 The worst defect in the repository, on one page. Two columns claim the same fact and disagree nine times out of ten. Step 5.1 settles which column is the authority; step 5.2 then corrects the answer, because the one agreement is also wrong.

Settle the master data question first

Step 5.1 — Decide which sheet is the authority on the golden source

Produce. A single written decision naming one authoritative column for the golden source, and a reconciliation of the two columns that disagree today.

You need. `O20 Data Entity`, `O43 Master Data Rule`, `R05 Application manages Data En`, Form C-3, and `GB-03 Data Governance Council` on the agenda.

Where it lands.

> **`O43 Master Data Rule` · columns `Ref`, `Entity`, `Rule type`, `Description`, `Golden source`**
>
> Ten rows covering `E01` through `E10`, so `E11 Quality Specification`, `E12 Non-conformance`, `E13 Inventory Position` and `E14 Customer Order` have no rule at all. `Rule type` holds `Validation`, `Golden source`, `Match rule` and `Survivorship`, and `Description` is a tautology on every row — *"Rule governing product mastering."* **A sheet called Master Data Rule contains no rule.** Decide that `O20.Master system` is the single authority, make `O43.Golden source` derived or delete it, and use `Description` for the actual rule text.

How.

1. Build the reconciliation on Form C-3, all fourteen entities, both columns side by side.
2. Mark the nine disagreements: `E02 PLM` against `Track & Trace`; `E03 Plant Analytics` against `Supplier Portal`; `E04 APS` against `Supplier Portal`; `E05 WMS` against

`Supplier Portal`; `E06` `APS` against `QMS`; `E07` `QMS` against `WMS`; `E08` `SCADA / Historian` against `MES`; `E09` `APS` against `QMS`; `E10` `ERP Core` against `QMS`.

3. Note that `R05`'s fourteen `Master` rows agree with `O20` exactly, so two of the three sheets agree and `O43` is the outlier. That does not make `O20` right — step 5.2 shows it is not.
4. Take the decision at `GB-03 Data Governance Council`, record it in `O60 Governance Decision`, and accept that `O60` has no date column so the date goes in the decision text.
5. Write real rule text into `O43.Description`: the match rule, the survivorship rule, the validation rule. One sentence each, testable.

Look it up. Volume 1 chapter 11, *One: what is the system of record for the item, the bill of material and the stock position?*, the question this step answers, and volume 1 chapter 20, *Map 4 — System class to what it masters*, the cross-map to check your answers against.

Signed by. `GB-03 Data Governance Council`, minuted.

Done when. One column holds the golden source, the other is derived or gone, and all fourteen entities have a row in whichever sheet now carries the rule.

Shows up in. `RPT05 Data Ownership`, which prints `Master system` and should now be the only place a reader finds it.

Goes wrong when. You reconcile the two columns without deciding which sheet owns the fact. In six months they will disagree again, on different rows.

Step 5.2 — Correct the master system on the entities that cannot be mastered where it says

Produce. A corrected `Master system` value for every entity, defensible against what each application actually does.

You need. The decision from step 5.1, `O23 Application Component`, `O39 Data Migration`, and the application owners.

Where it lands.

> **`O20 Data Entity` · column `Master system`**
>
> **`E01 Product` — the item master — is recorded as mastered in `SCADA / Historian`.** A historian is an append-only time-series store written by the control layer; it has no item hierarchy, no effectivity and no maintenance workflow, and it cannot master anything. `E06 Plant / Site`, `E09 Supplier` and `E11 Quality Specification` are all mastered in `APS`, a planning engine that consumes master data and creates none. `E13 Inventory Position` is mastered in `Supplier Portal`. Correct all five against what the application does, not against what the sheet says.

How.

1. For each entity, name the application that creates the record and the one that approves it. The master is the one that does both.

2. Correct `E01` to the system that runs item creation and approval — in this estate `ERP Core` or `PLM`, and the choice is an architecture decision, not a lookup.
3. Correct `E06`, `E09`, `E11` and `E13`. A planning engine and a supplier portal master nothing.
4. Check `O39 Data Migration` before you finish: `MIG-01` migrates `Product` **from `PLM` to `SCADA / Historian`**, `In progress`. A migration is under way that would implement the wrong answer. Stop it or record the decision to continue.
5. Re-read `REQ-007` through `REQ-014` and mark each one satisfied or not. Eight `Approved` `Must` requirements currently rest on values you have just changed.

Look it up. Volume 1 chapter 13, *Historians, and the number that changes itself*, and chapter 10, *Why a historian's data model is not relational, and should not be forced to be*. For the item master itself, chapter 9, *The item master, and why it is the hardest master data object in any industry*.

Signed by. `ROL-10 Domain Architect, Data`, with each application owner's acceptance recorded.

Done when. No `Master system` value names a historian, a planning engine or a portal, and every value has a named application owner who has accepted it.

Shows up in. `RPT05 Data Ownership`, and `RPT06 Requirements Traceability` once `REQ-007` onward are re-assessed.

Goes wrong when. You correct the sheet and leave the migration running. The repository then says one thing and the estate does another, which is worse than a wrong sheet.

Step 5.3 — Fix the key scheme in `R05`

Produce. A relationship sheet in which both ends of every row are named by reference, so the sheet can be joined.

You need. `R05 Application manages Data En` (28 instances) [3], `O23 Application Component`.

Where it lands.

> **`R05 Application manages Data En` · columns `Application`, `Application name`, `Data entity`, `Entity name`, `Role`**
>
> **The sheet uses two key schemes.** Its fourteen `Master` rows put the application *name* in the `Application` column — `SCADA / Historian`, `PLM`, `Plant Analytics`, `APS`, `WMS`, `QMS`, `ERP Core`, `Supplier Portal`, `Track & Trace` — where the fourteen `Consumer` rows correctly put a reference, `A01`, `A04`, `A07`, `A10`. A join on `Application` therefore returns half the sheet, and `00 Index` promises that *"every relationship row names both ends by reference and by name"*. Put the reference in `Application` on all twenty-eight rows.

How.

1. Replace the nine name values in `Application` with their `A`-references, leaving `Application name` as the human-readable column, and re-run the join: twenty-eight rows should resolve, not fourteen.
2. Look at what the `Consumer` rows actually say: only `A01 ERP Core`, `A04 SCADA / Historian`, `A07 QMS` and `A10 Supplier Portal` consume anything, cycling through the fourteen entities in order. Eight of the twelve applications consume no data at all, which no plant resembles.
3. Rebuild the consumer rows from the interface catalogue in `R06 Application interfaces with` and from the direct database reads `ZZ Checks` counts at four.

Look it up. Volume 1 chapter 12, *The three things it is always asked to do and always does badly*, for why the consumer list matters more than the master list in an ERP-centred estate.

Signed by. `ROL-10 Domain Architect, Data`.

Done when. A join from `R05.Application` to `O23.Ref` returns twenty-eight rows, and more than four applications appear as consumers.

Shows up in. `RPT13 Orphan Analysis`, which performs the only real join in the report set and will stop under-reporting once the keys are consistent.

Goes wrong when. You fix the keys and leave the consumer cycle. The sheet joins cleanly and still describes an estate in which eight applications read nothing.

Form C-3 — Master and golden-source reconciliation

This form feeds two sheets. Columns 2 and 3 are read from `O20.Master system` and `O43.Golden source`; column 4 is the decision you write back to whichever sheet step 5.1 made authoritative. Fourteen rows, one per entity in `O20`; `O43` holds only ten, so four will be blank there, and that absence is itself a finding. The first row is worked for you, and is not an example to delete.

Entity	O20 Master system	O43 Golden source	Decided master	Signed
E01 Product	SCADA / Historian	SCADA / Historian		

Form C-1 — Data entity, identify

Feeds `O20 Data Entity`, columns `Ref`, `Data entity`, `Class`, `Definition`. `Class` takes `Master`, `Transactional` or `Reference` — the repository uses only the first two, and a manufacturer needs the third for units of measure, reason codes and the downtime reason tree. `Definition` must be a sentence a plant manager and a financial controller would both sign.

The fourteen rows already there stay; a manufacturer needs twenty-five to thirty-five entities on a first pass. Signed by `ROL-10 Domain Architect, Data`.

Ref	Data entity	Class	Definition
E15	EXAMPLE — delete this row	Reference	Unit of measure and its conversions

Continue on a second copy of this form. Assess each entity on Form C-2 once it is defined.

Form C-2 — Data entity, assess

Feeds `O20 Data Entity`, columns `Classification`, `Owner`, `Master system`, `Lineage`. One row per entity identified on Form C-1, in the same reference order. `Owner` takes a **named person**, not a job title — step 5.4 explains why. `Lineage` takes `Full`, `Partial` or `Absent`, and the honest answer on a first pass is `Absent` more often than this repository admits.

Retention clock, personal-data flag and the applicable obligation go in three further columns typed into Excel; the repository has none. Signed by the named owner of each row.

Ref	Classification	Owner	Master system	Lineage
E15	Internal	EXAMPLE — delete this row	ERP Core	Absent

Continue on a second copy of this form.

Owners, classification and lineage

Step 5.4 — Name a person for every entity owner

Produce. An `Owner` value on every `O20` row that resolves to one identified person, with the accountability written down.

You need. `O20 Data Entity`, `O11 Role` as corrected in chapter 4, `O05 Principle` row `PRN-13`, and `O42 Metadata Element`.

Where it lands.

> **`O20 Data Entity` · column `Owner`**
>
> Ten distinct values today, of which **one resolves**: `Data Steward` matches `ROL-16`. The other nine are job titles that exist nowhere in the file — `Head of Finance` owns `E01 Product`; `Head of Sales` owns `E02 Bill of Material`; `Head of Quality Management` owns `E03 Routing / Recipe`; `Head of Health, Safety` owns both `E08 Equipment / Asset` and `E10 Purchase Order`; `Head of Manufacturing Operations` owns `E09 Supplier`; `Head of Warehouse` owns `E05 Batch / Lot` and `E14 Customer Order`. The assignments are also wrong: sales does not own the parts list and safety does not own the purchase order.

How.

1. For each entity, ask who signs off a change to it today. That person is the owner, whatever the sheet says.
2. Write a named person, or a named position held by one identified person. `PRN-13 Data Ownership is Individual` states it: *"An owner is a person, not a committee."*
3. Re-assign the wrong ones. The parts list belongs to engineering, the purchase order to procurement, the routing to manufacturing engineering, the quality specification to quality.
4. Check `O42 Metadata Element`: `MTD-02 Business owner` has `Coverage` = **`Absent`**, while `O20` has an owner on all fourteen rows. One of those two cells is false, and the honest reading is that `O20` holds a string rather than an owner.

Look it up. Volume 1 chapter 10, *Master data governance, and the item-creation workflow*, and chapter 9, *One row, six owners who disagree*, which is the argument you are about to have.

Signed by. `GB-03 Data Governance Council`, which `O57 Governance Body` mandates to *designate data owners and resolve disputes*.

Done when. Every `Owner` value resolves to a role in `O11 Role` and to one identified person, and each owner has accepted in writing.

Shows up in. `RPT05 Data Ownership`, whose `Owner` column becomes joinable for the first time.

Goes wrong when. You name a committee or a department. `O05 Principle` does exactly that — all twenty-four rows give `Owner` = `Architecture Board` — in breach of its own thirteenth principle.

BEFORE YOU CLASSIFY ANYTHING, ANSWER THESE

What is the most commercially sensitive document this group owns, and who may read it?

Why is `E11 Quality Specification` marked `Public`, and who decided?

Which entities hold employee or biometric data, and under what lawful basis?

Step 5.5 — Re-classify the entities, starting with the quality specification

Produce. A classification on every entity that survives a commercial and a regulatory challenge.

You need. `O20 Data Entity`, `O37 Data Attribute`, `O48 Environment`, the commercial director, and the quality manager.

Where it lands.

> **`O20 Data Entity` · column `Classification`**
>
> **Inverted exactly where it matters.** `E09 Supplier`, `E11 Quality Specification` and `E13 Inventory Position` all read **`Public`**, while `E04 Work Order` and `E07 Work Centre` read `Restricted`. A quality specification is the testable definition of the product: it is the most commercially sensitive document a manufacturer owns, and in a regulated plant it is also a compliance record an inspector may demand [12]. A work centre is a list of machines. The vocabulary — `Public`, `Internal`, `Confidential`, `Restricted` — is adequate; the assignments are not.

How.

1. Re-classify `E11` first, with the quality manager and the commercial director in the room. It sets the precedent for the rest.
2. Re-classify `E09 Supplier`: supplier terms, qualification status and empowerment data are not public in any manufacturer.
3. Re-classify `E13 Inventory Position`. Stock on hand by location is a competitive disclosure and, where material is held on consignment, somebody else's commercial information.
4. Push the classification down to the attribute. `O37 Data Attribute` classifies attributes independently of their entity — `Product.Identifier` is `Restricted` under an `Internal`

entity, and `Plant / Site.Owner` reads `Public` — so an attribute can be more open than its parent.

5. Flag the entities holding personal data. In a manufacturer that is overwhelmingly employee and biometric data, which makes it a special-information question rather than a marketing one [12].

Look it up. Volume 1 chapter 10, *Classification, and why an item has several*. For the regulatory weight of the specification, chapter 16, *Compulsory specifications, letters of authority, and section 20* [13], and *POPIA, where it actually bites a manufacturer*.

Signed by. The named owner of each entity, countersigned by `ROL-07 Chief Compliance Officer` for the personal-data rows.

Done when. No entity is classified more openly than its most sensitive attribute, and `E11` is not `Public`.

Shows up in. `RPT05 Data Ownership`, which prints `Classification`, and `RPT11 Security Position` indirectly through the controls that follow from it.

Goes wrong when. You classify in a workshop with no commercial representative. Everything becomes `Internal`, and a specification that an inspector can demand sits beside a price list.

Step 5.6 — Make lineage exist for the six entities that have none

Produce. A lineage statement for each entity, and a plan for the six that have none.

You need. `O20 Data Entity`, `O40 Source To Target Mapping`, `O42 Metadata Element`, `ZZ Checks`, and the reporting team.

Where it lands.

> **`O20 Data Entity` · column `Lineage`**
>
> Six of fourteen read **`Absent`**: `E03 Routing / Recipe`, `E04 Work Order`, `E05 Batch / Lot`, `E06 Plant / Site`, `E10 Purchase Order` and `E14 Customer Order`. `ZZ Checks` reports this correctly as *data entities without lineage 6*, one of the few lines on that sheet that holds. The six include the batch, the work order and the routing — the three things a traceability obligation turns on. `Lineage` as a three-value column is also too coarse: add a column naming the report that cannot be traced and the step where the trace breaks.

How.

1. Take the six `Absent` rows in traceability order — `E05`, then `E04`, then `E03` — and walk one number from a report back to the source record. Write down where the walk fails.
2. Look at `O40 Source To Target Mapping` [7]: all twelve rows hold placeholder text — `Source field` reads `PLM field`, `Target field` reads `MES field`, and `Rule` is only `Direct` or `Lookup`. **There is no mapping on the mapping sheet.** Fill it or mark it empty honestly.

3. Reconcile `O42 Metadata Element`, where `MTD-08 Lineage` reads `Partial` against six entities with none, and `MTD-11 Physical location` reads `Absent` while `O22 Physical Data Component` holds twelve rows [8]. The metadata register and the entity sheet disagree about the repository.
4. Record the retention clock for each entity while you are in the sheet. There is no column for it, and a single corporate retention period is always wrong in a manufacturer.

Look it up. Volume 1 chapter 10, *The reporting layer, and the reconciliation that is always wrong*, and *The plant-to-finance reconciliation*. For retention, chapter 16, *The retention clocks, and why a single corporate period is always wrong*.

Signed by. `ROL-05 Chief Data Officer`.

Done when. No entity carrying a regulatory obligation reads `Absent`, and every `Absent` row has a dated plan and a named owner.

Shows up in. `RPT01 Dashboard` line *Data entities with no lineage*, which reads 6 and should fall. It is one of the few figures on that sheet computed rather than typed.

Goes wrong when. You record `Partial` because a tool draws a diagram. Lineage means you can walk one number back to one record, with the transformation at each hop named.

The three sheets this industry needs and the file does not have

`EXT Industry Extension` declares `X03 Bill of Material`, `X04 Batch / Lot` and `X05 Quality Specification` and holds **no instances of any of them** [10]. All three are shadowed in `O20 Data Entity` as `E02`, `E05` and `E11` — one row each, four columns of description between them. One row cannot express four bill-of-material structures, a genealogy graph, or a specification with versions and test methods. The next three steps build the sheets, on the same convention the Preliminary-phase chapter used for `X02 Production Line`.

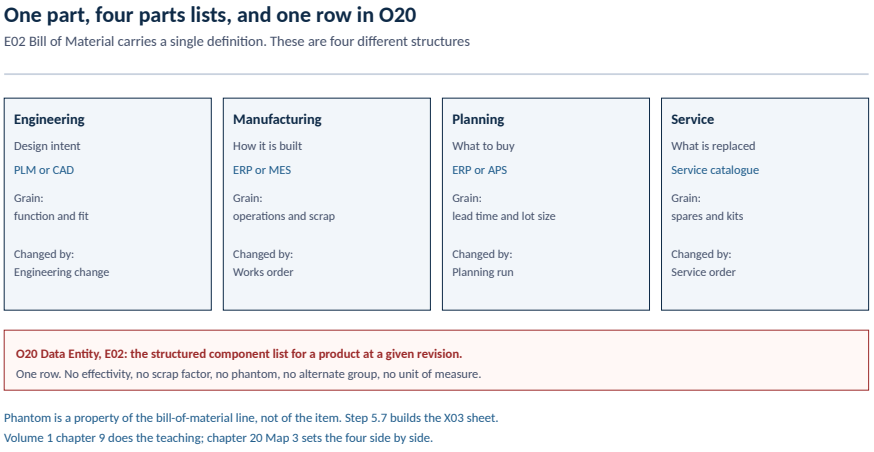


Figure 5.2 Four structures over the same parts. The engineering and manufacturing structures are the pair that most often never meet — volume 1 chapter 18, *Finding 2 — The engineering BOM and the manufacturing BOM do not meet*, is the worked case. Step 5.7 builds the sheet that can hold all four.

BEFORE YOU BUILD THE BILL-OF-MATERIAL SHEET, ANSWER THESE

How many bill-of-material structures does this group keep, and who owns each?

Is a phantom in this estate a property of the item or of the line it appears on?

Where does the engineering structure hand over to the manufacturing structure, and by what event?

Step 5.7 — Split the bill of material into its four structures

Produce. An `X03 Bill of Material` sheet holding header and line, with the structures the group actually keeps distinguished.

You need. `O20` row `E02`, `O37 Data Attribute` rows `ATR-007` to `ATR-012`, `A02 PLM`, `A03 MES`, and the engineering and manufacturing engineering leads.

Where it lands.

> **`X03 Bill of Material` · sheet to create · columns `Ref`, `Parent item`, `Structure`, `Revision`, `Effective from`, `Effective to`, `Component`, `Quantity`, `Unit of measure`, `Scrap factor`, `Phantom`, `Alternate group`**
>
> `Structure` takes `Engineering`, `Manufacturing`, `Planning` or `Service` — the four structures over the same parts. `Revision`, `Effective from` and `Effective to` carry the change event; `O20`'s single row for `E02` can express none of them, and `O37` gives the entity only `Identifier`, `Name`, `Status`, `Effective date`, `Owner` and `Classification`. **`Phantom` is a property of the line, not of the item**, so it belongs on this sheet and not on the item master. `Scrap factor` is per line too, and the arithmetic differs depending on whether it is applied to the component or the parent.

How.

1. Ask how many structures exist. The answer is rarely one: the common failure is an engineering structure in `PLM` and a manufacturing structure in `ERP Core`, never reconciled.
2. Create the sheet. One row per bill-of-material line, keyed on parent, structure, revision and component.
3. Settle effectivity before you load a row: date-effective, serial-effective, or both. Changing it later means reloading every line.

4. Record `Unit of measure` on every line and the conversion to the parent's unit. A quantity without a unit is the error that never announces itself.
5. Mark phantoms per line and record what the explosion is expected to do with them.

Look it up. Volume 1 chapter 9, *Several structures over the same parts*, then *Header and line*, *Effectivity*, *Phantom assemblies*, *Scrap, yield, and the arithmetic done properly* and the checklist in chapter 9, *Ten questions that settle a bill-of-material design*; then chapter 20, *Map 3 — The four bills of material, side by side*, to hand a sceptical engineer.

Signed by. The engineering manager and the manufacturing engineering manager jointly. Neither alone can sign a structure the other maintains.

Done when. Every structure the group keeps has rows, every line has a unit of measure, and effectivity is settled in writing.

Shows up in. No report, because no report knows the sheet exists. Register the absence in `O29 Gap` and extend the report set yourself.

Goes wrong when. You build one structure and call it the bill of material. Volume 1 chapter 18's second finding is exactly this failure, and it costs a year.

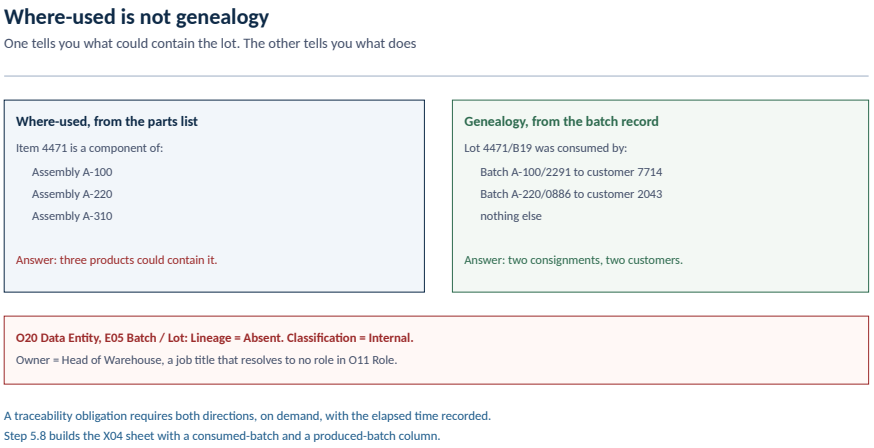


Figure 5.3 Where-used answers a different question. The parts list is a definition; genealogy is a record of what actually happened. Only the second answers a recall. The lot and consignment numbers here are illustrative.

Step 5.8 — Build the batch and genealogy sheet

Produce. An `X04 Batch / Lot` sheet that records genealogy as a graph, and a recall simulation that runs against it.

You need. `O20` row `E05`, `A11 Track & Trace` (which `O23` dispositions `Retire`), the quality manager, and the applicable traceability obligation.

Where it lands.

> **`X04 Batch / Lot` · sheet to create · columns `Ref`, `Batch`, `Item`, `Plant`, `Work order`, `Quantity`, `Unit of measure`, `Consumed batch`, `Produced batch`, `Status`, `Shelf life to`, `Retest date`**
>
> `Consumed batch` and `Produced batch` are what make this a graph rather than a list: one batch is produced from many, and feeds many. **Genealogy is not a bill of material.** Where-used on the parts list tells you which products *could* contain a lot; only genealogy tells you which *do*. `O20` gives `E05` one row, `Lineage` = `Absent`, `Classification` = `Internal`, owned by `Head of Warehouse` — against a traceability obligation that requires you to reconstruct both directions on demand.

How.

1. Decide the grain: batch, lot or serial, per product family. A product family may need two.
2. Create the sheet with both genealogy columns; a single parent column cannot hold a graph.
3. Load one real batch end to end, from goods receipt to despatch, recording how long it took and how many systems you touched.
4. Run a recall simulation against it: pick a received lot, list every despatched consignment that contains it, and time the answer. That number is the finding you report.

Look it up. Volume 1 chapter 9, *Batch, lot, serial, and genealogy as a graph*, then *Genealogy is a graph* and *Recall simulation is the real test*; chapter 12, *Traceability and serialisation*, for the systems; and chapter 18, *Finding 4 — The batch record is in a spreadsheet estate, against a traceability obligation*, for the worked case.

Signed by. The quality manager, countersigned by `ROL-07 Chief Compliance Officer`.

Done when. A recall simulation completes against the sheet and its elapsed time is recorded.

Shows up in. `RPT09 Regulatory Coverage`, once the obligation that compels genealogy is in `O63 Control Objective` — which, as chapter 4 notes, is currently populated for a different jurisdiction.

Goes wrong when. You record genealogy only forward. Backward tracing from a customer complaint to the received lot is the direction an inspector asks for first.

Step 5.9 — Build the quality specification sheet

Produce. An `X05 Quality Specification` sheet holding characteristics, limits, test methods and versions.

You need. `O20` row `E11`, `A07 QMS`, the quality manager, and the customer specifications held by commercial.

Where it lands.

> **`X05 Quality Specification` · sheet to create · columns `Ref`, `Specification`, `Item`, `Version`, `Characteristic`, `Unit of measure`, `Lower limit`, `Target`, `Upper limit`, `Test method`, `Sample plan`, `Applies from`**
>
> One row per characteristic per version, not one per specification. `Test method` matters as much as the limit: the same characteristic measured two ways gives two answers, and a customer dispute turns on which method was agreed. `Applies from` carries the version into effect. `O20` holds `E11` as one row, classified **`Public`**, mastered in `APS`, owned by a job title that resolves to nothing — and `043 Master Data Rule` has no row for it at all.

How.

1. List the specifications that bind: customer, internal, and any compulsory specification that applies [13].
2. Create the sheet, one row per characteristic per version, with the test method and sample plan against each. A limit with no method is unenforceable in a dispute.
3. Version properly. A specification change is a change event with a date and an approver, and the batch released yesterday was released against yesterday's version.

Look it up. Volume 1 chapter 9, *Quality*, for the object model, and chapter 16, *Compulsory specifications, letters of authority, and section 20* [13], for what a compulsory specification compels, and chapter 3, *Quality words, where the distinction has legal force*, to keep the vocabulary straight.

Signed by. The quality manager, countersigned by the commercial director for customer specifications.

Done when. Every characteristic has a unit, limits, a method and a version, and the current version of each specification is identifiable by date.

Shows up in. `RPT09 Regulatory Coverage` and `RPT12 Standards Conformance`, both of which are currently readable only against `O32 Standard`.

Goes wrong when. You hold one row per specification and keep the characteristics in an attached document. The limits then live outside the repository and change without anybody noticing.

Step 5.10 — Make ownership an attribute of inventory

Produce. An ownership attribute on the inventory entity, with the arrangements that make it necessary recorded.

You need. `O20` row `E13`, the contract sheet as corrected in chapter 4, the financial controller, and the warehouse manager.

Where it lands.

> **`O20 Data Entity` · row `E13 Inventory Position`**
>
> Defined as *"Quantity of an item at a location in a state."* That is three dimensions, and a manufacturer needs a fourth: **who owns it.** Material held under a tolling arrangement belongs to the customer; consignment stock belongs to the supplier until it is drawn; subcontract material sits at a third party and is still yours. Add `Ownership` as an attribute of the entity and as a column on every stock movement. Also correct the row's `Classification` from `Public`, and its `Master system` from `Supplier Portal`.

How.

1. Ask whether any tolling, consignment or subcontract arrangement exists, then ask the financial controller the same question.
2. Add `Ownership` to the entity, taking `Own`, `Customer`, `Supplier` or `Third party`, with the counterparty named.
3. Put the attribute on the movement, not only on the balance. A balance can be recalculated; a historic movement with no owner cannot be reconstructed.
4. Reconcile with the ledger: material you hold and do not own is not inventory.

Look it up. Volume 1 chapter 4, *Ownership is an attribute of inventory, and this one you cannot fix later*, and chapter 9, *Inventory, and ownership as a first-class attribute*; chapter 6, *Tolling, where you never own the material*, gives the accounting consequence.

Signed by. `ROL-02 Chief Financial Officer` and the warehouse manager jointly.

Done when. Every stock movement carries an owner, and the inventory balance reconciles to the ledger with third-party material excluded.

Shows up in. `RPT05 Data Ownership`, and in the Phase E business case, where tolled volume that was counted as own-account inventory changes the working-capital figure.

Goes wrong when. You add the attribute to the balance only. The first audit asks for the movement history, and it has no owner on it.

The model, the attributes and the physical layer

BEFORE YOU TOUCH THE ENTITY MODEL, ANSWER THESE

Which relationship in `036` would an engineer laugh at, and why?

Which entity in `020` participates in no relationship at all?

Where does a tag stop and a data entity start, in this estate?

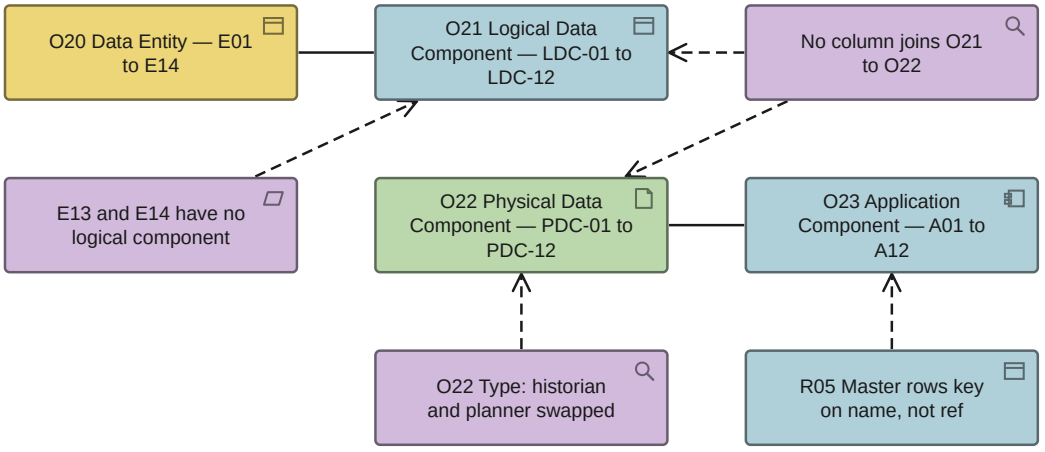


Figure 5.4 The join that does not exist. `O21 Logical Data Component` joins to `O20` on `Entity`; `O22 Physical Data Component` joins to `O23` on `Application`. Nothing joins `O21` to `O22`, so the logical-to-physical trace cannot be walked. Step 5.13 adds the column.

Step 5.11 — Rebuild the entity relationship model

Produce. An entity relationship model whose relationships and cardinalities are true, held in one sheet rather than two.

You need. `O36 Data Entity Relationship` (11 rows), `R19 Data Entity relates to Data` (11 rows) [6], and the rebuilt entity list.

Where it lands.

> **`O36 Data Entity Relationship` · columns `Ref`, `From entity`, `Relationship`, `To entity`, `Cardinality`**
>
> **`O36` and `R19` hold the same eleven rows**, with the same `REL-` references, differing only in column headers — and `R19` names both ends by name alone, with no reference, which breaks the index's own promise. The eleven rows are a positional chain: row *n* links `E` *n* to `E` *n+1*, which is how the file comes to assert *Product is derived from Bill of Material, 1:1*, *Equipment / Asset is composed of Supplier, 1:1* and *Supplier is derived from Purchase Order, 1:1*. `E13` and `E14` appear in no relationship at all. Keep one sheet; delete or derive the other.

How.

1. Read the eleven rows aloud. Three of them are backwards and two are absurd; that is the finding, and it takes four minutes to establish.
2. Rewrite each relationship with its true direction and cardinality. The parts list is derived from the product, one to many, not the reverse.
3. Add the relationships that matter and are missing: batch to work order, batch to batch, work order to routing, specification to item, inventory to batch.
4. Connect `E13` and `E14`. An entity in no relationship is either wrong or redundant.

Look it up. Volume 1 chapter 9, *The model on one page*, for the target, and chapter 17, *Information structure*, for the ArchiMate viewpoint this sheet feeds.

Signed by. `ROL-10 Domain Architect, Data`.

Done when. One sheet holds the model, every cardinality has been checked against a real record, and no entity sits outside the graph.

Shows up in. `RPT13 Orphan Analysis` once you extend it to data entities; today it looks only at applications and finds two.

Goes wrong when. You correct the cardinalities and keep both sheets. The duplicate will be edited in one place and read from the other.

Step 5.12 — Give every entity its attributes, with a unit of measure

Produce. An attribute list per entity that a developer could build from, with units, keys and mandatory flags.

You need. `O37 Data Attribute` (60 rows) [4], the rebuilt entity list, and the application owners.

Where it lands.

> **`O37 Data Attribute` · columns `Ref`, `Entity`, `Attribute`, `Type`, `Key`, `Classification`**
>
> Sixty rows, and they are ten entities multiplied by the same six attributes — `Identifier`, `Name`, `Status`, `Effective date`, `Owner`, `Classification`. **`E11 Quality Specification`, `E12 Non-conformance`, `E13 Inventory Position` and `E14 Customer Order` have no attributes at all.** `Key` reads `Yes` on two attributes per entity, `Identifier` and `Name`, so every entity has a composite key that includes its own name. There is no `Unit of measure`, no `Mandatory`, no `Length` and no `Precision` column — and not one attribute on the sheet is a manufacturing attribute. No quantity, no revision, no tolerance, no scrap factor.

How.

1. Add `Unit of measure`, `Mandatory`, `Precision` and `Source system` as columns. A quantity without a unit and a precision cannot be built.
2. Fix `Key`. One primary key per entity; a name is not part of it.
3. Write the real attributes, entity by entity, starting with the four that have none.
4. Keep the attribute classification consistent with the entity's, or state the rule that allows it to differ. Today `Product.Identifier` is `Restricted` under an `Internal` entity and `Plant / Site.Owner` is `Public`.

Look it up. Volume 1 chapter 9, *Units of measure, and the error that never announces itself*, and *Standard times, and why they are always wrong*, for the attributes a routing needs.

Signed by. The named entity owner for each block of rows.

Done when. Every entity has attributes, every quantity attribute has a unit, and no entity has two key attributes.

Shows up in. No report reads `O37`, which is worth stating: sixty rows that nothing consumes.

Goes wrong when. You model every attribute of every entity. Three hundred rows nobody reads is the same defect as sixty generated ones, at five times the cost.

Step 5.13 — Join the logical components to the physical ones

Produce. A traceable path from entity to logical component to physical component to application.

You need. `O21 Logical Data Component` (12 rows), `O22 Physical Data Component` (12 rows) [5], `O23 Application Component`, and `O38 Data Set`.

Where it lands.

> **`O21 Logical Data Component` · columns `Ref`, `Logical data component`, `Entity`, `Description`**
>
> `O21` joins to `O20` on `Entity`. `O22` joins to `O23` on `Application`. **Neither sheet has a column that joins to the other, so the logical-to-physical trace does not exist.** Add a `Logical component` column to `O22`. `O21` is also one row per entity, named *"Product store"*, *"Bill of Material store"*, and stops at `E12` — `E13 Inventory Position` and `E14 Customer Order` have no logical component. `O22.Type` is wrong where it matters: `PDC-01 ERP Core database` is typed `File`, `PDC-04 SCADA / Historian database` `Relational` and `PDC-05 APS database` `Time series`, which has the historian and the planner the wrong way round.

How.

1. Add the join column to `O22` and populate all twelve rows.
2. Add logical components for `E13` and `E14`, and stop naming components after entities — a logical component groups entities, which is the point of the layer.
3. Correct the four `Type` values that are wrong, and record what each store actually is: relational, time-series, document or file.
4. Check `O38 Data Set`: six rows, of which `DST-02 Customer analytics set` and `DST-05 Risk measurement set` — *"exposure and position data"* — belong to a different industry altogether. A manufacturer's sets are the batch record, production reporting, traceability and the cost roll-up.

Look it up. Volume 1 chapter 10, *Moving data between systems*, and chapter 12, *The shape of a South African estate*, for what the physical layer usually turns out to be.

Signed by. `ROL-10 Domain Architect, Data` and `ROL-11 Domain Architect, Application` jointly.

Done when. A single query joins `O20` to `O23` through `O21` and `O22` with no manual step.

Shows up in. `RPT13 Orphan Analysis`, which can then find data components that belong to no application.

Goes wrong when. You add the column and populate it from row position. That is how the thirteen diagonal relationship sheets in this file were made.

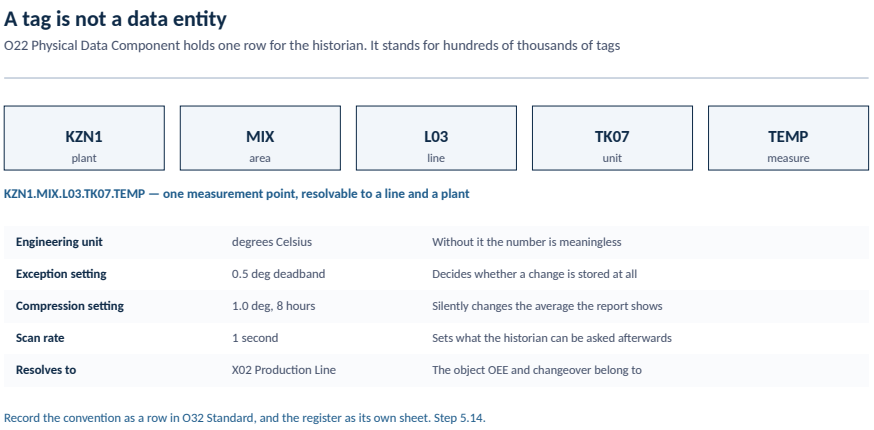


Figure 5.5 The namespace `022` cannot express. Volume 1 chapter 13, *Why a tag is not a data entity*, and *Exception and compression, and how they silently change a reported number*. Phase D needs this register; the tag and plant codes here are illustrative.

Step 5.14 — Declare the tag namespace

Produce. A written tag-naming convention and a register of what a tag is, where it lives, and how it relates to the entities in `O20`.

You need. `O22 Physical Data Component`, `A04 SCADA / Historian`, `A12 Plant Analytics`, the control engineer, and `X01` / `X02` from chapter 4.

Where it lands.

> **`O22 Physical Data Component` · columns `Ref`, `Physical data component`, `Application`, `Type`**
>
> Four columns, and **nothing that can express a tag namespace**. `PDC-04 SCADA / Historian database` is one row standing for hundreds of thousands of tags. A tag is not a data entity: it is a named measurement point with a plant, an area, a line, a unit, an engineering unit and a compression setting, and two of those settings silently change the number it reports. Record the convention as a `Standard` row in `O32 Standard` and the register as a separate sheet keyed on `plant.area.line.unit.measure`.

How.

1. Get the existing convention from the control engineer. There is always one, and it is usually inconsistent across plants, each commissioned by a different integrator.
2. Write the convention down: the segments, their order, the allowed values, and who may create a tag.
3. Record the engineering unit on every tag, and the exception and compression settings, stating plainly that the latter two change the reported value. Phase D will need this.
4. Map the tag namespace onto `X01` and `X02` so a tag resolves to a line and a line to a plant.

Look it up. Volume 1 chapter 13, *Why a tag is not a data entity*, then *Exception and compression, and how they silently change a reported number*; chapter 10, *Time-series and tag data, which no other*

industry in this series has, is the teaching half.

Signed by. The control engineer at each plant, countersigned by `ROL-12 Domain Architect, Technology`.

Done when. A tag name parses into its segments by rule, every tag has an engineering unit, and the convention is a row in `O32 Standard` with a conformance position.

Shows up in. `RPT12 Standards Conformance`, once the convention is a standard rather than a habit.

Goes wrong when. You model tags as data entities in `O20`. Fourteen entities become two hundred thousand, and the sheet that held the item master becomes unusable.

Closing Phase C for data

This phase produced one authoritative column for the golden source, a corrected master system on five entities, owners who are people, a classification that survives a commercial challenge, three new sheets and a tag convention. It did not produce a change log: there is no version column anywhere in this repository and no baseline to compare against. Raise that in Phase H.

Carry two numbers into the application chapter. `R05` says four applications consume everything; `ZZ Checks` says four read another's database directly. Those are the same fact seen from two sheets, and Phase C for applications is where they get reconciled.

THE REVIEWER'S ASK — BEFORE YOU SIGN THE DATA ARCHITECTURE OFF

Name the one system of record for the item master, and the person who approved it.

How long did the recall simulation take, and against which sheet?

Which entities still read `Absent` for lineage, and why is that acceptable?

Which attribute carries a quantity with no unit of measure?

What would catch it if `O20.Master system` and `O43.Golden source` diverged again?

THE CHAPTER IN TEN LINES

- O20 Master system and O43 Golden source name the same ten entities and disagree on nine of them.
- The one row they agree on, E01 Product, says the item master is mastered in SCADA / Historian, which an append-only historian cannot do.
- E06, E09 and E11 are all recorded as mastered in APS, a planning engine, and E13 in Supplier Portal.
- O39 Data Migration has MIG-01 migrating Product from PLM to SCADA / Historian, in progress.
- Nine of the ten distinct Owner values in O20 resolve to nothing; only Data Steward matches ROL-16.
- O20 Classification reads Public on E09 Supplier, E11 Quality Specification and E13 Inventory Position, and Restricted on E04 Work Order and E07 Work Centre.
- Six of fourteen entities read Lineage Absent, and they include the batch, the work order and the routing.
- R05 puts the application name in the Application column on its Master rows and a reference on its Consumer rows, so a join returns half the sheet.
- O36 and R19 hold the same eleven rows, a positional chain asserting that Equipment / Asset is composed of Supplier, one to one.
- O37 Data Attribute is ten entities by the same six attributes, with no unit of measure column, and E11 to E14 have no attributes at all.
- Nothing joins O21 to O22, so the logical-to-physical trace does not exist; O40 holds placeholder field names on all twelve rows.
- EXT declares X03 Bill of Material, X04 Batch / Lot and X05 Quality Specification and holds no instances, so three steps build sheets.
- Phantom is a property of the bill-of-material line, not of the item, and ownership of inventory cannot be retrofitted onto historic movements.

SOURCES FOR THIS CHAPTER

- **1.** Manufacturing_TOGAF_Metamodel.xlsx, sheet O20 Data Entity (fourteen rows; columns Ref, Data entity, Class, Definition, Classification, Owner, Master system, Lineage)
- **2.** Manufacturing_TOGAF_Metamodel.xlsx, sheet O43 Master Data Rule (ten rows, E01 to E10; column Golden source)
- **3.** Manufacturing_TOGAF_Metamodel.xlsx, sheet R05 Application manages Data En (28 instances; fourteen Master rows keyed on application name, fourteen Consumer rows keyed on reference)
- **4.** Manufacturing_TOGAF_Metamodel.xlsx, sheet O37 Data Attribute (sixty rows: ten entities by six attributes)
- **5.** Manufacturing_TOGAF_Metamodel.xlsx, sheets O21 Logical Data Component and O22 Physical Data Component (twelve rows each, no column joining them)
- **6.** Manufacturing_TOGAF_Metamodel.xlsx, sheets O36 Data Entity Relationship and R19 Data Entity relates to Data (the same eleven REL- rows)
- **7.** Manufacturing_TOGAF_Metamodel.xlsx, sheets O39 Data Migration and O40 Source To Target Mapping
- **8.** Manufacturing_TOGAF_Metamodel.xlsx, sheet O42 Metadata Element (fourteen rows; column Coverage)
- **9.** Manufacturing_TOGAF_Metamodel.xlsx, sheet O06 Requirement, rows REQ-007 to REQ-014, category Data, priority Must, status Approved
- **10.** Manufacturing_TOGAF_Metamodel.xlsx, sheet EXT Industry Extension (six declared types X01 to X06, no instances)
- **11.** Manufacturing_TOGAF_Metamodel.xlsx, sheets RPT05 Data Ownership, RPT13 Orphan Analysis and ZZ Checks
- **12.** Protection of Personal Information Act 4 of 2013, section 26 — prohibition on processing special personal information https://popia.co.za/section-26-prohibition-on-processing-of-special-personal-information/
- **13.** National Regulator for Compulsory Specifications Act 5 of 2008, consolidated text (SAFLII), section 20 https://www.saflii.org/za/legis/consol_act/nrfcsa5o2008543/

CHAPTER 6

Application Architecture

Twelve applications, four of them owned by the functions that must audit them, and one row describing two different machines.

Phase C's application half answers four questions: what applications exist, what each one is for, which of them is the system of record for what, and how they are wired together. The repository has sheets for all four. Three of the four are populated by a rule rather than by evidence, and the fourth — `O23 Application Component` — is populated by evidence that was entered into the wrong columns.

Kestrel Industrial Group is an invention, as is volume 1's Karoo Industrial Group. Spell `R04 Business Service supported ` with the trailing space the tab actually carries; this chapter says so once.

Twelve applications is a small estate for a three-plant manufacturer. Read it as the list somebody could name in a meeting, not as the estate. The gap between those two lists is step 6.6, and it is usually the most expensive finding in the chapter.

THE TWELVE CLASSES AGAINST THE TWELVE O23 ROWS

Volume 1 ch 11	Application capability	O23 row
M-01	Transactional and financial record	A01 ERP Core
M-02	Item and bill-of-material mastering	A02 PLM
M-03	Demand and supply planning	A05 APS
M-04	Detailed scheduling	A05 APS, same row
M-05	Production execution	A03 MES
M-06	Process control	A04, half a row
M-07	Process data historisation	A04, the other half
M-08	Product lifecycle and change	A02 PLM / A09 CAD
M-09	Quality management and laboratory	A07 QMS, no LIMS
M-10	Warehouse and logistics execution	A06 WMS, no TMS
M-11	Asset and maintenance management	A08 CMMS / EAM
M-12	Shop-floor data capture	**NO ROW**
--	Integration platform	**NO ROW**
--	Label and marking printing	**NO ROW**
--	Time and attendance, access	**NO ROW**

Four classes have no row at all. Three more share a row with another class.

Figure 6.1 Volume 1 chapter 11's twelve application capabilities against the twelve rows of `O23 Application Component`. Four classes have **no row at all** and three share a row with something else. The absence is the finding, not the count [1].

Start with the thing that cannot be delegated

Step 6.1 — Give every application a named individual owner

Produce. `O23` with an `Owner` value that is a person's role, not an assurance function.

You need. `O09 Organization Unit`, `O11 Role`, `O05 Principle` (`PRN-13`), and forty minutes with the Chief Information Officer. Volume 1 chapter 5 for who exists in a plant.

Where it lands.

> **`O23 Application Component`**
>
> Ten columns: `Ref`, `Application`, `Type`, `Description`, `Disposition`, `Criticality`, `Health`, `In service`, `Users`, `Owner`. `Owner` holds organisation-unit names from `O09`, not roles: `A01 ERP Core` reads `Legal`, `A04 SCADA / Historian` and `A07 QMS` read `Internal Audit`, `A05 APS` reads `Risk & Compliance`, `A02 PLM` and `A11 Track & Trace` read `Data & Analytics`. Overwrite the column with a role from `O11 Role`, one row at a time.

Three of those twelve rows name a function whose job is to audit or govern the system it is recorded as owning. Internal audit cannot own the quality management system and then test it. And `O05 PRN-13 Data Ownership is Individual` states that an owner is a person and not a committee — so each row is wrong twice: wrong class of thing, and against an approved principle. `O05` does the same to itself, carrying `Architecture Board` in all twenty-four of its own `Owner` cells [1].

How.

1. Write the real owners out on paper first, from `O11 Role`. Nine of the eighteen roles in `O11` are architects and executives; none runs an application.
2. Where no role in `O11` fits — and for `A03 MES` and `A04 SCADA / Historian` none does — add it in the same sitting. A missing role is the finding, not an obstacle.
3. Replace the value. Do not add a second column; `Owner` beside `Real owner` will be read in the wrong order within a month.
4. Record the three audit-conflict rows separately for the Architecture Board. Reversing them needs a decision, not a correction.
5. Ask each named owner to confirm in writing. An owner who has not agreed is a label.

Signed by. Chief Information Officer (`ORG-05`), countersigned by the named owner per row.

Done when. Every `Owner` value resolves to a row in `O11 Role`, and no assurance function appears.

Shows up in. `RPT02 Application Portfolio`, which reprints `Owner` without change. If the step went well the column reads as eleven or twelve distinct roles, with no function appearing twice.

Goes wrong when. Somebody fills the column with the name of the vendor's account manager, or with the delivery lead who implemented the system in 2014 and left.

Look it up. Volume 1 chapter 5, *Who to see first*, for the roles a plant actually has.

BEFORE YOU TOUCH O23, ANSWER THESE

Who, by name, decides whether MES is upgraded or replaced? If two people can answer yes, neither of them owns it.

`A07 QMS` is owned by `Internal Audit`. Who signed that off, and what did they think the word owner meant?

There are eighteen roles in `O11 Role` and no plant manager among them. Which of your twelve applications has an owner who has stood on a production line this year?

Step 6.2 — Rebuild Disposition against Health and In service, and name the successor

Produce. A `Disposition` value per application that a funded work package can be traced to, and a named successor for every `Replace` and every `Consolidate`.

You need. Step 6.1. `O30 Work Package`, `O54 Architecture Alternative`, `O49 Resilience Tier`. Volume 1 chapter 12 for what each class is for.

Where it lands.

> **`O23 Application Component`**
>
> `Disposition` holds `Retain` (4), `Replace` (4), `Invest` (2), `Consolidate` (1), `Retire` (1). `Health` holds `Poor` (4), `At risk` (4), `Adequate` (4) — an exact three-way split, which is a generation pattern and not an assessment, and the scale has no healthy value in use at all. There is no `Successor`, `Target application` or `Decision` column, so `Replace` is an intention with nowhere to point. Add `Successor` and `Decision ref` and fill them.

The two rows that should stop the room:

Ref	Application	In service	Health	Criticality	Disposition
`A04`	SCADA / Historian	2012	Adequate	**Critical**	**Replace**
`A03`	MES	**2010**	**Poor**	Medium	**Retain**
`A02`	PLM	2020	Poor	Medium	Consolidate
`A06`	WMS	2017	Poor	High	Replace

`A04` is the only `Critical` application in the file and the repository proposes to replace it with nothing named. `A03` has run the shop floor since 2010, is assessed `Poor`, and is to be retained with no work package against it. Those are the same defect: `Disposition` was filled without reading `Health`, and `Criticality` without reading either.

How.

1. Take the five dispositions in the sheet as the allowed values. Do not invent a sixth.
2. For each of the four `Replace` rows and the one `Consolidate` row, write the successor's name in the new `Successor` column, or change the disposition. An unnamed replacement is a `Retain` with an opinion attached.
3. Re-score `Criticality` by asking one question per row: if this stops at 06:00 on a Monday, does product stop leaving the plant? `A03 MES` answers yes. `A05 APS` does not.
4. Reconcile against `O30`. Four `Replace` rows and no work package naming an application is a roadmap that does not deliver the portfolio.
5. Put `A03` on the Architecture Board agenda as a single item. Sixteen years, `Poor`, `Retain` is a decision somebody has to own out loud.

Signed by. Architecture Board (`GB-01`), on the recommendation of the Domain Architect, Application (`ROL-11`).

Done when. No `Replace` or `Consolidate` row has an empty `Successor`, and every `Criticality` value has a one-line stoppage test written beside it.

Shows up in. `RPT02 Application Portfolio`. If the step went well, `Critical` appears on three or four rows, not one, and `Retain` never sits beside `Poor`.

Goes wrong when. The portfolio is re-scored to match the funding that already exists, so the four unfunded `Replace` rows quietly become `Retain` and the finding disappears.

Look it up. Volume 1 chapter 12, *Six questions, asked of every class*.

Step 6.3 — Replace the Users column with a business-value score

Produce. A scored portfolio position per application, on Form 6-A, that does not depend on `Users`.

You need. Step 6.2. `O15 Business Service`, `O13 Business Capability`, `O04 Measure`.

Where it lands.

> **`O23 Application Component`**
>
> `Users` holds a headcount per application. It cannot carry business value. `A05 APS` reads 266 against `A01 ERP Core` at 196 and `A03 MES` at 157 — and no plant resembles that. A finite-capacity planning engine is used by a handful of planners; an execution system is used by every operator on every shift. The twelve values also sum to 2,494, against 1,896 people in `O09 Organization Unit` and 1,265 in `O12 Location` — two totals that do not agree with each other either. Leave the column where it is, stop reading it, and score value on Form 6-A instead.

How.

1. Score on two axes only: business value, and technical fitness. Four to six criteria per axis, each scored 1 to 5, each with a written definition.
2. Draw business value from `O15 Business Service` and `O13 Business Capability` criticality, not from headcount.
3. Draw technical fitness from `Health`, `In service`, the `O49` row, the count of `Unmanaged` rows in `R06` touching the application, and the `O27` support position of what it runs on.
4. Score in a room with the owner from step 6.1 present. A score made by one architect alone gets argued at the gate instead of at the workshop.
5. Keep the sheet. The score is a dated artefact and next year's comparison is the point of it.

Signed by. Domain Architect, Application (`ROL-11`), with each application owner on their row.

Done when. Every application has both scores and a written criterion behind each, and no score cites `Users`.

Shows up in. Nowhere yet — no report reads a score, because no column holds one. It feeds step 6.2's dispositions and the Phase E alternatives in `O54`.

Goes wrong when. The axes become cost and risk, which produces a chart nobody can act on because every system is expensive and every system is risky.

Look it up. Volume 1 chapter 7, *So the first deliverable is a measure definition register*.

Form 6-A — Application portfolio score

Two scores per application, 1 to 5 on each axis, with the criterion that produced the score. One row per `023` row; twelve rows for this estate. Does not transcribe into any existing column — `023` has no score column, so keep the form and take the disposition from it. Signed by the Domain Architect, Application (`ROL-11`).

Ref	Application	Business value 1-5	Criterion used	Technical fitness 1-5	Criterion used
A03	EXAMPLE — delete this row	5	Plant cannot dispatch an order without it	1	2010, Poor, 7 unmanaged interfaces

The two manufacturing arguments

Step 6.4 — Split A04 SCADA / Historian into two rows

Produce. Two `023` rows where there is one, and the relationship rows that follow.

You need. Step 6.1. `027 Physical Technology Compone` , `022 Physical Data Component` , `049 Resilience Tier` . The process control engineer, who is not in `011 Role` .

Where it lands.

> **O23 Application Component**
>
> `A04 SCADA / Historian` is one row, `Type` = `Platform`, `Criticality` = `Critical`, `Disposition` = `Replace`. Add a row — the next free ref is `A13` — and split: supervisory control on one, process data historisation on the other. Then split `A04`'s dependants: `ISS-04` in `O24`, `PDC-04` in `O22`, `ABB-04` in `O45`, `RES-04` in `O49`, `PTC-01` and `PTC-03` in `O27`, and every `R05`, `R06`, `R07`, `R18`, `R20` and `R25` row naming `A04`.

A supervisory control system and a process historian are different classes: different grain — a tag position against a time series; different licensing — tag count against point count and retention; different failure modes — a historian outage loses history, a control outage stops the plant; and different owners, because process control engineering owns the first and will not let anyone else near it. One row makes every number about either of them wrong, including the one that matters most: `RES-04` is the only internally coherent row in `O49 Resilience Tier` (`Tier 1`, one hour, `Tested`), and it is coherent for the control half and indefensible for the historian half [1].

How.

1. Create `A13` and name it for the historian, keeping `A04` as the supervisory control system.
2. Set `Criticality` separately. Control is `Critical`; a historian that is down for an hour costs reporting, not production — unless the plant is regulated, in which case say so in `Description`.
3. Set `Disposition` separately. The existing `Replace` almost certainly belonged to one half and was applied to both.
4. Repoint the dependent rows listed above. Expect twelve to fifteen edits.
5. Write one line in `Description` on each row saying what the other row is, so nobody re-merges them.

Signed by. Architecture Board (`GB-01`). This is a model change, not a data correction.

Done when. No sheet in the repository carries a row describing control and historisation together, and `O49` holds two rows with different recovery objectives.

Shows up in. `RPT02 Application Portfolio` gains a row; `RPT13 Orphan Analysis` should not change, because both halves are reachable from `R04` once step 6.7 is done.

Goes wrong when. The split is made in `O23` and nowhere else, so the twelve relationship rows still name `A04` and the historian becomes an orphan.

Look it up. Volume 1 chapter 13, *DCS, SCADA, and the difference that survives*, and *Historians, and the number that changes itself*.

Step 6.5 — Write the ERP/MES boundary paper and get it signed

Produce. A one-page signed statement of which system is the record for what, across the transactional and execution boundary, on Form 6-B.

You need. Steps 6.1 and 6.4. `O20 Data Entity`, `O43 Master Data Rule`, `R05`. The plant manager, the production planner and the finance controller, in one room.

Where it lands.

> **`O20 Data Entity`**
>
> Eight columns: `Ref`, `Data entity`, `Class`, `Definition`, `Classification`, `Owner`, `Master system`, `Lineage`. `Master system` is where the boundary decision is recorded, and it currently reads `APS` for `E04 Work Order`, `E06 Plant / Site`, `E09 Supplier` and `E11 Quality Specification`, and `SCADA / Historian` for `E01 Product`. `O43 Master Data Rule`'s `Golden source` disagrees with `O20` on nine of the ten entities they share. One of the two is the register; decide which and strike the other, or the boundary paper has no landing place.

This is the most common integration argument in manufacturing architecture, and the repository has `A01 ERP Core`, `A03 MES`, `A05 APS` and `A12 Plant Analytics` with no sheet recording where the line was drawn. It is not abstract. It decides whether the work order is released by finance or by the plant, whether stock moves on confirmation or on receipt, whether the routing lives with engineering or with execution, and who owns the standard time the cost of goods is built on. Settle it once, in writing, and re-open it only by change request.

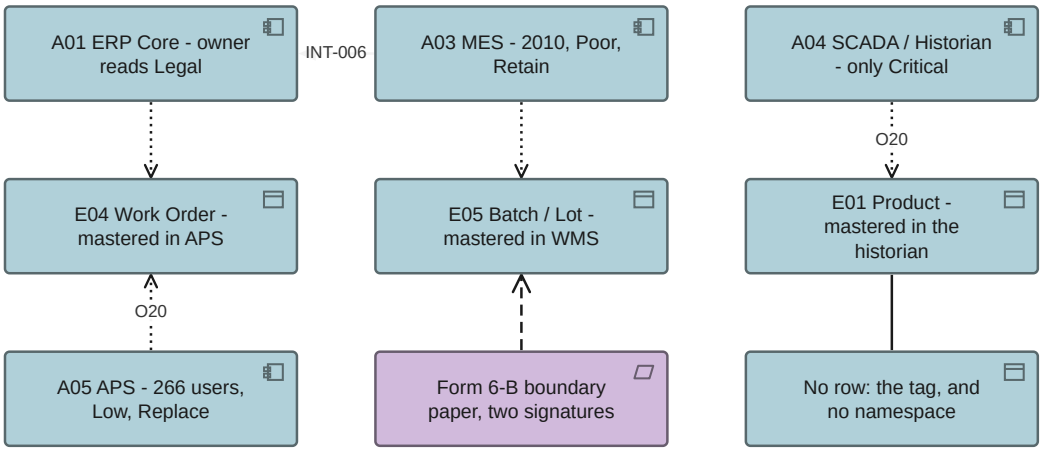


Figure 6.2 The ERP/MES boundary as the repository records it. `R06` holds **one** interface between `A01` and `A03`, running MES to ERP only, and `O20.Master system` puts the work order in the planning engine and the item master in the historian. Form 6-B is the signed statement that replaces all three [1].

How.

1. Take the fourteen entities in `O20` and ask, per entity: which system creates it, which amends it, which is quoted when two disagree. Write the third answer into `Master system`.

2. Settle the six boundary entities first — `E01 Product`, `E02 Bill of Material`, `E03 Routing / Recipe`, `E04 Work Order`, `E05 Batch / Lot`, `E13 Inventory Position`. The rest follow.
3. Strike `SCADA / Historian` from `E01 Product` in the same sitting. A historian is append-only; it cannot master anything. Step 6.9 does the rest of that repair.
4. Record the decision in `O60 Governance Decision` with the body that took it. `O60` has no date column — put the date in the decision text.
5. Publish the one page to the plant, not to the architecture team. The people who will breach it are on shift.

Signed by. Architecture Board (`GB-01`) and the Chief Operating Officer (`ORG-07`). Without operations, the paper is an opinion.

Done when. Every entity in `O20` has one `Master system`, `O43`'s `Golden source` agrees with it or has been struck, and the page carries two signatures.

Shows up in. `RPT05 Data Ownership`, which reprints `Master system` unchanged. If the step went well, no planning engine and no historian appears in that column.

Goes wrong when. The paper is written by the architecture team and circulated for comment. Nobody comments, nobody is bound, and the boundary is re-litigated at the first integration design.

Look it up. Volume 1 chapter 11, *The two questions that decide every manufacturing application architecture*, and chapter 12, *The boundary war, and how it actually gets settled*. Volume 1 chapter 17 names the view to draw: *The two you have to invent: the ERP/MES boundary view*.

THE BOUNDARY, BEFORE YOU WRITE THE PAPER

When the plant and finance disagree on how much was made last Tuesday, which system is quoted, and who decided that?

`E01 Product` is mastered in a historian. How did that row get written, and what else was written the same way?

Form 6-B — The ERP/MES boundary paper

One row per data entity across the transactional and execution boundary. Transcribes into `020 Data Entity` — `Ref`, `Data entity` and `Master system` are that sheet's own columns; the other three are the working columns and stay on the form. Fourteen rows for this estate. Signed by the Architecture Board and the Chief Operating Officer.

Ref	Data entity	Creates it	Amends it	Master system	Quoted when they disagree
E04	EXAMPLE — delete this row	ERP Core	MES		

Step 6.6 — Log the application classes the estate has no row for

Produce. An undeclared-systems log, on Form 6-C, naming every class volume 1 identifies that `023` has no row for, with a decision against each.

You need. Step 6.1. The twelve classes in volume 1 chapter 11. Two hours on each plant floor, and the shift supervisor's own spreadsheets.

Where it lands.

> **`O23 Application Component`**
>
> Twelve rows. Against volume 1 chapter 11's twelve application capabilities, `O23` has no row for shop-floor data capture at all, and no row for a laboratory information management system — `A07 QMS` is a quality management system, which is a different class. There is also no transport management, no label and marking printing, no time-and-attendance or access control, no integration platform, and no simulation or twin. `O46 Technology Building Block` holds `TBB-02 Industrial edge compute`, `TBB-04 Unified namespace / MQTT` and `TBB-05 Digital twin` as aspirations with no application row beneath them. Add a row per class the walkdown finds; the next free ref after step 6.4 is `A14`.

The integration absence is the sharpest of them. Eight rows in `R06` read `Managed`, `O28 Platform Service` holds `PSV-02 API gateway`, and `O65 Enforcement Point` records that point's position as *Product not yet selected* — so eight interfaces are managed by a platform that does not exist, and `O41 SEC-05 Interface logging` reads `In place` at the same unchosen gateway.

The time-and-attendance absence matters for a different reason: in a manufacturer, the personal data worth protecting is overwhelmingly employee and biometric, and the system holding it is not in this portfolio. Step 7.10 in the next chapter picks that up.

How.

1. Walk the three production locations in `O12 Location` and the two logistics ones. Ask one question at each station: what do you type into, and what do you print?
2. Write down every spreadsheet with a macro, every label printer with its own software, and every machine with a vendor's own client on a terminal.
3. For each, decide: bring into `O23`, replace with an existing row's function, or retire. Record the decision and a date.
4. Count what you found. Fewer than ten on a three-plant estate means the walkdown was too short.
5. Hand the vendor-remote-access findings to chapter 7. They are a zone question, not a portfolio one.

Signed by. Chief Information Officer (`ORG-05`) and the plant manager per site — a role you will have had to add to `O11 Role` in step 6.1.

Done when. Form 6-C has a decision and a date against every row, and `O23`'s row count has risen.

Shows up in. `RPT02 Application Portfolio` and `RPT13 Orphan Analysis`. Expect the orphan count to rise before it falls; a newly declared system has no relationships yet.

Goes wrong when. The log is built from a software asset-management export. That tool sees what corporate IT installed, which is the half of the estate that was never the problem.

Look it up. Volume 1 chapter 11, *The twelve*, and chapter 12, *Shop-floor data collection* and *LIMS and QMS*. Chapter 12's *The shape of a South African estate* says what to expect to find.

Form 6-C — Undeclared systems log

One row per system found on a plant walkdown that has no row in `O23 Application Component`. Feeds `O23` — `Application`, `Type` and `Description` are its columns; `Where found`, `Who runs it` and `Decision` stay on the form until the row is created. Ten to twenty rows on a first pass; continue on a second copy of this form. Signed by the Chief Information Officer and the plant manager.

Where found	Application	Type	Who runs it	What it holds	Decision and date
Northern Plant, line 2	EXAMPLE — delete this row	Custom	Shift supervisor	Downtime reasons	Bring into O23, Nov

The relationship sheets, rebuilt

Step 6.7 — Rebuild R04 and R07 from evidence

Produce. Two relationship sheets whose rows survive a question.

You need. Steps 6.1 to 6.6. `O15 Business Service`, `O27 Physical Technology Compone`, `O48 Environment`. The run team's deployment list.

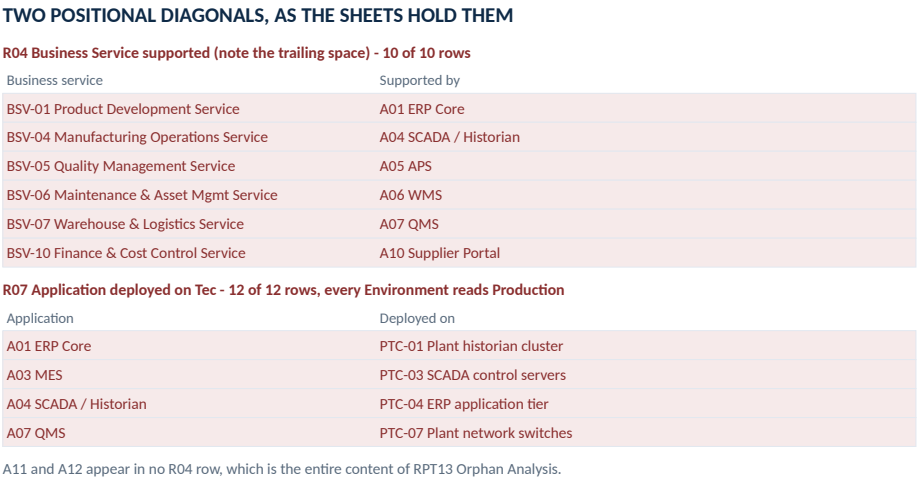
Where it lands.

R04 Business Service supported

Four columns: `Business service`, `Service name`, `Application`, `Application name`. Ten rows, and the join is positional — `BSV-01` to `A01`, `BSV-02` to `A02`, onward to `BSV-10` to `A10`. So `BSV-05 Quality Management Service` is supported by `A05 APS`, `BSV-06 Maintenance & Asset Management Service` by `A06 WMS`, and `BSV-04 Manufacturing Operations Service` by `A04 SCADA / Historian` rather than by the execution system. `A11` and `A12` appear in no row, which is the whole content of `RPT13 Orphan Analysis`. Delete all ten rows and rebuild.

R07 Application deployed on Tec

Five columns: `Application`, `Application name`, `Platform component`, `Component name`, `Environment`. Twelve rows, the same positional join — `A01 ERP Core` deployed on `PTC-01 Plant historian cluster`, `A03 MES` on `PTC-03 SCADA control servers`, and `A04 SCADA / Historian` on `PTC-04 ERP application tier`. Every `Environment` reads `Production` while `048 Environment` holds four environments. Delete all twelve and rebuild.



TWO POSITIONAL DIAGONALS, AS THE SHEETS HOLD THEM

R04 Business Service supported (note the trailing space) - 10 of 10 rows

Business service	Supported by
BSV-01 Product Development Service	A01 ERP Core
BSV-04 Manufacturing Operations Service	A04 SCADA / Historian
BSV-05 Quality Management Service	A05 APS
BSV-06 Maintenance & Asset Mgmt Service	A06 WMS
BSV-07 Warehouse & Logistics Service	A07 QMS
BSV-10 Finance & Cost Control Service	A10 Supplier Portal

R07 Application deployed on Tec - 12 of 12 rows, every Environment reads Production

Application	Deployed on
A01 ERP Core	PTC-01 Plant historian cluster
A03 MES	PTC-03 SCADA control servers
A04 SCADA / Historian	PTC-04 ERP application tier
A07 QMS	PTC-07 Plant network switches

A11 and A12 appear in no R04 row, which is the entire content of RPT13 Orphan Analysis.

Figure 6.3 `R04` and `R07` are row-number joins, not evidence. The quality service comes to be supported by the planning engine, the execution system runs on the control servers, and the historian runs on the ERP tier. Six and four rows of each are shown [1].

How.

1. Take `015`'s ten business services one at a time and ask the owner which applications are needed to deliver each. Expect two to four per service, not one.
2. Expect `A03 MES` against three or four services and `A09 CAD / CAE` against one. A one-to-one map means the sheet was generated.
3. For `R07`, take the deployment per application from the run team and expect several rows per application.
4. Fill `Environment` from `048`'s four values. Twelve production-only rows in a four-environment estate means three environments nobody can see.

5. Flag every application whose component reads `Past support` in `027`. That list opens chapter 7.

Signed by. Domain Architect, Application (`ROL-11`) for `R04` ; Chief Information Officer (`ORG-05`) for `R07` .

Done when. No row in either sheet survives because its row number matched, and `R04` 's row count has risen above ten.

Shows up in. `RPT13 Orphan Analysis` , which is the only report in the file that performs a join, and tests applications against `R04` alone. If the step went well, `RPT13` reports no application orphans.

Goes wrong when. `R04` is rebuilt one-to-one with better pairings. The improvement is real and the shape is still wrong: a business service that needs one application was never a service.

Look it up. Volume 1 chapter 17, *Application co-operation*, for the view these two sheets feed.

Step 6.8 — Rebuild R20 and R25, and let an application sit at more than one site

Produce. `R20` and `R25` rebuilt, and a written note on what `O12 Location` cannot say.

You need. Step 6.7. `O12 Location` , `O18 Product` .

Where it lands.

> **`R20 Application located at Loca`**
>
> Four columns: `Application` , `Application name` , `Location` , `Location name` . Twelve rows against eight locations, so the join does not merely run down the diagonal — it wraps. `A01` to `LOC-01` , `A08` to `LOC-08` , then `A09` back to `LOC-01` and `A12` to `LOC-04` . Every application sits at exactly one place, which for `A01 ERP Core` at `Head Office` and `A03 MES` at `Coastal Plant` alone is false: two of the three plants in `O12` then run no execution system. Delete and rebuild with one row per application per site.

`R25 Application supports Produc` holds four rows on a stride of three — `PRD-01` to `A01` , `PRD-02` to `A04` , `PRD-03` to `A07` , `PRD-04` to `A10` — so `PRD-02 Spare Parts` is supported by the historian and `PRD-03 Service Contracts` by the quality system. Four products and twelve applications cannot produce a four-row sheet honestly.

`O12 Location` has no `Line count` , no `Shift pattern` and no `Ownership` column, so it cannot say how many lines a plant runs or on what pattern — the fact every capacity, changeover and effectiveness number depends on. It also gives `LOC-08 Cloud Region` 261 people, more than any plant. Write both as findings against the Phase B owner.

How.

1. Rebuild `R20` from the run team's deployment list and the walkdown of step 6.6. Expect thirty to forty rows for twelve applications and five physical sites.

2. Drop `LOC-08 Cloud Region` from `R20` unless the row is about a hosting region, and say so in the row.
3. Rebuild `R25` by asking which applications a product cannot be made or sold without. Expect three to six per product.
4. Record `O12` 's four missing columns in `O29 Gap` as one gap, domain `Business` .

Signed by. Chief Information Officer (`ORG-05`), with the plant manager per site.

Done when. Every production location in `O12` has at least one execution-system row in `R20` , and `R25` 's row count exceeds the number of products.

Shows up in. No report reads `R20` or `R25` . Both feed the Phase E and Phase F work, so an error here surfaces as a scoping miss months later.

Goes wrong when. The cloud region is treated as a site and the architecture ends up with an application estate that runs nowhere physical.

THE JOINS YOU WERE HANDED

How many of the ten `R04` rows survive contact with the service owner?

`A03 MES` sits at one of the three plants. What runs the other two, and is it in `O23` at all?

Step 6.9 — Repair R05's two key schemes

Produce. `R05` with one key scheme, and `E01 Product` no longer mastered in a historian.

You need. Step 6.5's boundary paper. `O20` , `O43` .

Where it lands.

> **`R05 Application manages Data En`**
>
> Five columns: `Application` , `Application name` , `Data entity` , `Entity name` , `Role` . Twenty-eight rows, fourteen `Master` and fourteen `Consumer` . The `Master` rows put the application **name** in the `Application` column — `SCADA / Historian` , `PLM` , `Plant Analytics` , `APS` — where the `Consumer` rows correctly put a ref. A join on `Application` therefore returns half the sheet, and the half it returns is the consumer half. Overwrite the fourteen `Master` rows with the ref.

This is the defect that breaks `00 Index`'s own promise, which reads: every relationship row names both ends by reference and by name, so a broken link is visible without a lookup. Fourteen rows name one end by name twice.

The content is as wrong as the keys. `E01 Product` is mastered in `SCADA / Historian`; `E03 Routing / Recipe` in `Plant Analytics`, a reporting platform; `E04`, `E06`, `E09` and `E11` in `APS`, a planning engine. And the `Consumer` half is a stride of three — `A01`, `A04`, `A07`, `A10` repeating down fourteen rows — so `E14 Customer Order` is consumed by the historian and nothing else, and `A03 MES` consumes nothing at all.

How.

1. Overwrite the fourteen `Master` rows' `Application` column with the ref from `O23`. Five minutes.
2. Replace the master assignments with step 6.5's boundary paper, entity by entity.
3. Delete the fourteen `Consumer` rows and rebuild from the interface catalogue: if an application reads an entity across an interface in `R06`, it is a consumer of it.
4. Expect forty to seventy rows when done. Fourteen consumers across fourteen entities is a stride, not an estate.

Signed by. Chief Data Officer (`ORG-06`) through the Data Governance Council (`GB-03`).

Done when. Every `Application` cell in `R05` matches a `Ref` in `O23`, and no historian or planning engine holds a `Master` row.

Shows up in. `RPT05 Data Ownership` reads `O20` rather than `R05`, so the repair is invisible in the reports until `O20.Master system` is changed with it. Change both.

Goes wrong when. The keys are fixed and the content is not, so the sheet joins cleanly to the wrong answer — which is worse, because now a report can be built on it.

Look it up. Volume 1 chapter 9, *The item master, and why it is the hardest master data object in any industry*.

Step 6.10 — Triage the thirty interfaces, and fix the rows where Type fights Frequency

Produce. `R06` with every row's carrier and cadence consistent, and a triage decision against each of the twenty-two unmanaged rows.

You need. Step 6.7. `O28 Platform Service`, `O41 Security Control`, `O65 Enforcement Point`.

Where it lands.

R06 Application interfaces with

Nine columns: `Ref`, `Source`, `Source name`, `Target`, `Target name`, `Type`, `Frequency`, `Managed`, `Logged`. Thirty rows, `INT-001` to `INT-030` — the only reference space in the file living on a relationship sheet. `Type` holds `Batch` (8), `Event` (8), `API` (6), `File` (4) and `Direct DB read` (4), which mixes four carriers with one anti-pattern. `ZZ Checks` reports 22 unmanaged, 21 unlogged and 4 direct reads, and all three are correct.

Six rows have a `Type` that contradicts their `Frequency`:

Ref	Source	Target	Type	Frequency
`INT-003`	PLM	CMMS / EAM	Batch	Real time
`INT-004`	PLM	CAD / CAE	File	Real time
`INT-008`	MES	APS	Event	Nightly
`INT-018`	QMS	WMS	Batch	Real time
`INT-024`	CAD / CAE	APS	API	Nightly
`INT-030`	Plant Analytics	Track & Trace	Event	Nightly

A batch is not real time and an event is not nightly. One of the two cells is wrong in each row, and which one is wrong decides whether there is a latency problem or a reporting error.

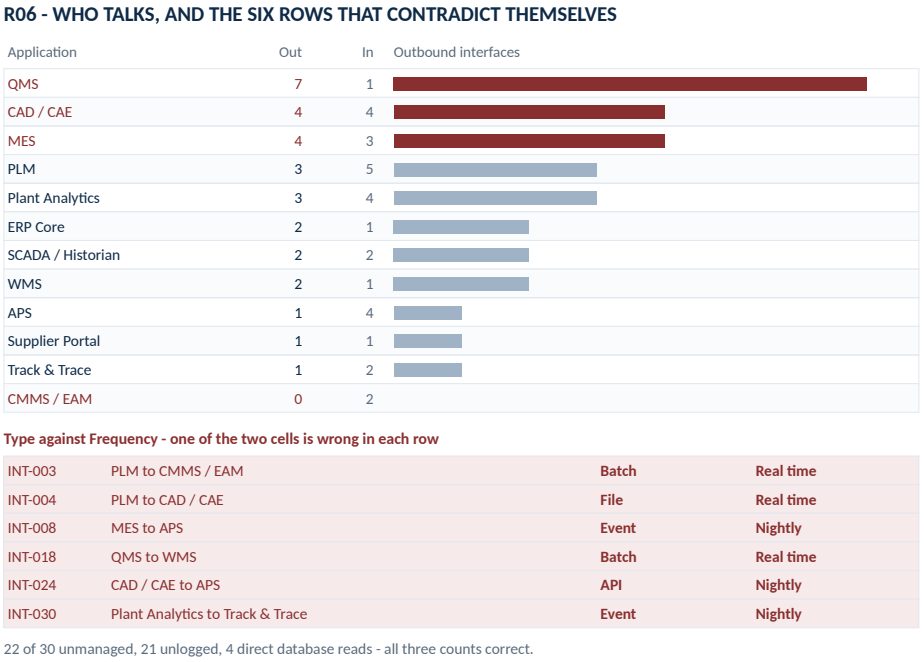


Figure 6.4 `R06 Application interfaces with`, by out-degree. **`A07 QMS` is the busiest source in the estate with seven outbound interfaces, every one unmanaged**, and `A08 CMMS / EAM` has none. In a plant the execution system is the hub. Six rows carry a `Type` that cannot be true at their `Frequency` [1].

The topology is as telling as the rows. `A07 QMS` has seven outbound interfaces and one inbound — the busiest source in the estate, all seven unmanaged — while `A01 ERP Core` has three in total and `A08 CMMS / EAM` none outbound at all. There is exactly one interface between the ERP and the execution system, `INT-006`, running MES to ERP only: nothing records order release in the other direction. And `O06 Requirement` carries `REQ-015` to `REQ-020`, *integration managed*, for six applications — and `QMS` is not among them.

How.

1. Fix the six contradiction rows first, with the team that runs each interface. Ten minutes each.
2. Reclassify the four `Direct DB read` rows as what they are — an unmanaged dependency on another application's internal schema — and raise one `O29 Gap` row covering all four.
3. Triage the twenty-two unmanaged rows into three buckets: route through the gateway, replace with a published service, retire. Write the bucket into a new `Decision` column.
4. Resolve the gateway question before you promise anything to bucket one. Eight rows already read `Managed` against an enforcement point whose product has not been chosen.
5. Add the missing integration requirement for `A07 QMS` to `O06`, and for every application added in step 6.6.

Signed by. Domain Architect, Application (`ROL-11`), with the Chief Information Officer (`ORG-05`) on the gateway decision.

Done when. No row has a `Type` and `Frequency` that cannot both be true, and every `Unmanaged` row carries a bucket and a date.

Shows up in. `RPT04 Interface Catalogue` and `ZZ Checks`. If the step went well the unmanaged count falls and the footer, which is a typed cell and not arithmetic, has to be retyped — which is step 6.13's point.

Goes wrong when. The twenty-two unmanaged rows are all assigned to bucket one and a platform is bought. Six months later they are unmanaged and there is a platform.

Look it up. Volume 1 chapter 13, *Integration, as it really is*, and within it *Polling, events and the queue that must not fill*.

Step 6.11 — Publish the information system services

Produce. `O24` with a `Style` and `Published` value per row that describes a real interface, and a consumer named for each.

You need. Step 6.10. `O28 Platform Service`, `O45 Application Building Block`.

Where it lands.

> **`O24 Information System Service`**
>
> Six columns: `Ref`, `Information system service`, `Application`, `Description`, `Style`, `Published`. Twelve rows, one per application, every name formed as the application's name plus *Service* and every `Description` reading *Service exposed by* the application. `Style` holds `File`, `Batch`, `Event` and `API`; `Published` holds `Published` on five rows and `Unpublished` on seven. No column records the consumer, the contract, the schema or the owning team, so a published service here is an assertion with nothing behind it.

The sheet is also unreachable: `O24` appears in no relationship sheet, so no information system service connects to a business service, a building block or a requirement. One application, one service is the wrong grain anyway — `A01 ERP Core` exposes dozens.

How.

1. Replace the twelve generated rows with the services that actually have a consumer. Take them from `R06`'s targets after step 6.10.
2. Name each service for what it does, not for the system that exposes it: *order release*, *goods receipt confirmation*, *specification lookup*.
3. Set `Published` from one test: is there a schema a consuming team can read without asking anybody? If not, it is `Unpublished`, whatever the sheet says.
4. Add a `Consumer` column and fill it. A service with no consumer is a plan.
5. Bind each row to an `O45` application building block, so the acceptance criterion has something to test.

Signed by. Domain Architect, Application (`ROL-11`).

Done when. Every row names a consumer, and `Published` is true only where a schema exists.

Shows up in. No report reads `O24`. It feeds the design gate `GT-1`, so an unpublished service recorded as published is the sort of thing that passes a gate and fails an integration.

Goes wrong when. The sheet is kept as it is because twelve rows look complete. Twelve rows that each describe one system are a list of systems with the word service appended.

Step 6.12 — Complete O45 and make the acceptance criteria testable

Produce. `O45` covering every application, with acceptance criteria a gate can fail.

You need. Steps 6.4 and 6.6. `O55 Building Block`, `O58 Governance Gate`, `O59 Compliance Assessment`.

Where it lands.

> **O45 Application Building Block**
>
> Five columns: `Ref`, `Application building block`, `Application`, `What it must do`, `Acceptance criteria`. Ten rows, `ABB-01` to `ABB-10`, covering `A01` to `A10` — so `A11 Track & Trace` and `A12 Plant Analytics` have no building block, and neither will the rows added in steps 6.4 and 6.6. Every `What it must do` reads *at target disposition*; every `Acceptance criteria` reads *No unmanaged interface to or from* the application. Add the missing rows and rewrite both columns.

The criterion is not useless — it is testable against `R06`, and it is the only acceptance criterion in the application domain that is. But one criterion repeated ten times tests one thing ten times. A building block needs a criterion derived from the requirement that called for it, and `R09 Requirement satisfied by Bu` is a positional diagonal, so `REQ-021 Product Development measurable` is satisfied by `ABB-01 ERP Core target` and the five technology requirements `REQ-027` to `REQ-031` are satisfied by application building blocks [1].

How.

1. Add `ABB-11` onward for every application without a row, including the historian from step 6.4.
2. Rewrite `What it must do` as a behaviour: what the application must be able to do that it cannot do today.
3. Write two to four acceptance criteria per block, each one a test with a pass and a fail. Keep the unmanaged-interface criterion as one of them.
4. Rebuild `R09`'s application rows from the requirements, not from row order.
5. Note that `O59 Compliance Assessment` is empty and `R24` is empty in consequence, so no criterion in this sheet has ever been tested at a gate. That is chapter 7's step 7.11.

Signed by. Architecture Function (`GB-05`), which `O57` says publishes criteria and assesses compliance.

Done when. Every `O23` row has at least one `O45` row, and every criterion can be answered yes or no by somebody who was not in the room when it was written.

Shows up in. No report reads `O45` directly. `RPT06 Requirements Traceability` reads `R09`, so step 4 of this step moves that report.

Goes wrong when. The criteria are written as *implemented to the agreed design*, which cannot fail.

TWENTY-TWO UNMANAGED INTERFACES

Of the twenty-two, how many carry data a traceability obligation turns on? Name them.

Eight rows read `Managed` and the gateway product has not been chosen. What is managing them?

Read the reports before you trust a number

Step 6.13 — Read all thirteen reports and write down what each one computes

Produce. A one-line statement per report of what it computes, what it reprints, and whether its footer is arithmetic or typed.

You need. An hour, and the thirteen `RPT` tabs.

Where it lands.

> **`RPT01 Dashboard`**
>
> Three columns: `Measure`, `Value`, `What it means`. Eighteen rows of typed numbers, and a footer reading *Every red figure is computed from this workbook, not asserted*. There are **no formulas anywhere in the workbook** — not one cell in any of the 105 sheets. Every figure on this sheet is typed, including the four that are wrong.

THE THIRTEEN REPORTS - WHAT EACH ONE ACTUALLY COMPUTES

Report	Computes	Reprints	Footer
RPT01 Dashboard	nothing	18 typed figures	prose, 4 wrong
RPT02 Application Portfolio	nothing	O23 less Description	none
RPT03 Capability Heatmap	Gap = Target - Current	O13 level 1 only	none
RPT04 Interface Catalogue	nothing	R06 less both ref cols	22/30, 21 - right
RPT05 Data Ownership	nothing	O20 less Definition	6 of 14 - right
RPT06 Requirements Traceab.	nothing	O06 + R09 Traced	prose
RPT07 Gap Analysis	nothing	O29 + R12 package	none
RPT08 Roadmap	nothing	O30 less Domains	4 - impossible
RPT09 Regulatory Coverage	nothing	O63 entire	5 of 6 - right
RPT10 Technology Currency	nothing	O27 entire	4 - really 2
RPT11 Security Position	nothing	O41 entire	1 - right
RPT12 Standards Conform.	nothing	O32 entire	none
RPT13 Orphan Analysis	O23 joined to R04	nothing	prose

No cell in any of the 105 sheets holds a formula. Every footer is typed.

RPT13 is the only join, and it tests applications against R04 alone. Across every relationship sheet, 32 of the 64 populated object sheets are reachable from none.

Figure 6.5 All thirteen reports, read against the sheets they claim to compute over. Twelve are column-subset reprints; `RPT13 Orphan Analysis` is the only join and `RPT03` the only derived column that is arithmetically right. **The workbook contains no formulas at all**, so every footer is a typed cell [1].

Twelve of the thirteen are column-subset reprints of one object sheet. `RPT13 Orphan Analysis` is the only one that performs a join, and it joins `O23` against `R04` alone — which is why it reports two orphans. Run the join across every relationship sheet instead and **thirty-two of the sixty-four populated object sheets are reachable from no relationship sheet at all**, with 372 of 813 rows referenced by nothing [1]. Three of the thirty-two are this chapter's: `O21 Logical Data Component`, `O22 Physical Data Component` and `O24 Information System Service`.

Two reports do add a derived column. `RPT03 Capability Heatmap` adds `Gap` = `Target` − `Current`, arithmetically correct on all ten rows and typed; it also drops `O13`'s ninety sub-rows. `RPT06 Requirements Traceability` adds `Reaches a block`, which is `R09`'s `Traced` column copied — and `R09` names a building block on every one of its thirty-one rows, so eight rows assert `No` while naming the block they reach.

And `RPT04 Interface Catalogue` drops `R06`'s `Source` and `Target` ref columns and keeps the names only, so no row in the interface catalogue can be traced back to `O23` without matching on a string containing a slash and an ampersand.

How.

1. Open each report beside the sheet it reads. Note which columns were dropped and whether any column was added.
2. Check the footer by hand. Six report footers carry a number; four of the six are right.
3. Write the thirteen lines down. Keep them with the repository — this is the page a new architect needs on day one.
4. Mark `RPT10 Technology Currency` and `RPT08 Roadmap` as unreliable. Chapter 7 step 7.1 takes `RPT10` apart.

Signed by. Chief Enterprise Architect (`ROL-08`).

Done when. Thirteen lines exist, each naming the source sheet and the footer's status.

Shows up in. Nowhere. This is the step that stops the reports from showing up in your conclusions.

Goes wrong when. A board paper quotes a dashboard figure. The number is typed, the date it was typed is nowhere, and the correction lands after the decision.

THE REVIEWER'S ASK — BRING THESE ANSWERS TO THE PHASE C REVIEW

Which application is the system of record for the item master, and who signed that?

How many `O23` rows did not exist when you started, and what was the largest one?

`A04` was the only `Critical` application and was to be replaced by nothing. What is it now, and what replaces it?

Twenty-two interfaces are unmanaged and the gateway product has not been chosen. Which of those two facts are you fixing first, and why that one?

Name the one number in the thirteen reports you would let a board paper quote, and say why only that one.

What leaves this chapter

Phase D needs four things: the corrected application list, the split of `A04`, the rebuilt `R07`, and the triage buckets from step 6.10. Phase E needs the portfolio scores from Form 6-A and the undeclared-systems log from Form 6-C, because both change what the roadmap pays for. Requirements Management needs the integration requirements added in step 6.10 and the rebuilt `R09` rows.

The one thing not to carry forward is any number from `RPT01 Dashboard`. There are no formulas in the file.

THE CHAPTER IN TEN LINES

- O23 Owner names assurance functions; replace every value with a role from O11 and have the owner confirm it.
- A04 SCADA / Historian is the only Critical application, is to be Replaced, and nothing is named to replace it.
- A03 MES has run the shop floor since 2010, is assessed Poor, and is to be Retained with no work package.
- O23 Health splits exactly four, four, four across Poor, Adequate and At risk, and has no healthy value in use.
- Users cannot score business value: APS reads 266 against the MES at 157, and the column sums to 2,494.
- Split A04 into control and historisation, then repoint every dependent row in twelve other places.
- Write the ERP/MES boundary paper on Form 6-B and have operations sign it, not comment on it.
- Four application classes volume 1 names have no O23 row at all, shop-floor data capture among them.
- R04 and R07 are row-number joins; R20 wraps modulo eight locations; R25 runs on a stride of three.
- R05 puts the application name in the key column on its fourteen Master rows, so a join returns half the sheet.
- Six R06 rows carry a Type that cannot be true at their Frequency; QMS has seven unmanaged outbound interfaces.
- Twelve of the thirteen reports are column-subset reprints; RPT13 is the only join and it tests R04 alone.
- There are no formulas anywhere in the workbook, so every reported figure is a typed cell.

SOURCES FOR THIS CHAPTER

- **1.** Manufacturing_TOGAF_Metamodel.xlsx, Kestrel Industrial Group — sheets O09, O12, O15, O18, O20, O21, O22, O23, O24, O28, O41, O45, O55, O58, O59, O65, R04, R05, R06, R07, R09, R12, R13, R18, R20, R25, RPT01–RPT13 and ZZ Checks
- **2.** The Open Group, TOGAF Standard, 10th Edition — ADM Phase C, Application Architecture https://pubs.opengroup.org/architecture/togaf10-doc/adm/chap08.html

CHAPTER 7

Technology Architecture

The chapter that has to carry the IT/OT boundary — and ten sheets that record the control servers in the data centre.

Phase D answers four questions: what the estate offers, what kind of thing offers it, which product on which contract, and where it sits. In a manufacturer a fifth sits above all of them — **which side of the IT/OT boundary is this on** — and the repository has no column for it.

That is the difference between this chapter and the same chapter in any other industry. A plant must keep producing when the corporate network is down. Nothing in these ten sheets records that requirement, and `027 Physical Technology Compone` records the opposite — it puts the SCADA control servers in the data centre. Volume 1 chapter 13 is this chapter's teaching half; read it first.

Two tab names are cut at Excel's thirty-one characters — `026 Logical Technology Componen` and `027 Physical Technology Compone`. Spell them as the tabs do. Kestrel Industrial Group is an invention, as is volume 1's Karoo Industrial Group.

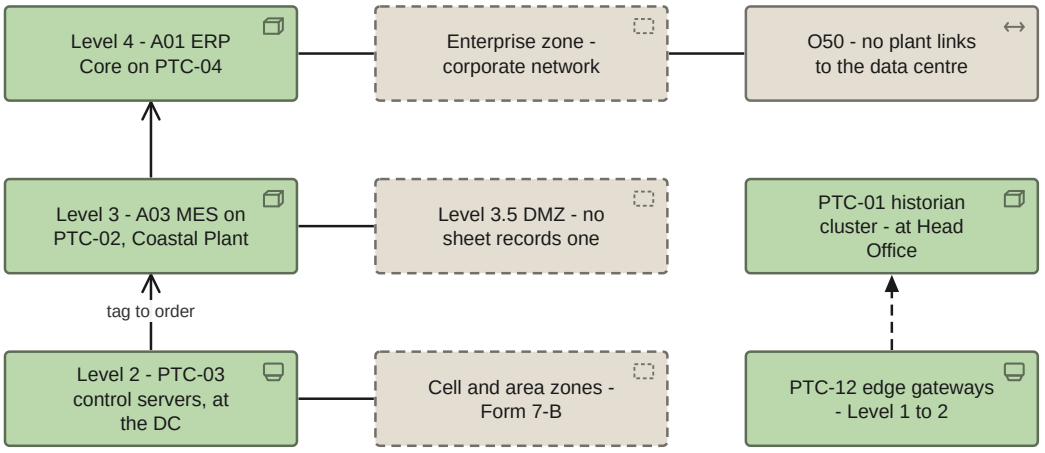


Figure 7.1 The IT/OT boundary, which no column in the repository records. Nothing in the file says which side a component is on, and `027.Location` puts `PTC-03 SCADA control servers` in the data centre — the one row that has to be wrong for the plant to keep producing. Step 7.6 adds the level, step 7.7 adds the zone [1].

Currency, which is a day-one argument

Step 7.1 — Recompute Position from Support end, and write the currency paper

Produce. `027` with a `Position` value that agrees with the date in the cell beside it, and Form 7-A stating the position per component.

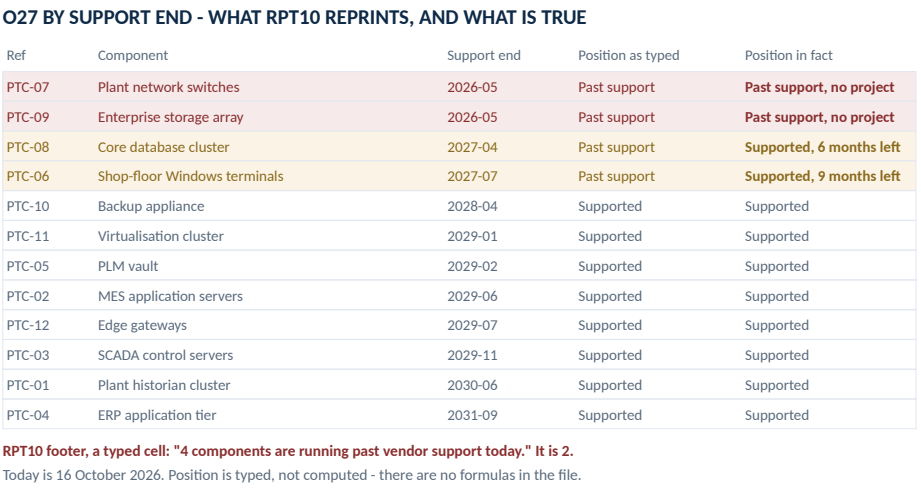
You need. Today's date. `O27`, `RPT10 Technology Currency`, `ZZ Checks`. The vendor support contracts, which nobody will be able to find for four of the twelve.

Where it lands.

> **`O27 Physical Technology Compone`**
>
> Seven columns: `Ref`, `Physical technology component`, `Vendor`, `Class`, `Support end`, `Position`, `Location`. Twelve rows. `Position` holds `Past support` on four — `PTC-06`, `PTC-07`, `PTC-08`, `PTC-09` — and `Supported` on the other eight. Two of those four dates are in the future: `PTC-06 Shop-floor Windows terminals` ends 2027-07 and `PTC-08 Core database cluster` ends 2027-04. The column is typed, not computed, and `RPT10` reprints the error and adds a footer reading *4 components are running past vendor support today*.

Ref	Component	Support end	Position as typed	Position in fact
`PTC-07`	Plant network switches	2026-05	Past support	Past support
`PTC-09`	Enterprise storage array	2026-05	Past support	Past support
`PTC-08`	Core database cluster	**2027-04**	Past support	**Supported, 6 months left**
`PTC-06`	Shop-floor Windows terminals	**2027-07**	Past support	**Supported, 9 months left**

Do this on day one, and not because two is a nicer number than four. A column that disagrees with the column beside it tells you how the whole file was populated — and the two rows genuinely past support are the two nobody is talking about, a switch estate and a storage array, neither with a project. There are **no formulas anywhere in the workbook**, so every such column is a typed opinion.



O27 BY SUPPORT END - WHAT RPT10 REPRINTS, AND WHAT IS TRUE

Ref	Component	Support end	Position as typed	Position in fact
PTC-07	Plant network switches	2026-05	Past support	Past support, no project
PTC-09	Enterprise storage array	2026-05	Past support	Past support, no project
PTC-08	Core database cluster	2027-04	Past support	Supported, 6 months left
PTC-06	Shop-floor Windows terminals	2027-07	Past support	Supported, 9 months left
PTC-10	Backup appliance	2028-04	Supported	Supported
PTC-11	Virtualisation cluster	2029-01	Supported	Supported
PTC-05	PLM vault	2029-02	Supported	Supported
PTC-02	MES application servers	2029-06	Supported	Supported
PTC-12	Edge gateways	2029-07	Supported	Supported
PTC-03	SCADA control servers	2029-11	Supported	Supported
PTC-01	Plant historian cluster	2030-06	Supported	Supported
PTC-04	ERP application tier	2031-09	Supported	Supported

RPT10 footer, a typed cell: "4 components are running past vendor support today." It is 2.

Today is 16 October 2026. Position is typed, not computed - there are no formulas in the file.

Figure 7.2 `O27 Physical Technology Compone` sorted by `Support end`, which is what `RPT10 Technology Currency` reprints before adding its one footer line. **Two of the four rows typed `Past support` carry dates in the future**, and the two that are genuinely past support have no work package naming them [1].

How.

1. Recompute `Position` from `Support end` against today's date, with three values and not two: `Past support`, `Within twelve months`, `Supported`.
2. Add a `Support reference` column and put the contract number in it. A support date with no contract behind it is a recollection.
3. Separate extended support from standard support. `PTC-06` is a shop-floor Windows estate, where extended support is usually bought and rarely recorded.
4. Retype `RPT10`'s footer and `ZZ Checks` row 13. Both are cells, not sums.
5. Put `PTC-07` and `PTC-09` into `O29 Gap` with a date. They are the sheet's only two true findings and neither is named in `O30 Work Package`.

Signed by. Domain Architect, Technology (`ROL-12`), with the Chief Information Officer (`ORG-05`) on the two gap rows.

Done when. Every `Position` value follows from its `Support end` date and today's date, and `RPT10`'s footer number matches the sheet.

Shows up in. `RPT10 Technology Currency` and `ZZ Checks`. If the step went well, `RPT10` reads 2 and not 4, and two components move onto the roadmap.

Goes wrong when. All four `Past support` rows are pushed into one work package called technology currency — which is what `WP-04` is, at $3.2m, `Not started` — so the two real exposures are funded at the same urgency as the two with nine months left.

Form 7-A — The technology currency paper

One row per physical technology component. Transcribes directly into `027 Physical Technology Compone` — `Ref`, `Physical technology component`, `Support end` and `Position` are that sheet's own columns in sheet order; `Support reference` and `Extended support` are the two columns you are adding and must create in Excel. Twelve rows for this estate. Signed by the Domain Architect, Technology.

Ref	Physical technology component	Support end	Position	Support reference	Extended support
PTC-07	EXAMPLE — delete this row	2026-05	Past support		No

BEFORE YOU TRUST A CURRENCY REPORT

Which two components are genuinely out of support today, and what stops when either fails?

`Position` is typed and not computed. Name two other columns in the repository you now suspect for the same reason.

Step 7.2 — Put real vendors into the Vendor column

Produce. `027.Vendor` with a vendor per row, and a named account contact per vendor.

You need. Step 7.1. Procurement's supplier list. `O19 Contract`.

Where it lands.

> **`O27 Physical Technology Compone`**
>
> `Vendor` holds the literal string `Vendor` in **seven of twelve cells**. The five that are populated read `OSIsoft-class`, `Microsoft`, `Cisco-class`, `Oracle-class` and `VMware-class` — placeholders naming a market position rather than a vendor, with one exception. Overwrite all twelve.

Seven cells holding the word *Vendor* is the clearest signal that `O27` was generated and never reviewed. It also makes the sheet useless for the question a technology architect is asked most often — how many suppliers are we exposed to, and on what terms. `O19 Contract`'s ten rows carry a `Counterparty` and an `Expiry` and no link to any component.

How.

1. Fill `Vendor` from procurement's supplier master, not from the engineer's memory of who installed it.
2. Where the component is a class rather than one product — edge gateways, shop-floor terminals — record the vendor per instance in a child sheet and leave `Vendor` as the dominant one.
3. Add a `Contract` column holding the `O19` ref, and raise the missing relationship sheet as a gap.
4. Flag every component whose vendor appears nowhere in `O19`. Those are the ones found out of support in two years.

Signed by. Chief Information Officer (`ORG-05`).

Done when. No cell in `Vendor` reads `Vendor`, and every populated cell names a legal entity.

Shows up in. `RPT10 Technology Currency`, which reprints `Vendor` unchanged.

Goes wrong when. The placeholders are replaced with the same market-position labels in prose — *an industrial historian vendor* — which reads better and answers nothing.

Look it up. Volume 1 chapter 15 for who the providers actually are in this market.

Step 7.3 — Relocate the control servers, and record why

Produce. `O27.Location` corrected, and a one-paragraph design rule stating where control sits and why.

You need. Steps 7.1 and 7.2. `O12 Location`, `O50 Network Link`, `R20`. The process control engineer and the plant electrical engineer.

Where it lands.

> **O27 Physical Technology Compone**
>
> `Location` holds values from `O12 Location`. Three rows invert the physical reality of a plant: `PTC-03 SCADA control servers` reads `Primary Data Centre`, `PTC-07 Plant network switches` reads `Central Distribution Centre`, and `PTC-01 Plant historian cluster` reads `Head Office` — while `PTC-02 MES application servers` correctly reads `Coastal Plant`. Four more rows place application, database and storage components at `Components Plant` and `Regional Warehouse`, which is a corporate estate scattered across sites that have no data halls.

Control servers sit at the plant they control. That is the central claim of volume 1's OT section and it is not a preference: every component production depends on has to be inside the plant's own power, network and failure domain. Put the supervisory system in a data centre and a wide-area link becomes a production dependency — and in this file that link is the worst of the ten in `O50 Network Link`, because no plant has a direct link to the `Primary Data Centre` at all.

The plant network switches are worse. A switch estate is where the plant is, by definition; a row placing it at a distribution centre means either the row is wrong or the plant has no network of its own, and the second answer would be the finding of the engagement.

How.

1. Walk the three production sites and write down where each component physically is. Two hours per site, with the engineer who holds the keys.
2. Correct `Location` and expect to add rows: a control system at three plants is three rows.
3. Write the design rule into `O32 Standard`, or beside `O05 PRN-04 Business Continuity`, whose statement reads *operations continue despite the loss of* and then does not say of what.
4. For every component that must stay central, write the degraded mode: what the plant does with no path above Level 2.
5. Record the vendor remote-access paths the walkdown finds. They belong in step 7.7's register.

Signed by. Chief Operating Officer (`ORG-07`) and Domain Architect, Technology (`ROL-12`). Operations signs this one because operations carries the consequence.

Done when. No component production depends on sits outside the plant it serves, and the degraded mode is written for every exception.

Shows up in. `RPT10 Technology Currency`, which reprints `Location`. The report will show plant names where it showed data-centre names.

Goes wrong when. The locations are corrected and the dependency is not. Moving a row does not move a server.

Look it up. Volume 1 chapter 13, *The Purdue model as it is actually used*, and within it *Where the diagram and the plant diverge*. Also *The test*, which states the design rule as a requirement.

WHERE THINGS ACTUALLY ARE

Stand in the control room at the Northern Plant. Which of the twelve `027` rows can you point at?

The corporate network goes down at 02:00 and stays down until 06:00. What stops, in what order?

The three layers, and the one nobody filled

Step 7.4 — Rebuild O25's Category column, and add the services the list omits

Produce. `O25` with a `Category` value drawn from a published scheme, and rows for the plant services the list has none for.

You need. Step 7.3. The infrastructure service catalogue if one exists.

Where it lands.

> **`025 Technology Service`**
>
> Four columns: `Ref`, `Technology service`, `Description`, `Category`. Twelve rows — Compute, Storage, Network, Identity, Integration, Database, Backup and Recovery, Monitoring, Data Platform, End User Compute, Security Operations, Automation. The names are the right grain. `Category` is drawn from the wrong list: it holds only `Infrastructure`, `Data` and `Security`, which are domain names, and it assigns them by rotation. `TSV-02 Storage` reads `Security`; `TSV-05 Integration` and `TSV-07 Backup and Recovery` read `Data`; `TSV-09 Data Platform` reads `Security`; `TSV-11 Security Operations` reads `Infrastructure`.

The omission matters more than the error. There is no technology service for process control, plant data collection, the shop-floor network or time synchronisation — the sheet describes the corporate estate and stops at the boundary this chapter exists to cross. A plant that loses its clock produces unusable product records, and no row in the file mentions one.

How.

1. Keep the twelve names. They are what a consuming team can request.
2. Publish a category scheme and rewrite the column from it. `Infrastructure`, `Platform`, `Data`, `Security`, `Workplace`, `Operational technology` works and takes ten minutes.
3. Add the plant services: process control, plant data collection, shop-floor network, time synchronisation. Four rows, `TSV-13` onward.

4. Record that `O25` appears in no relationship sheet, so nothing in the file consumes a technology service.

Signed by. Chief Information Officer (`ORG-05`), with the process control engineer on the four new rows.

Done when. Every `Category` value traces to the published scheme and the plant services exist.

Shows up in. No report reads `O25` . It feeds `O46 Technology Building Block` and the standards catalogue, so an error here travels silently.

Goes wrong when. The list turns into a product list. A technology service is what is offered; step 7.2 is where products go.

Step 7.5 — Reconcile O26, O47 and O46, and resolve the second Ref column

Produce. One register of technology patterns with one status per pattern, and a decision on the `T01` – `T07` key space.

You need. Step 7.4. `O26` , `O47` , `O46` , `O06 Requirement` , `R13` .

Where it lands.

> **`O26 Logical Technology Componen`**
>
> Four columns: `Ref` , `Logical technology component` , `Rationale` , `Status` . Seven rows, and they are not component kinds — they are seven manufacturing patterns that `O47 Reference Model Entry` holds a second time as `RME-01` to `RME-07` and `O46 Technology Building Block` holds a third time as `TBB-01` to `TBB-07` , with identical names and identical `Rationale` text. The base platform — compute, storage, network, identity, database — has no logical layer at all.

The same seven patterns, three times, with disagreeing statuses:

Pattern	`O26` `Status`	`O47` `Status`	In `O46`
OT/IT segmentation	Emerging	Adopted	`TBB-01`
Industrial edge compute	Current	Adopted	`TBB-02`
Time-series data platform	Current	Adopted	`TBB-03`
Unified namespace / MQTT	Current	Adopted	`TBB-04`
Digital twin	Emerging	Adopted	`TBB-05`
Computer vision inspection	Target	Adopted	`TBB-06`
Deterministic shop-floor network	Current	Adopted	`TBB-07`

`O47` reads `Adopted` on all eleven of its rows, which makes the column decoration. `O26` carries three values across seven rows. And step 7.7 will show there is no zone register behind `TBB-01`

`OT/IT segmentation` at all — so a building block the roadmap pays for through `WP-07` has no design artefact beneath it.

`O46` also has **two columns headed `Ref`**. The first holds `TBB-01` to `TBB-07`; the second holds `T01` to `T07`, a key space nothing else in the file uses and nothing resolves to. Call it `Ref (T)` when you must name it, and delete it.

How.

1. Choose one sheet to be the register. `O47 Reference Model Entry` is the right one — it is the house reference model.
2. Move the status there and leave the other two pointing at it. Status in three places is status in none.
3. Rewrite `O26` as what it is named for: logical component kinds behind the twelve services in `O25`.
4. Delete `O46`'s second `Ref` column. Seven refs resolving to nothing are seven invitations to a wrong join.
5. Check requirement coverage. `O06` carries `REQ-027` to `REQ-031` for five of the seven patterns; `TBB-06` and `TBB-07` have none, and `R09`'s positional join satisfies `REQ-031 Adopt Digital twin` with `TBB-01 OT/IT segmentation`.

Signed by. Chief Enterprise Architect (`ROL-08`). This is a model decision and it outranks all three sheets.

Done when. Each pattern has one status in one named sheet, and `O46` has one `Ref` column.

Shows up in. No report reads any of the three. The error surfaces at the `GT-1` design gate, where a design is tested against a reference model that says three things.

Look it up. Volume 1 chapter 14, *The constructive answer: build the house reference model*, and within it *Then declare it*.

The boundary, written down

Step 7.6 — Add the ISA-95 equipment-level column

Produce. `O27` with a level per component, from the ISA-95 equipment hierarchy, and the plant-hierarchy reconciliation written down.

You need. Step 7.5. `O12 Location`, `O20 Data Entity` (`E06`, `E07`, `E08`). The process control engineer and whoever maintains the ERP plant structure.

Where it lands.

> **`O27 Physical Technology Compone`**
>
> No column records which level a component sits at, and `Class` holds a technology kind (`Historian`, `Control`, `Application`, `Network`, `Storage`, `Compute`, `Database`, `End user`) rather than a level. Add a `Level` column and a `Zone` column — the second is step 7.7's — and fill `Level` for all twelve rows.

Two corrections before you fill it, because almost every secondary account gets both wrong. **This is not a seven-level stack.** The useful content is an equipment hierarchy — enterprise, site, area, and then, depending on the kind of manufacturing, process cell and unit, or production line and work cell, or storage zone and storage unit. The enumeration as published carries fifteen values, several of which are **alternatives selected by the kind of process rather than levels stacked on each other**, with work centre and work unit as the generic terms. And **equipment module and control module are ISA-88 concepts that ISA-95 references**, not ISA-95's own.

The second correction is the expensive one. The hierarchy **does not match an ERP's plant and storage-location model**, and reconciling the two is a project in its own right — as is reconciling the production and maintenance hierarchies, two trees over one set of steel. `O12 Location` cannot help: eight rows, no `Line count`, no `Shift pattern`, and 261 people at `LOC-08 Cloud Region`.

How.

1. Write the plant hierarchy out per site, to the level the plant actually manages — usually area and line, sometimes cell.
2. Assign a level to every `O27` row. Expect `PTC-03` and `PTC-12` below the boundary and the rest above it, once step 7.3 has moved them.
3. Record the ERP structure beside it and mark every disagreement. That list is a work package, not a note.
4. Do not reproduce the standard's table. Name the hierarchy in your own words and cite it.

Signed by. Domain Architect, Technology (`ROL-12`), with the Chief Operating Officer (`ORG-07`) on the hierarchy itself.

Done when. Every `O27` row has a level, and the ERP-to-plant reconciliation list exists with a named owner.

Shows up in. `RPT10 Technology Currency` gains a column. Nothing else reads it, which is why the hierarchy has to be published separately.

Goes wrong when. Somebody draws the famous pyramid, labels it levels 0 to 4 and treats it as the standard's content. The pyramid is the picture; the hierarchy is the standard.

Look it up. Volume 1 chapter 13, *The equipment hierarchy, and the correction that matters most*, and chapter 9, *The plant hierarchy, and the reconciliation nobody budgets for*. Chapter 14's *ISA-95 and IEC 62264 — the one everybody has heard of* has the version facts.

Step 7.7 — Build the zone and conduit register

Produce. A zone and conduit register on Form 7-B, with a target security level per zone and every crossing named.

You need. Step 7.6. `041 Security Control`, `050 Network Link`, `065 Enforcement Point`, and the firewall rule base. The OT asset inventory, which will not exist.

Where it lands.

> **`046 Technology Building Block`**
>
> `TBB-01 OT/IT segmentation` exists as a building block with `What it must do` reading *Purdue-model network zones between plant and enterprise* and an acceptance criterion reading *OT/IT segmentation in production for at least one site*. There is no zone register, no conduit list and no zone column anywhere in the 105 sheets, so the acceptance criterion cannot be tested and `059 Compliance Assessment` has never been asked to try.

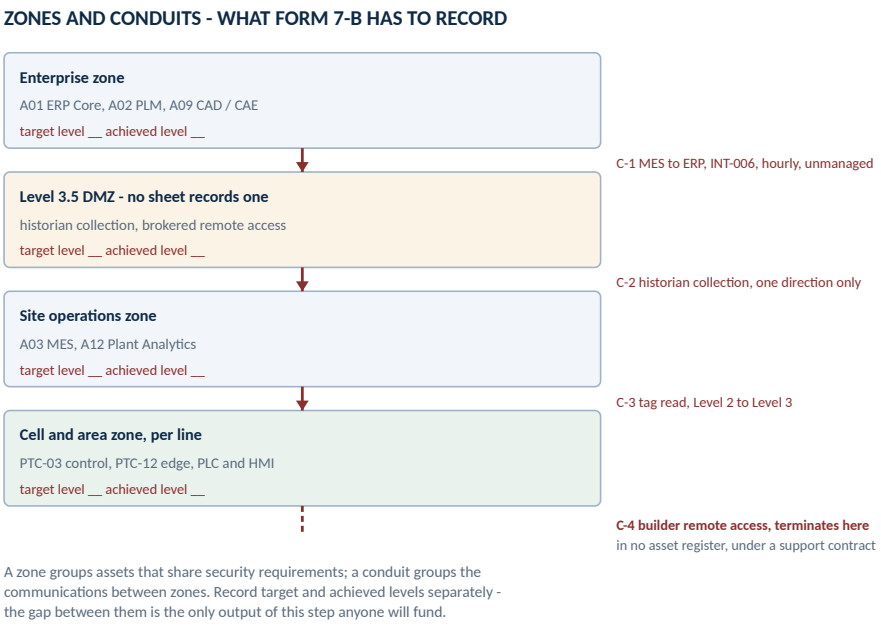


Figure 7.3 The zone and conduit register the repository has no sheet for. `TBB-01 OT/IT segmentation` exists as a building block with an acceptance criterion and nothing beneath it. The machine builder's remote-access path terminates in the lowest zone, below every control the enterprise built [1].

A zone groups assets that share security requirements; a conduit groups the communications between zones. Do not reproduce the standard's text — describe the method and cite it. Three distinctions to hold: the target level set by risk assessment, the level achieved as built, and the level a supplier can demonstrate. The gap between the first two is the remediation programme, and it is the only output of this step anyone will fund.

Two things the walkdown will find, because they are found almost everywhere. The dual-homed engineering workstation, one card on the plant network and one on corporate. And the machine builder's remote-access path, wired into the cell zone at commissioning, terminating

below every control the enterprise built, in no asset register. There are commonly dozens in one plant.

How.

1. Draw the zones from step 7.6's plant hierarchy, not from the network diagram. Zones follow consequence, not cabling.
2. List every conduit — every path crossing a zone boundary — with its protocol, direction and enforcement point from `O65`.
3. Set a target level per zone and record the achieved level separately. The two columns are the point of the register.
4. Build the asset inventory by passive discovery, walkdown and configuration extraction. Not by an active scan, which can stop a line.
5. Add a `Zone` column to `O27` and fill it, so the register and the component sheet agree.

Signed by. Chief Information Officer (`ORG-05`) and the Chief Operating Officer (`ORG-07`), jointly. A zone register signed by one of them will be ignored by the other's staff.

Done when. Every conduit has a named enforcement point, every zone has a target and an achieved level, and the vendor-access paths from step 7.3 appear as conduits.

Shows up in. `RPT11 Security Position` once the controls in step 7.11 are rewritten against zones. Today it reads `O41`'s eight rows and nothing else.

Goes wrong when. The register is built from the firewall rule base, which describes what was configured. The register has to describe what is required, and the difference is the work.

Look it up. Volume 1 chapter 13, *The standards, with their uneven dates*, and *The air gap, and where the real path is*. Chapter 17's *Zones and conduits* says how to model it, and *And the IT/OT zone view* names the view.

Form 7-B — Zone and conduit register

One row per zone, then one row per conduit on a second copy. Feeds no existing sheet — the repository has no zone register, so create a sheet with these columns and the `Zone` column you added to `027 Physical Technology Compone` in step 7.7. Six to twelve zones and twenty to sixty conduits on a three-plant estate; continue on a second copy of this form. Signed jointly by the Chief Information Officer and the Chief Operating Officer.

Zone	Level	What is in it	Target level	Achieved level	Conduits out
Northern Plant cell 2	Area / line	EXAMPLE — delete this row	SL-T 2		3

Step 7.8 — Record the historian's exception, compression and namespace settings

Produce. A statement, per tag class, of the exception and compression settings, the step flag, the retention period and the aggregation method reports must use.

You need. Steps 7.4 and 7.6. `022 Physical Data Component`, `037 Data Attribute`, `042 Metadata Element`. The historian administrator.

Where it lands.

> **`022 Physical Data Component`**
>
> Four columns: `Ref`, `Physical data component`, `Application`, `Type`. Twelve rows, one per application. `Type` is drawn from the wrong row: `PDC-04 SCADA / Historian database` reads `Relational` and `PDC-05 APS database` reads `Time series`, which is exactly backwards, and `PDC-01 ERP Core database` reads `File`. The sheet has no `Retention`, no `Volume`, no `Grain` and no namespace column, so it cannot express a tag namespace at all — and `037 Data Attribute` is a cross-product of six attribute names against ten entities, with no unit of measure, no sample interval and no deadband.

`A04` / `PTC-01` is the plant historian cluster, and a historian changes the number it reports. The reference implementation applies two reduction mechanisms at two points: **exception reporting** at the interface, before data is sent, and **compression** at the archive, using a swinging-door test that removes points which can be rebuilt by interpolation. The magnitudes are not marginal. A 0-or-1 status tag over 48 hours at one-second scans is **172,800 events with no compression and 15 with minimal compression**; a cited pilot reports exception alone cutting data over the wire by about **70%** and exception plus compression cutting archived data by about **94%** [1]. The defaults are non-zero, so compression is on unless somebody turned it off.

So the reported number and the recorded number differ, and nobody usually knows by how much. The amount is what this step produces.

How.

1. Get the exception and compression deviation per tag class from the historian administrator, in engineering units and as a percentage of span, with the maximum interval.
2. Get the `step` flag per tag class. A discrete tag wrongly flagged as analogue returns impossible intermediates — a valve 37% open when it has two positions.
3. State the aggregation method reports must use, once. A time-weighted average over archived points is not the arithmetic mean of the original scans, and the two diverge most during the transients that matter.
4. Put the settings under change control. Changing a deviation changes all future data for that tag and not the past, so the series has a discontinuity nobody records.
5. Fix `O22.Type` and add `Retention` and `Grain`. A historian recorded as relational will be migrated as relational.

Signed by. Chief Data Officer (`ORG-06`) through the Data Governance Council (`GB-03`), with the process control engineer.

Done when. Every tag class has settings on paper, a step flag, a retention period and one named aggregation method, and `O22.Type` is right.

Shows up in. No report reads `O22` . The consequence surfaces in whatever reports effectiveness — `A12 Plant Analytics` — where two correct reports of the same thing differ.

Goes wrong when. A data scientist is handed historian extracts and told they are measurements. They are the smallest set of points from which the curve can be rebuilt to a configured deviation.

Look it up. Volume 1 chapter 13, *Exception and compression, and how they silently change a reported number*, and *Why a tag is not a data entity*. Chapter 10's *Time-series and tag data, which no other industry in this series has* has the longer treatment.

Resilience, and the power the plant actually gets

Step 7.9 — Rebuild O49 against what stops production, and record dip tolerance per line

Produce. `O49` with tiers that follow from production consequence, and Form 7-C recording each line's dip tolerance and restart sequence.

You need. Steps 7.3 and 7.6. `O49`, `O23 Application Component` as corrected in chapter 6. The plant electrical engineer, the process engineer and the line supervisor.

Where it lands.

> **`O49 Resilience Tier`**
>
> Five columns: `Ref`, `Application`, `Tier`, `Recovery objective`, `Tested`. Twelve rows. Tier and recovery objective are uncorrelated: `RES-01 ERP Core` is `Tier 3` at 24 hours; `RES-02 PLM` is `Tier 1` at 24 hours; `RES-06 WMS` and `RES-12 Plant Analytics` are both `Tier 2` at 15 minutes. Eight of twelve read `Never tested`. The only internally coherent row is `RES-04 SCADA / Historian` — `Tier 1`, one hour, `Tested` — and chapter 6 step 6.4 splits that row in two, after which only its control half stays coherent.

O49 - TIER AGAINST RECOVERY OBJECTIVE, AND WHAT WAS TESTED

Ref	Application	Tier	Recovery objective	Tested
RES-01	ERP Core	Tier 3	24 hours	Never tested
RES-02	PLM	Tier 1	24 hours	Never tested
RES-03	MES	Tier 2	1 hour	Never tested
RES-04	SCADA / Historian	Tier 1	1 hour	Tested
RES-05	APS	Tier 1	1 hour	Never tested
RES-06	WMS	Tier 2	15 min	Never tested
RES-07	QMS	Tier 2	24 hours	Tested
RES-08	CMMS / EAM	Tier 1	4 hours	Never tested
RES-09	CAD / CAE	Tier 2	24 hours	Tested
RES-10	Supplier Portal	Tier 2	4 hours	Tested
RES-11	Track & Trace	Tier 1	24 hours	Never tested
RES-12	Plant Analytics	Tier 2	15 min	Tested

One coherent row in twelve - RES-04, Tier 1, one hour, Tested - and chapter 6 splits it in two.
Eight of twelve never tested. No report in the file reads O49 at all.

Figure 7.4 `O49 Resilience Tier`: tier and recovery objective are uncorrelated. `Tier 1` appears at 24 hours twice and `Tier 2` at 15 minutes twice. **Eight of the twelve rows have never been tested**, and no report reads the sheet, so the resilience position of the estate appears nowhere [1].

Then the design input nobody budgets for. Load-shedding has ended — 504 consecutive days as at 2 October 2026, with the energy availability factor at 68.11% year to date [2] — but **load reduction is a separate, continuing programme and must not be conflated with it**: 202,590 customers at the same date, down from 1.69 million, with elimination targeted for March 2027 [2]. It is predominantly residential, so a manufacturer is less likely to be directly affected, while its workforce is and a municipal site may be.

Voltage quality is the continuing problem, and it is written down. **NRS 048-2:2025, Edition 5, approved by NERSA on 6 May 2025** [3], defines a dip, classifies dips into windows, and

assigns **window-Y immunity to the customer** — the national quality-of-supply standard tells you that a class of dip is yours to survive. It publishes characteristic unplanned interruption counts and attaches **no compliance criteria** to them. What breaks, in order: drives trip, contactors drop out, controllers survive and therefore believe a pump is running that has stopped, operator stations reboot, slow instruments take minutes to read validly, and the process itself is damaged — a set batch, a solidified extruder, a furnace needing a purge. Those are hours-to-days recovery events and no tier in `049` is expressed in those terms.

How.

1. Re-tier against one question per application: does the plant stop, and how fast? Keep the sheet's own tiers and recovery objectives.
2. Make tier and recovery objective agree. `Tier 1` at 24 hours is not a tier.
3. Schedule a test for each of the eight untested rows, with a date. An untested recovery objective is a target.
4. Record, per line, the deepest dip it survives, what trips first, and the restart sequence as a designed and tested artefact.
5. Specify uninterruptible-supply and generator scope here, and design the degraded mode for a dark plant and not only a disconnected one.

Signed by. Chief Operating Officer (`ORG-07`), with the Domain Architect, Technology (`ROL-12`) on the application tiers.

Done when. No row pairs a tier with an incompatible recovery objective, every row has a test date, and every production line has a dip tolerance and a restart sequence on paper.

Shows up in. No report reads `049` , which is itself a finding — the resilience position of the estate appears in none of the thirteen reports.

Goes wrong when. Power is written into `062 Delivery Risk` as a risk with a likelihood and an impact. It is not a risk; it is a design input, and the correct response is a specification.

Look it up. Volume 1 chapter 13, *Voltage quality is the design input, and it is written down*, then *What breaks, in order*, then *Load-shedding ended; the discipline did not*.

POWER AS A DESIGN INPUT

The plant loses supply for 90 seconds at 03:10. Which line is still making conforming product at 03:20, and which one needs stripping?

Eight of twelve resilience rows have never been tested. Which test are you running first, and on what date?

Form 7-C — Line dip tolerance and restart

One row per production line. Feeds no existing sheet — `O12 Location` holds sites and `O20 E07 Work Centre` holds resources, and a production line is neither — so this feeds the line register you create. `049 Resilience Tier`'s own five columns are few enough to type straight into Excel; add a `Test date` column there while you are in it. One row per line; continue on a second copy of this form. Signed by the Chief Operating Officer and the plant electrical engineer.

Plant	Line	What trips first	Deepest dip survived	Restart sequence written	Tested on
Northern Plant	EXAMPLE — delete this row	Extruder drive		No	

Security, the design gate and the network

THREE CELLS, ONE OF THEM FALSE

`ENV-01 Production` reads `No` for personal data. What does the access-control system at the gate hold, and which environment is it in?

`SEC-07` says masking is in place and `EP-05` says most refreshes are unmasked. Which one was measured?

Step 7.10 — Settle O48's personal-data answer

Produce. `O48` with an answer per environment that is true, and the lawful basis recorded for the data it actually holds.

You need. Step 7.9. `O41 SEC-07`, `O64 Security Policy`, `O65 EP-05`, `O42 Metadata Element`. The privacy officer, and the chapter 6 step 6.6 log.

Where it lands.

> **`O48 Environment`**
>
> Four columns: `Ref`, `Environment`, `Description`, `Personal data present`. Four rows. `ENV-01 Production` reads `No`; `ENV-02 Pre-production` and `ENV-04 Development` read `Yes`; `ENV-03 Test` reads `No`. Meanwhile `O41 SEC-07 Non-production masking` reads `In place`, enforced at `Provisioning`, and `O65 EP-05 Non-production provisioning` records its own position as *Most refreshes are unmasked*. At least one of those cells is false and probably three are.

Production holding no personal data while development does is the answer inverted. And in a manufacturer the personal data in question is overwhelmingly **employee and biometric** — access control, time and attendance, biometric clocking at the gate, occupational health records, injury reports. Biometrics and health data are special personal information under POPIA section 26, which makes this a prohibition-with-exceptions question and not a marketing one.

The repository cannot answer it: chapter 6 step 6.6 showed there is no `O23` row for time and attendance or access control at all. So either `ENV-01` is wrong, or the system holding the data is outside the portfolio. Both are findings, and they are not exclusive.

How.

1. Ask the privacy officer what production holds, by category, and write it into `Description`.
2. Correct `Personal data present` per row and add a `Special information` column.
3. Reconcile `SEC-07` against `EP-05`. One says masking is in place, the other that most refreshes are unmasked; the second is a measurement and the first an intention.
4. Record the lawful basis for the biometric processing, per purpose. `O42 MTD-07 Consent basis` reads `Recorded, not held`, which means nowhere.
5. Raise the missing application row from step 6.6 as a gap in `O29`.

Signed by. Chief Compliance Officer (`ROL-07`), with the Chief Data Officer (`ORG-06`).

Done when. Every `Personal data present` value has a category list behind it, and the masking contradiction is resolved in one direction with a date.

Shows up in. `RPT11 Security Position`, which reprints `O41` including `SEC-07`. The report will carry whichever answer you resolved to.

Goes wrong when. The sheet is corrected and the environments are not. A cell reading `Yes` does not mask anything.

Look it up. Volume 1 chapter 16, *POPIA, where it actually bites a manufacturer*.

Step 7.11 — Close the design gate

Produce. `O59 Compliance Assessment` with rows in it, and `SEC-02` moved off `Absent` with evidence.

You need. Step 7.10. `O41`, `O58 Governance Gate`, `O59`, `R24`, `O64`, `O65`, `O63`, `R16`.

Where it lands.

> **`O59 Compliance Assessment`**
>
> Seven columns: `Ref`, `Building block`, `Block name`, `Acceptance criterion`, `Gate`, `Verdict`, `Evidence`. **Zero rows — header only.** `R24 Compliance Assessment tests` is empty in consequence. Every building block in `O35`, `O44`, `O45` and `O46` carries an acceptance criterion and not one has ever been tested at a gate. One row per criterion per gate crossing; expect thirty to seventy.

This is the sharpest internally consistent finding in the file. `O41 Security Control` reads `Absent` on exactly one row — `SEC-02 Design review`, enforced at the `Design gate` — which `ZZ Checks` reports correctly as 1. `O65 EP-01 Design review gate` records its position as *Newly established*. `O59` and `R24` are empty, so the gate has produced zero records. Four sheets agree.

The fifth disagrees, and it is the one to take to the board. `O64 POL-04` reads *Security is designed in, not added afterwards*, scope *Enforced at the design gate*, `Position` = **`Met`**. A policy cannot be met at a gate whose control is absent and whose record set is empty. `POL-03 Material activity is logged and observable` also reads `Met`, against 21 of 30 `R06` interfaces carrying no logging.

How.

1. Take the gates from `O58` as they are — refs `GT-0` to `GT-3`, named `G0` to `G3`, a mismatch worth fixing in the same sitting.
2. Create one `O59` row per acceptance criterion per gate crossing, with a verdict and a pointer to the evidence. Backfill the delivered packages first.
3. Change `SEC-02` from `Absent` only when `O59` has rows. A control is in place when it produces records.
4. Retype `O64 POL-04` and `POL-03` to `Breached` or `Partial`, with the reference that proves it.
5. Rebuild `R16`. Step 7.13 — its `Position` column reports the wrong sheet.

Signed by. Architecture Function (`GB-05`), which `O57` says assesses compliance, with the Architecture Board (`GB-01`), which holds `G1` and `G2`.

Done when. `059` has a row per criterion per gate crossing, `R24` resolves, and no policy in `064` reads `Met` without a record set behind it.

Shows up in. `RPT11 Security Position`, whose footer reads *1 controls are absent* — a typed cell, correct, and ungrammatical. Fix the grammar while you are in there.

Goes wrong when. `SEC-02` is changed to `In place` because a design review now happens. It happens; it is not evidenced; and the gate exists to produce evidence.

Step 7.12 — Rebuild O50 Network Link

Produce. `O50` with one row per real link, a carrier per row, and the plant-to-data-centre paths that the sheet currently has none of.

You need. Step 7.3. `O12 Location`, `O27`, and the network team's circuit inventory.

Where it lands.

> **`O50 Network Link`**
>
> Five columns: `Ref`, `From`, `To`, `Bandwidth`, `Diverse path`. Ten rows. Two location pairs appear twice with conflicting answers: `Northern Plant` to `Coastal Plant` is `NWL-01` at 10 Gbps, `Diverse path` `No`, and `NWL-09` at 10 Gbps, `Diverse path` `Yes`; `Coastal Plant` to `Components Plant` is `NWL-02` at 10 Gbps `Yes` and `NWL-10` at 100 Mbps `No`. There is no `Carrier`, no `Latency`, no `Link type` and no `Provider` column, so the sheet cannot say whether two links on one pair are actually diverse.

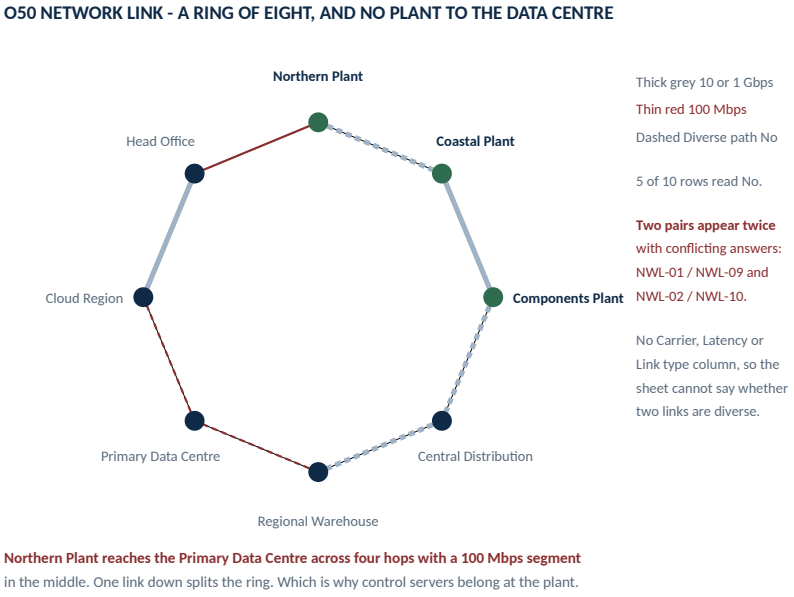


Figure 7.5 The ten rows of `O50 Network Link` drawn as the topology they describe: **a single ring of eight sites with no plant-to-data-centre link at all**. Two location pairs appear twice with conflicting bandwidth and diverse-path answers, and the sheet has no `Carrier` column to settle them [1].

The topology is a single ring of eight, Northern Plant round to Head Office and back. Two consequences. **No plant has a direct link to the Primary Data Centre**, so a plant reaches it across three or four hops with a 100 Mbps segment in the middle — which is why step 7.3's relocation is not negotiable. And **a ring with one link down splits**, with five of the ten rows reading `Diverse path` = `No`.

How.

1. Resolve the two duplicate pairs with the network team. One of each pair is wrong, or they are two circuits and the sheet needs both rows with carriers named.
2. Add `Carrier`, `Latency` and `Link type`. Two circuits on one carrier are not a diverse path, whatever the column says.
3. Add the links the ring omits — plant to data centre, plant to cloud region — or record that they do not exist, which is the finding.
4. Mark which links production depends on after step 7.3. The answer should be none.

Signed by. Domain Architect, Technology (`ROL-12`).

Done when. No location pair appears twice with conflicting answers, every row names a carrier, and no row carries a production dependency.

Shows up in. No report reads `O50`. It is one of the thirty-two object sheets no relationship sheet reaches.

Goes wrong when. `Diverse path` = `Yes` is taken as read. Diversity is a property of the route and the carrier, not of a column.

Step 7.13 — Fix the Position columns in R16 and R17

Produce. Two relationship sheets whose status column reports the end it claims to.

You need. Steps 7.10 and 7.11. `O41`, `O63`, `O64`, `O65`.

Where it lands.

> **`R16 Control Objective met by Co`**
>
> Five columns: `Control objective`, `Obligation`, `Control`, `Control name`, `Position`. Six rows, positional — `CO-01` to `SEC-01`, onward — so `ISO 9001` is evidenced by `SEC-01 Identity lifecycle`, product conformity by `SEC-03 Privileged access management`, and substance restriction by `SEC-04 Attributable access`. And the `Position` column carries the **obligation's** `Evidence` value from `O63`, not the control's `Position` from `O41`: row one reads `No mechanism` where `SEC-01` reads `Partial`, and row two reads `In breach` where `SEC-02` reads `Absent`.

`R17 Policy enforced at Enforcem` does the same thing in the other direction. Its `Position` column carries the **enforcement point's** position from `O65`, not the policy's from `O64`: `POL-03` reads `Met` in `O64` and `Product not yet selected` in `R17`. And the diagonal misassigns the

pairings it most needed to get right — `POL-04`, whose scope says *enforced at the design gate*, is mapped to `EP-04 Data platform`, while `EP-01 Design review gate` is mapped to `POL-01`, the personal-information policy.

So two relationship sheets report one end's status against the other end's identity, in opposite directions. Decide once what a `Position` column on a relationship sheet means — the status of the relationship, not of either end — and apply it to both.

How.

1. Rebuild `R16` from the obligations as rewritten in Phase B. `O63`'s six are for a European manufacturer; the South African set is different, and this sheet has to follow it.
2. Rebuild `R17` by asking, per policy, where it is actually enforced. Expect several points per policy.
3. Redefine `Position` as the state of the relationship — evidenced, partial, no mechanism — and write the definition into the sheet's row-2 subtitle.
4. Read `RPT12 Standards Conformance` while you are here. It reprints all five columns of `O32 Standard`, computes nothing, and its subtitle promises *whether anything conforms to it* while no column answers that. `R18 Standard applies to Applica` holds the answer, and its sixteen rows reach only the odd-numbered applications `A01`, `A03`, `A05`, `A07`, `A09` and `A11`. Six have no standard applied, including `A04`, the only `Critical` one.

Signed by. Chief Compliance Officer (`ROL-07`) for `R16`; Chief Information Officer (`ORG-05`) for `R17`.

Done when. Neither sheet's `Position` column reproduces a value from the sheet at the other end of the relationship, and every application in `O23` appears in `R18`.

Shows up in. `RPT09 Regulatory Coverage` and `RPT12 Standards Conformance`. Both are reprints with no join, so neither will show the repair; the repair shows up at the gate.

Goes wrong when. The `Position` column is deleted instead of defined. It is the only place in the file where the state of a relationship is recorded at all.

THE REVIEWER'S ASK — BRING THESE ANSWERS TO THE PHASE D REVIEW

How many components are genuinely past vendor support, and how did you work it out?

Where are the SCADA control servers now, and what did it cost to move them?

How many remote-access paths into the cell zone did the walkdown find, how many were in an asset register, and which conduit does each cross?

The plant loses the corporate network at 02:00. For how long does it keep producing conforming, traceable product, and who signed that number?

`064 POL-04` says security is designed in and reads `Met`, and `059 Compliance Assessment` is empty. Which one are you changing?

What leaves this chapter

Phase E needs the two genuine currency exposures from step 7.1, the zone register's target-against-achieved gap from step 7.7, and the resilience test schedule from step 7.9 — those three are most of the technology roadmap. Phase G needs `059` populated from step 7.11. Requirements Management needs the two technology building blocks that no requirement reaches.

And carry one sentence into every design review: a plant must keep producing when the corporate network is down. Nothing in these ten sheets said it. After this chapter, one of them does.

THE CHAPTER IN TEN LINES

- O27 Position is typed, not computed: two of the four rows marked Past support carry dates in the future.
- O27 Vendor holds the literal string Vendor in seven of twelve cells, and the other five are market labels.
- Control servers sit at the plant they control. O27 puts them in the Primary Data Centre; move them.
- O25 Category holds domain names assigned by rotation, and there is no technology service for the plant floor.
- The same seven patterns appear in O26, O47 and O46 with three statuses; O46 has two columns headed Ref.
- ISA-95 publishes an equipment hierarchy of fifteen values with alternatives, not a seven-level stack.
- The equipment hierarchy does not match an ERP plant structure, and reconciling them is a work package.
- TBB-01 OT IT segmentation has an acceptance criterion and no zone register behind it; build one.
- A status tag over 48 hours at one-second scans is 172,800 events uncompressed and 15 compressed.
- O49 pairs Tier 1 with 24 hours twice; eight of twelve rows have never been tested and no report reads it.
- Load-shedding ended but load reduction continues, and NRS 048-2 Edition 5 assigns window-Y dips to you.
- O48 says production holds no personal data and development does, while SEC-07 masking reads In place.
- SEC-02 Design review is Absent, O59 and R24 are empty, and POL-04 reads Met across those five sheets.

SOURCES FOR THIS CHAPTER

- **1.** Manufacturing_TOGAF_Metamodel.xlsx, Kestrel Industrial Group — sheets O05, O12, O19, O22, O25, O26, O27, O28, O29, O32, O37, O41, O42, O46, O47, O48, O49, O50, O57, O58, O59, O63, O64, O65, R07, R09, R16, R17, R18, R24, RPT10, RPT11, RPT12 and ZZ Checks
- **2.** Eskom, “More than 500 days without loadshedding, EAF improves to 68.11%”, media statement, 2 October 2026 https://www.eskom.co.za/more-than-500-days-without-loadshedding-eaf-improves-to-68-11-dies-irae/
- **3.** NERSA, NRS 048-2:2025 Edition 5, Electricity supply: Quality of supply, Part 2 (approved by NERSA 6 May 2025, superseding the 2015 fourth edition) https://www.nersa.org.za/files/files/2025/05/NRS-048-2-Edition-5.pdf
- **4.** The Open Group, TOGAF Standard, 10th Edition — ADM Phase D, Technology Architecture https://pubs.opengroup.org/architecture/togaf10-doc/adm/chap09.html

CHAPTER 8

Opportunities and Solutions

Twelve work packages, three transition states and two gaps — and the gaps were written from the packages.

What Phase E in this repository has already decided

Phase E turns four domain architectures into work packages somebody will fund [2]. In the repository you have been handed it has already been done: twelve packages, three transition states, five alternatives, thirty-seven building blocks and two gaps. Almost none of it was derived from the gaps. It was derived from the building blocks, and the gaps were written afterwards to match. Everything below either proves that or repairs it.

Kestrel Industrial Group, whose data fills the repository, is an invention, as is Karoo Industrial Group, volume 1's group, which appears here only as a pointer back into that book. Never mix their numbers.

Two things to settle first. **The whole file is in dollars** — `030 Work Package` reads `$5.5m`, `054 Architecture Alternative` reads `$46m` — and there is no exchange rate in it [1]. Read the magnitudes, not the currency, and put the rand figure in your own plan with the rate and the date. And **there is no change control anywhere in the repository**: no version column, no change log, no baseline [1]. A roadmap with no baseline cannot be compared against the one you publish next quarter. Steps 8.11 and 8.12 are the substitute.

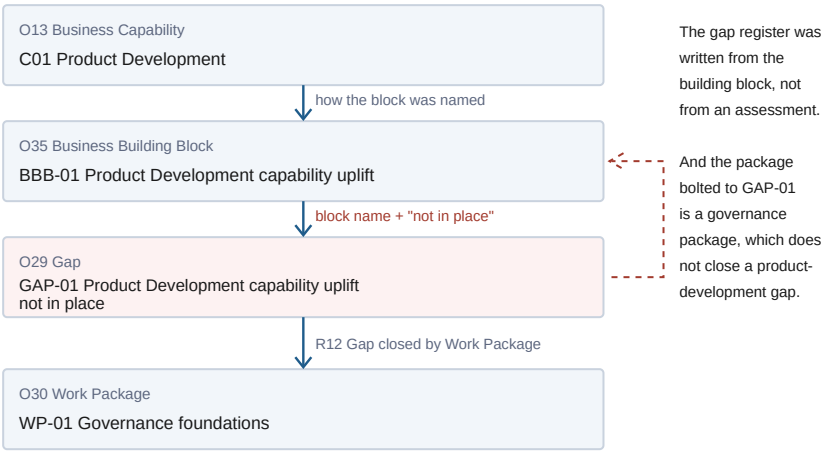


Figure 8.1 How `029 Gap` was actually written. The gap text is the building block's own name with **not in place** appended, and `R12` then bolts it to a package that does not close it.

BEFORE YOU OPEN O29 GAP, ANSWER THESE

Say in one sentence what a gap is, such that a delivery manager could tell a gap from a building block nobody has built yet.

Who here is allowed to say a gap is closed, and what do they look at to say it?

The repository holds two gaps and thirty-seven building blocks. Which of those two numbers is wrong, and how would you prove it?

The gap register

Step 8.1 — Read both rows of O29 Gap and establish which direction the register was written in

Produce. A one-page note stating, with cell values, whether the gap register was derived from the architecture or back-filled from the solution.

You need. `O29 Gap`, `O35 Business Building Block`, `R12 Gap closed by Work Package`, `RPT07 Gap Analysis`.

Where it lands.

> **`O29 Gap` — seven columns, two rows**
>
> - `Ref` `GAP-01`, `GAP-02`; `Building block` `BBB-01`, `BBB-02`; `Status` `Open` and `Overdue`; `Due` `2029-12` and `2028-05`.
> - `Gap` reads `Product Development capability uplift not in place` and `Demand & Supply Planning capability uplift not in place`.
> - `Domain` reads `Business` on both rows — no `Data`, `Application` or `Technology` gap exists — and `Severity` reads `High` on both, so the column holds one value.

How.

1. Open `O35`. `BBB-01` is `Product Development capability uplift`; `BBB-02` is `Demand & Supply Planning capability uplift`.

2. Put each beside the `Gap` text in `O29`. The gap text is the block's own name with `not in place` appended — both rows, no exceptions. Quote it; do not paraphrase.
3. Check `R12`: two instances, `GAP-01` to `WP-01 Governance foundations` and `GAP-02` to `WP-02 Data ownership and mastering`. `BBB-01` is a product-development capability and `WP-01` is a governance package. The package does not close the gap it is bolted to.
4. Check `RPT07 Gap Analysis`: it reprints the two rows with `Building block` dropped and `Work package` added, so the report hides the column that gives the game away.

Signed by. Chief Enterprise Architect (`ROL-08`), who owns the repository through `GB-05`.

Done when. The note names both `Gap` strings, both `BBB` names, and states the derivation direction in a sentence a non-architect can read.

Shows up in. `RPT07 Gap Analysis`. `ZZ Checks` does not count gaps at all: an estate with twenty-two unmanaged interfaces and six entities without lineage [1] has more than two, and nothing notices.

Goes wrong when. You accept the two rows and plan against them. Two `High` gaps over a four-domain estate is a placeholder, and Phase F schedules against the answer rather than the assessment.

Look it up. Volume 1 chapter 18, *The findings* — eight findings from two weeks on a plant floor, each with the check that evidences it.

Step 8.2 — Rebuild the register from the four domain assessments you already hold

Produce. A gap register with one row per assessed shortfall, each traceable to a cell in a domain sheet.

You need. Chapters 4 to 7 of this workbook; `R06`, `O20 Data Entity`, `O49 Resilience Tier`, `O63 Control Objective`; Form E-1.

Where it lands.

> **`O29 Gap` — the same seven columns, filled properly**
>
> - `Ref` continues `GAP-03` onward. Do not renumber the two existing rows; you want the contrast on the record.
> - `Gap` states the shortfall, not the solution. `Twenty-two of thirty interfaces are unmanaged` is a gap; `Integration platform not in place` is a work package wearing a gap's clothes.
> - `Domain` must stop reading `Business` on every row: four domains produced findings, four domains owe gaps. `Building block` may be blank while nothing closes it. `Severity` needs more than one value.

How.

1. Pull what exists. `R06` holds 30 interface rows: 22 `Unmanaged`, 21 with `Logged` = `No`, four of `Type` = `Direct DB read`. That is four gaps, not one.
2. `O20 Data Entity`: six of fourteen read `Lineage` = `Absent`, including `E03 Routing / Recipe`, `E04 Work Order` and `E05 Batch / Lot` — the three a traceability obligation turns on.
3. `O49 Resilience Tier`: eight of twelve read `Never tested`. `O63 Control Objective`: five of six obligations have no evidence, and the obligations are written for a European manufacturer.
4. Add the organisation gap from chapter 4: `O10 Actor` and `O11 Role` hold the same eighteen corporate titles and no plant role at all.
5. Fill Form E-1 in the room with the architect who produced each number, then type it in the same day.

Signed by. Chief Enterprise Architect (`ROL-08`), countersigned by the domain architect for each row's `Domain`.

Done when. Every row's `Gap` text traces to a cell in a named sheet, and no row is another sheet's name with a negation bolted on.

Shows up in. `RPT07` grows from two rows to between fifteen and thirty. If it does not, you have not finished.

Goes wrong when. You write one gap per domain to keep the page tidy. A gap is the unit the board votes on: four gaps, four votes, forty packages.

Look it up. Volume 1 chapter 19, *What to count before you model anything* — the counts that turn an impression into a gap.

Form E-1 — Gap register

Feeds `029 Gap`. Six of its seven columns; `Status` is typed straight into Excel, because you set it from the package. A first pass over four domains needs fifteen to thirty rows. Signed by the Chief Enterprise Architect and the domain architect named in `Domain`.

Ref	Gap	Domain	Building block	Severity	Due
GAP-03	EXAMPLE — delete this row. 22 of 30 interfaces unmanaged	Application		High	

Continue on a second copy. Keep the two original rows at the top of the sheet: when somebody asks why the register tripled, they are the answer.

The money

Twelve packages in `030 Work Package` sum to **$65.0m** against `ALT-B Foundations first` **at $46m**, `Recommended`. Nineteen million dollars of work is in the plan and in no alternative [1].

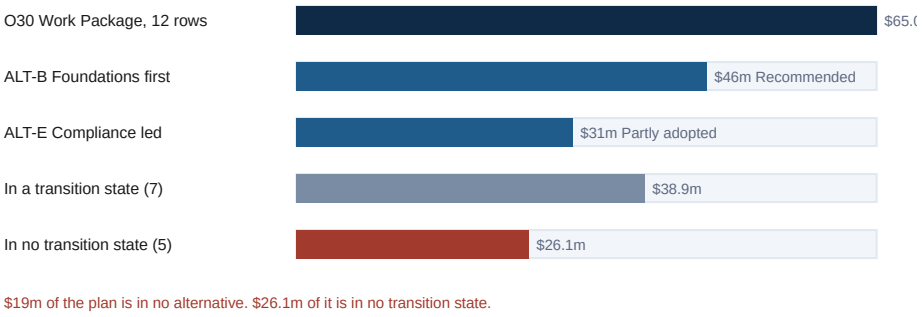


Figure 8.2 The three numbers that do not reconcile. The plan is **$65.0m**, the recommended alternative is **$46m**, and $26.1m of the plan sits in no transition state.

TAKE THESE TO THE INVESTMENT COMMITTEE

What does the recommended alternative buy that the rejected ones do not? The sheet has no scope column, so the answer is not in the file.

Which packages would you stop to bring the plan back to the funded number?

Step 8.3 — Reconcile the package total against the recommended alternative

Produce. A reconciliation that either closes the $19m or records, line by line, what the difference is made of.

You need. `O30` (`Cost`), `O54`, `O60 Governance Decision`, the CFO's delegate from `GB-02 Investment Committee`.

Where it lands.

> **`O54 Architecture Alternative` — five rows, and the columns it does not have**
>
> - `ALT-A` `$24m` `Rejected`; `ALT-B Foundations first` `$46m` `Recommended`; `ALT-C` `$52m` `Rejected`; `ALT-D` `$88m` `Rejected`; `ALT-E Compliance led` `$31m` **`Partly adopted`**.
> - **No `Scope`, `Assumption`, `Date` or `Decision` column**, so the sheet cannot say what $46m buys or when the figure was struck. Put the reconciliation in `Description`, and raise the missing columns in `O61 Governance Change Request`.

How.

1. Total `O30.Cost` in the sheet, with a formula, and leave the formula there: 5.5, 8.8, 3.1, 3.2, 3.1, 3.9, 9.5, 3.4, 7.4, 4.2, 5.7, 7.2 — **$65.0m**.
2. Set that against `ALT-B` at $46m. The difference is $19m, 29 per cent of the plan.
3. Test the two explanations usually offered. Is $46m the transition states only? `R15` covers seven packages worth $38.9m, not $46m. Is it net of a withdrawn package? None is marked withdrawn and no status value exists for it.
4. `ALT-E` at $31m is `Partly adopted`. $46m plus some part of $31m overshoots or undershoots $65.0m depending which part, and nothing says which. That is a range.
5. Give the committee three lines: what is in the plan, what is in the recommendation, what the difference is. Add the rand equivalent and the conversion date, because the committee funds in rands.

Signed by. Investment Committee (`GB-02`), which holds G0 funding under `058`.

Done when. One number is the plan. Today there are two and neither is marked superseded.

Shows up in. `RPT08 Roadmap` reprints the twelve costs, so the total is derivable. Nothing in `ZZ Checks` tests it, which is why it survived.

Goes wrong when. The committee approves `ALT-B` and delivery runs `O30`. Nineteen million dollars proceeds unfunded, and it will be the packages nobody is watching — step 8.8 says which five.

Look it up. Volume 1 chapter 4, *Maintenance capital versus growth capital* — the distinction that decides which half of a difference like this is discretionary.

Step 8.4 — Get the alternative decided, and record what Partly adopted means

Produce. A dated decision in `O60` adopting one alternative, with the part-adopted scope of any other written out.

You need. `O54`, `O60` (seven rows), `O58`, `O61`, `O52 Implementation Factor`.

Where it lands.

> **`O60 Governance Decision — four columns, seven rows, no date`**
>
> - `DEC-01 Adopt the recommended architecture alternative`, `Body` `Investment Committee`, `Status` **`Outstanding`** — as are `DEC-02 Select the integration product` and `DEC-03 Fund compliance mechanisms`. `DEC-05` to `DEC-07` read `Approved`.
> - **No date column**, so a decision cannot be aged and `Outstanding` cannot be shown overdue. Record the date inside `Decision` until a column exists.
> - No decision covers `ALT-E`, which nonetheless reads `Partly adopted`.

How.

1. Read `DEC-01`. `O54` recommends and the decision to adopt is `Outstanding`: the sheet recommends and nobody has accepted.
2. Find the part of `ALT-E` that was adopted. If none can be named, set the cell to `Rejected`.
3. Put `DEC-01`, `DEC-02` and `DEC-03` on one Investment Committee agenda. Everything in Phase F waits on those three, and `O52` already records two as blockers — `IF-01 Vendor selection not made` and `IF-02 Data owners not designated`.
4. Take the decision against step 8.3's reconciled number, not against `$46m`. Write the date into the `Decision` text, raise the missing date column in `O61`, and give every `Outstanding` row an owner and a committee date.

Signed by. Investment Committee (`GB-02`), escalating to `GB-04 Executive Committee`, whose stated mandate is bodies that cannot agree.

Done when. `DEC-01` reads `Approved` with a date, and no `O54.Recommendation` value is unsupported by a decision.

Shows up in. Nothing. No report covers `O54` or `O60`, so a decision is visible only in the sheet. Say that to the committee; it changes how carefully they read it.

Goes wrong when. Delivery treats `Recommended` as approval. Six months on, the committee asks when it agreed to $65.0m and no row says it did.

Look it up. Volume 1 chapter 18, *The decisions that must be taken before anything can start*.

The building blocks

`O55 Building Block` is not a sheet. It is a composed filter over four others: ten `BBB-` rows from `O35`, ten `DBB-` from `O44`, ten `ABB-` from `O45`, seven `TBB-` from `O46`, with a `Domain` column added and the parent reference dropped. Thirty-seven rows, every one of which exists twice in the file [1]. Draw the repository and every building block appears twice.

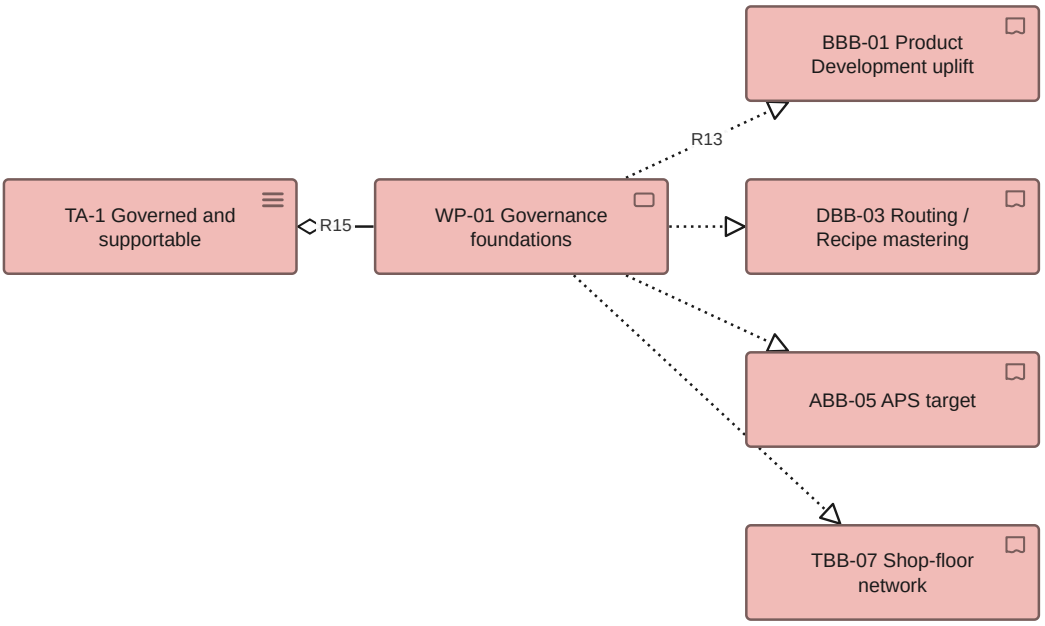


Figure 8.3 Everything `R13` says `WP-01 Governance foundations` delivers. Four building blocks drawn from four domains by row position, with nothing in common.

Step 8.5 — Test all four acceptance-criteria families and mark which a gate could use

Produce. A verdict per family — testable, or generated template — written into the sheet.

You need. `O35`, `O44`, `O45`, `O46`, `O55`; `R06`; `O04 Measure`; `O58`; Form E-2.

Where it lands.

> **`O55 Building Block — the Acceptance criteria column, 37 rows`**
>
> - `BBB-01` – `BBB-10`: `<Capability> measured and owned.` **Template** — names no measure and no owner, and `O04 Measure` holds no capability measure to point at.
> - `DBB-01` – `DBB-10`: `No <entity> record without a named owner.` **Well-formed query, untestable here** — `O20` carries `Owner` per entity, not per record, so nothing can be counted.
> - `ABB-01` – `ABB-10`: `No unmanaged interface to or from <application>.` **Testable** against `R06.Managed` today.
> - `TBB-01` – `TBB-07`: `<Block> in production for at least one service.` **Template** — true the moment anything ships, so it gates nothing.

How.

1. Run the `ABB` criterion. Join `R06` on `Source` and `Target` and count `Unmanaged`: `A01 ERP Core` 3 of 3, `A02 PLM` 7 of 8, `A03 MES` 5 of 7, `A04 SCADA / Historian` 4 of 4, `A05 APS` 3 of 5, `A06 WMS` 1 of 3, `A07 QMS` 8 of 8, `A08 CMMS / EAM` 1 of 2, `A09 CAD / CAE` 4 of 8, `A10 Supplier Portal` 2 of 2. All ten fail. That is what a testable criterion looks like.

2. Note what the family misses: `O23` holds twelve applications against ten `ABB` rows, so `A11 Track & Trace` and `A12 Plant Analytics` have no block and carry six unmanaged interfaces between them. `RPT13 Orphan Analysis` already lists both as orphans.
3. Try the `BBB` criterion against `O04 Measure`. Sixteen measures, none a capability measure, and two — `MSR-05 Single customer view coverage`, `MSR-06 Cross-sell ratio` — from another industry entirely. No OEE, no first-pass yield, no on-time-in-full, no scrap rate.
4. Try the `DBB` criterion. `O20.Owner` holds job titles — `Head of Finance`, `Head of Sales`, `Head of Health, Safety` — of which only `Data Steward` resolves against `O11 Role`. No per-record owner exists to count.
5. Rewrite the three failing families so each criterion names a sheet, a column and a threshold. Form E-2 gives the shape, and the verdicts go in `O59 Compliance Assessment` — empty, header row only: `Ref`, `Building block`, `Block name`, `Acceptance criterion`, `Gate`, `Verdict`, `Evidence`.

Signed by. Architecture Function (`GB-05`), whose mandate is publishing criteria and assessing compliance.

Done when. All 37 rows carry a criterion naming a sheet, a column and a number, with the verdict in `O59`.

Shows up in. `O59` goes from zero rows to 37 and `R24 Compliance Assessment tests` stops being header-only, so `ZZ Checks` can claim 65 of 65 object types populated truthfully.

Goes wrong when. You rewrite the criteria and leave them in a document. `G2 Build complete` tests that every criterion is evidenced, and a criterion not in the repository is not tested.

Look it up. Volume 1 chapter 7, *So the first deliverable is a measure definition register* — why `measured and owned` is unenforceable until the measure has a source and a denominator.

Form E-2 — Building-block acceptance criteria

Feeds `O55 Building Block` and, through it, `O35`, `O44`, `O45` and `O46`, because `O55` is a filter and the real rows live on the parents. Four of five columns; `What it must do` is typed into Excel. Thirty-seven rows, so expect four copies. Signed by the Architecture Function.

Ref	Domain	Building block	Acceptance criteria
ABB-01	Application	ERP Core target	EXAMPLE — delete. R06.Managed shows 0 Unmanaged rows for A01

Step 8.6 — Collapse the duplication, and decide what O46's second Ref column is for

Produce. A statement of which sheet is authoritative, and a resolution for the `T01`–`T07` key space.

You need. `O55`, `O35`, `O44`, `O45`, `O46`, `R13 Work Package delivers Build`, `00 Index`.

Where it lands.

> **`O46 Technology Building Block` — five headers, two of them identical**
>
> - Headers read `Ref`, `Technology building block`, **`Ref`**, `What it must do`, `Acceptance criteria`. The second `Ref` holds `T01` to `T07`, and nothing else in the file uses a `T` prefix, so they resolve to nothing [1]. Call it `Ref (T)` in every document you write.
> - `O55` reproduces `O46`'s rows but not its second `Ref`, so the only place the `T` space exists is a column with a duplicate header.

How.

1. Decide the direction of truth. `O55` adds `Domain` and drops the parent reference — `Capability` on `O35`, `Entity` on `O44`, `Application` on `O45` — so the parents hold more.

Make them authoritative and `O55` a view.

2. Say so in `O55`'s subtitle row, which reads only `37 records`.
3. Resolve `T01`–`T07`. Either map them to `O25 Technology Service` refs and keep the column, or delete it. Do not leave a key space resolving to nothing.
4. Check `R13`, whose 37 rows are one per `O55` row, so it is bound to the view. If `O55` becomes a view, `R13` must point at the parents.
5. Raise both in `O61`, which already carries `CR-G03` and `CR-G04`, and record the count consequence: 37 of the 813 object rows are the same blocks twice.

Signed by. Chief Enterprise Architect (`ROL-08`).

Done when. A reader can say which sheet to edit, and no header appears twice on one sheet.

Shows up in. `RPT01 Dashboard` and `ZZ Checks` both carry record counts. Both should move; the direction says whether anybody recomputed or retyped.

Goes wrong when. Somebody edits `O55` and the parent does not change. Two versions of one block, no change log to tell them apart, and `G1 Design` assessing the wrong one.

Look it up. Volume 1 chapter 12, *IIoT platforms, unified namespace, edge and the digital twin* — what `TBB-02` to `TBB-05` are, and why the seven technology blocks are the most credible rows on any building-block sheet here.

The transition states

Three states in `O53 Transition Architecture`, seven assignments in `R15`, twelve packages in `O30`. The arithmetic does not reach.



Figure 8.4 The transition-state overlap. `R15` carries seven assignments over twelve packages, so **five packages worth $26.1m** belong to no plateau and therefore have no exit criterion.

ON THE TRANSITION STATES, BEFORE YOU WRITE A SINGLE CRITERION

What would you have to see, in which sheet and column, to say TA-1 is complete?

Five packages sit in no state. For each, is that an omission or a decision?

Step 8.7 — Write the exit criteria, because the column currently holds a count

Produce. Named, testable exit criteria for `TA-1`, `TA-2` and `TA-3` — eighteen of them, on a sheet that holds one per row.

You need. `O53`, `R15`, `O04 Measure`, `O58`, `O59`, Form E-3.

Where it lands.

> **`O53 Transition Architecture — five columns, three rows`**
>
> - `TA-1 Governed and supportable`, `Work packages` `WP-01, WP-02, WP-04, WP-05`, `Exit criteria` `5 exit criteria`
> - `TA-2 Integrable`, `WP-03`, `6 exit criteria`
> - `TA-3 Business visible`, `WP-07, WP-11`, `7 exit criteria`

The column states how many criteria there are. It does not state one. Eighteen are claimed; none is written down [1].

How.

1. Read the three `Description` cells, the only real content on the sheet: `Estate governable and within vendor support`; `Managed integration carrying production traffic`; `First capability the customer notices`.
2. Turn each into criteria naming a sheet, a column and a threshold. For `TA-1`: `R06.Managed` shows zero `Unmanaged` rows for `A01`, `A03` and `A04`; `O27.Support end` is in the future on all twelve rows; `O20.Owner` resolves against `O11 Role` on all fourteen; `O11` carries a plant manager, a production planner and a quality manager, each `Filled` in `R21`.
3. For `TA-2` write against `R06.Logged` and `R06.Type`, where four rows read `Direct DB read`. For `TA-3` pick one business service from `O15` and a measure that exists — if none does, step 8.5's problem has arrived.
4. Keep the counts of five, six and seven if the criteria support them; change them if not.

5. `O59` takes a criterion per building block, not per transition state, so create `O53E Transition Exit Criterion` with Form E-3's columns and say in its subtitle row that it is an addition to the metamodel. Replace the `Exit criteria` cell with a pointer to it, not with a count.

Signed by. Architecture Board (`GB-01`), which holds `G1` and `G2` .

Done when. Eighteen rows exist, each naming a sheet, a column and a threshold, and no cell reads `n exit criteria` .

Shows up in. Nothing, because no report covers `O53` . Add the criteria count and pass rate to `RPT01 Dashboard` , the only report with room for a computed line.

Goes wrong when. `G2` arrives and the board is asked to confirm a state against a count. It either refuses, which costs a quarter, or waves it through, which costs the gate.

Look it up. Volume 1 chapter 10, *The measure definition register* — the artefact that makes a threshold arguable rather than negotiable.

Form E-3 — Transition-state exit criteria

This form feeds a sheet the repository does not have. `053` holds one `Exit criteria` cell per state and that cell holds a count. Create `053E Transition Exit Criterion` with the columns below, headers in row 4 and data from row 5, as every sheet in the file is laid out. Eighteen rows across three states. Signed by the Architecture Board.

Ref	Transition	Criterion	Sheet and column	Threshold	Evidence
TAE-01	TA-1	EXAMPLE — delete. ERP, MES and SCADA interfaces managed	R06.Managed	0 Unmanaged	

Step 8.8 — Put the five orphaned work packages into a transition state, or out of the plan

Produce. An `R15` with twelve assignments, or a written decision that five packages sit outside the transition sequence, and why.

You need. `R15`, `053`, `030`, `066 Architecture Increment`, the reconciled number from step 8.3.

Where it lands.

> **`R15 Transition contains Work Pa — four columns, seven instances`**
>
> - `TA-1` holds `WP-01`, `WP-02`, `WP-04`, `WP-05` — $20.6m
> - `TA-2` holds `WP-03` — $3.1m
> - `TA-3` holds `WP-07`, `WP-11` — $15.2m
> - No state holds `WP-06`, `WP-08`, `WP-09`, `WP-10`, `WP-12` — **$26.1m**

Forty per cent of the plan by value is in no transition state, including `WP-09 Resilience uplift` at $7.4m and `WP-12 Supplier and partner integration` at $7.2m — the largest

packages outside `WP-07`.

How.

1. Confirm the sums: 20.6 plus 3.1 plus 15.2 plus 26.1 is 65.0, which ties to `O30`. Then take the orphans one at a time and ask which `Description` in `O53` they serve. `WP-09 Resilience uplift` and `WP-10 Compliance mechanisms` belong in `Governed and supportable`; `WP-12` belongs in `Integrable`.
2. `WP-06 Analytics and reporting` and `WP-08 Process automation` fit none of the three. That is a signal the file is short a transition state — `O66` already names `Compliant by mechanism`, `Analytics and evidence` and `Core renewal`, three plateaus `O53` does not have.
3. Either add the missing states to `O53` or move the packages. An unassigned package has no exit criterion and so no gate.
4. Write the new rows into `R15`: `Transition`, `Name`, `Work package`, `Package name` — both ends by reference and by name, the file's own convention. Then hand the result to Phase F; chapter 9 step 9.10 picks it up.

Signed by. Architecture Board (`GB-01`), with the Investment Committee notified: $26.1m moves into or out of a funded plateau.

Done when. Every `WP` ref in `O30` appears in `R15`, and the per-state costs sum to the plan total.

Shows up in. `RPT08 Roadmap` lists all twelve packages and shows no transition state, so the orphaning is invisible. Add a `Transition` column or it stays hidden.

Goes wrong when. The programme tells the board a plateau-by-plateau story and delivers the orphans in parallel: three states governed, $26.1m not.

Look it up. Volume 1 chapter 18, *The target and the transition* — three states with verifiable exit criteria, and the point that the first year buys agreement rather than software.

The traceability

Step 8.9 — Break the two positional diagonals before Phase F inherits them

Produce. An `R13` and an `R09` whose rows mean something, and a note stating what the originals were.

You need. `R13` (37 rows), `R09 Requirement satisfied by Bu` (31 rows), `O06 Requirement`, `O55`, `RPT06 Requirements Traceability`.

Where it lands.

> **R13 and R09 — the two sheets Phase E depends on most**
>
> - `R13` cycles the twelve packages against the 37 building blocks in order, so `WP-01 Governance foundations` delivers `BBB-01 Product Development capability uplift`, `DBB-03 Routing / Recipe mastering`, `ABB-05 APS target` and `TBB-07 Deterministic shop-floor network` — four things with nothing in common.
> - `R09` pairs `REQ-001` to `BBB-01`, `REQ-002` to `BBB-02` onward, so a requirement is satisfied by whichever block shares its row number. `REQ-001 Evidence ISO 9001` is satisfied by a product-development capability uplift.
> - `R09.Traced` reads `No` on `REQ-001`, `-005`, `-009`, `-013`, `-017`, `-021`, `-025`, `-029`: eight rows, **every fourth row from the first**. That is a generation pattern. No assessor produces that.

How.

1. Prove the stride before you argue about it. List the eight `No` rows, then check `RPT06`, whose `Reaches a block` column reprints the same eight.
2. Rebuild `R09` from `O06`'s own `Requirement` text. `REQ-015 ERP Core integration managed` belongs to `ABB-01 ERP Core target`; `REQ-011 Batch / Lot mastered once` to `DBB-05 Batch / Lot mastering`. Most of the 31 map obviously once read.
3. Leave `Traced` = `No` where nothing satisfies the requirement and let the number be ugly. `REQ-003 Evidence Product safety and CE / UKCA marking` has no South African application at all, which chapter 12 of this workbook deals with.
4. Rebuild `R13` from the parents. `WP-02 Data ownership and mastering` should carry the ten `DBB-` rows, not three rows drawn by position.
5. Recount. `R13` stops having 37 rows; some packages deliver many blocks and some none, and a package delivering none is a finding rather than an error.
6. Record what each sheet used to be in its subtitle row — without a change log that line is the only history there is.

Signed by. Chief Enterprise Architect (`ROL-08`), with the Head of Delivery (`ROL-13`) on `R13`, which becomes the project scope.

Done when. No Phase E relationship sheet resolves by row position, and `R09.Traced` has no arithmetic stride.

Shows up in. `RPT06` and its footer. Expect `Reaches a block` = `No` to rise sharply, then fall as the work is done.

Goes wrong when. You fix `R09` and leave `R13`. Phase F then charters twelve projects against blocks assigned by row number.

Look it up. Volume 1 chapter 17, *Requirements realization* — the viewpoint that makes a diagonal visible the moment you draw it.

Step 8.10 — Turn the eight implementation factors into dated, owned constraints

Produce. Eight factors each carrying an owner, a date and the packages it binds, ready for Phase F.

You need. `O52`, `O07 Constraint`, `O30`, `O62 Delivery Risk`, `O11 Role`.

Where it lands.

> **`O52 Implementation Factor` — four columns, eight rows, the only planning constraint in the file**
>
> - `IF-01 Vendor selection not made`, `Dependency` — *Blocks integration work until resolved*
> - `IF-02 Data owners not designated`, `Dependency` — *Blocks mastering work*
> - `IF-03 Peak trading freeze`, `Constraint` — *No change in the peak window*
> - `IF-04 Regulatory deadline`, `Constraint` — *Fixed external date*
> - `IF-05 Scarce legacy skills`, `Resource` — *Limits parallel change on the core*
> - `IF-06 Concurrent capacity unknown`, `Assumption` — *Plan assumes capacity that is unrecorded*
> - `IF-07 Licence model change`, `Risk` — *Vendor commercial change mid-programme*
> - `IF-08 Site access windows`, `Constraint` — *Physical work limited to shutdowns*

How.

1. Read `IF-08`: `Physical work limited to shutdowns`. That is the one row written by somebody who has been in a plant, and chapter 9 hangs off it. The file names the constraint and records no shutdown date anywhere.
2. Read `IF-03`: `Peak trading freeze`. `Trading` is retail language, and `O07`'s `CON-06 Change freeze windows` repeats it. A manufacturer's freeze is a seasonal production peak, a model-year changeover or a customer's shutdown — three different dates. Rewrite the row.
3. Add the three columns the sheet lacks: `Owner`, `Date`, `Binds`. With no owner column, no factor has anybody to resolve it.
4. Tie `IF-01` and `IF-02` to `DEC-02` and `DEC-04` in `O60`, both `Outstanding`, and to `RSK-01` and `RSK-02` in `O62`, both `High` impact. The same two facts sit in three sheets and are joined in none.
5. Tie `IF-05` and `IF-06` to `CON-02 Delivery capacity unrecorded` and `CON-03 Legacy platform skills scarce`: four rows, two facts. Decide which sheet owns each.
6. Hand `IF-03` and `IF-08` to chapter 9 steps 9.4 and 9.6 with dates filled in. They are the two that stop a plan.

Signed by. Head of Delivery (`ROL-13`) for `IF-05` and `IF-06`; Chief Enterprise Architect (`ROL-08`) for the rest.

Done when. Every factor has a named person from `O11 Role`, a date, and the `WP` refs it binds.

Shows up in. Nothing. No report reads `O52` — which, for the only sheet recording a planning constraint, is itself the finding.

Goes wrong when. Phase F plans against a twelve-month year because `IF-08` carried no dates. Chapter 9 is almost entirely about that failure.

Look it up. Volume 1 chapter 7, *Time fences, and why nobody respects them* — why a freeze window that is not in the plan will be broken and then enforced.

The comparison tool

BEFORE YOU ASK ANYONE TO BUILD AN EXTRACT

What are you comparing against, given that this repository has no baseline and no version column?

Who will read the diff, and what will they do with a row marked UNGOVERNED CHANGE?

This set ships `Manufacturing_Architecture_Comparison_Tool_TEMPLATE.xlsx` beside the repository [3]. It exists because the repository has no change control: with no version column and no change log, the only way to know what moved between two points is to extract both and difference them. The contract matters more than the workbook, because you will be told to re-run it against a later extract and the columns have to line up.

THE COMPOSITE KEY, WHICH IS THE WHOLE CONTRACT IN ONE LINE

Every attribute row is keyed `Sheet|Ref|Attribute` — `O13|C02.03.01|Current`, or `O30|WP-05|Approval`. The record key is the same string without the third part: `O30|WP-05`. Pipe-delimited, no spaces around the pipes, `Ref` spelled exactly as the repository spells it.

Step 8.11 — State the comparison-tool contract before anyone extracts anything

Produce. A one-page contract note, agreed with whoever builds the extract, covering all ten points.

You need. The repository, the tool template, and the person who will generate the extract.

Where it lands.

The tool's sheets and the columns each one carries

Sheet	Columns
`01 Config`	`Snapshot A label`, `Snapshot A date`, `Snapshot B label`, `Snapshot B date`, `Comparison mode`, `Prepared by`, `Prepared on`
`10` / `11`	`Key`, `Kind`, `Type`, `Type Name`, `Ref`, `Name`, `Attribute`, `Value`; `10R` / `11R` the same, less `Attribute` and `Value`
`12 Input Change Log`	Matched on the composite key; supplies `Change Request` and `Logged new value`
`30 Diff Records`	`Scope`, `Key`, `Kind`, `Type`, `Type Name`, `Ref`, `Name`, `In A`, `In B`, `Status`
`32 Diff Attributes`	`Scope`, `Key`, `Type`, `Type Name`, `Ref`, `Name`, `Attribute`, `Value in A`, `Value in B`, `Differs`, `Logged change`, `Change Request`, `Logged new value`, `Classification`
`50 Review`	`Key`, `Type Name`, `Ref`, `Attribute`, `Tool classification`, `Your classification`, `Rationale`, `Owner`, `Action`
`60 Summary`	The `01 Config` fields, plus `Records in A`, `Records in B`, `Attribute rows compared`, `Change log entries supplied`, `Records added`, `Records removed`, a six-value classification split and a four-value review split
`70 Gap Candidates`	`O29 Gap`'s seven columns, then `Source difference` and `Owner`

How.

1. Fix the key: `Sheet|Ref|Attribute` for attribute rows, `Sheet|Ref` for records. Nothing downstream works if two extracts key differently.
2. Fix the enumerations, because the formulas switch on them. `Kind` is `Object` or `Relationship`; `Type` is the sheet prefix, `O30`, `R15`, `EXT`; `Comparison mode` is `Baseline vs Target` or `Snapshot vs Snapshot`; `Scope` on `30` and `32` is `From A` or `From B`; `Status` on `30` is `Present both`, `REMOVED in B` or `ADDED in B`.
3. Fix the direction of the two typed sheets. `12 Input Change Log` is an input: it joins on the key and feeds `Change Request` and `Logged new value` into `32`. `50 Review` is the only sheet a person types into, and `Your classification` drives `70`.
4. Fix `70 Gap Candidates`: only rows a human marked `Gap` on `50`, carrying `O29 Gap`'s seven columns in `O29`'s order so a row transcribes without translation, then `Source difference` and `Owner`. `Severity` arrives as a placeholder for a person to set; `Status` defaults to `Open`.
5. Fix the header name. The key column is `Ref` on every sheet, matching the repository's own first column, so nothing is renamed either way. Get the note signed by the extract's author: a contract agreed after the extract exists is a rework order.

Signed by. Chief Enterprise Architect (`ROL-08`), with the extract author named on the note.

Done when. All ten points are written down and the extract author has read them back to you.

Shows up in. `60 Summary`, whose figures are formulas over the diff sheets. If `60` disagrees with `30` and `32`, the contract broke upstream.

Goes wrong when. The second extract spells an attribute differently — `Approval` against `Approval status` — and every row in `32` reports as both added and removed. The tool tells you the whole estate changed.

Look it up. Volume 1 chapter 10, *Moving data between systems* — why the key, not the payload, is what breaks.

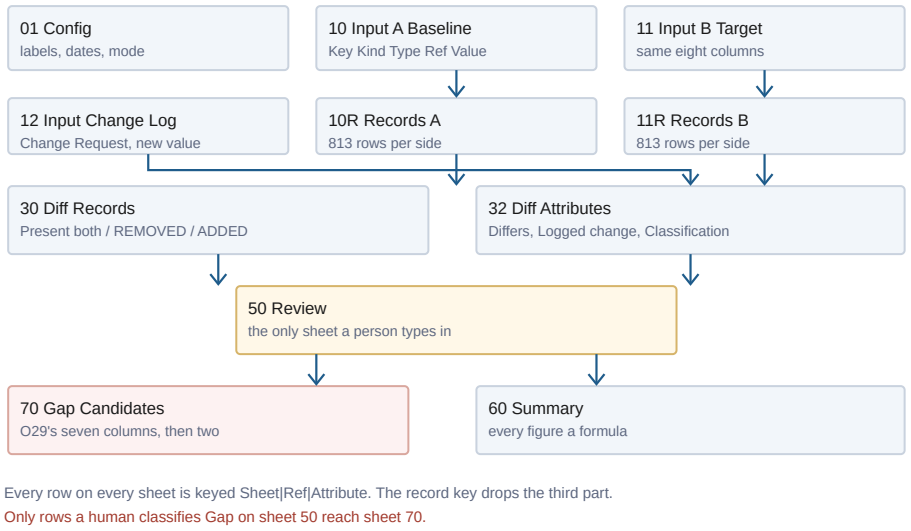


Figure 8.5 The comparison tool as a data flow. Two extracts in, two diff sheets computed, one human review sheet, and **only reviewed gaps** reaching `70 Gap Candidates`.

Step 8.12 — Size the extract from the counted rows, not from ZZ Checks

Produce. A stated extract size per side, with the count you used and why you did not use the file's own figure.

You need. The repository, `ZZ Checks`, `00 Index`, and openpyxl or a count by hand.

Where it lands.

> **`10R Records A and 11R Records B — the record lists, one row per record`**
>
> - **813 records per side**, counted across the 64 object sheets plus `EXT Industry Extension`.
> - `00 Index` and `ZZ Checks` both say **843**. The difference is exactly 30, the row count of `R06 Application interfaces with` — the only relationship sheet with a `Ref` column. The total reconciles only if an interface counts as a record, which concedes that an interface is an object with no sheet of its own.
> - Relationship rows total **430**, the one count in `ZZ Checks` that holds.

How.

1. Count. Iterate the 64 object sheets plus `EXT`, data from row 5, counting rows with a non-empty first cell. The answer is 813.
2. Quote 813 everywhere, and say in the same sentence that the file claims 843 and why. A reader who finds the discrepancy after you stops trusting the rest.
3. Size the attribute extract. Treating every column except `Ref` and the name column as an attribute gives about **3,000 attribute rows per side**. `O13 Business Capability` alone contributes about 700 — 100 rows by seven attributes — and 90 of those rows appear in no report.
4. So decide the attribute subset before extracting. Take all of `O13` and a quarter of the diff is a generated cross-product nobody reads.
5. Set `01 Config` honestly: `Snapshot A label` names a baseline that does not exist here, so say so rather than inventing one. And note that `EXT` holds six type definitions and no instances, so it contributes six records and almost no attributes.

Signed by. Chief Enterprise Architect (`ROL-08`).

Done when. The note states 813, states 843, gives the 30-row reason, and names the sheets excluded from the attribute extract.

Shows up in. `60 Summary`'s `Records in A`. If it reads 843, the extract counted something twice.

Goes wrong when. You take 843 from `ZZ Checks` and the tool reports 30 records removed on the first run. A day goes on hunting a deletion that never happened.

Look it up. Volume 1 chapter 10, *Duplicate parts as a measurable cost* — the same arithmetic on an item master, and the reason a generated row is not free.

Closing the phase

Step 8.13 — Take Phase E through G0 and hand the sequencing to Phase F

Produce. A pack the Investment Committee can vote on: reconciled cost, decided alternative, rebuilt register, testable criteria, real exit criteria, twelve packages assigned.

You need. Steps 8.1 to 8.12; `O31 Deliverable`, `O33 Architecture Contract`, `O58`, `O61`.

Where it lands.

> **O31 Deliverable — twenty rows, and the three that are yours**
>
> - `DLV-13 Consolidated Gaps`, `Phase` `E`, `Status` **`Approved`** — approved over a register of two rows. `DLV-14 Architecture Roadmap` and `DLV-15 Work Packages` are both `Draft`.
> - Thirteen of twenty rows read `Approved`, every one from Phase A to Phase D plus `DLV-13`. The sheet has no date, no approver and no version column, so an approval cannot be aged or attributed.

How.

1. Move `DLV-13` back to `Draft` until the register has been rebuilt.
2. Assemble the pack in the order the committee reads it: the reconciled number, the decision being asked for, the five orphaned packages and the $26.1m, the exit criteria.
3. Check `O33 Architecture Contract`: eight contracts against twelve packages, so `WP-09` to `WP-12` have none and cannot pass `G0 Charter`.
4. Take the two reading `Non-compliant` — `ACO-02` against `WP-02`, `ACO-03` against `WP-03` — to the board as contract breaches rather than roadmap items.
5. Close out `CR-G03 Raise unfunded building blocks as gaps`, already `Raised` in `O61`, which is step 8.2's work, and hand chapter 9 the assigned `R15`, the dated `O52` and the reconciled total.

Signed by. Investment Committee (`GB-02`) at `G0`, on the Architecture Board's recommendation.

Done when. `DEC-01` is `Approved` with a date, `O29` has more than two rows, `R15` covers twelve packages, and each of the 37 acceptance criteria names a sheet and a threshold.

Shows up in. `RPT07`, `RPT08` and `RPT06` all move, and `ZZ Checks` can report 65 of 65 object types populated truthfully because `O59` is no longer empty.

Goes wrong when. You hand Phase F the roadmap and not the arithmetic. It then schedules $65.0m against $46m of funding, five packages with no plateau, and eighteen exit criteria that do not exist.

Look it up. Volume 1 chapter 14, *Maturity models, and what they are actually for* — a score at a gate is a conversation opener, not a pass mark.

THE REVIEWER'S ASK — ANSWER THESE BEFORE THE PACK LEAVES THE ROOM

State the plan total and the funded total, in rands, with the date of the rate you used.

Name the five work packages in no transition state and the value they carry.

Which of the four acceptance-criteria families can be tested against the repository as it stands, and against which column?

How many records are in the repository, and what does the file say? Explain the difference in one sentence.

Which Phase E deliverable in `031` is marked Approved and should not be?

THE CHAPTER IN TEN LINES

- O29 Gap holds two rows. Both read High, and both are a building block's own name with not in place appended.
- R12 bolts GAP-01 to WP-01 Governance foundations, which does not close a product-development gap.
- O30 sums to 65.0 million against a recommended alternative of 46 million. Nineteen million is unexplained.
- ALT-E Compliance led reads Partly adopted at 31 million, and no decision in O60 records it.
- DEC-01, the decision to adopt the recommended alternative, is still Outstanding.
- O53 states a count where a criterion belongs: five, six and seven exit criteria, none of them written down.
- R15 covers seven of twelve work packages. Five, worth 26.1 million, sit in no transition state at all.
- O55 Building Block is a view over O35, O44, O45 and O46, so every one of its 37 rows exists twice in the file.
- Only the ABB acceptance criteria are testable. They run against R06.Managed, and all ten fail today.
- R13 cycles twelve packages against 37 blocks by position, and R09 pairs requirements to blocks by row number.
- R09.Traced reads No on every fourth row. That is generation, not assessment.
- O52 Implementation Factor holds the only planning constraint in the file, and IF-08 limits physical work to shutdowns.
- The repository holds 813 records, not the 843 it claims. Size the comparison extract on 813.

SOURCES FOR THIS CHAPTER

- **1.** Manufacturing_TOGAF_Metamodel.xlsx, Kestrel Industrial Group — sheets 00 Index, O04, O07, O11, O20, O23, O29, O30, O31, O33, O35, O44, O45, O46, O49, O52, O53, O54, O55, O58, O59, O60, O61, O62, O63, R06, R09, R12, R13, R15, R21, RPT01, RPT06, RPT07, RPT08, RPT13, ZZ Checks
- **2.** The Open Group, TOGAF Standard, Architecture Development Method — Phase E, Opportunities and Solutions https://pubs.opengroup.org/architecture/togaf-standard/adm/
- **3.** Manufacturing_Architecture_Comparison_Tool_TEMPLATE.xlsx — sheets 01 Config, 10 Input A Baseline, 11 Input B Target, 10R Records A, 11R Records B, 12 Input Change Log, 30 Diff Records, 32 Diff Attributes, 50 Review, 60 Summary, 70 Gap Candidates

CHAPTER 9

Migration Planning

A delivery year in a plant is about thirty-five usable weeks, and nothing in this repository records a single one of them.

Why this chapter is not the one in the other volumes

Phase F takes the work packages Phase E agreed and turns them into a dated, resourced, sequenced plan [2]. In most industries that is an exercise in dependency and capacity. In a manufacturer it is an exercise in **plant access**, and nothing in this repository records plant access at all.

`O52 Implementation Factor` comes closest: `IF-08 Site access windows` reads `Physical work limited to shutdowns` [1]. That is the right constraint, and there is not one shutdown date in the 105 sheets. So the first half of this chapter repairs what `O30 Work Package` and `O56 Project` say about themselves; the second adds the four things a manufacturing migration plan needs and this file lacks — the window and freeze calendar, the cost of a cutover in lost production, the parallel-run design, and the change-control evidence a system carrying a quality record attracts.

Kestrel Industrial Group is an invention, as is Karoo Industrial Group, volume 1's group, which appears here only as a pointer back into that book. Never mix their numbers.



Figure 9.1 `O30`'s twelve packages placed on `Approval` against `Delivery`. The three outlined cells cannot be true, and the worst of them is **`WP-05` being built with approval `Not started`**.

BEFORE YOU OPEN O30 WORK PACKAGE, ANSWER THESE

Which of your twelve packages needs the line stopped, and for how long?

How many usable weeks are in next year for work that needs plant access? Show the arithmetic.

Who signs off that a cutover may proceed — and is it the same person who signs that the plan is realistic?

What the repository says about itself, and why it cannot be true

Step 9.1 — Reconcile Approval against Delivery, and stop the package nobody approved

Produce. A twelve-row status reconciliation, four contradictions named, one escalated the same day.

You need. `O30 Work Package`, `RPT08 Roadmap`, `O33 Architecture Contract`, `O60 Governance Decision`, the Head of Delivery.

Where it lands.

> **`O30 Work Package` — eight columns, twelve rows, two status columns that disagree**
>
> - `Approval`: `Approved` on seven rows, `In flight` on two (`WP-03`, `WP-07`), `Not started` on three (`WP-04`, `WP-05`, `WP-11`). `Delivery`: `Delivered` on two, `Past end date` on four, `In flight` on three, `Not started` on three.
> - **`WP-05 Access and security`**, **$3.1m: `Approval` = `Not started`, `Delivery` = `In flight`** — a package being built that nobody approved.
> - `WP-07 Core platform renewal`, $9.5m: `Approval` = `In flight`, `Delivery` = `Delivered`, so delivered before approval.
> - `WP-01 Governance foundations` reads `Delivered` with an `End` of `2030-01`. `WP-08 Process automation` reads `Approved` and `Not started`, the one coherent pair.

How.

1. Build the grid above from the sheet yourself, putting each `WP` ref in its cell. The empty cells matter as much as the full ones.

2. Mark the three impossible cells: approval behind delivery on `WP-05` and `WP-07`, and `Delivered` against a future end date on `WP-01`.
3. Escalate `WP-05` the same day. Build without approval is a governance incident, not a data-quality item, and `O33` records `ACO-05 Access and security architecture contract` as `Not yet assessable` — so there is no contract behind it either.
4. `WP-01` reads `Delivered` with thirty-seven months still to run, so somebody typed a word rather than reading a date.
5. Add the `As at` column the sheet lacks: two status columns with no date between them can be reconciled only by their author. Then take the four contradictions to the Architecture Board as one item.

Signed by. Head of Delivery (`ROL-13`), countersigned by the Chief Enterprise Architect (`ROL-08`) on the escalation.

Done when. No approval state sits behind its delivery state, and every row carries the date its status was set.

Shows up in. `RPT08 Roadmap`, which reprints `Approval` and `Delivery` unchanged and adds a footer. Expect the footer to stop matching once the columns are real.

Goes wrong when. `WP-05` keeps running. Access and security touches every system in the estate, and the first audit finding will be that it was built outside the gate regime.

Look it up. Volume 1 chapter 18, *What it costs to be wrong* — the arithmetic that turns a governance finding into a number an executive reacts to.

Step 9.2 — Replace the typed Past end date with a comparison the sheet performs

Produce. A computed overdue flag, and a correction to `ZZ Checks` and `RPT08`.

You need. `O30` (`Start`, `End`, `Delivery`), `RPT08 Roadmap`, `ZZ Checks`, today's date.

Where it lands.

> **`O30.End` and the four rows that claim to be overdue**
>
> - `WP-02` `End` `2030-10`, `WP-03` `2031-06`, `WP-09` `2029-07`, `WP-12` `2030-01` — all four read `Delivery` = `Past end date`, and not one of the dates is in the past.

Every `End` in `O30` falls between `2029-07` and `2031-10`. All twelve are in the future. `ZZ Checks` reports `Work packages past end date 4` against a threshold of `0`, and `RPT08`'s footer reads that four work packages are past their end date [1]. **Both are counting a typed word, not comparing a date.**

How.

1. Sort `O30` on `End`: the first and last values are `2029-07` and `2031-10`, so no package can be overdue.

2. The footer counts the four rows whose `Delivery` cell holds the string `Past end date`. The arithmetic is right; the input is a label.
3. Replace `Delivery` on those four rows with a real state, and add an `Overdue` column comparing `End` against today.
4. Correct the `ZZ Checks` line — one of four wrong lines on that sheet, alongside the record count, the platform-support count and the claim that every object type carries records — and rewrite `RPT08`'s footer to read from the computed column.
5. Note the pattern for Phase G: `027.Position` types `Past support` the same way, against two support-end dates also in the future.

Signed by. Chief Enterprise Architect (`ROL-08`), who owns `ZZ Checks` through `GB-05`.

Done when. The overdue count is a formula, and running it today returns zero.

Shows up in. `ZZ Checks` and `RPT08`. A zero where a four used to be is the right answer, and you will explain it twice.

Goes wrong when. Somebody plans a recovery for four packages that are not late, and capacity goes to the wrong place.

Look it up. Volume 1 chapter 7, *On-time-in-full, and its four date bases* — why a date comparison has to say which date, against which promise.

Step 9.3 — Decide whether O56 Project is a sheet or a transcription, and fix the leads

Produce. Either a project sheet that differs from the package sheet, or a decision to delete it.

You need. `O56 Project`, `O30`, `R14 Project delivers Work Packa`, `O11 Role`, `O09 Organization Unit`.

Where it lands.

> **`O56 Project` — seven columns, twelve rows, eleven of them copied**
>
> - `Project` names are the twelve `O30` package names in order, with identical `Cost`, `Start` and `End`; `R14` pairs `PRJ-01` to `WP-01` onward, one to one.
> - The only column `O56` adds is `Lead`, and every value resolves against `O11 Role` — more than most columns in this file manage.
> - `PRJ-05 Access and security` and `PRJ-08 Process automation` are both led by **`Internal Auditor`** (`ROL-18`): the assurance role leads $6.5m of build.
> - `PRJ-11 Customer experience` is led by `Chief Compliance Officer`; `PRJ-09 Resilience uplift`, a technology package, by `Domain Architect, Data`; `PRJ-06` and `PRJ-12` both by `Data Steward`, $11.1m between them; `PRJ-01` by `Chief Executive`.

How.

1. Diff the two sheets column by column: eleven of twelve `O56` values are `O30` values, and only `Lead` is new.

2. Decide. Either `O56` carries what a project has and a package does not — sponsor, delivery method, team size, benefit date — or it goes and `Lead` moves onto `O30`.
3. Fix the two assurance leads first. An internal auditor leading the access-and-security build cannot then audit it, and `O58`'s `G3 Benefit` depends on somebody independent confirming the benefit.
4. Fix `PRJ-09`: a data architect leading a resilience programme will not be the person who tests a recovery objective, and `O49` shows eight of twelve never tested. Check the `Data Steward` load too — two projects, $11.1m, on the only resolving data owner in the file.
5. Note what `O09 Organization Unit` offers instead: twelve units, `Executive` through `Legal`, and **not one plant, production, quality or maintenance unit**. There is nobody in the organisation model to lead a plant project.

Signed by. Chief Operating Officer (`ROL-03`) on the leads; Chief Enterprise Architect (`ROL-08`) on the sheet decision.

Done when. No project is led by the role that assures it, and `O56` either differs from `O30` or no longer exists.

Shows up in. Nothing. No report reads `O56`, so the leads are visible only in the sheet — which is how an auditor came to be leading a build.

Goes wrong when. The leads stay, and at `G3` the Investment Committee asks internal audit whether the benefit landed on a project internal audit ran.

Look it up. Volume 1 chapter 5, *Who to see first* — who actually has the authority to commit a plant to a date.

The plant year

A delivery year in a manufacturer is not twelve months. It is bounded by three things this repository records nowhere: the **shutdown and maintenance windows**, the **planning and sales-and-operations cycle**, and in a food or agricultural business the **seasonal peak**. `O30` has a `Start` and an `End` and nothing else.

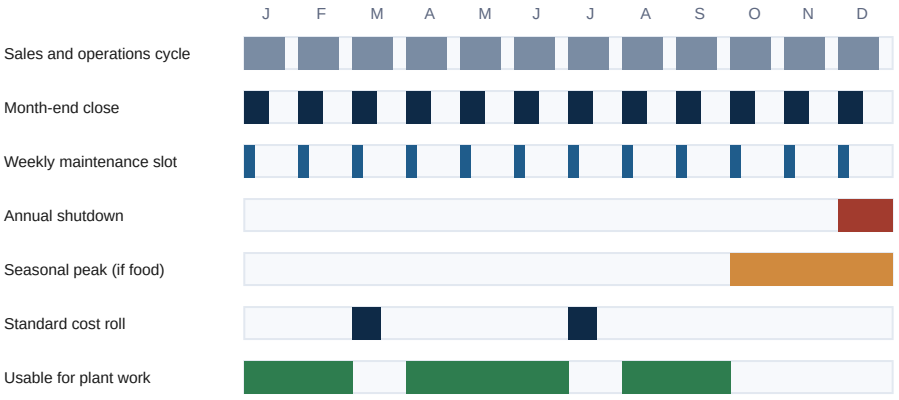


Figure 9.2 The calendar a manufacturing roadmap is actually bounded by: the planning cycle, the close, the maintenance slot, the shutdown, the seasonal peak and the cost roll. **The green band is what is left.**

Step 9.4 — Build the window and freeze calendar before you date anything

Produce. A dated calendar, per plant, of every window in which change may happen and every period in which it may not.

You need. `012 Location`, the plant managers — who do not exist in `010 Actor`, so find them in the business — the maintenance planner, the sales-and-operations chair, `052`, Form F-1.

Where it lands.

> **`012 Location` — eight rows, and what it cannot tell you**
>
> - `LOC-02 Northern Plant` 198 people, `LOC-03 Coastal Plant` 259, `LOC-04 Components Plant` 111, all `Type` = `Production`.
> - There is **no `Line count`, `Shift pattern`, `Format` or `Ownership` column**, so the sheet cannot say how many lines a plant runs or on what pattern — the fact everything in a manufacturing plan turns on.
> - `LOC-08 Cloud Region` carries `261 people`, more than any plant, which tells you how the `Scale` column was filled.
> - Create `012W Site Window` with Form F-1's columns and tie it to `LOC-02`, `LOC-03`, `LOC-04`.

How.

1. Ask each plant for planned shutdown dates for the next three years, not one: `030`'s packages run to `2031-10`.
2. Record the routine windows separately from the annual ones. A weekly sanitation slot and a Saturday maintenance slot are where most control-layer work actually fits; the annual shutdown is the one everybody plans around and the hardest to get on.
3. Record the freeze periods: the seasonal production peak, a customer's model-year changeover, the standard-cost roll and the month-end close. Each closes a different set of weeks to a different set of people.

4. Record the sales-and-operations cycle as a blackout on people, not systems. Its demand and supply reviews occupy most of the month and need exactly the people a workshop needs.
5. Do the arithmetic and write the number down: fifty-two weeks, less the shutdown and its ramp, less the cost-roll blackouts, less the first fortnight of each month for anybody needing the plant controller.
6. Rewrite `IF-03 Peak trading freeze` in `O52` with the real dates and the real name. A manufacturer has no trading peak.

Signed by. Chief Operating Officer (`ROL-03`), with each plant manager signing their own site's rows.

Done when. Every production site in `O12` has a dated calendar for three years, and the usable-weeks figure is written on the plan's front page.

Shows up in. Nothing yet. Add the calendar to `RPT08 Roadmap` as a band above the package bars: a roadmap that hides the closed weeks is one nobody at the plant will believe.

Goes wrong when. You plan against forty-eight weeks, run eight weeks late by March, and have it recorded as a delivery failure rather than an arithmetic error.

Look it up. Volume 1 chapter 18, *The roadmap and what the year allows* — the worked version of this arithmetic, with the number it lands on.

Form F-1 — Site window and freeze calendar

This form feeds a sheet the repository does not have. `O12 Location` has no window, shift or line columns and `O30` has no window column. Create `O12W Site Window`, headers in row 4, data from row 5. One row per window per site, so three production sites need twenty-five to forty rows. Signed by the Chief Operating Officer and the plant manager.

Ref	Site	Window type	Opens	Closes	Change permitted
SW-01	LOC-03	EXAMPLE — delete. Annual shutdown	2027-12-18	2028-01-08	Control layer and network

Step 9.5 — Test every package's start and end against the calendar

Produce. A twelve-row test, package against window, with every collision named and re-dated.

You need. `O30` (`Start`, `End`), `O56`, the calendar from step 9.4, `R15 Transition contains Work Pa`, `O66 Architecture Increment`.

Where it lands.

> **`O30` and the two columns it needs**
>
> - `Start` and `End` are month strings, `2027-02` to `2031-10`. A month is the wrong grain: a shutdown is a fortnight, a sanitation slot is a shift.
> - Add `Needs plant access`, yes or no per package, and `Window`, the `SW` ref it depends on. Without the first you cannot tell which packages the calendar applies to.
> - `WP-04 Technology currency` and `WP-05 Access and security` need plant access — `O27` places `PTC-03 SCADA control servers` and `PTC-07 Plant network switches` in the estate — and `WP-07 Core platform renewal` needs it at every site.

How.

1. Mark each of the twelve packages yes or no for plant access; expect five or six yes.
2. For each yes, find the windows its `Start` and `End` fall inside. A package whose end date falls in a freeze period has no cutover date at all.
3. Count the windows each package needs against the windows in its span. Three sites with one annual shutdown each gives three windows a year, and `WP-04` touches all three.
4. Re-date on the window, not on the plan. A package that misses its shutdown slips a year, not a month, and the plan has to say so.
5. Re-check the dependencies: `R15` sequences packages into transition states and `O66` dates the increments, so moving one package by a year moves a plateau. Put the collisions to the Investment Committee as a list of dates, not as a risk.

Signed by. Head of Delivery (`ROL-13`), with the plant manager signing each dated window.

Done when. Every package needing plant access names a window, and no end date falls in a freeze period.

Shows up in. `RPT08 Roadmap` . Expect the plan to get longer. A plan that gets shorter after this step was not tested.

Goes wrong when. Two packages book the same shutdown. The plant gives the window to the one with the production case, and the architecture package waits a year.

Look it up. Volume 1 chapter 7, *Sales and operations planning* — the monthly cycle that owns the people you need, and the step in it you must not be the reason for skipping.

You cannot stop a process plant to do a cutover

Asset class	Shortest safe stop	What the stop costs
Discrete assembly line	hours	Lost output only. Recoverable.
Batch plant, mid-batch	the whole batch	Batch cannot be quality-released. Destroyed material at full standard cost.
Kiln or dryer	days	Cool-down and re-heat. Start-up scrap and off-spec material on restart.
Furnace or smelter	cannot be stopped	Hearth freeze can mean permanent loss of the asset.
Cold chain	minutes	The product is gone. Loss, not delay.

And a restart produces transition material that is neither raw material nor product.

Figure 9.3 What a cutover costs, by asset class. The shortest safe stop is a plant number, not an IT number, and for two of these classes **there is no safe stop at all**.

Step 9.6 — Cost a cutover per plant, in lost production and in the shortest safe window

Produce. A per-site cutover cost: what stopping costs, the shortest safe stop, and what cannot be stopped at all.

You need. `O12`, the process engineer and plant manager at each site, `O39 Data Migration`, `O49 Resilience Tier`, volume 1 chapter 2.

Where it lands.

> **`O12W Site Window, extended — three columns a cutover plan needs`**
>
> - `Shortest safe stop`, in hours, from the plant and not from IT.
> - `Cost of a stop`: lost production plus restart scrap, which is not proportional to the length of the stop.
> - `Cannot be stopped`: yes for any asset where an interruption risks the asset or the batch.

How.

1. Ask it in the plant's terms: not how long the system is down, but how long the line is down and what comes off it when it restarts.
2. Record the mechanism per asset class, because it decides the answer. A kiln means a cool-down and a re-heat measured in days. A furnace risks freezing the hearth, which can mean losing the asset. A batch interrupted mid-process frequently cannot be quality-released, and that is destroyed material at full standard cost, not deferred output.
3. Record the restart output separately. A grade transition or a restart produces material that is neither raw material nor product; it has a mass, a cost and a disposition, and no discrete-manufacturing system models it.
4. Get the shortest safe window per site in writing from the person who will be on site. A plan that assumes a maintenance window exists is a plan that will be refused.
5. Set the number against the package: `WP-07 Core platform renewal` at $9.5m across three sites will not fit unless it is staged site by site.
6. Carry the services with you. Compressed air, cooling water and effluent handling restart on a sequence, and losing any of them trips everything downstream.

Signed by. Chief Operating Officer (`ROL-03`), on the plant manager's and process engineer's figures.

Done when. Each production site has a stated shortest safe stop in hours, a cost of stopping, and a list of what cannot be stopped.

Shows up in. Nothing in the repository. Put it on the Phase F cover page, because it is the number that decides whether the plan is real.

Goes wrong when. The plan assumes a weekend. The plant says no the week before go-live, and the programme discovers in public that it never had a window.

Look it up. Volume 1 chapter 2, *The non-stoppable asset, and why power interruption hit manufacturing hardest* — the four mechanisms, and why the cost of an interruption is not proportional to its length.

ON THE CUTOVER, BEFORE YOU WRITE A DATE

For each production site, what is the shortest stop the plant will agree to, and who said so?

Which of your migrations cannot be reversed once started, and what is the fallback?

Step 9.7 — Design the parallel run per application, and name who is on shift

Produce. A parallel-run design per migrating application: what runs twice, for how long, keyed by whom, and reconciled against what.

You need. `O23 Application Component`, `O39`, `R06 Application interfaces with`, the shift pattern from step 9.4, Form F-2.

Where it lands.

> **`O23 Application Component` — twelve rows, five of them Replace**
>
> - `A04 SCADA / Historian` is the **only** `Critical` application and its `Disposition` is `Replace`, with nothing named to replace it. `A03 MES` has been in service since `2010`, `Health` `Poor`, `Disposition` `Retain`.
> - `A02 PLM`, `A05 APS`, `A06 WMS` and `A12 Plant Analytics` are the other dispositions implying a cutover, and there is no `Parallel run`, `Cutover`, `Rollback` or `Reconciliation` column on the sheet.
> - `O49` gives `RES-04 SCADA / Historian` one hour, `Tested` — its only coherent row — and `RES-03 MES` one hour, `Never tested`.

How.

1. State what people get wrong: a parallel run in a plant is not a weekend of double entry in an office. It is two systems live while the line runs, with somebody on every shift keying into both.
2. Name that somebody, per shift, per site. A twelve-hour two-shift pattern over a two-week parallel run on three sites is a staffing requirement out of production headcount.

3. Set the duration by what has to reconcile, not by comfort. A production confirmation reconciles daily, a stock position at a count, a batch genealogy only when a batch completes — which may be weeks.
4. Decide the authority for each parallel pair. Two systems holding the same work order is a mastering question, and `R05 Application manages Data En` puts application names where refs belong on its `Master` rows, so the join will not help.
5. Design the interface behaviour. `R06` holds 30 interfaces, 22 `Unmanaged` and four `Direct DB read`, and a direct read against a system being replaced breaks silently during a parallel run.
6. Write the rollback: what returns the plant to the old system, how long it takes, and who decides.

Signed by. Head of Operations (`ROL-14`) on the shift commitment; Head of Service Management (`ROL-15`) on the rollback.

Done when. Each migrating application has a parallel-run duration, a reconciliation, a named person per shift per site, and a rollback with a decision owner.

Shows up in. `RPT02 Application Portfolio` and `RPT04 Interface Catalogue` carry the dispositions and the interfaces. Neither shows a cutover, so the design lives in the plan.

Goes wrong when. Nobody is on the night shift. The parallel run has a twelve-hour hole in it daily, and the reconciliation meant to prove the migration proves nothing.

Look it up. Volume 1 chapter 7, *Shifts, crews, and what a pattern costs* — what you are actually asking for when you ask for a person on every shift.

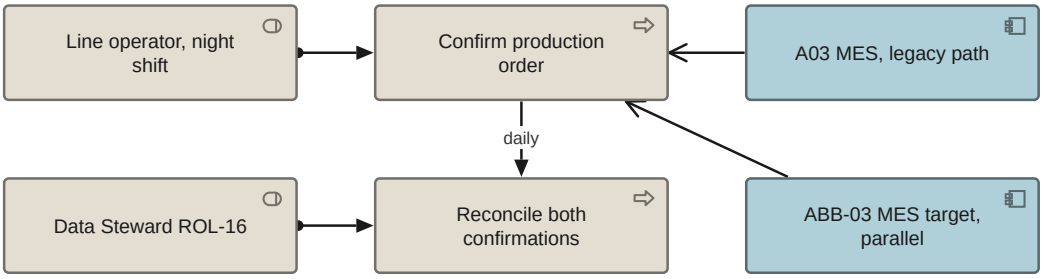


Figure 9.4 A parallel run as it actually has to be designed: two applications serving one process, somebody on every shift keying into both, and a named role reconciling them daily.

Step 9.8 — Record the change-control obligation where a system carries a quality record

Produce. A per-application statement of whether a change is a change-control event, and what evidence it owes.

You need. `023`, `063 Control Objective`, `059 Compliance Assessment`, the quality manager — who appears in no sheet in this file.

Where it lands.

> **`O59 Compliance Assessment — seven columns, zero rows`**
>
> - Headers read `Ref`, `Building block`, `Block name`, `Acceptance criterion`, `Gate`, `Verdict`, `Evidence`. The sheet is empty and `R24 Compliance Assessment tests` is header-only in consequence [1].
> - **So there is no compliance evidence anywhere in this repository.** For a migration that touches a system holding a quality record, that is the whole finding.
> - `O63` names `ISO 9001` and `ISO 14001 / 45001` as obligations with `Evidence` reading `No mechanism` or `In breach` on five of six rows.

How.

1. Decide, per application, whether it holds a record a regulator or customer audit would ask for. `A07 QMS` holds specifications, inspection results and non-conformance; `A11 Track & Trace` holds serialisation and genealogy; `A03 MES` holds the production confirmation.
2. Where it does, treat the change as an event in the plant's own quality system, not only the programme's: a written change request, an impact assessment, a test record and a release authorisation, each retained.
3. In a food or pharmaceutical plant the obligation is explicit, and the authority is the international good-manufacturing-practice guide the national regulator adopts rather than the national guideline, which does not carry the computerised-systems detail. Cite the right one or a quality manager will correct you in the first review.
4. Write the obligation into `O59` as rows: one per acceptance criterion, `Gate` set to `G2`, `Evidence` naming the artefact.
5. Add the retention clock. A quality record's retention period is set by the product and the scheme, not by one corporate period, so a migration that drops history may breach it.
6. Put the obligation on the critical path. Validation evidence is produced before release, and it is the step most often discovered late.

Signed by. Chief Compliance Officer (`ROL-07`), with the quality manager named once the role exists in `O11`.

Done when. Every migrating application is marked as carrying a quality record or not, and every one that does has rows in `O59` naming the evidence and the gate.

Shows up in. `RPT09 Regulatory Coverage` and, once `O59` has rows, `ZZ Checks`' control-objective line. Five of six obligations without evidence should start falling.

Goes wrong when. The migration completes and the next audit asks for the change record. There is none, and the finding lands on the quality system, not the programme.

Look it up. Volume 1 chapter 16, *Pharmaceutical* — the sourcing trap, and why the national guideline is the wrong citation for change control and computerised systems.

The migration itself

ON THE PARALLEL RUN, WITH THE OPERATIONS MANAGER IN THE ROOM

Which two systems will be live at once, on which lines, for how many weeks?

What reconciles, how often, and who signs that it reconciled?

Step 9.9 — Rebuild O39 Data Migration so a migration can be planned from it

Produce. A migration register with a volume, a cutover date, a reconciliation test and a fallback per entity.

You need. `O39 Data Migration`, `O20 Data Entity`, `O40 Source To Target Mapping`, `O43 Master Data Rule`, Form F-2.

Where it lands.

> **`O39 Data Migration` — five columns, six rows, and three missing**
>
> - `MIG-01` Product, `PLM` to `SCADA / Historian`, `In progress`
> - `MIG-02` Bill of Material, `MES` to `APS`, `Planned`
> - `MIG-03` Routing / Recipe, `SCADA / Historian` to `WMS`, `Planned`
> - `MIG-04` Work Order, `APS` to `QMS`, `Planned`
> - `MIG-05` Batch / Lot, `WMS` to `CMMS / EAM`, `In progress`
> - `MIG-06` Plant / Site, `QMS` to `CAD / CAE`, `Not started`

Three `Status` values — `In progress`, `Planned`, `Not started` — and **no `Volume`, no `Cutover date`, no `Reconciliation`**. `From` and `To` are a positional shift: the source is the next application in `O23`'s order, the target the one after. So `Product` migrates **into a historian**, an append-only time-series store that cannot master anything, and `Plant / Site` migrates into `CAD / CAE`.

How.

1. Read the six rows against `O23`'s descriptions. All six source-and-target pairs are wrong, in the same direction, which is how you know the column was generated.
2. Rebuild the pairs from `O20.Master system` and `O43.Golden source` — then stop, because those two disagree on nine of the ten entities they share. Chapter 5 of this workbook has that step.

3. Add `Volume`, in records, from the source system. A migration with no row count has no duration.
4. Add `Cutover date`, tied to a window reference from step 9.4, and `Reconciliation`, naming the count or total that must match on both sides.
5. Add `Fallback`. For `Batch / Lot` and `Routing / Recipe` there may not be one, because a batch genealogy partially migrated is a traceability break.
6. Extend the register past six rows. `O20` holds fourteen entities, six reading `Lineage` = `Absent`, including the batch, the work order and the routing.

Signed by. Chief Data Officer (`ROL-05`) through `GB-03 Data Governance Council`, with the Data Steward (`ROL-16`) per entity.

Done when. Every row has a volume, a dated cutover tied to a window, a reconciliation test, and a fallback or a statement that there is none.

Shows up in. `RPT05 Data Ownership`. Expect it to move once the mastering argument is settled, and the register to grow past six rows.

Goes wrong when. The cutover runs with no reconciliation. The counts are never compared, and the gap is found eighteen months later by a recall simulation that cannot complete.

Look it up. Volume 1 chapter 9, *Batch, lot, serial, and genealogy as a graph* — why a partially migrated genealogy is worse than none, and what a recall simulation actually tests.

Form F-2 — Cutover and reconciliation register

This form feeds columns the repository does not have. `039 Data Migration` carries only `Ref`, `Entity`, `From system`, `To system` and `Status`. Add the columns below to `039`, or create `039C Cutover` if you may not alter it. One row per entity migrated, and `020` holds fourteen, so expect twelve to sixteen rows. Signed by the Chief Data Officer and the Data Steward for the entity.

Ref	Entity	Volume	Cutover window	Reconciliation test	Fallback
MIG-05	Batch / Lot	EXAMPLE — delete	SW-01	Batch count and open quantity	None — genealogy break

Step 9.10 — Assign the five orphaned packages and re-date the six increments

Produce. A sequence in which every package belongs to a plateau and every increment lands inside its delivering project.

You need. `R15`, `053 Transition Architecture`, `066 Architecture Increment`, `R23 Increment delivered by Proj`, `056`, step 8.8's output.

Where it lands.

O66 Architecture Increment and R23 — six increments, six pairings

- `INC-1 Governed and supportable` Month 10; `INC-2 Integrable` Month 16; `INC-3 Compliant by mechanism` Month 22; `INC-4 Business visible` Month 30; `INC-5 Analytics and evidence` Month 42; `INC-6 Core renewal` Month 56.
- `R23` pairs them to `PRJ-01`, `PRJ-03`, `PRJ-05`, `PRJ-07`, `PRJ-09`, `PRJ-11` — **every second project**, which is a pattern rather than a dependency.
- Three of the six increment names — `Compliant by mechanism`, `Analytics and evidence`, `Core renewal` — exist in no row of `O53`, so three plateaus have no transition architecture and therefore no exit criteria.

How.

1. Set month 1 at the earliest `Start` in `O56`, `2027-02`. `INC-1` then lands `2027-11`, `INC-2` `2028-05`, `INC-3` `2028-11`, `INC-4` `2029-07`, `INC-5` `2030-07`, `INC-6` `2031-09`.
2. Compare each against its project's dates. `INC-1` lands 26 months before `PRJ-01` ends, `INC-2` 37 months before `PRJ-03` ends, `INC-3` 35 months before `PRJ-05` ends, `INC-4` 10 months before `PRJ-07` ends. Four of six increments land before the project delivering them has finished.
3. Then read the other two. `INC-5` lands 12 months **after** `PRJ-09` ends; `INC-6` lands 14 months after `PRJ-11` ends. Those two have no delivery mechanism at all in the window they claim.
4. Read `INC-3` once more. It lands `2028-11`; `PRJ-05 Access and security` starts `2028-12`. **The increment lands a month before the project that delivers it begins.**
5. Replace `R23`'s one-to-one pairing: `INC-1` corresponds to `TA-1`, which `R15` says contains `WP-01`, `WP-02`, `WP-04` and `WP-05` — four projects, not one.
6. Carry across the five packages step 8.8 assigned, and give `INC-3`, `INC-5` and `INC-6` a transition architecture in `O53` or drop them.

Signed by. Architecture Board (`GB-01`), with the Head of Delivery (`ROL-13`) on the dates.

Done when. Every increment lands inside the span of every project contributing to it, and every `WP` ref appears in `R15`.

Shows up in. `RPT08 Roadmap`, which shows packages and not increments. Add the increment dates as milestones or the arithmetic stays invisible.

Goes wrong when. The programme reports `INC-5` at month 42 against a project that closed at month 30, and it is the architect who is asked at the steering committee.

Look it up. Volume 1 chapter 7, *The planning cadence* — the rhythm an increment has to land inside if anyone is to notice it landing.

ON THE SEQUENCE

Which increment lands before its own project starts, and by how long?

Three increments have no transition architecture. Are they plateaus or are they milestones?

Risk, and handing over

Step 9.11 — Re-own the eight delivery risks, starting with the owner who does not exist

Produce. An `O62` in which every owner resolves against `O11 Role`, and every `Certain` risk has a dated mitigation.

You need. `O62 Delivery Risk`, `O11 Role`, `O09`, `O52`, `O60`.

Where it lands.

> **`O62 Delivery Risk` — five columns, eight rows**
>
> - Three rows read `Likelihood` = `Certain`: `RSK-01 Integration product not selected` (`High`, `Chief Information Officer`), `RSK-07 Benefit not measurable at G3` (`Medium`, `Chief Financial Officer`) and `RSK-08 Unmanaged interfaces proliferate` (`Medium`, `Chief Enterprise Architect`).
> - **`RSK-06 Key supplier withdraws capability` is owned by `Head of Procurement`, which appears nowhere** — not in `O11 Role`, not in `O10 Actor`, not in `O09.Head`. Seven of eight owners resolve; that one does not.
> - No `Mitigation`, `Date`, `Status` or `Residual` column, so a risk cannot be worked, only listed.

How.

1. Resolve `RSK-06`: add the role to `O11`, the actor to `O10` and the join to `R21`, or move the risk to a role that exists. Do not leave a risk owned by a title nobody holds.
2. Deal with the three `Certain` rows first. A certain risk is a condition, not a risk, and two already have decisions waiting in `O60` — `DEC-02 Select the integration product` for `RSK-01`, and the measure problem behind `RSK-07`.
3. Add `RSK-09` and `RSK-10` for what this chapter found: no plant access recorded anywhere, no compliance evidence in `O59`.
4. Add the four columns the sheet lacks. A register with a likelihood, an impact and no mitigation is a list of worries.

5. Cross-check `O52`: `IF-07 Licence model change` is categorised `Risk` there and appears nowhere in `O62`. Decide which sheet owns it.
6. Give `RSK-04 Legacy platform failure before renewal` a date, because `A03 MES` has been in service since 2010 with `Health` = `Poor` and `Disposition` = `Retain`.

Signed by. Chief Risk Officer (`ROL-06`), with the Head of Delivery (`ROL-13`) on the mitigations.

Done when. Every owner resolves against `O11`, every `Certain` row has a dated action, and the register has a mitigation column.

Shows up in. `RPT01 Dashboard`. Nothing else reads `O62`, which for a delivery risk register is itself worth reporting.

Goes wrong when. `RSK-06` sits unowned until the supplier withdraws. The register showed a name, so nobody checked that it was real.

Look it up. Volume 1 chapter 5, *Maintenance* — who actually carries the consequence when a legacy platform fails before its renewal lands.

Step 9.12 — Set O31 Deliverable honestly and hand Phase G something it can govern

Produce. A Phase F pack: dated plan, window calendar, cutover costs, parallel-run designs, migration register, re-dated increments, re-owned risks.

You need. Steps 9.1 to 9.11; `O31 Deliverable`, `O33`, `O58 Governance Gate`, `O61`.

Where it lands.

> **`O31 Deliverable` — twenty rows, and what Approved currently means**
>
> - `DLV-16 Migration Plan` and `DLV-17 Project Charters`, both `Phase F`, both `Draft`. Those two are yours.
> - Thirteen of twenty rows read `Approved` — every deliverable from Phase A to Phase D, plus `DLV-13 Consolidated Gaps`. Nothing from Phase E onward is approved.
> - Set that against `ZZ Checks`: 22 unmanaged interfaces against a threshold of 0, 21 with no logging against 0, four direct database reads against 0, six entities without lineage against 0, five control objectives without evidence against 0 [1]. **Thirteen deliverables approved over an estate failing five thresholds**, and the sheet has no approver, date or version to say who did it.

How.

1. Do not approve `DLV-16` until the window calendar exists. A migration plan with no plant access in it is the deliverable this chapter exists to prevent.
2. Add `Approver`, `Date` and `Version` to `O31`. Without them an approval is an assertion.
3. Reconcile the thirteen approvals against the five failing thresholds in one table and take it to `GB-01`. You are not asking for them to be withdrawn, only for what they were approved against to be recorded.

4. Complete `033 Architecture Contract`: eight contracts against twelve packages, so `WP-09` to `WP-12` cannot pass `G0`. Then close `CR-G04 Record benefit realisation dates per package`, already `Approved` in `061`, by putting the benefit date on each package — what `G3 Benefit` tests.
5. Hand Phase G four things: the dated plan, the exit criteria from chapter 8, the `059` rows from step 9.8, and the list of what is still typed rather than computed.

Signed by. Investment Committee (`GB-02`) on the funding; Architecture Board (`GB-01`) on the plan.

Done when. `DLV-16` and `DLV-17` are `Approved` with a date and an approver, every package has a window and a benefit date, and `059` has rows.

Shows up in. `RPT08 Roadmap` and `RPT01 Dashboard`. The plan will be longer and the package count unchanged: that is what a tested plan looks like.

Goes wrong when. Phase G inherits a plan never tested against the plant year, and then spends its gates re-dating work instead of assessing it.

Look it up. Volume 1 chapter 13, *What you are actually expected to produce* — the artefacts a plant engagement is actually judged on, which is the standard Phase G will hold you to.

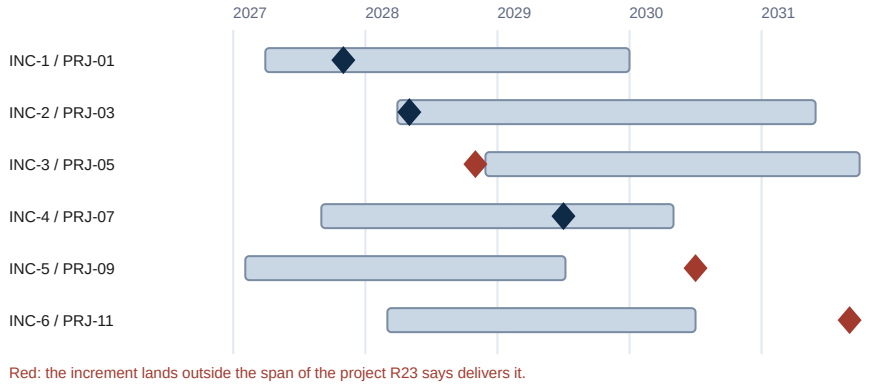


Figure 9.5 The six increments of `066` against the spans of the six projects `R23` says deliver them. **`INC-3` lands a month before its project starts; `INC-5` and `INC-6` land after theirs have closed.**

THE REVIEWER'S ASK — ANSWER THESE BEFORE THE PLAN IS PUBLISHED

How many usable weeks does next year hold for work needing plant access, and what did you subtract?

Which work package is being built without approval, at what value, and who was told?

Name the increment that lands after its delivering project has closed, and the one that lands before its project opens.

For each migrating application, who is on the night shift during the parallel run?

Which delivery risk has an owner who does not exist in the repository?

THE CHAPTER IN TEN LINES

- O30 holds two status columns that disagree. WP-05 Access and security, at 3.1 million, is being built with approval Not started.
- WP-07 reads In flight on approval and Delivered on delivery. WP-01 reads Delivered against an end date of 2030-01.
- Every end date in O30 falls between 2029-07 and 2031-10, so the four work packages past their end date cannot exist.
- ZZ Checks and RPT08 both count a typed label rather than comparing a date.
- O56 Project is O30 with one column added. Only Lead is new, and every lead resolves against O11 Role.
- The Internal Auditor leads two projects worth 6.5 million, one of them Access and security. The Data Steward leads two worth 11.1 million.
- O09 Organization Unit holds twelve units and not one plant, production, quality or maintenance unit.
- A delivery year in a plant is bounded by shutdown windows, the planning cycle and the seasonal peak. The file records none of them.
- IF-08 Site access windows limits physical work to shutdowns, and there is no shutdown date anywhere in the 105 sheets.
- O39 Data Migration has six rows and no volume, cutover date or reconciliation column. Product is being migrated into a historian.
- Four of the six increments land before the project delivering them ends. INC-5 lands twelve months after, INC-6 fourteen months after.
- INC-3 lands one month before PRJ-05, the project R23 says delivers it, even begins.
- RSK-06's owner, Head of Procurement, exists in no sheet in the repository, and O59 Compliance Assessment is empty.

SOURCES FOR THIS CHAPTER

- **1.** Manufacturing_TOGAF_Metamodel.xlsx, Kestrel Industrial Group — sheets O09, O10, O11, O12, O20, O23, O27, O30, O31, O33, O39, O43, O49, O52, O53, O56, O58, O59, O60, O61, O62, O63, O66, R05, R06, R14, R15, R23, RPT01, RPT02, RPT04, RPT05, RPT08, RPT09, ZZ Checks
- **2.** The Open Group, TOGAF Standard, Architecture Development Method — Phase F, Migration Planning https://pubs.opengroup.org/architecture/togaf-standard/adm/

CHAPTER 10

Implementation Governance

The gate regime is fully designed, the contracts are issued, and the gate has produced zero records.

What Phase G is for, and what this repository says about it

Phase G is where the architecture stops describing and starts testing. Something is built, somebody says whether it conforms, the verdict is written down with the evidence behind it. Read the governance sheets in that order and the regime looks finished.

`058 Governance Gate` holds four rows — `GT-0 G0 Charter` before work starts, `GT-1 G1 Design` before build, `GT-2 G2 Build complete` before release, `GT-3 G3 Benefit` after operation — each with a `When` and a `What it tests` [1]. `057 Governance Body` holds five bodies with a `Cadence`, a `Membership` and a `Mandate`, and `GB-01 Architecture Board` is recorded as holding G1 and G2 and approving dispensations with an expiry [2]. `060 Governance Decision` holds seven decisions. `061 Governance Change Request` holds five, one of them a dispensation expiring `2029-06`. `033 Architecture Contract` holds eight, one each for `WP-01` to `WP-08` [3].

`059 Compliance Assessment` holds nothing. Header row only — `Ref`, `Building block`, `Block name`, `Acceptance criterion`, `Gate`, `Verdict`, `Evidence` — and its own subtitle says *0 records* [4]. `R24 Compliance Assessment tests Building Block` is empty in consequence. Meanwhile `ZZ Checks` line one reads *Object types populated · 65 · Every type carries records*, which is false: it is 64 of 65, and the one is this one.

So the gates are designed, the bodies named, the contracts issued, the dispensation granted — and **the gate has produced zero records**. That is not a gap to tidy up. It is your first step.

> **THE FILE CORROBORATES ITSELF ON THIS, WHICH IS RARE**
>
> `041 Security Control` row `SEC-02 Design review` reads `Position = Absent` and `Enforced at = Design gate` [5]. The control that is absent is the control the gate would be. `RPT11 Security Position` reprints the same eight rows and adds a typed footer — *1 controls are absent. A control that is absent is not a control.* Two sheets, written independently, agree: there is no design review here and nothing has been assessed. Open with that and the repository will not argue back.

Kestrel Industrial Group, the manufacturer in this repository, is an invention. So is Karoo Industrial Group in volume 1. Keep their data apart.

The part of Phase G that no other industry has

A manufacturer runs two change-control regimes at once, and they are not the same regime. The architecture gate governs the model: a building block, an acceptance criterion, a verdict. The plant's **management-of-change** process governs a physical change to a plant — a re-rate, a new line, a

modified recipe, a changed interlock — and in a regulated plant it carries an evidence obligation of its own, enforced by somebody who does not report to the CIO and will not accept a repository row as a record. **Nothing in this file records the second regime**: no sheet, no column, no body in `057` that owns it.

Two more facts belong only here. **Engineering change arrives from outside**: in a tier-supplier an engineering change request comes from the customer and is not negotiable. And **the plant's calendar outranks the architecture board**: a shutdown window, a seasonal peak and a sales-and-operations lock date are dates on which a gate cannot sit.

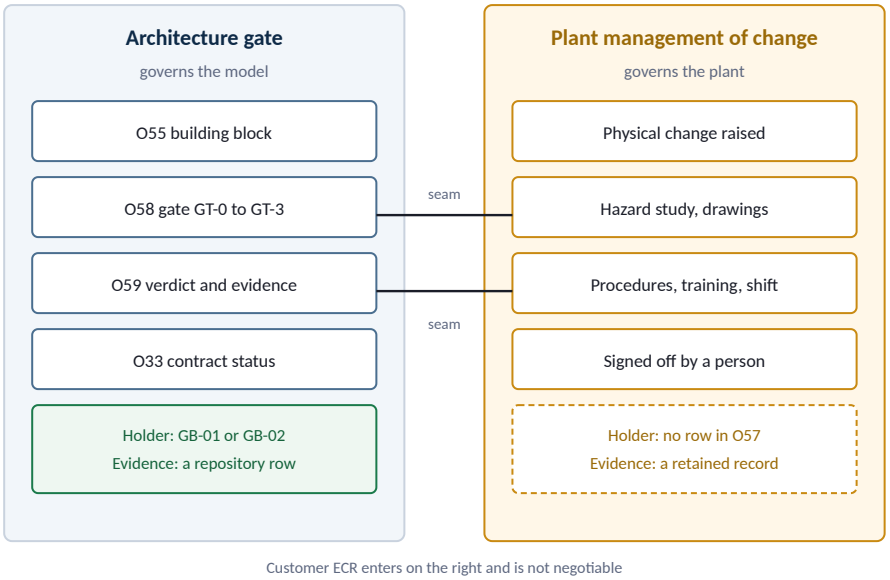


Figure 10.1 Two change-control regimes, side by side. The left-hand column exists in the repository; the right-hand one does not. The seams are the only two places a project has to satisfy both, and in a regulated plant the right-hand register wins.

BEFORE YOU OPEN THE GOVERNANCE SHEETS

Who, by name, chaired the last gate that sat in this organisation, and what did it decide?

If `059` is empty because no gate has sat, who stopped it sitting — and what did they fear?

Who owns the plant's management-of-change process at each of the three plants in `012 Location`?

The eleven steps

Step 10.1 — Write the first compliance assessment, and say out loud that it is the first

Produce. The first populated rows of `059 Compliance Assessment`, and a one-page note recording that before them the sheet was empty.

You need. `055 Building Block` for the 37 refs and their `Acceptance criteria`; `058 Governance Gate` for the gate each criterion is tested at; `033 Architecture Contract` for which work package the block belongs to; whoever built something you can actually look at.

Where it lands.

> **`059 Compliance Assessment` — all seven columns, from row 5**
>
> Fill `Ref` as `CA-001` onward; the sheet has no reference space yet because it has no rows, so you are setting the convention. `Building block` takes a ref from `055` — `BBB-`, `DBB-`, `ABB-` or `TBB-`. `Block name` is the name as `055` spells it, so a reader need not look it up. `Acceptance criterion` is copied from `055`, not paraphrased. `Gate` takes `GT-0` to `GT-3`. `Verdict` takes one of `Conformant`, `Conformant with dispensation`, `Non-conformant`, `Not assessable`. `Evidence` names the artefact, its date and where it is held — not the word "yes".

How.

1. Pick the blocks with something built against them. On this file that is thin: `WP-01` and `WP-07` read `Delivered` in `030`, so start with the blocks `R13` ties to them.
2. Copy the criterion from `055` verbatim. If it is not testable as written, record `Not assessable` and raise a change request in Phase H rather than inventing a criterion.
3. Assess against the evidence in front of you. No evidence means `Not assessable`, never `Conformant`.
4. Write `Evidence` as a locator: document, version, date, repository or share.
5. Resolve the row into `R24` so the join exists both ways.
6. Correct `ZZ Checks` line one from *65* to *64 of 65*, with the date.

Signed by. `GB-05 Architecture Function` writes it; `GB-01 Architecture Board` accepts the first batch at its monthly meeting.

Done when. `059` has at least three rows, each with a verdict and a locatable evidence cell, and `R24` resolves them.

Shows up in. Nothing yet — there is no compliance report among the thirteen. `RPT11 Security Position` is the nearest, and `SEC-02` should move off `Absent` only once the design gate has an assessment behind it.

Goes wrong when. Somebody back-fills `Conformant` across all 37 blocks in an afternoon to clear the `ZZ Checks` line. That is 37 assertions with no evidence, which is worse than an empty sheet, because an empty sheet is honest.

Look it up. Volume 1, chapter 16, *What a standard is, and what it is not*.

Step 10.2 — Give each gate a holder, an evidence demand and testable criteria

Produce. Two new columns on `O58 Governance Gate`, filled for all four gates, and a criteria set of four to eight lines per gate published before the next gate sits.

You need. `O57 Governance Body`'s `Mandate` column, which already assigns G0 and G3 to `GB-02 Investment Committee` and G1 and G2 to `GB-01 Architecture Board`; `O05 Principle` (24 rows); `O32 Standard` (16 rows); `O55 Building Block`'s `Acceptance criteria`.

Where it lands.

> **`O58 Governance Gate` — add `Holder` and `Evidence required`; create `O68 Gate Criterion`**
>
> `O58` has four columns: `Ref`, `Gate`, `When`, `What it tests`. There is **no holder column and no evidence column**, so as the sheet stands a gate cannot say who holds it or what it demands to see. Add `Holder` in column E, taking a `GB-` ref rather than a committee name in prose, and `Evidence required` in column F, naming the artefact class the gate will not sit without — a design document for `GT-1`, a populated `O59` row set for `GT-2`, a dated benefit measurement for `GT-3`. The criteria themselves do not fit a one-row-per-gate sheet, so create `O68 Gate Criterion` on the file's convention — title row 1, subtitle row 2, headers row 4, data row 5 — with `Ref` (`GC-001` onward), `Gate`, `Criterion`, `Tested against`, `Evidence`, `Owner`. `Tested against` takes a `PRN-`, `STD-` or `REQ-` ref, so a criterion is never a private opinion. `O68` does not exist in the file you were handed; say so when you publish it.

How.

1. Derive each gate's holder from `O57`'s five `Mandate` cells. Two resolve cleanly; `GT-0` splits between funding and scope, so name one holder and the other a consultee.
2. Fill `Evidence required` with one named artefact class per gate. Cross-check `O31 Deliverable`: `DLV-18 Architecture Contract` and `DLV-19 Compliance Assessment` both read `Phase G`, `Status` = `Draft`, and those are your `GT-2` evidence classes.
3. Write criteria only where something in the repository can be pointed at. No `PRN-`, `STD-` or `REQ-` behind it, no criterion. Then test each one by asking what evidence would fail it; if nothing can fail it, it is a sentiment.
4. Handle `O05` with care. It has no `Tier`, `Rationale`, `Implications`, `Approved` or `Review date` column, every row is owned by `Architecture Board` — against `PRN-13 Data Ownership is Individual`, whose statement is that an owner is a person, not a committee — and `Status` = `Draft` falls on every fifth row, which is a generation pattern and not an assessment.
5. Circulate the set for two weeks before it binds anything.

Signed by. `GB-05 Architecture Function` drafts; `GB-01 Architecture Board` approves, as holder of two of the four gates.

Done when. Every `O58` row names a `GB-` ref and an artefact class, and every `O68` criterion names a ref in `Tested against`.

Shows up in. `RPT12 Standards Conformance`, where seven of sixteen standards read `Not assessed` or `Breach` against a `Current` status. A criterion citing `STD-03 Data classification standard` inherits that `Breach`. No report reads `O58` at all, which is worth saying: the gate regime is invisible to all thirteen.

Goes wrong when. The criteria read as a list of good intentions, the first project refused appeals to `GB-04 Executive Committee` — which `O57` names as the escalation point — and wins. The gate is decorative from then on.

Look it up. Volume 1, chapter 14, *The constructive answer: build the house reference model*. A gate tests against a declared reference, and manufacturing does not hand you one.

Form G-1 — Compliance assessment

Feeds `O59 Compliance Assessment`, empty in the file you were handed. One row per acceptance criterion assessed, never per building block; a first pass needs eight to twelve. Signed by `GB-05 Architecture Function`, accepted by `GB-01 Architecture Board`. `Block name` is a lookup from `O55` — type it straight into Excel.

Ref	Building block	Acceptance criterion	Gate	Verdict	Evidence
CA-001	EXAMPLE — delete this row	Criterion copied from O55 verbatim	GT-2	Not assessable	No evidence offered, 2026-10-16

Form G-2 — Gate criteria

Feeds `068 Gate Criterion`, a sheet the repository does not have and which you create on the file's convention. One copy per gate; four to eight rows each. `Tested against` must hold a `PRN-`, `STD-` or `REQ-` ref. Signed by `GB-01 Architecture Board`.

Ref	Gate	Criterion	Tested against	Evidence	Owner
GC-001	GT-1	EXAMPLE — delete this row	STD-11	Segmentation design, signed	

Step 10.3 — Date every decision and every dispensation, and clear the four that block the roadmap

Produce. A `Decided` column on `O60 Governance Decision` and three new columns on `O61 Governance Change Request`, all filled, with a dated route to resolution for every open row.

You need. `O60`'s seven rows; `O61`'s five; `O54 Architecture Alternative`; `O30 Work Package`; `O62 Delivery Risk`; the two bodies named in `O60`'s `Body` column.

Where it lands.

> **`O60` — add `Decided` and `Decision date due`; `O61` — add `Raised`, `Owner`, `Condition`**
>
> `O60` has four columns only: `Ref`, `Decision`, `Body`, `Status`. **There is no date column**, so a decision cannot be aged and nobody can be late. Add `Decided` (blank while outstanding) and `Decision date due`, and keep `Status` to `Outstanding`, `Approved`, `Refused`, `Deferred`. `O61` has `Ref`, `Change request`, `Against`, `Type`, `Status`, `Expires` — the `Expires` column is right and almost unused. Add `Raised` so a request can be aged, `Owner` so it has a person, and `Condition` so a dispensation says what has to become true for it to lapse rather than be renewed.

Read the seven decisions before you add anything. Four are `Outstanding`, three `Approved`:

Ref	Decision	Body	Status
DEC-01	Adopt the recommended architecture alternative	Investment Committee	Outstanding
DEC-02	Select the integration product	Investment Committee	Outstanding
DEC-03	Fund compliance mechanisms	Investment Committee	Outstanding
DEC-04	Designate data owners	Data Governance Council	Outstanding
DEC-05	Approve the governance framework	Architecture Board	Approved
DEC-06	Adopt the four-gate regime	Architecture Board	Approved
DEC-07	Grant dispensation for point-to-point transfer	Architecture Board	Approved

The three `Approved` rows are what the Architecture Board could take on its own authority. The four `Outstanding` rows are every decision needing money or an owner, and they are what the whole roadmap waits on. `DEC-01` gates `ALT-B Foundations first` at $46m, `Recommended`, against an `O30` total of $65.0m. `DEC-02` is `RSK-01`, `Likelihood` = `Certain`. `DEC-04` is `RSK-02` and `OBJ-10`. `DEC-03` funds `WP-10 Compliance mechanisms` at $4.2m, which `O30` already shows `In flight` without it.

Then the five change requests. `CR-G01 Dispensation for point-to-point file transfer`, `WP-04`, `Granted`, **expires 2029-06** — the only populated `Expires` cell, matching `DEC-07`, and nearly three years out, which for a point-to-point transfer is a permanent exception with a date

on it. `CR-G02`, `WP-03`, `Scope`, `Approved`, no expiry. `CR-G03` and `CR-G04`, both `Repository`, `Correction`. And `CR-G05 Refuse or re-baseline the integration package`, `WP-03`, `Compliance`, **board decision required** — the sharpest row on the sheet, because `WP-03` reads `Approval = In flight` and `Delivery = Past end date` while `DEC-02` is still `Outstanding`.

How.

1. Add the columns. Back-fill `Decided` and `Raised` from the minutes; no minutes means `Undated`, and the approval is unproven.
2. Set a `Decision date due` on each `Outstanding` row, from the earliest `Start` in `O30` it blocks, minus one month, and tie it to its risk in `O62` so decision and risk age together.
3. Set an expiry and a `Condition` on every `Dispensation` row. For `CR-G01` the condition is the integration product in production, which depends on `DEC-02`.
4. Take the four decisions to the bodies `O60` names, and `CR-G05` to the board with the three facts above in one sentence.
5. Record a refusal as a refusal. `Deferred` three times is a refusal nobody signed.
6. Move `CR-G03` and `CR-G04` into chapter 11's change request sheet, where corrections belong.

Signed by. `GB-02 Investment Committee` for three decisions, `GB-03 Data Governance Council` for one, `GB-01 Architecture Board` for the dispensations — whose `O57` mandate already says it approves them with an expiry.

Done when. No `Outstanding` row lacks a `Decision date due`, no `Approved` row lacks a date, and no `Dispensation` row lacks both an expiry and a condition.

Shows up in. `RPT08 Roadmap`, which footers four packages past their end date against `End` dates between 2029-07 and 2031-10 — all future, so the column is typed and not computed. A dated `O60` and `O61` are the first time-aware things in the file. Nothing reads `O61` at all, so publish the open count and the oldest in your own pack.

Goes wrong when. The four decisions are carried as a standing item and "noted" each month, and a dispensation is renewed three times without its condition being tested, becoming the de facto standard. `STD-01 API design standard` already reads `Deprecated` with `Conformance = Conformant`, which is what that looks like at the end.

Step 10.4 — Re-assess every contract in `O33`, because nothing was assessed

Produce. A corrected `Compliance status` on all eight rows of `O33 Architecture Contract`, each traceable to an `O59` row.

You need. `O33`; the newly populated `O59`; `O30` for delivery state.

Where it lands.

> **`O33 Architecture Contract` — `Compliance status`, plus `Assessment ref` to add**
>
> `O33` has four columns: `Ref`, `Architecture contract`, `Work package`, `Compliance status`. Add `Assessment ref` so each status names the `CA-` row that produced it. Without that column the status is an assertion, and six of the eight currently are.

Read what is there. `ACO-02` and `ACO-03` read `Non-compliant`; the other six read `Not yet assessable`. There are **no assessments in the file**, so `Non-compliant` on those two is an assertion with nothing behind it — a verdict without a test. The direction of the error is instructive: it is the harsher verdict that was invented, and the six `Not yet assessable` rows are, by accident, the only honest cells on the sheet.

How.

1. Change `ACO-02` and `ACO-03` to `Not yet assessable` the same day, and log the change — the cleanest demonstration of why Phase H's change log exists.
2. Assess the two against criteria from step 10.2, write the `O59` rows, then set the status from the verdict.
3. Fill `Assessment ref` on every row. A blank there with a status other than `Not yet assessable` is a defect.
4. Check the contract-to-package mapping: `O33` stops at `WP-08`, so `WP-09` to `WP-12` — $24.5m across resilience, compliance mechanisms, customer experience and supplier integration — have **no architecture contract at all**. Raise four.
5. Give every contract an expiry, read across from `O19 Contract`, whose ten vendor agreements carry an `Expiry`, four of them in 2027.

Signed by. `GB-01 Architecture Board`, with the work package lead from `O56 Project`.

Done when. Eight rows become twelve, every one carries an `Assessment ref` or reads `Not yet assessable`, and no row asserts non-compliance without a `CA-` row behind it.

Shows up in. Nothing. No report reads `O33`. Put the twelve-row table in your own pack and say the repository cannot show it.

Goes wrong when. You defend the two `Non-compliant` cells because somebody senior believes them. A verdict with no assessment behind it loses the first time a supplier disputes it, and the gate's credibility goes with it.

Step 10.5 — Complete `O57`, because there is no plant in it

Produce. `O57 Governance Body` extended from five rows to the bodies a manufacturer has.

You need. `O57`; `O12 Location`; `O09 Organization Unit`; `O07 Constraint` row `CON-08 Union and works council consultation`; volume 1 chapter 5.

Where it lands.

> **`O57 Governance Body — Ref`, `Governance body`, `Cadence`, `Membership`, `Mandate`**
>
> Continue the `GB-` space from `GB-06`. Fill `Cadence` with a real interval — `Weekly`, `Per shutdown`, `On demand within 48 hours` — not `Continuous`, which `GB-05` uses and which cannot answer whether a body can sit before Friday. `Mandate` says what the body can refuse, not what it is interested in.

The five rows are `GB-01 Architecture Board` (monthly), `GB-02 Investment Committee` (quarterly), `GB-03 Data Governance Council` (monthly), `GB-04 Executive Committee` (monthly) and `GB-05 Architecture Function` (continuous). Now check what is missing.

- **No board of directors and no audit committee.** `GB-04` is management. Nothing in the file is the body a statutory obligation reports to.
- **No change advisory board.** Release of a production change has no owner anywhere in `O57`.
- **No body owns the plant's own change freeze.** `CON-06 Change freeze windows`, `IF-03 Peak trading freeze` and `IF-08 Site access windows` all exist as constraints, and no body in `O57` can grant or refuse an exception to any of them.
- **No plant-level body of any kind.** Three plants in `O12`, 568 people between them, and every membership list in `O57` is drawn from the eighteen corporate names shared by `O10 Actor` and `O11 Role`.
- **No works council or union consultation route**, against `CON-08`.

How.

1. Add a management-of-change committee per plant, `Cadence` = `Weekly`, with a mandate that includes refusing a physical change.
2. Add a change advisory board for release, and say in its mandate which gate it sits behind.
3. Add the audit committee, with a mandate limited to receiving evidence, not designing it.
4. Add the consultation route required by `CON-08`, with its notice period in `Cadence`.
5. Name people, not seats, in `Membership` — and accept that this exposes the deeper defect: there is no plant manager, production planner, maintenance planner, quality manager or process engineer anywhere in `O10` or `O11`, so most of your new members do not exist in the repository yet.

Signed by. `GB-04 Executive Committee`, the only body with standing to create another.

Done when. Every gate in `O58` and every freeze constraint in `O07` and `O52` names a body in `O57` that can grant an exception to it.

Shows up in. `ZZ Checks` line *Vacant architecture roles · 0*, true only by construction because all eighteen rows of `R21 Role performed by Actor` read `Filled`. Real bodies with real seats turn that zero into a number.

Goes wrong when. You add the bodies and leave the membership as job titles. `O20 Data Entity` already names owners like `Head of Health, Safety` that resolve against nothing in `O11`.

Look it up. Volume 1, chapter 5, *Safety, improvement, and the industrial-relations layer*, and chapter 5, *The plant*.

Step 10.6 — Record the plant's management-of-change regime and where it meets your gate

Produce. A one-page seam document per plant: who owns management of change, what it requires, which changes need both regimes, and which register wins when they disagree.

You need. The three plants in `012 Location`; the plant's own procedure, which is a paper or quality-system document and not in this repository; volume 1 chapters 13 and 16.

Where it lands.

> **A sheet you must create — `069 Change Regime`**
>
> Nothing in the 105 sheets records a second change-control regime. Create `069 Change Regime` with `Ref` (`CRG-01` onward), `Regime`, `Scope`, `Owner`, `What it requires`, `Evidence held in`, `Seam with the architecture gate`. One row per regime per plant. Reference it from `058`'s new `Evidence required` column so a gate knows when a plant record is part of its evidence.

A management-of-change process asks a different question from yours. Yours asks whether a design conforms. The plant's asks whether a physical change has been assessed for hazard, whether the affected procedures, interlocks, relief devices, training records and drawings have been updated, and whether the people on shift have been told. In a regulated plant those answers are records with retention clocks, and the inspector asks for them by name.

How.

1. Get the procedure. Ask the plant's safety, health and environment lead, not the CIO.
2. Record its trigger list. Expect a re-rate, a material-of-construction change, a new or modified interlock, a recipe change, a new raw-material supplier and any change to a relief path.
3. Record what it demands: a hazard study, an updated drawing set, a procedure revision, a training record and a sign-off naming a person.
4. Name the seam. Most changes need one regime; the ones needing both are changes to a computerised system that also change plant behaviour — a new recipe download path, a changed setpoint range, a historian tag feeding a control decision.
5. Decide which register is authoritative on a conflict and write it down. In a regulated plant it is the plant's, every time.
6. Record that control-layer change happens in shutdown windows, so gate dates follow the plant's calendar and not the reverse.

Signed by. The plant's management-of-change owner, counter-signed by `GB-01 Architecture Board`.

Done when. Each of the three plants has a row in `069`, and the seam is written as a rule a project manager can apply without asking you.

Shows up in. Nowhere in the thirteen reports. Say so: the repository is blind to the regime that can stop a change physically.

Goes wrong when. You discover the second regime at a gate, when a release is refused by somebody who was never on your stakeholder list.

Look it up. Volume 1, chapter 13, *What you are actually expected to produce*, and chapter 13, *Why patching a controller is not like patching a server*.

Step 10.7 — Record the engineering-change path, with the effectivity decision named

Produce. The engineering-change path across the applications in `O23`, with the five hand-offs named and the effectivity decision and stock disposition recorded as fields on the change.

You need. `O23 Application Component` — `A02 PLM`, `A01 ERP Core`, `A03 MES`, `A10 Supplier Portal`; `R06 Application interfaces with`; `O20 Data Entity` rows `E02 Bill of Material` and `E03 Routing / Recipe`; volume 1 chapters 8, 9 and 12.

Where it lands.

> **`O16 Business Process` and `R06 Application interfaces with`**
>
> `O16` holds 20 processes with `Ref`, `Business process`, `Value stream`, `Description`, `Automation`. Engineering change is a process and belongs here as a `BPR-` row with its value stream. The hand-offs are interfaces and belong in `R06`, which carries `INT-001` to `INT-030` — the only reference space in the file living on a relationship sheet. `O20` has no effectivity column at all, so the effectivity rule is recorded against the change process in `O16`'s `Description` until Phase C adds one.

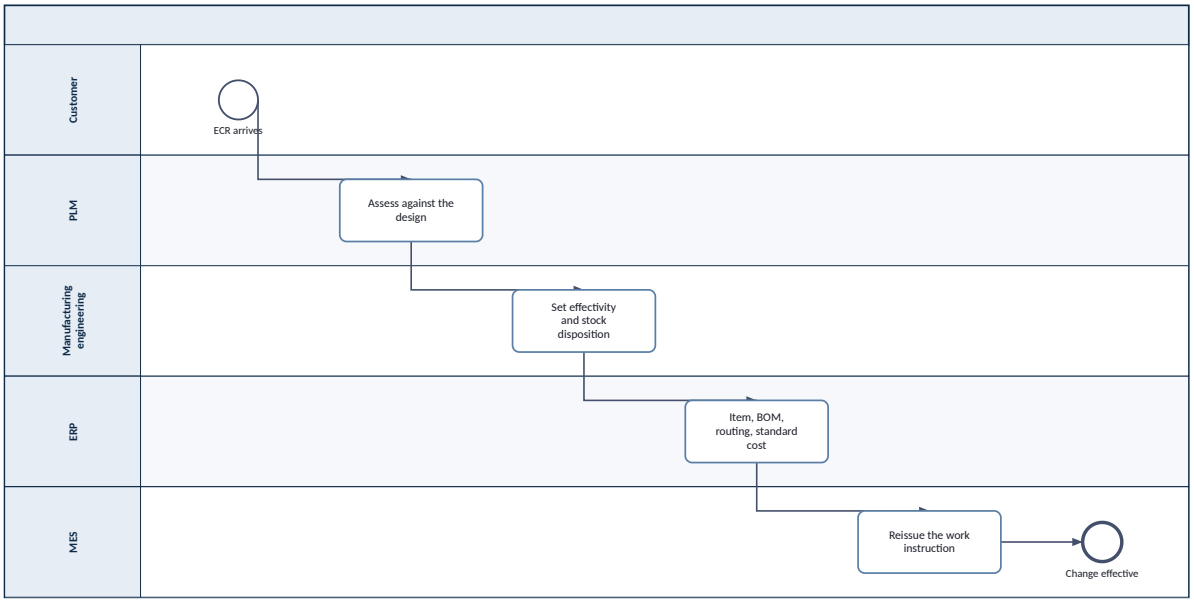


Figure 10.2 The engineering-change path, five hand-offs. A customer ECR is not negotiable, so the decisions are effectivity and stock disposition. The supplier notification, drawn nowhere here, is almost always the late one. See volume 1 chapter 8, *Engineering change*.

An engineering change request can arrive from the customer, and in a tier-supply arrangement it is not negotiable. You are not deciding whether. You are deciding when it takes effect, what happens to stock and work in progress, and which five systems have to agree.

How.

1. Establish the organisation's own words first. Request, order and notice are used inconsistently between vendors, and two workshops disappear into the argument.
2. Draw the path as it is: `A02 PLM` to manufacturing engineering to `A01 ERP Core` to `A03 MES`, plus the supplier notification through `A10 Supplier Portal`, which is almost always the late one.
3. Record the effectivity regime — date, serial, lot or change-number — per product family, and the date basis where it is date, because systems default differently.
4. Record the disposition of existing stock and work in progress as a field on the change: use as is, rework, scrap, return to supplier, segregate.
5. Check `R06` for the interfaces that should carry this and do not. 22 of 30 are unmanaged and 21 have no logging, so the fan-out is largely invisible.
6. Record which regulatory artefacts a change can void — a letter of authority is bound to an item revision.

Signed by. The engineering change board where one exists; otherwise `GB-01 Architecture Board` until step 10.5 creates one.

Done when. A `BPR-` row exists for engineering change, the five hand-offs are named as `INT-` refs or recorded as missing, and the effectivity regime is written down per product family.

Shows up in. `RPT04 Interface Catalogue`, where the hand-offs should appear with `Managed` and `Logged` populated. Most will not.

Goes wrong when. The chain has no single status, so "is change 4471 implemented?" means checking five systems. Mature manufacturers build a change dashboard; immature ones hold a weekly meeting, and the supplier ships the old part after the effectivity date.

Look it up. Volume 1, chapter 8, *Engineering change*; chapter 9, *Revision, and engineering change as a data event*; chapter 9, *Effectivity*; chapter 12, *Revision, effectivity and change*.

BEFORE YOU WRITE A GATE CRITERION

Which criterion on your list could a project actually fail, and what would the evidence be?

Which of your criteria rests on a principle in `005` whose `Status` reads `Draft`?

When the plant refuses a release your gate approved, who decides?

Step 10.8 — Build the inspector-evidence register

Produce. One register tying each inspecting body to what it asks to see, the record class that answers it, the repository column that holds the evidence, and the retention clock on it.

You need. `063 Control Objective`; `020 Data Entity`; `042 Metadata Element`; `012 Location`; volume 1 chapter 16 throughout and chapter 20's two maps.

Where it lands.

> **`A sheet you must create — 070 Inspection Demand`**
>
> No sheet in the file holds an inspecting body. `063` holds obligations, not the bodies that inspect them, and `057 Governance Body` holds internal committees only. Create `070 Inspection Demand` with `Ref` (`INS-01` onward), `Body`, `What it inspects`, `What it asks to see`, `Record class`, `Repository column`, `Retention clock`, `Position`. `Repository column` takes a sheet-and-column pair — `020.Lineage`, `037.Classification` — or the word `None`, which is the answer you will write most often.

The inspector is a governance stakeholder with a data demand, and the demand is specific. This is the register, with the column that would hold the answer in this file:

Body	What it asks to see	Column in this file
Employment and Labour, health and safety	Legal appointments; incident records with investigations; two-yearly risk assessments; monitoring by approved authorities	**None.** No appointments, incident or exposure sheet
Employment and Labour, pressure equipment	Inspection and test history on a **36-month** interval; the 12-month item is external inspection of fire-tube steam generators only	**None.** `027` has no inspection column
Employment and Labour, lifting equipment	**Three clocks at once** — 12 months on the machine, 6 on ropes and safety devices, 3 per identified item of tackle	**None**
Employment and Labour, chemical agents	Exposure assessments and air monitoring, retained a **minimum of 30 years**	**None**
Employment and Labour, noise	Assessments, monitoring and audiometry including baseline, retained **40 years**	**None**
Air quality licensing authority	The standard applying to this plant now; **at least 80% valid hourly values**; a **SANAS-accredited audit at least every two years**; the annual report	`023.A04 SCADA / Historian` — marked `Replace` with nothing named to replace it
Water department	Which uses are exercised; abstraction and discharge volumes; effluent quality with method and laboratory	**None**
Compulsory specifications regulator	The specification and version; the letter of authority with scope and expiry. **The instrument, not the SANS standard, is the citation**, and **section 20 makes a test result generally valid for the whole batch**	**None.** `020 E11 Quality Specification` reads `Classification = Public`
Legal metrology division	Type approval and verification history with marks; net-content control. **The Act fixes no verification interval**	**None**
Revenue service, customs	Classification with its basis; origin with supplier declarations; declarations linked to the commercial documents; **drawback lodged within six months of export**	**None**
Information Regulator, under POPIA	The lawful basis for biometric processing; the processing register; operator agreements; deletion evidence	`048.Personal data present` — which says production holds none and development does

Eight of the eleven demands have **no column anywhere in this file**. `063 Control Objective` covers none of them, because it is written for a European manufacturer: its obligations are `ISO 9001`, `ISO 14001 / 45001`, `Product safety and CE / UKCA marking`, `REACH / RoHS`, `Customs and origin rules` and `Extended producer responsibility`, and **CE and UKCA marking have no application in South Africa**.

How.

1. Build it from volume 1 chapter 16's own table and chapter 20's two maps. Do not re-derive the regulation.
2. Fill `Repository column` honestly. `None` is the right answer for most rows, and writing it is the deliverable.
3. Put the retention clock on the row, because it decides the design. **A single corporate retention period is always wrong**: five years breaches the Companies Act, seven breaches four occupational-health instruments. Flag the two clocks that outlive any application — 30 and 40 years, in systems replaced every seven to ten years, so what is needed is an export format with its reference data, readable without the source application.
4. Note the sequencing trap on drawback: the export declaration must already carry the correct procedure code and drawback item before the goods leave, so an error forfeits the claim irrecoverably. And **section 20 makes batch size an exposure decision**.
5. Record **biometric data as special personal information under section 26**, and that the longest-retained special information a manufacturer holds is its occupational health file.

Signed by. The Chief Compliance Officer, who owns all six drivers in `O01`, counter-signed per plant.

Done when. Every row names a body, a demand, a record class and either a sheet-and-column pair or `None`, and the `None` count is reported as a number.

Shows up in. `RPT09 Regulatory Coverage`, which reprints `O63`'s six rows and footers *5 obligations have no mechanism or a mechanism in breach*. Your register shows that 5 understates it, because the sheet is missing the obligations entirely.

Goes wrong when. The register is built as a compliance deliverable and filed. It is an architecture deliverable: each `None` is a sheet or column somebody has to build.

Look it up. Volume 1, chapter 16, *Who inspects, and what they ask for*; chapter 16, *The retention clocks, and why a single corporate period is always wrong*; chapter 20, *Map 7 — Regulator to evidence*; chapter 20, *Map 8 — Retention clock by record class*.

Form G-3 — Inspector evidence

Feeds `070 Inspection Demand`, a sheet the repository does not have. One row per demand, not per body. A first pass needs twenty to thirty rows, so continue on a second copy of this form. Signed by the Chief Compliance Officer and the plant's safety, health and environment lead.

Ref	Body	What it asks to see	Record class	Repository column	Retention clock
INS-01	EXAMPLE — delete this row	Audiometry including baseline	Employee-to-zone assignment	None	40 years

BEFORE YOU MEET AN INSPECTOR

Which body inspects each of the three plants in `012 Location`, and when did it last call?

Name one record with a retention clock longer than the life of the system holding it.

Which demand on your register has no column anywhere in the 105 sheets?

Step 10.9 — Stop `O63` 's tautology at the gate

Produce. Every `Control objective` cell in `O63 Control Objective` rewritten as a testable statement, and the jurisdiction finding recorded in writing, held for the chapter 12 rebuild.

You need. `O63`; `R16 Control Objective met by Control`; `RPT09`; volume 1 chapter 16.

Where it lands.

> **`O63 Control Objective` — `Ref`, `Obligation`, `Source`, `Control objective`, `Evidence`**
>
> Rewrite `Control objective` only. Leave `Obligation` and `Source` for the full rebuild in chapter 12 so the two edits do not collide. The cell must name the mechanism, the record it produces and the gate that tests it — not restate the obligation.

Step 10.8 established the jurisdiction error. The domestic equivalents of the European marking schemes are the **compulsory specification and the letter of authority**, and REACH and RoHS bind on export to the markets that impose them, not on domestic supply.

And `Control objective` is a tautology on every row. `CO-01` reads *"Control that evidences ISO 9001."* `CO-02` reads *"Control that evidences ISO 14001 / 45001."* Six for six, the obligation name substituted into one sentence. A tautology cannot be tested, which is why five of six `Evidence` cells read `No mechanism` or `In breach`, and the sixth, `CO-05`, reads `Evidenced` with nothing on the sheet to support it.

How.

1. Write the jurisdiction finding as one sentence naming both marking schemes, in the Phase G pack rather than a cell comment.
2. Rewrite each `Control objective` as mechanism plus record plus gate. For `CO-01`: a documented-information register with internal audit and corrective-action records, tested at `GT-2`.
3. Check `R16`, which pairs `CO-01` to `SEC-01 Identity lifecycle` and runs down the diagonal from there. **`ISO 9001` is evidenced by an identity lifecycle control** in this file. Mark all six instances unusable rather than re-pointing them now.
4. Leave `Evidence` alone until the rebuild. Five of six are already honest.

Signed by. Chief Compliance Officer, with `GB-01 Architecture Board` noting the correction.

Done when. No `Control objective` cell contains the obligation's own name.

Shows up in. `RPT09 Regulatory Coverage`, which reprints `O63` column for column and carries the tautology into the report. Fixing the sheet fixes the report — the only time in this repository that is true.

Goes wrong when. Somebody deletes the European rows to tidy up. Two have real domestic analogues, and deleting them loses the obligation along with the wrong citation.

Look it up. Volume 1, chapter 16, *Compulsory specifications, letters of authority, and section 20*, and chapter 16, *What a standard is, and what it is not*.

Step 10.10 — Put the plant's calendar on the gate

Produce. A column on `058 Governance Gate` recording when each gate cannot sit, and a window register behind it.

You need. `007` rows `CON-06 Change freeze windows` and `CON-08`; `052` rows `IF-03 Peak trading freeze` and `IF-08 Site access windows`; the plant's shutdown calendar; the sales-and-operations lock date; volume 1 chapter 7.

Where it lands.

> **`058 Governance Gate` — add `Cannot sit when`; and create `071 Change Window`**
>
> Add `Cannot sit when` to `058` in column G, taking `CW-` refs. Create `071 Change Window` with `Ref` (`CW-01` onward), `Window`, `Type`, `Plant`, `From`, `To`, `Owner`, `What it blocks`. `Type` takes `Shutdown`, `Seasonal peak`, `Planning lock`, `Stock count`, `Audit`. There is no calendar sheet in the file and no `Shift pattern` or `Line count` column on `012 Location`, so the repository cannot say when a plant is available.

The constraints are already in the file. `CON-06` says peak periods are closed to change. `IF-03` says no change in the peak window. `IF-08` says physical work is limited to shutdowns. **Not one of them is attached to a gate, a work package date or a project.** `030`'s twelve packages carry `Start` and `End` months with no regard for any of it, and `056 Project` repeats the same dates.

Three window types, three owners, all three outranking you. A **shutdown window** is when control-layer and physical work can happen at all, once or twice a year. A **seasonal peak** is when the plant will not accept a release. A **planning lock** is the date the sales-and-operations plan freezes, after which a change to planning master data invalidates it.

How.

1. Get each plant's shutdown dates for the next 24 months from the maintenance planner. They are fixed years out and everyone else works around them.
2. Get the seasonal peak from the demand planner and the lock date from the planning cycle owner.
3. Write them into `071` as dated windows per plant, with an owner who can grant an exception.
4. Fill `Cannot sit when` on all four rows of `058`. `GT-1 Design` can sit any time; `GT-2` cannot sit inside a peak, because nobody who must attend will be free.
5. Re-read every `Start` and `End` in `030` against the windows and report the collisions as a number.

Signed by. The plant manager for shutdown and peak; the planning cycle owner for the lock date.

Done when. Each plant has 24 months of windows in `O71`, every `O58` row reads a `CW-` ref or `Any time`, and the `O30` collision count is written down.

Shows up in. `RPT08 Roadmap`, where starts and ends can now be read against real windows. Expect collisions on `WP-04 Technology currency` and `WP-05 Access and security`, which both touch equipment that only moves in a shutdown.

Goes wrong when. A gate is scheduled into a shutdown because the diary was free. It was free because everyone who matters is on the floor in overalls.

Look it up. Volume 1, chapter 7, *Shifts, crews, and what a pattern costs*; chapter 7, *Sales and operations planning*; chapter 7, *Time fences, and why nobody respects them*.

Step 10.11 — Publish the monthly governance pack

Produce. A four-page pack, same shape every month, that a body can act on without asking you a question first.

You need. `RPT01 Dashboard`, `RPT09 Regulatory Coverage`, `RPT11 Security Position`, `RPT12 Standards Conformance`, `RPT08 Roadmap`, `RPT13 Orphan Analysis`; your own `O59`, `O60`, `O61`, `O70` and `O71`.

Where it lands.

> **`The reports, read in a fixed order`**
>
> `RPT01 Dashboard` is a 17-line reprint of `ZZ Checks` with two extra lines and a typed footer, so treat its figures as claims. `RPT11` reprints `O41`'s eight rows with a typed footer. `RPT09` reprints `O63`. `RPT12` reprints `O32`'s sixteen rows with no footer at all. `RPT08` reprints `O30` with a footer repeating the uncomputable *4 work packages past their end date*. Only `RPT13 Orphan Analysis` performs a join, and it returns two rows.

How.

1. Page one, the gate: assessments written this month, verdicts by type, the count of `Not assessable` rows and why.
2. Page two, the decisions: `O60` outstanding count, age of the oldest, what each blocks in dollars from `O30`.
3. Page three, the obligations: `O70` rows whose `Repository column` reads `None`, and the retention clocks longer than any system in `O27 Physical Technology Component`.
4. Page four, windows and dispensations: next shutdown per plant, gate dates against it, open dispensations with their expiry dates.
5. Footnote the four wrong `ZZ Checks` lines every month until somebody fixes them — records 843 against a counted 813; object types 65 against 64; platform components past

support 4 against 2; work packages past end date 4 against 0 computable.

6. Keep it to four pages. A longer one is read by nobody and the gate reverts to a conversation.

Signed by. `GB-05 Architecture Function` publishes; `GB-01 Architecture Board` receives it as a standing item.

Done when. The pack has gone out three months running in the same shape, and a body has refused something on the strength of it.

Shows up in. The pack is the output. A month's pack with no new `O59` rows means the gate did not sit.

Goes wrong when. It becomes a status report. A governance pack asks for decisions; a status report asks for nothing, and after four months nobody opens it.

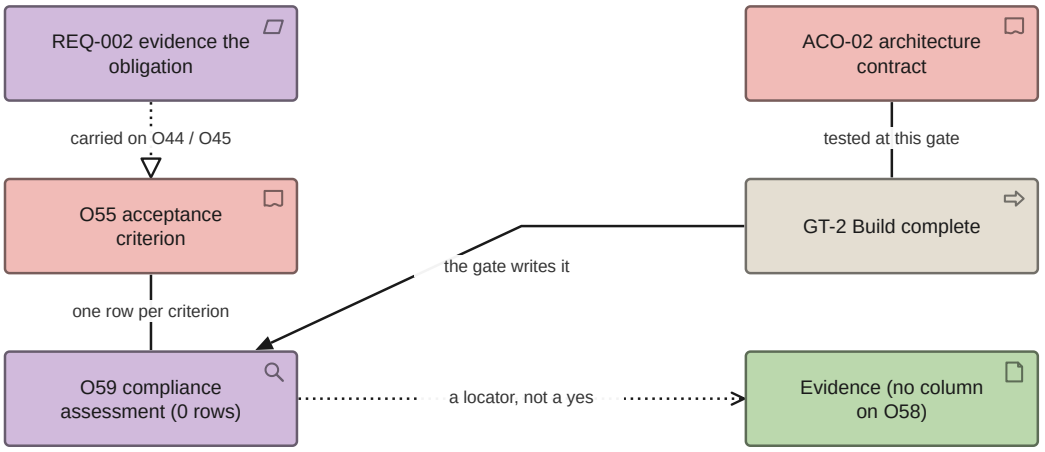


Figure 10.3 What a gate is supposed to produce. `O59` sits at the centre of the chain and holds nothing, so the contract status on the left is an assertion and the evidence on the right has no column to live in.

What good looks like at the end of Phase G

Three numbers say whether this chapter worked. `O59 Compliance Assessment` is not empty and every row has a locatable evidence cell. `O60 Governance Decision` has no `Outstanding` row without a due date. And the `None` count in your inspection register is falling month on month, because each one is a sheet or a column somebody built.

THE REVIEWER'S ASK — ANSWER THESE BEFORE YOU CALL PHASE G DONE

Point at one thing this gate refused. What was the evidence, and who appealed?

Which register would an inspector be shown, and would it answer the question asked?

Which of your gate dates falls inside a shutdown window, a peak or a planning lock?

If the plant's management-of-change process and your gate disagree tomorrow, which wins, and where is that written down?

THE CHAPTER IN TEN LINES

- O59 Compliance Assessment is empty, so R24 is empty, so the gate has produced zero records. That is the chapter's first finding.
- O41 SEC-02 Design review reads Absent, enforced at the Design gate. The file corroborates its own gap.
- ZZ Checks claims every one of 65 object types carries records. It is 64 of 65.
- O58 Governance Gate has no holder column and no evidence column. Add both before a gate sits.
- O60 Governance Decision has no date column. Four of its seven rows read Outstanding, and they are what the roadmap waits on.
- O33 asserts Non-compliant on two contracts with no assessment anywhere in the file behind either.
- O57 has five bodies and no plant in them: no audit committee, no change advisory board, nobody who owns the plant change freeze.
- A manufacturer runs two change-control regimes. The plant's management-of-change process is in no sheet, no column and no governance body here.
- An engineering change request can arrive from the customer and is not negotiable. The decisions are effectivity and stock disposition.
- Eight of eleven inspector demands have no column anywhere in the 105 sheets.
- O63 Control Objective is written for a European manufacturer. CE and UKCA marking have no application in South Africa.
- O63's Control objective column is a tautology on all six rows: Control that evidences ISO 9001.
- A shutdown window, a seasonal peak and a planning lock are dates on which a gate cannot sit. None of them is attached to a gate in this file.

SOURCES FOR THIS CHAPTER

- **1.** Manufacturing_TOGAF_Metamodel.xlsx, sheet O58 Governance Gate
- **2.** Manufacturing_TOGAF_Metamodel.xlsx, sheet O57 Governance Body
- **3.** Manufacturing_TOGAF_Metamodel.xlsx, sheets O33 Architecture Contract, O60 Governance Decision and O61 Governance Change Request
- **4.** Manufacturing_TOGAF_Metamodel.xlsx, sheets O59 Compliance Assessment, R24 Compliance Assessment tests Building Block and ZZ Checks
- **5.** Manufacturing_TOGAF_Metamodel.xlsx, sheets O41 Security Control and RPT11 Security Position
- **6.** The Open Group, TOGAF Standard — ADM Phase G, Implementation Governance https://pubs.opengroup.org/architecture/togaf-standard/adm/chap11.html

CHAPTER 11

Architecture Change Management

The repository has no version column, no change log, no change requests and no baselines. This chapter builds all four.

There is no change control anywhere in this repository

Search the 105 sheets for a version column. There is none. For a change log: none. For a change request register: `061 Governance Change Request` holds five rows about work packages and the repository, which is a governance inbox, not a change log. For a baseline: nothing. For a date on which anything was agreed: `060 Governance Decision` has no date column at all [1].

So the file cannot answer four questions a reader will ask within a week. *Which version am I looking at? What changed since I last read it? Who changed it, and under what authority? What was the state of the model when the board approved the roadmap?*

This chapter is different from the others. The rest tell you how to work the sheets you have. This one tells you to build the four sheets and the five columns the file is missing, and then gives you the pipeline that fills them.

> **THE ONE NUMBER TO ESTABLISH FIRST**
>
> `00 Index` and `ZZ Checks` and `RPT01 Dashboard` all state **843 records**. Count the rows yourself and the 64 object sheets plus `EXT Industry Extension` hold **813**. The difference is exactly 30 — the row count of `R06 Application interfaces with` , the only relationship sheet with a `Ref` column. The total reconciles only if an interface counts as a record, which concedes that an interface is an object with no object sheet of its own. **Back-fill 813, not 843**, and put the reconciliation in the first version note [2].

Kestrel Industrial Group, the manufacturer in the repository, is invented; so is volume 1's Karoo Industrial Group. Do not mix their data.

What you are building

Four sheets, in this order, placed before `00 Index` so a reader meets them first, each on the file's own convention — title in row 1, subtitle in row 2, headers in row 4, data from row 5, `Ref` first:

Sheet	Grain	Answers
`01 Version Control`	One row per published version	Which version am I reading
`02 Change Log`	**One row per field changed**	What changed, from what, to what
`03 Change Requests`	One row per request	Who asked, who decided, when it expires
`04 Baselines`	One row per baseline cut	What the model was when somebody agreed to it

Then five columns appended to every object sheet, and a pipeline that keeps them true.

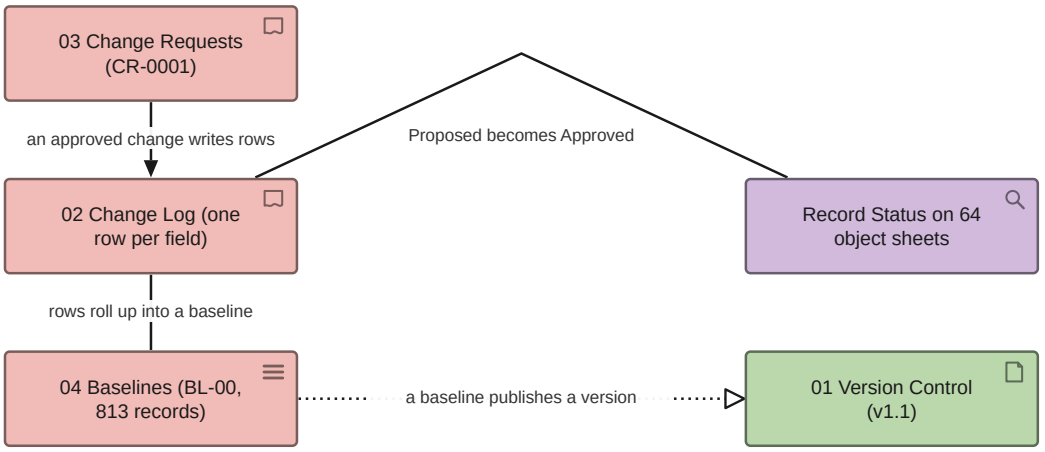


Figure 11.1 The four sheets you create, and how they hold each other up. A change request authorises log rows; log rows roll into a baseline; a baseline publishes a version. None of these five objects exists in the file you were handed.

BEFORE YOU CREATE A SINGLE SHEET

Who, today, can change a cell in this repository without telling anyone?

Which version of the model did the board approve when it adopted the four-gate regime at `DEC-06`?

If you had to roll the model back to last quarter, what would you roll it back from?

The thirteen steps

Step 11.1 — Count the records before you version anything

Produce. A counted record inventory, sheet by sheet, and the reconciliation note that explains the 30-row difference.

You need. The workbook; a script, because 65 sheets by hand is a day you do not have.

Where it lands.

> **`ZZ Checks` and the note that will become `01 Version Control` row 5**
>
> `ZZ Checks` lines read *Object types populated · 65 · Every type carries records*, *Records · 843*, *Relationship instances · 430*. Two of those are wrong and one is right. Objects populated is **64 of 65**, because `059 Compliance Assessment` holds zero rows. Records is **813** counted. Relationship instances is **430** and checks out. Write the counted figures into the version note, not over the `ZZ Checks` cells, until a change request authorises the edit.

How.

1. Count rows from row 5 down on every sheet whose name starts `0` plus `EXT Industry Extension`. Remember `034` and `051` do not exist, so a loop from `001` to `066` fails on the holes.
2. Record the per-sheet counts. You will need them for `04 Baselines`'s `Record count` column.
3. Reconcile against 843 and name the 30 rows of `R06`.
4. Check `059` separately and record it as the one empty object sheet.
5. Keep the script. You will run it at every baseline.

Signed by. `GB-05 Architecture Function`.

Done when. You can state the record count of any sheet from a file rather than from memory, and the 843 figure has a written explanation.

Shows up in. `RPT01 Dashboard` line *Records · 843*, which you now know to footnote.

Goes wrong when. Somebody edits `ZZ Checks` to say 813 without recording why, and the next reader finds three different totals in three places with no note between them.

Look it up. Volume 1, chapter 19, *What to count before you model anything*.

Step 11.2 — Create `01 Version Control` and publish version 1.0

Produce. The first sheet of the repository, with one row describing the state you received.

You need. Step 11.1's counts; the names of whoever has edited the file; `GB-01 Architecture Board` for the approval.

Where it lands.

> **`01 Version Control` — create it, first tab, before `00 Index`**
>
> Columns, in order: `Version`, `Date`, `Author`, `Approved by`, `Scope of change`, `Change refs`, `Baseline ref`. `Version` is `major.minor` — major on a baseline, minor on anything else. `Date` is ISO `YYYY-MM-DD`, never a locale format. `Author` is a person, not a team. `Approved by` is a `GB-` ref from `057 Governance Body`. `Scope of change` is one sentence naming the sheets touched. `Change refs` is a comma-separated list of `CR-` refs from `03 Change Requests`. `Baseline ref` is a `BL-` ref where this version cut one, otherwise blank.

How.

1. Create the sheet, title row `Version Control`, subtitle the organisation name, matching every other sheet.
2. Write row 5 as version `1.0`, dated the day you received the file, `Author` = whoever handed it over, `Scope of change` = *as received, 813 records counted against 843 stated*.
3. Write row 6 as version `1.1` for your own first edits, and never edit row 5 again.
4. Put the sheet first in tab order, and add a one-line pointer to it in `00 Index`, which describes structure but not state.

Signed by. `GB-01 Architecture Board` from version `1.1` onward.

Done when. Every published copy of the file carries a version number that appears in `01 Version Control`, and no two copies in circulation share a version with different content.

Shows up in. Nowhere in the thirteen reports, which is itself a finding: add the version to the footer of your own packs instead.

Goes wrong when. Versions are tracked in filenames. Within a month there are four files called *final*, and the one on the shared drive is not the one the board saw.

Step 11.3 — Create `02 Change Log`, one row per field

Produce. The sheet that makes every edit to the repository reversible.

You need. `01 Version Control`; `03 Change Requests` once step 11.4 is done; the discipline to write a row before you make an edit rather than after.

Where it lands.

> **`02 Change Log — create it, second tab`**
>
> Columns, in order: `Log ref`, `Date`, `Sheet`, `Record ref`, `Field`, `Old value`, `New value`, `Change Ref`, `Changed by`, `Reason`. The grain is **one row per field**, not per record and not per change request. Changing `Disposition`, `Health` and `Criticality` on `A04 SCADA / Historian` is three rows. `Log ref` runs `LOG-0001`. `Sheet` is the tab name spelled exactly — remember `R04 Business Service supported ` ends in a space. `Old value` is copied before the edit, which is the only reason this sheet works. `Reason` is a sentence, not the word *correction*.

How.

1. Create the sheet. Freeze rows 1 to 4 and set `Old value` and `New value` to text format, so a date or a currency string is not silently reformatted.
2. Write the first rows from chapter 10's step 10.4: `033.ACO-02.Compliance status` from `Non-compliant` to `Not yet assessable`, and the same for `ACO-03`. Two rows, one change request.
3. Make the rule explicit: no edit without a log row, no log row without a `Change Ref`. Write down the one exception — typographical fixes go in as a single row per sheet with `Change Ref` = `HOUSE`.
4. Review the log monthly against `01 Version Control`, because a version with no log rows did not happen.

Signed by. `GB-05 Architecture Function` maintains it; the reviewer at each baseline tests it.

Done when. Every cell that differs between version `1.0` and today has a log row, and a reader can rebuild version `1.0` from the current file and the log alone.

Shows up in. Your own monthly pack: rows logged, rows with no `Change Ref`, oldest unlogged difference.

Goes wrong when. The log is kept at record grain. `A04 SCADA / Historian changed` tells a reader nothing, and the one column that mattered — `Disposition` = `Replace` with nothing named to replace it — is invisible.

Form H-2 — Change log

Feeds `02 Change Log`, a sheet the repository does not have and which you create. **One row per field.** Fill it at the keyboard, before the edit, with `Old value` copied not remembered. `Change Ref`, `Changed by` and `Reason` go in straight to Excel. Twelve rows is a normal week; continue on a second copy. Signed by `GB-05 Architecture Function`.

Log ref	Date	Sheet	Record ref	Field	Old value
LOG-0001	2026-10-16	O33 Architecture Contract	ACO-02	Compliance status	Non-compliant

Step 11.4 — Create `03 Change Requests`

Produce. The register that decides what gets changed, by whom, and until when.

You need. `057 Governance Body` for the decision bodies; `061 Governance Change Request`, whose five rows migrate into this sheet; `058 Governance Gate`.

Where it lands.

> **`03 Change Requests` — create it, third tab**
>
> Columns, in order: `Change ref`, `Raised`, `Raised by`, `Against`, `Type`, `Description`, `Classification`, `Impact`, `Decision body`, `Decided`, `Status`, `Expires`, `Decision ref`. `Change ref` runs `CR-0001` — do not reuse `CR-G`, which belongs to `061`. `Against` names a sheet, a record ref, a work package or `Repository`. `Type` takes `Correction`, `Scope`, `Standard`, `Dispensation`, `Compliance`. `Classification` takes `Clarify`, `Extend`, `Re-architect`. `Expires` is mandatory on a `Dispensation` and blank otherwise. `Decision ref` points at a `DEC-` row in `060 Governance Decision`.

How.

1. Create the sheet and migrate `061`'s five rows, keeping their original refs in `Description` so the history survives.
2. Set the rule that anybody may raise a request and only a body in `057` may decide one.
3. Make `Impact` a count, not an adjective: sheets touched, records touched, work packages affected, cost where there is one.
4. Make `Expires` mandatory on `Dispensation`. `061`'s `CR-G01` already shows what good looks like, with an expiry of `2029-06`; `CR-G02` shows what it looks like without one.
5. Age the open rows monthly. Report the count, the oldest and the number expiring within 90 days.

Signed by. `GB-01 Architecture Board` for architecture change; `GB-02 Investment Committee` where money moves; `GB-03 Data Governance Council` for ownership and mastering.

Done when. No edit in `02 Change Log` lacks a `Change Ref` resolving here, and no `Dispensation` row lacks an `Expires`.

Shows up in. Nothing reads it, so publish the open count, the oldest and the 90-day expiry list monthly.

Goes wrong when. The register becomes a request queue with no decider. Requests accumulate, the architecture function becomes the bottleneck everyone routes around, and the sheets drift anyway.

Step 11.5 — Create `04 Baselines`

Produce. The sheet that lets somebody say what the model was on the day they agreed to it.

You need. Step 11.1's per-sheet counts; `01 Version Control`; `060 Governance Decision`; `053 Transition Architecture` and `066 Architecture Increment` for the natural baseline points.

Where it lands.

> **`04 Baselines` — create it, fourth tab**
>
> Columns, in order: `Baseline ref`, `Date`, `Label`, `Scope`, `Record count`, `Approved by`, `Decision ref`, `Supersedes`. `Baseline ref` runs `BL-00` for the received state, then `BL-01` onward. `Scope` says which sheets the baseline covers — `All`, or a domain list. `Record count` is the counted figure from step 11.1, never the stated one. `Decision ref` is the `DEC-` row that approved it. `Supersedes` is the previous `BL-` ref, so the chain is walkable in either direction.

How.

1. Create the sheet and write `BL-00` for the file as received: `Record count` = 813, the rest blank.
2. Decide what triggers a baseline. Three are enough: a gate decision in `058`, a transition state reached in `053`, and the start of each `066` increment.
3. Snapshot the file at each baseline. A baseline that is only a row in a sheet is a promise — keep the file beside it.
4. Tie each baseline to a decision. A baseline nobody approved is a save point, not a baseline.
5. Never edit a baseline row. Supersede it.

Signed by. `GB-01 Architecture Board`, with `GB-02 Investment Committee` where the baseline supports a funding decision.

Done when. `BL-00` exists with a counted record total, and every subsequent baseline names a `DEC-` ref and the version it was cut from.

Shows up in. `RPT01 Dashboard`, whose figures now have a date and a baseline behind them instead of floating free.

Goes wrong when. Baselines are cut monthly out of habit. Twenty baselines, none of them tied to a decision, and the question *what did the board approve* still has no answer.

BEFORE YOU BACK-FILL 813 RECORDS

Which records in this file do you believe a governance body genuinely approved, and where is the decision written down?

Who will object to seeing the whole repository marked `Proposed`, and what will they say?

Step 11.6 — Append five columns to all 64 object sheets and back-fill `Proposed`

Produce. Every record in the repository carrying a version, a status, an author, a change reference and an approver.

You need. Step 11.1's inventory; `03 Change Requests` for the back-fill reference; a script.

Where it lands.

> **`All 64 object sheets plus EXT Industry Extension — append five columns`**
>
> In this order, after the last existing column of each sheet: `Version`, `Record Status`, `Changed By`, `Change Ref`, `Approved By`. `Version` is the `01 Version Control` version in which the row last changed. `Record Status` takes exactly one of `Proposed`, `Approved`, `Superseded`, `Retired`. `Changed By` is a person. `Change Ref` is a `CR-` ref. `Approved By` is a `GB-` ref. Note the column letters differ per sheet — `O08 Assumption` has three columns and `O23 Application Component` has ten, so append by position, not by letter.

Back-fill every record as `Proposed`. That is 813 records, not the 843 the index claims. Nothing in this file has been approved by a body that left a record, so `Approved` would be an invention. The back-fill is the single most informative thing you will do in Phase H: it turns a repository that reads as settled into one that reads as proposed, which is what it is.

How.

1. Write the loop over sheet names, not over a numeric range. `O34` and `O51` do not exist, so `O01`–`O66` fails on the holes; there are **64** object sheets plus `EXT`.
2. Append the five headers in row 4 of each sheet, matching the file's header formatting.
3. Fill `Version` = `1.1`, `Record Status` = `Proposed`, `Change Ref` = the `CR-` row that authorises the back-fill, `Changed By` = you, `Approved By` = blank, on all 813 rows.
4. Write one `02 Change Log` row per sheet for the structural change, not one per record. 65 rows, not 4,065.

5. Then walk the sheets a domain at a time and promote rows to `Approved` only where a body decided and the decision is in `O60`.
6. Re-run the count. 813 rows should carry a status; a sheet returning a different number has a blank row or a stray value below the data.

Signed by. `GB-01 Architecture Board` approves the structural change; `GB-05 Architecture Function` executes it.

Done when. `Record Status` is populated on all 813 rows, takes only the four permitted values, and the count of `Approved` rows is a number you can defend row by row.

Shows up in. Every report, indirectly: a report over `Proposed` records is a report over a proposal. Say so on the face of each pack until the promotions are done.

Goes wrong when. Somebody back-fills `Approved` to avoid the conversation. You now have 813 approvals nobody gave, and the status column is worthless from day one.

Step 11.7 — Cut baseline `BL-00` and publish it

Produce. A frozen, counted, dated copy of the repository as received, with the version note and the four new sheets in place.

You need. Steps 11.1 to 11.6; somewhere immutable to keep the file.

Where it lands.

> **`04 Baselines` row 5 and `01 Version Control`**
>
> `BL-00` in `04 Baselines`, `Label` = *as received, back-filled Proposed*, `Record count` = 813, `Date` = the day you cut it, `Scope` = `All`. Then a `01 Version Control` row whose `Baseline ref` is `BL-00`. The pair is what makes the baseline findable: a reader coming from either sheet reaches the other.

How.

1. Freeze the file. Copy it somewhere nobody can edit, named with the version and the baseline ref.
2. Write the counted per-sheet record totals into a tab or an attachment, because the next baseline's diff is computed against them.
3. Record the four known `ZZ Checks` errors in the baseline note: records 843 against 813; object types 65 against 64; platform components past support 4 against 2; work packages past end date 4 against 0 computable.
4. Publish the baseline reference to everyone who holds a copy, and ask them to delete theirs.
5. Set the next baseline trigger from step 11.5's three rules.

Signed by. `GB-01 Architecture Board`.

Done when. The frozen copy exists, its record count is written down, and both sheets point at each other.

Shows up in. Every later diff. Without `BL-00` there is nothing to diff against, and the comparison tool has no left-hand column.

Goes wrong when. The baseline is cut but not published, so three people keep working on their own copies and the first diff is between two files that were never the same file.

Form H-3 — Baseline

Feeds `04 Baselines`, a sheet the repository does not have and which you create. One row per baseline cut; three or four in a first year, so this form lasts. `Decision ref` and `Supersedes` go in straight to Excel. Signed by `GB-01 Architecture Board`.

Baseline ref	Date	Label	Scope	Record count	Approved by
BL-00	2026-10-16	EXAMPLE — delete this row	All	813	

Step 11.8 — Write the trigger list

Produce. A published list of what makes a change request mandatory, with the register each trigger lands in.

You need. `O58 Governance Gate`; `O60`; `O30 Work Package`; `O27 Physical Technology Component`; `O19 Contract`; the plant's management-of-change register from chapter 10's step 10.6.

Where it lands.

`03 Change Requests` — the `Type` and `Against` columns, governed by this list

The trigger list does not get a sheet of its own. It governs what may appear in `Against` and `Type`, so publish it beside the register and keep it to one page. Every trigger must name a register, and where the answer is more than one register, say which is authoritative.

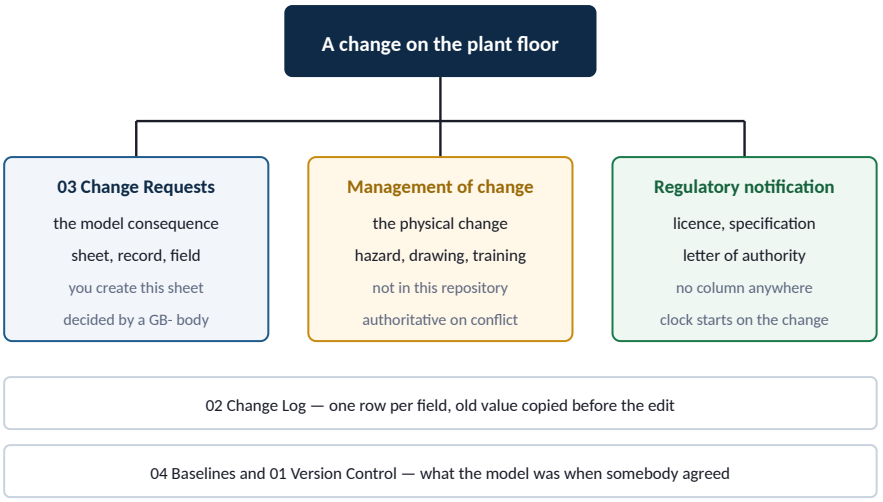


Figure 11.2 One plant change, three registers. The middle column is not in the repository and is authoritative on anything physical. The right-hand column has no column anywhere in the 105 sheets, and its clock starts on the day of the change.

Trigger	Architecture register	Plant register	Regulatory notification
New production line	`03 Change Requests`, `Type = Scope`	Management of change	Likely — emission licence and health and safety
Line re-rate or capacity change	`03 Change Requests`, `Type = Scope`	Management of change	Possible — licence condition is capacity-bound
Recipe or formula change	`03 Change Requests`, `Type = Scope`	Management of change	Yes where the product is regulated
Bill-of-material revision	`03 Change Requests`, `Type = Correction`	Engineering change	Possible — a letter of authority binds to a revision
New raw-material supplier	`03 Change Requests`, `Type = Scope`	Management of change	Possible — substance and origin
Vendor support date passed in `027`	`03 Change Requests`, `Type = Standard`	None	No
Contract expiry in `019`	`03 Change Requests`, `Type = Scope`	None	No
Gate verdict of `Non-conformant` in `059`	`03 Change Requests`, `Type = Compliance`	None	No
Dispensation expiry in `03 Change Requests`	Same sheet, new row	None	No
Change to a computerised system holding a quality record	`03 Change Requests`, `Type = Compliance`	Management of change	Yes in a regulated plant

How.

1. Start from the ten rows above and add the triggers your plants actually raise.
2. For each, name the authoritative register. Where both apply, the plant register wins on anything physical.
3. Attach a date source to every trigger that has one, so the trigger fires without a person remembering. `027`'s `Support end` and `019`'s `Expiry` are both dated columns already.
4. Publish the list with the register, and review it every six months.

Signed by. `GB-01 Architecture Board`, with the plant management-of-change owners.

Done when. Every trigger names a register and an authoritative one, and the dated triggers have a source column named.

Shows up in. `RPT10 Technology Currency`, which repeats `027`'s typed `Position` column and so reports four components past support when two of the four dates — `PTC-06` at 2027-07 and `PTC-08` at 2027-04 — are in the future. A dated trigger would compute it instead.

Goes wrong when. The list is written for the architecture register only. A new line then goes in with a change request, no hazard study and no notification, and the first person to notice is an

inspector.

Look it up. Volume 1, chapter 8, *Engineering change*, for the trigger list a manufacturer actually raises; chapter 16, *Who inspects, and what they ask for*, for which of them notify.

Step 11.9 — Classify a change and route it

Produce. A classification rule a project manager can apply without you, and the routing that follows from it.

You need. `03 Change Requests`; `057`; `053 Transition Architecture`; `054 Architecture Alternative`.

Where it lands.

> **`03 Change Requests` — `Classification`, `Impact`, `Decision body`**
>
> `Classification` takes `Clarify` (the model was unclear, the architecture is unchanged), `Extend` (something new that fits the current architecture) or `Re-architect` (the current architecture cannot carry it). `Impact` is counted, not described. `Decision body` follows from the classification, which is the whole point of having one.

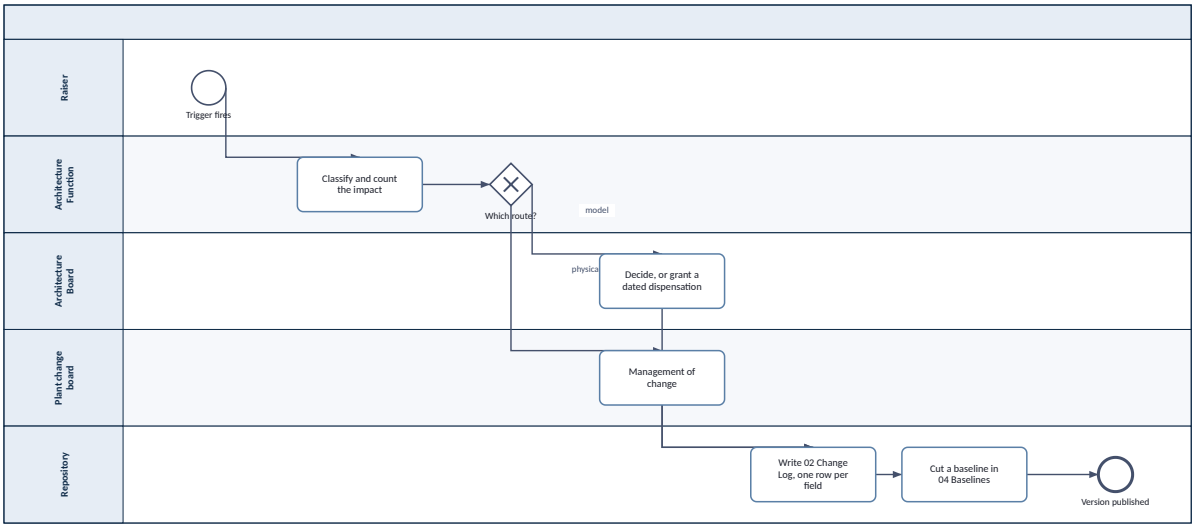


Figure 11.3 The change pipeline. Classification decides the route, and the route decides the body. Both routes end in the same two repository actions, which is what keeps the model and the plant from drifting apart.

How.

1. Apply the test in one question: does the target architecture still hold after this change? Yes and it was unclear, `Clarify`. Yes and it is new, `Extend`. No, `Re-architect`.
2. Route on the classification. `Clarify` goes to `GB-05 Architecture Function` and is logged. `Extend` goes to `GB-01 Architecture Board`. `Re-architect` goes to `GB-02 Investment Committee`, because it moves money.

3. Count the impact: sheets, records, work packages, dollars. `O30` sums to $65.0m across twelve packages against `ALT-B`'s $46m, so a `Re-architect` request has $19m of unexplained scope to argue with before it starts.
4. Refuse to classify your own request. The raiser proposes, the architecture function classifies.
5. Publish the classification mix monthly. A register that is 90% `Clarify` means the model was written badly; one that is 30% `Re-architect` means the target architecture is not agreed.

Signed by. `GB-05 Architecture Function` classifies; the routed body decides.

Done when. Every open row has a classification, and the classification determines the decision body without a conversation.

Shows up in. Your monthly pack as a three-way mix, and in `RPT07 Gap Analysis` when a `Re-architect` request lands as a new gap.

Goes wrong when. Everything is classified `Clarify` to avoid a board meeting. The architecture then changes by accumulation, and the baseline nobody updated stops describing anything.

Step 11.10 — Grant, date and expire a dispensation

Produce. A dispensation that ends.

You need. `03 Change Requests`; `O61`'s `CR-G01` as the worked example; `O32 Standard`; `O59`.

Where it lands.

> **`03 Change Requests` — `Type` = `Dispensation`, `Expires`, plus `Condition` to add**
>
> `Expires` is already in the column list and is mandatory on this type. Add `Condition` as a fourteenth column: what has to become true for the dispensation to lapse rather than be renewed. Without it, every expiry date is a renewal date.

A dispensation is a standard temporarily not applying to one thing. It is not a decision that the standard is wrong — that is a `Standard` request against `O32`. The distinction matters because the first expires and the second does not.

How.

1. Name the standard being set aside, by `STD-` ref, and the one thing it is set aside for.
2. Set an expiry. Eighteen months is long; three years, which is what `CR-G01` carries, is a permanent exception wearing a date.
3. Write the `Condition`. For `CR-G01` it is the integration product in production, which depends on `DEC-02 Select the integration product` — still `Outstanding`.
4. Record the risk of the dispensation holding past its date, in `O62 Delivery Risk`, with an owner.

5. Review expiries every month at 90 days out, and either close, extend with a new decision, or escalate.

Signed by. `GB-01 Architecture Board`, whose `O57` mandate already names this power.

Done when. No `Dispensation` row lacks an expiry and a condition, and the 90-day list is published monthly.

Shows up in. `RPT12 Standards Conformance`. `STD-01 API design standard` reads `Deprecated` with `Conformance` = `Conformant`, which is what a dispensation looks like once it has outlived the standard it suspended.

Goes wrong when. The dispensation is renewed without the condition being tested, three times, and the exception becomes the pattern everybody copies.

Step 11.11 — Route a plant-floor change into the right register

Produce. A written routing for the five plant changes that arrive most often, each naming the registers it lands in and the order they are filled.

You need. Chapter 10's step 10.6 seam document; `O69 Change Regime`; `O12 Location`; the plant's management-of-change owners; volume 1 chapters 9, 13 and 16.

Where it lands.

> **`03 Change Requests`, `O69 Change Regime` and the plant's own register**
>
> A change to the plant floor is not a change to a spreadsheet. The architecture register records the model consequence; the plant register records the physical change; neither substitutes for the other. `O69 Change Regime` is the sheet you created in chapter 10 and is where the seam rule lives.

Five changes, worked through:

- **A new line.** Architecture: a new record in whatever sheet holds a production line — and there is none, because `X02 Production Line` in `EXT Industry Extension` is shadowed nowhere, `O12 Location` holds sites and `E07 Work Centre` holds resources. Plant: management of change, with hazard study and commissioning records. Regulatory: likely an emission licence variation and a health and safety notification.
- **A re-rate.** Architecture: capacity is an attribute of a line, so it has nowhere to live either. Plant: management of change. Regulatory: possible, where a licence condition is capacity-bound.
- **A recipe change.** Architecture: `E03 Routing / Recipe`, whose `Lineage` reads `Absent` and whose golden source `O20` and `O43` disagree on — `Plant Analytics` against `Supplier Portal`. Plant: management of change, with a released formula version frozen at batch creation. Regulatory: yes where the product is regulated.
- **A new supplier.** Architecture: `E09 Supplier`, classified `Public` in `O20`, mastered in `APS`. Plant: supplier qualification, and management of change where a material of construction

changes.

- **A bill-of-material revision.** Architecture: `E02 Bill of Material`, `O20` saying `PLM` and `043` saying `Track & Trace`. Plant: engineering change with an effectivity rule and a stock disposition. Regulatory: possible, because a letter of authority binds to an item revision.

How.

1. Write the routing for these five first. They cover most of what arrives.
2. For each, name which register is filled first. The plant's, every time, where the change is physical.
3. Name the fields the architecture register needs from the plant register — the change number, the effectivity date and the disposition at minimum.
4. Record the three objects that have no sheet to land in: the production line, the line's capacity and the changeover. Raise them as a `Re-architect` request rather than inventing columns.

Signed by. The plant management-of-change owner and `GB-01 Architecture Board` jointly.

Done when. Each of the five has a written routing, and a project manager can route a sixth by analogy without asking you.

Shows up in. `RPT05 Data Ownership` and `RPT13 Orphan Analysis`, where the entities these changes touch are the ones with absent lineage and disputed golden sources.

Goes wrong when. The architecture register is filled and the plant register is not, because the project team are IT people and nobody told them the second one exists.

Look it up. Volume 1, chapter 9, *Effectivity*; chapter 9, *Revision, and engineering change as a data event*; chapter 13, *What you are actually expected to produce*.

BEFORE YOU GRANT YOUR FIRST DISPENSATION

Which standard are you setting aside, by `STD-` ref, and for what one thing?

What has to become true for this dispensation to lapse rather than be renewed?

Who carries the risk if it is still open on its expiry date, and is that person in `062` ?

Step 11.12 — Govern a change to a computerised system holding a quality record

Produce. A change route for systems whose records are compliance records, with the evidence obligation stated before the change is made.

You need. `O23 Application Component`; `O48 Environment`; `O49 Resilience Tier`; `O41 Security Control`; the quality scheme that applies to the plant; volume 1 chapters 14 and 16.

Where it lands.

> **`03 Change Requests` with `Type` = `Compliance`, and `O69 Change Regime`**
>
> Mark the applications whose records are compliance records. In `O23` that is at least `A07 QMS`, `A03 MES` and `A04 SCADA / Historian` — the batch record, the executed order and the time series an emissions return is computed from. Add a `Compliance record` column to `O23` taking `Yes`, `No` or `Unknown`, because the sheet cannot currently express it.

In a regulated plant a change to a system holding a quality record carries an evidence obligation of its own: what changed, who authorised it, what was tested, and that the record before the change is still readable afterwards. That is not the same as an architecture verdict, and it is not satisfied by one.

How.

1. Mark the compliance-record systems in `O23` and get the marking agreed by the quality manager, not by IT.
2. For each, record what a change must produce: an authorisation, a test record, a record-integrity check and a trained-user record.
3. Check the three sheets that contradict each other here. `O48 Environment` says `ENV-01 Production` holds no personal data while `ENV-02` and `ENV-04` do, with `SEC-07 Non-production masking` reading `In place`. One of those three cells is false, and in a manufacturer the personal data in question is overwhelmingly employee and biometric.
4. Check `O49`: `RES-04 SCADA / Historian` is the only internally coherent row — `Tier 1`, one hour, `Tested`. Eight of twelve are never tested, so a change to any of them has no recovery evidence behind it.
5. Route these changes through both registers, always, with the plant's authoritative.

Signed by. The quality manager, with `GB-01 Architecture Board`.

Done when. Every compliance-record system is marked in `O23`, and a change to one cannot be closed without its four records.

Shows up in. `RPT02 Application Portfolio` and `RPT11 Security Position`. `SEC-02 Design review` reads `Absent` at the design gate, so the authorisation record this step needs currently has no mechanism.

Goes wrong when. The change is treated as an IT release. It passes the architecture gate, the quality manager finds out at the next audit, and the system is taken out of use until the record

can be reconstructed.

Look it up. Volume 1, chapter 14, *What each scheme compels, which is what makes it architecture*; chapter 16, *POPIA, where it actually bites a manufacturer*.

Form H-1 — Change request

Feeds `03 Change Requests`, a sheet the repository does not have and which you create. One row per request. `Description`, `Impact`, `Decision body`, `Decided`, `Status`, `Expires` and `Decision ref` are typed straight into Excel after the decision — this form captures what the raiser knows. Eight to twelve rows is a month. Signed by `GB-05 Architecture Function` on receipt.

Change ref	Raised	Raised by	Against	Type	Classification
CR-0001	2026-10-16	EXAMPLE — delete this row	O33 ACO-02	Correction	Clarify

Step 11.13 — Measure whether change management is working

Produce. Six numbers, published monthly, that say whether the pipeline is running or being bypassed.

You need. `01 Version Control`; `02 Change Log`; `03 Change Requests`; `04 Baselines`; the `Record Status` column on all 64 object sheets.

Where it lands.

> **`Your monthly pack, computed over the four new sheets`**
>
> No report in the thirteen reads any of them, so these numbers live in your pack. Compute them, do not estimate them: each is a count over a sheet you built, and a number you cannot compute is a number you should not publish.

Measure	What it says	Unhealthy looks like
Unlogged differences	Cells that differ from the last baseline with no `02 Change Log` row	Any number above zero
Records still `Proposed`	How much of the 813 has been through a body	Flat for three months
Open requests, oldest	Whether the register decides anything	Oldest above 90 days
Classification mix	Whether the model or the target is the problem	90% `Clarify`, or 30% `Re-architect`
Dispensations past expiry	Exceptions becoming the standard	Any
Days since last baseline	Whether anything is agreed	Above 180

How.

1. Compute all six from the sheets. Re-run step 11.1's script for the first two.
2. Publish them in the same order every month. A changing shape hides a trend.
3. Set one threshold per measure and say what happens when it is crossed.
4. Report unlogged differences to the Architecture Board by name, not by count, the first three times. After that a count will do.

Signed by. `GB-05 Architecture Function` publishes; `GB-01 Architecture Board` receives.

Done when. The six have been published three months running and one of them has caused a decision.

Shows up in. Your own pack. Nothing in the repository computes any of it, which is the condition this chapter exists to end.

Goes wrong when. The measures are published and never breached, because the thresholds were set where the current numbers already sit.

Look it up. Volume 1, chapter 7, *So the first deliverable is a measure definition register* — the same discipline applies to a governance measure as to a plant one.

What you have at the end of Phase H

Four sheets the file did not have, five columns on all 64 object sheets, 813 records back-filled as `Proposed`, one baseline, and a register that decides. None of it is visible in the thirteen reports, so the monthly pack is the only place it shows. That is worth saying out loud each time you publish it.

THE REVIEWER'S ASK — ANSWER THESE BEFORE YOU CALL PHASE H DONE

Reconstruct `BL-00` from the current file and `02 Change Log` alone. Did it work?

How many of the 813 records are still `Proposed`, and which domain is the worst?

Name a change that went through the plant register and never reached `03 Change Requests`.

Which dispensation expires next, and what is its condition?

THE CHAPTER IN TEN LINES

- There is no change control anywhere in this repository: no version columns, no change log, no change requests, no baselines.
- Create four sheets before 00 Index: 01 Version Control, 02 Change Log, 03 Change Requests, 04 Baselines.
- 02 Change Log is at field grain. One record changed in three columns is three rows.
- Append five columns to all 64 object sheets: Version, Record Status, Changed By, Change Ref, Approved By.
- Record Status takes Proposed, Approved, Superseded or Retired, and nothing else.
- Back-fill every record as Proposed. That is 813 records counted, not the 843 the index claims.
- The 30-row difference is exactly the row count of R06, the only relationship sheet with a Ref column.
- O34 and O51 do not exist, so a loop from O01 to O66 fails on the holes. There are 64 object sheets plus EXT.
- A dispensation without an expiry and a condition is a change of standard that nobody approved.
- A change on the plant floor lands in up to three registers. The plant register is authoritative on anything physical.
- A new line, a re-rate and a changeover have no sheet to land in, because X02 Production Line is shadowed nowhere.
- In a regulated plant a change to a system holding a quality record carries its own evidence obligation, which an architecture verdict does not satisfy.
- No report among the thirteen reads any of the four new sheets, so the monthly pack is the only place change management shows.

SOURCES FOR THIS CHAPTER

- **1.** Manufacturing_TOGAF_Metamodel.xlsx, sheets O60 Governance Decision and O61 Governance Change Request
- **2.** Manufacturing_TOGAF_Metamodel.xlsx, sheets 00 Index, ZZ Checks, RPT01 Dashboard and R06 Application interfaces with
- **3.** The Open Group, TOGAF Standard — ADM Phase H, Architecture Change Management https://pubs.opengroup.org/architecture/togaf-standard/adm/chap12.html

CHAPTER 12

Requirements Management

Nothing in the twenty-five relationship sheets joins a driver to a goal, or a requirement to an objective. The spine has two breaks in it.

Requirements Management is the one activity that never stops

Every other chapter in this volume belongs to a phase. This one runs underneath all of them, and its test is simple: pick any requirement in the repository and walk it outward in both directions. Up to the business reason it exists, down to the thing that will deliver it. If the walk breaks, the requirement is an opinion.

Walk `REQ-002 Evidence ISO 14001 / 45001` and see where it stops.

The motivation sheets are well populated. `O01 Driver` holds 12 rows, six of them `Regulatory` and owned by the Chief Compliance Officer, six `Business` and owned by `Group Executive`. `O02 Goal` holds 6, each `Active` with a named executive owner. `O03 Objective` holds 14, each naming its `Goal` and a `Horizon`. `O04 Measure` holds 16. `O06 Requirement` holds 31. `O07 Constraint` holds 8 and `O08 Assumption` holds 6 [1].

Three relationship sheets join them. `R08 Requirement derives from Driver` carries 31 instances, one per requirement, and it is the healthiest join in the whole file — every requirement names the driver it came from and the names resolve. `R10 Objective measured by Measure` carries 14. `R11 Goal realised by Objective` carries 14, in blocks of two or three per goal [2].

Now the breaks. **Nothing in the 25 relationship sheets joins a driver to a goal.** And **nothing joins a requirement to an objective.** So the chain runs driver → requirement in one place and goal → objective → measure in another, and the two halves never meet. You can say why `REQ-002` exists and you can say how `GOL-01 Protect the licence to operate` will be measured, but you cannot say that `REQ-002` moves `MSR-01 Obligations with evidence`, because no row anywhere asserts it.

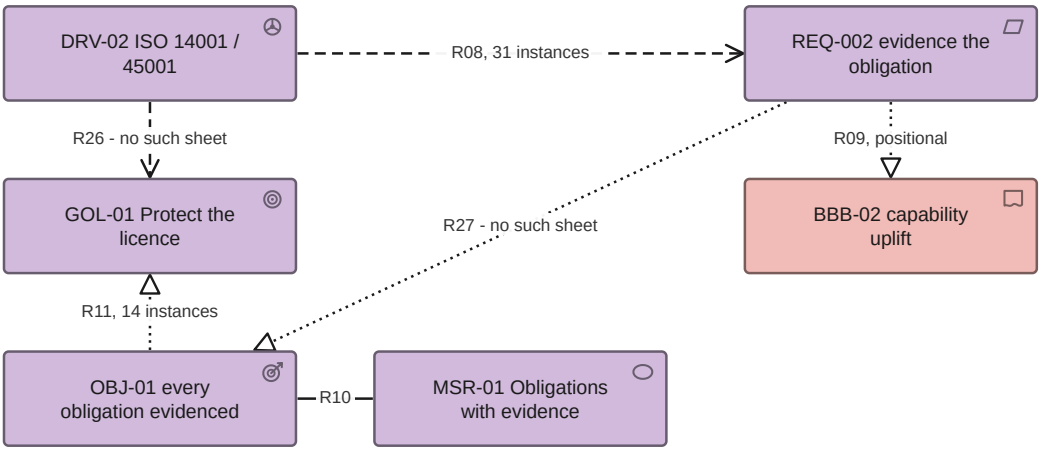


Figure 12.1 The motivation spine, with its two breaks. The solid joins exist and carry 31, 14 and 14 instances. The two labelled *no such sheet* are the ones you create as `R26 Goal addresses Driver` and `R27 Requirement realises Object`. `R10` and `R09` exist but are positional.

Kestrel Industrial Group, the manufacturer in this repository, is invented, as is volume 1's Karoo Industrial Group. Their data never mixes.

BEFORE YOU TOUCH THE MOTIVATION SHEETS

Take one requirement. Name the driver, the objective and the measure it moves. Which of the three could you not find?

Who would be asked to defend this requirements catalogue, and to whom?

The twelve steps

Step 12.1 — Walk the spine and record where it breaks

Produce. A one-page spine diagnostic: each join that exists, its instance count, and each join that does not.

You need. `O01` to `O08`; `R08`, `R09`, `R10`, `R11`; the full sheet list, because the absence of a sheet is the finding.

Where it lands.

> **The relationship sheets, read as a set**
>
> `R08 Requirement derives from Dr` — 31 instances, `Requirement`, `Requirement name`, `Driver`, `Driver name`. `R09 Requirement satisfied by Bu` — 31, with a `Traced` column. `R10 Objective measured by Measu` — 14. `R11 Goal realised by Objective` — 14. Note the tab names are cut at Excel's 31 characters: only `R11` among these four is intact.

How.

1. List the eight motivation sheets with their row counts, from the file and not from the index.
2. List every relationship sheet whose two ends are both motivation sheets. There are four.
3. Write down the two joins that are missing by name: driver to goal, and requirement to objective.
4. Test `R11`'s shape. It pairs `GOL-01` to `OBJ-01` and `OBJ-02`, `GOL-02` to `OBJ-03` and `OBJ-04`, and so on in blocks — which is plausible and may even be right, but it is a pattern, so check two blocks against someone who would know.
5. Test `R10`'s shape. `OBJ-01` to `MSR-01`, `OBJ-02` to `MSR-02`, down to `OBJ-14` to `MSR-14`. A positional diagonal, and it leaves `MSR-15 Architecture compliance rate` and `MSR-16 Initiatives through a design gate` measuring nothing.

Signed by. `GB-05 Architecture Function`.

Done when. The diagnostic names four existing joins with counts, two missing joins, and the two orphaned measures.

Shows up in. `RPT06 Requirements Traceability`, which reports traceability to a building block and says nothing about traceability to an objective — because there is nothing to report.

Goes wrong when. The diagnostic becomes a complaint. It is a work list: two sheets to create and two measures to attach.

Look it up. Volume 1, chapter 17, *The practical discipline*.

Step 12.2 — Create the two relationship sheets the spine needs

Produce. Two new relationship sheets, named after their meaning, populated and resolved.

You need. Step 12.1's diagnostic; `O01`, `O02`, `O03`, `O06`; the goal owners in `O02`.

Where it lands.

> **Two sheets you must create — `R26 Goal addresses Driver` and `R27 Requirement realises Object`**
>
> `R26 Goal addresses Driver`: columns `Goal`, `Goal name`, `Driver`, `Driver name`, `Strength`, on the file's convention, headers row 4, data row 5. A goal may address several drivers and a driver may be addressed by several goals, so this is many-to-many and there is no diagonal to fall back on. `R27 Requirement realises Object`: columns `Requirement`, `Requirement name`, `Objective`, `Objective name`, `Contribution`. The second name is **cut at Excel's 31 characters** — the full phrase is *Requirement realises Objective* — which is the same truncation the file already carries on twenty of its own tabs, so it reads as native rather than as an error.

How.

1. Build `R26` first; it is the shorter argument. Twelve drivers, six goals. Expect `GOL-01 Protect the licence to operate` to address all six `Regulatory` drivers.
2. Use `Strength` to record `Primary` or `Contributing`, so a goal addressing six drivers is not read as six equal commitments.
3. Build `R27` from the requirement side. Each of the 31 requirements names an objective or is recorded as contributing to none, which is a finding and not a blank.
4. Check the arithmetic both ways. A driver addressed by no goal is a pressure nobody accepted; an objective realised by no requirement is an intention nothing will deliver.
5. Expect the technology requirements to fail. `REQ-027` to `REQ-031` all derive from `DRV-09 Legacy estate` and match no objective in `O03` at all.

Signed by. `GB-01 Architecture Board`, with each goal's owner from `O02` confirming their own rows.

Done when. `R26` and `R27` exist, every driver and every objective appears in at least one row or is listed as deliberately unattached, and the counts are in your pack.

Shows up in. Nothing, until somebody builds a report over them. Say so, and put the two counts in the monthly pack beside `RPT06`'s.

Goes wrong when. `R27` is built as a diagonal to save a workshop. The file already has thirteen positional diagonals; a fourteenth is not a traceability model.

Step 12.3 — Repair `R09`, and stop `RPT06` reporting a pattern as an assessment

Produce. A corrected `R09 Requirement satisfied by Bu`, with `Traced` assessed row by row, and a note for every reader of `RPT06`.

You need. `R09`; `O55 Building Block` and its 37 refs; `O35`, `O44`, `O45`, `O46`; `RPT06`.

Where it lands.

> **`R09 Requirement satisfied by Bu — Requirement, Requirement name, Building block, Block name, Traced`**
>
> `R09` pairs `REQ-001` to `BBB-01`, `REQ-002` to `BBB-02`, down to `REQ-010` to `BBB-10`; then `REQ-011` to `DBB-01` through `REQ-020` to `DBB-10`; then `REQ-021` to `ABB-01` through `REQ-030` to `ABB-10`; then `REQ-031` to `TBB-01`. It is a positional walk down the 37 refs of `O55`, in `O55`'s order, not a satisfaction model.

Two defects, and the second is the one to take to the board.

The pairing is positional, and the right answer is sitting on the sheet. `REQ-011 Batch / Lot mastered once` is satisfied by `DBB-01 Product mastering`. `DBB-05` is called *Batch / Lot mastering* and is four rows below. Every data requirement is off by four, and `REQ-001 Evidence ISO 9001` — a regulatory requirement — is satisfied by `BBB-01 Product Development capability uplift`.

And `Traced` is generated, not assessed. It reads `No` on `REQ-001`, `REQ-005`, `REQ-009`, `REQ-013`, `REQ-017`, `REQ-021`, `REQ-025` and `REQ-029` — **every fourth row, starting at the first**, eight of thirty-one. No assessment produces that. `RPT06 Requirements Traceability` copies the column into `Reaches a block` row for row, including the mechanical pattern, and footers *A requirement that reaches no building block is a requirement nothing will deliver* — which is true, and is not what the column measures.

How.

1. Re-point the data requirements first. `REQ-007` to `REQ-014` map to `DBB-01` to `DBB-08` by entity name, which the sheet already carries in `Block name`.
2. Re-point the regulatory requirements. `REQ-001` to `REQ-006` have no building block that evidences an obligation, so record them as unsatisfied rather than pairing them with a capability uplift.
3. Clear the `Traced` column entirely and re-assess it. Assessing 31 rows takes an afternoon.
4. Annotate `RPT06` where it is published, naming the pattern, until `R09` is repaired.
5. Re-count. The number of requirements reaching a block will fall, and that fall is the finding.

Signed by. `GB-05 Architecture Function` assesses; `GB-01 Architecture Board` receives the corrected count.

Done when. No four-row periodicity remains in `Traced`, and every `No` has a reason recorded beside it.

Shows up in. `RPT06`, whose `Reaches a block` column becomes an assessment instead of a copy. Expect the `Yes` count to drop from 23.

Goes wrong when. The diagonal is kept because it looks complete. A business service then comes to be satisfied by whichever block shares a row number, which is the same defect that gives this file `ISO 9001` evidenced by an identity lifecycle control.

Look it up. Volume 1, chapter 20, *Map 4 — System class to what it masters*, for the pairing the data requirements should carry.

Step 12.4 — Give every requirement an acceptance criterion and a verification method

Produce. Two new columns on `O06 Requirement`, filled for all 31 rows.

You need. `O06`; `O55`'s `Acceptance criteria` column; `O58 Governance Gate`; `O59 Compliance Assessment`.

Where it lands.

> **`O06 Requirement` — add `Acceptance criterion`, `Verification`, `Objective`**
>
> `O06` has seven columns: `Ref`, `Requirement`, `Statement`, `Category`, `Priority`, `Driver`, `Status`. There is no acceptance criterion, no verification method and no objective link. Add `Acceptance criterion` (what a gate would test), `Verification` (`Inspection`, `Test`, `Demonstration`, `Record`) and `Objective` (an `OBJ-` ref, mirroring `R27`). `Statement` stays as written; a criterion is not a restatement of it.

Read the statements before you write criteria, because twenty-six of the thirty-one are template instances. `REQ-001` to `REQ-006` all read *"The architecture must produce evidence that X is met, without manual reconciliation."* `REQ-007` to `REQ-014` all read *"X is mastered in one system and consumed elsewhere through a published interface."* `REQ-015` to `REQ-020` all read *"Every interface to or from X is managed, published and logged."* `REQ-021` to `REQ-026` all read *"X exposes measures that the executive can act on."* The template is not wrong, but it is not testable either.

How.

1. Write a criterion that names a number, a record or a named artefact. *Managed, published and logged* becomes *appears in `R06` with `Managed` and `Logged` populated*.
2. Fix the measurable set. *Exposes measures the executive can act on* is untestable until the measure definition register in step 12.5 exists, so write the criterion against the register.
3. Choose a verification method per row and accept that `Record` is the honest answer for most regulatory requirements.
4. Fill `Objective` and keep it identical to `R27`. Two places, one answer, checked at every baseline.
5. Hand the criteria to Phase G. A criterion that never appears in `O59` is a criterion nobody tested.

Signed by. The requirement's own `Driver` owner — Chief Compliance Officer for the six regulatory, `Group Executive` for the rest.

Done when. All 31 rows carry a criterion naming something checkable, a verification method and an objective ref.

Shows up in. `RPT06`, and then `O59` in Phase G, where the criteria become verdicts.

Goes wrong when. The criterion restates the requirement in different words. You then have two statements, neither testable, and a gate that cannot refuse anything.

Look it up. Volume 1, chapter 17, *Requirements realization* — and note there is no Constraint element in ArchiMate 4, so a constraint is expressed as a Requirement.

Form R-1 — Requirement

Feeds `006 Requirement`. `Acceptance criterion` is a column you must add; so are `Verification` and `Objective`. `Statement` and `Status` go in straight to Excel. A first pass over one domain needs eight to twelve rows. Signed by the owner of the requirement's driver in `001`.

Ref	Requirement	Category	Priority	Driver	Acceptance criterion
REQ-032	EXAMPLE — delete this row	Regulatory	Must	DRV-13	80% valid hourly values per period

BEFORE YOU BUILD THE MEASURE REGISTER

Pick one measure. What is its denominator, and who chose it?

Which of your plants would compute it differently, and by how much?

What target has the objective owner actually agreed to?

Step 12.5 — Build the measure definition register against `O04`

Produce. A measure definition register: one row per measure, with the columns `O04 Measure` does not have.

You need. `O04`; `R10`; `O03`; the plant's reporting owners; volume 1 chapters 3, 7 and 10.

Where it lands.

> **`O04 Measure` — add `Definition`, `Denominator convention`, `Target`, `Owner`, `Source systems`, `Comparability scope`**
>
> `O04` has six columns: `Ref`, `Measure`, `Unit`, `Domain`, `Baseline`, `Cadence`. It carries a `Baseline` and a `Cadence` and **no target and no owner**, so sixteen measures are reported on a cadence with nothing to report against and nobody to ask. `Comparability scope` takes `Within a line`, `Within a plant`, `Across plants` or `Not comparable`, and it is the column everyone forgets.

In a manufacturer this register is not a tidy-up. **It is the first deliverable, before any dashboard**, and the reason is arithmetic rather than preference.

There is no OEE standard. The site publishing the most-used definition says so itself. The numerator is uncontested and the denominator is a choice: the research behind volume 1 found **seven availability conventions in circulation**, so two physically identical plants can report availability thirty points apart with neither of them lying. The seventh is the subtle one — downtime shorter than a company-set threshold, typically one to five minutes, is conventionally counted under performance rather than availability, so **two plants with identical behaviour and different micro-stop thresholds report different availability and different performance with the same OEE.** And **OEE is explicitly documented as unsuitable for comparing**

machines, for benchmarking, and for aggregating to department or plant level, because a line is a serial system whose output is governed by its constraint.

How.

1. Add the six columns and fill `Owner` first. A measure with no owner cannot be defended, and ten of `O04`'s sixteen are plant or data measures with nobody named.
2. Write `Definition` as a formula with its denominator stated, not as a phrase. `MSR-07 Components within vendor support` is computable from `O27`'s `Support end`; most of the rest are not.
3. Set `Comparability scope` honestly, and mark anything built on an availability denominator `Not comparable` until the micro-stop threshold is harmonised across the three plants.
4. Set targets with the objective owner from `O03`, not with the reporting team. `MSR-01` has a `Baseline` of 58 and `OBJ-01` says every obligation has a named control and evidence — which is a target of 100, and nobody has written it down.
5. Attach `MSR-15` and `MSR-16` to objectives, since `R10`'s diagonal leaves both measuring nothing.
6. Never publish a benchmark figure you cannot source. Targets are set against this plant's own baseline.

Signed by. The objective owner in `O03` for the target; the named `Owner` for the definition.

Done when. All 16 rows carry a definition, a denominator convention, a target, an owner, a source and a comparability scope, and no two plants compute the same measure differently without the register saying so.

Shows up in. `RPT01 Dashboard`, which currently reports counts rather than measures. The register is what lets a measure appear in a board pack without being argued about.

Goes wrong when. The dashboard ships first. The first time two plants' numbers meet in a board pack somebody finds the definitions differ, and the dashboard loses its credibility in one meeting and never recovers it.

Look it up. Volume 1, chapter 7, *Why OEE is not comparable, stated hard*; chapter 7, *So the first deliverable is a measure definition register*; chapter 10, *The measure definition register*; chapter 3, *OEE, and the three traps inside it*.

Form R-2 — Measure definition

Feeds `O04 Measure`. `Target` and `Comparability scope` are columns you must add, along with `Definition`, `Denominator convention`, `Owner` and `Source systems`, which go in straight to Excel because they do not fit six columns. Sixteen rows covers the whole sheet. Signed by the objective owner in `O03`.

Ref	Measure	Unit	Baseline	Target	Comparability scope
MSR-01	EXAMPLE — delete this row	%	58	100	Across plants

Step 12.6 — Re-express the constraints as requirements

Produce. Eight constraints restated as requirements where they bind, with the ones that are really assumptions moved.

You need. `O07 Constraint`; `O06`; `O08 Assumption`; `O52 Implementation Factor`.

Where it lands.

> **`O07 Constraint` and `O06 Requirement`**
>
> `O07` has four columns: `Ref`, `Constraint`, `Description`, `Type`. It is a sheet in the repository, but **ArchiMate 4 has no Constraint element** — a constraint is expressed as a Requirement, so any view you draw has to carry these as requirements anyway. Keep the sheet, because `CON-` refs are cited elsewhere, and add an `Expressed as` column holding the `REQ-` ref that carries it into the model.

Read the eight. `CON-01 Flat technology budget`, `CON-02 Delivery capacity unrecorded`, `CON-03 Legacy platform skills scarce`, `CON-04 Regulatory change in flight`, `CON-05 Data residency`, `CON-06 Change freeze windows`, `CON-07 Vendor roadmap dependency`, `CON-08 Union and works council consultation`. Four of them are genuine constraints that bind a design: residency, freeze windows, consultation and the vendor roadmap. Two are

measurements nobody has taken — `CON-02` says concurrent delivery capacity is not known, which is a task, not a constraint. And `CON-04` is an assumption wearing a constraint's clothes; `O08`'s `ASM-05 Regulatory interpretation remains stable` already says the same thing, `Status` = `Open`.

How.

1. For each of the four real constraints, write the requirement it implies and record the `REQ-` ref in `Expressed as`.
2. Move `CON-02` to a work item and `CON-04` to `O08`, logging both in `02 Change Log`.
3. Cross-check `O52 Implementation Factor`, which carries the same ideas again as `IF-03 Peak trading freeze`, `IF-08 Site access windows` and `IF-06 Concurrent capacity unknown`. Three sheets, one fact, no join between them.
4. Decide which sheet is authoritative and say so. `O07` is the motivation record; `O52` is the planning deduction.

Signed by. `GB-01 Architecture Board`.

Done when. Every row of `O07` either names a `REQ-` ref in `Expressed as` or has been moved, and the duplication with `O52` is documented.

Shows up in. `RPT06` once the new requirements appear, and in Phase F planning once `O52` stops disagreeing with `O07`.

Goes wrong when. The constraints are left as a list. They then bind nothing, and the first roadmap draft schedules work into a freeze window that `CON-06` already prohibited.

Step 12.7 — Rebuild `O63 Control Objective` for the jurisdiction the plant is in

Produce. A replacement `O63` whose obligations are the instruments that inspect this plant, each with a testable control objective and a named evidence source.

You need. `O63`; `R16`; chapter 10's inspection register `O70`; `O01 Driver`; volume 1 chapter 16 and chapter 22's South Africa section.

Where it lands.

> **`O63 Control Objective` — `Ref`, `Obligation`, `Source`, `Control objective`, `Evidence`, plus `Gate` to add**
>
> Replace the rows; keep the columns and add `Gate`, taking a `GT-` ref, so an obligation says where it is tested. Continue the `CO-` space rather than renumbering, so `R16`'s existing refs remain resolvable while you re-point them. Then add the matching `DRV-` rows to `O01`, because each new obligation is a regulatory driver and `R08` expects one.

What is there now is written for a European manufacturer: `ISO 9001`, `ISO 14001 / 45001`, `Product safety and CE / UKCA marking`, `REACH / RoHS`, `Customs and origin rules`, `Extended producer responsibility`. **CE and UKCA marking have no application in South**

Africa. `Control objective` is a tautology on all six rows — *"Control that evidences ISO 9001."* And five of six `Evidence` cells read `No mechanism` or `In breach`, which `ZZ Checks` and `RPT09 Regulatory Coverage` both report correctly.

The domestic instruments that belong on the sheet:

Obligation	Instrument	What the control objective must produce
Occupational health and safety	OHSA and its regulations	Appointments, incident, exposure and equipment registers with per-class clocks
Atmospheric emissions	Air Quality Act, atmospheric emission licence	The standard applying to this plant now, and monitoring that meets the availability test
Water use	National Water Act, water use licence	Which uses are exercised, with abstraction, discharge and effluent records
Product conformity	NRCS compulsory specification and letter of authority	A register at product-and-version grain, bound to the item revision
Measurement	Legal Metrology Act	An instrument register with a dated instrument-to-purpose assignment
Carbon	Carbon tax	Activity data with source measurements, emission factors with versions
Personal information	POPIA	A lawful basis for biometric processing, and deletion evidence
Sector scheme	Whichever applies — food, pharmaceutical, automotive, chemicals	Whatever that scheme compels, marked `basis = contractual` where it is not statutory

How.

1. Replace the obligations with the instruments that inspect your plants, taking the list from `O70` rather than a template.
2. Keep customs, which is real here, and keep the two export regimes as export-conditional rows rather than deleting them.
3. Write every `Control objective` as mechanism plus record plus gate. No sentence may contain the obligation's own name. Fill `Gate` — mostly `GT-2`, with the design-time ones at `GT-1`.
4. Add a `DRV-` row per new obligation and an `R08` row per new requirement, so the spine stays whole.
5. Re-point `R16`, which pairs `CO-01` to `SEC-01 Identity lifecycle` and walks the diagonal. All six instances are unusable.

Signed by. Chief Compliance Officer, who already owns all six existing regulatory drivers in `O01`, with the plant safety, health and environment lead per site.

Done when. Every row names a domestic instrument or an export-conditional one, no `Control objective` cell contains the obligation's name, and every row has a `Gate`.

Shows up in. `RPT09 Regulatory Coverage`, which reprints `O63` column for column. The report is only as good as the sheet, which here is the good news.

Goes wrong when. The European rows are deleted and nothing replaces them. The sheet then reports full coverage of zero obligations, which is worse than reporting a breach.

Look it up. Volume 1, chapter 16, *Compulsory specifications, letters of authority, and section 20*; chapter 16, *The Act, and why the regulations are the architecture*; chapter 22, *South Africa*.

BEFORE YOU REBUILD `O63`

Which of the six existing obligations survives, and on what basis?

Which instrument inspects your plant that is on no sheet in the file?

For each new obligation, who in `O01` will own the driver?

Step 12.8 — Work one obligation end to end

Produce. One obligation taken through driver, control objective, requirement, acceptance criterion and gate, with every ref written down, as the pattern for the rest.

You need. The rebuilt `O63`; `O01`; `O06`; `O44 Data Building Block`; `O58`; `O59`; `O04`.

Where it lands.

> **`Seven sheets, one chain`**
>
> `O01` for the driver, `O63` for the obligation, `R08` for the derivation, `O06` for the requirement and its new `Acceptance criterion`, `O44` for the data building block that carries it, `O58` for the gate, `O59` for the verdict. Write the refs as you go; a chain with one unwritten ref is not a chain.

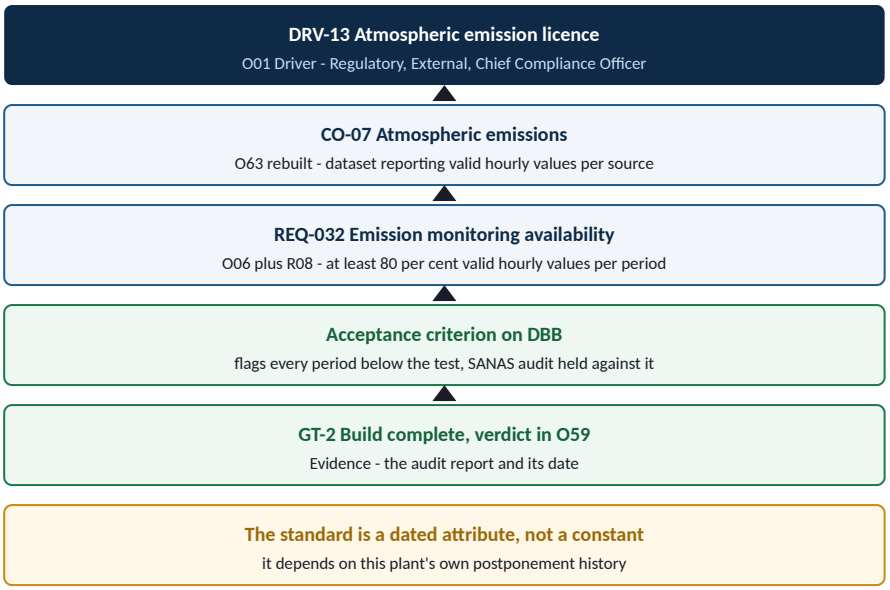


Figure 12.2 One obligation, end to end, with every reference written down. Six links, seven sheets. The bottom band is the detail that makes this a dated attribute rather than a constant, and it is the one most often hard-coded.

Take the atmospheric emission licence, because it is the cleanest testable obligation a South African plant carries.

- **Driver.** `DRV-13 Atmospheric emission licence`, `Type` = `Regulatory`, `Origin` = `External`, `Owner` = Chief Compliance Officer, `Pressure` = `High`.
- **Obligation.** `CO-07 Atmospheric emissions`, `Source` = the Air Quality Act and the licence conditions, `Gate` = `GT-2`. The control objective: *a monitoring dataset that reports valid hourly values per source per averaging period, retained with its audit record.*
- **Requirement.** `REQ-032 Emission monitoring availability`, `Category` = `Regulatory`, `Priority` = `Must`, `Driver` = `DRV-13`, with an `R08` row to match. Statement: *the emissions dataset reports at least 80% valid hourly values for each averaging period, computed from the historian.*
- **Acceptance criterion.** *For any licence period, the dataset reports the valid-hourly-value percentage per source and flags every period below 80%, and the SANAS-accredited audit at least every two years is held against it.*
- **Gate and verdict.** `GT-2 Build complete`, with an `O59` row naming the audit report and its date as `Evidence`.
- **Measure.** `MSR-17 Valid hourly values`, `%`, `Domain` = `Compliance`, with a target and an owner from the start, which is more than any of `O04`'s existing sixteen rows has.

Two details decide whether this works. **The standard that applies to this plant is a function of its own postponement history**, not of a published date, so it is a dated attribute and not a constant. And this is a historian problem wearing a reporting problem's clothes: the availability test, bad-quality intervals and the averaging rules are time-series concerns, and an estate that builds emissions reporting on a relational extract of hourly values cannot reproduce last year's numbers. In this file that lands on `A04 SCADA / Historian`, the only `Critical` application, whose `Disposition` is `Replace` with nothing named to replace it.

How.

1. Write the chain for this obligation first, refs and all, and keep it to one page.
2. Check that each ref resolves in both directions before you move on.
3. Repeat it for the obligation with the longest retention clock, which will be the occupational health file, and then for the one with the shortest notice period.
4. Publish the three chains as the pattern, and require it of every new obligation.

Signed by. Chief Compliance Officer, with the data owner for the monitoring dataset.

Done when. Three obligations have a complete written chain, and no link in any of them is unwritten.

Shows up in. `RPT09 Regulatory Coverage` and `RPT06 Requirements Traceability` together, which is the first time in this repository those two reports describe the same thing.

Goes wrong when. The chain is drawn as a diagram and never written as refs. A diagram cannot be joined, and the next architect rebuilds it.

Look it up. Volume 1, chapter 16, *Air quality — and the deadline that is not final*; chapter 10, *Time-series and tag data, which no other industry in this series has*.

Form R-3 — Obligation to evidence

Feeds the rebuilt `063 Control Objective`. `Gate` is a column you must add. One row per obligation; eight to twelve covers a single-site manufacturer, more where sector schemes apply. `Control objective` may not contain the obligation's own name. Signed by the Chief Compliance Officer.

Ref	Obligation	Source	Control objective	Evidence	Gate
CO-07	EXAMPLE — delete this row	Emission licence conditions	Dataset reporting valid hourly values per source	Historian plus audit report	GT-2

Step 12.9 — Set priority and status so they mean something

Produce. `Priority` and `Status` on `O06` defensible row by row, with the generated pattern removed.

You need. `O06`; `R27` from step 12.2; `O30 Work Package`; `O60 Governance Decision`.

Where it lands.

> **`O06 Requirement — Priority and Status`**
>
> `Priority` currently runs `Must` on `REQ-001` to `REQ-014`, `Should` on `REQ-015` to `REQ-026` and `Could` on `REQ-027` to `REQ-031` — three contiguous blocks, which is a category mapping rather than a prioritisation. `Status` runs `Approved` on the first twenty and `Proposed` on the last eleven, breaking exactly where `Category` changes from `Integration` to `Business`.

Neither column was assessed. Both patterns follow the `Category` column precisely, which tells you they were derived from it. That matters because `Must` is the word a project manager uses to refuse descoping, and twenty `Approved` rows imply a body approved them — and there is no decision in `O60` that did.

How.

1. Re-assess `Priority` against the objective in `R27`. A requirement realising no objective is not a `Must`, whatever its category.
2. Re-assess `Status`, and write `Proposed` wherever no `O60` decision approved it. On this file that is most of the sheet.
3. Check the five technology requirements, `REQ-027` to `REQ-031`, which read `Could` and `Proposed` and include `REQ-027 Adopt OT/IT segmentation` — the one requirement on the sheet that a plant network depends on, and the only one that touches the IT/OT boundary.
4. Log every change at field grain in `02 Change Log`, because this is the step most likely to be argued about later.

Signed by. `GB-01 Architecture Board`, with each driver owner for their own rows.

Done when. Neither column follows `Category` in contiguous blocks, and every `Approved` row names a `DEC-` ref.

Shows up in. `RPT06`, whose `Priority` column is reprinted from `O06` and currently inherits the blocks.

Goes wrong when. `Must` is left on all fourteen data and regulatory requirements. Everything is mandatory, so nothing is, and the first budget cut is made by somebody who does not read the sheet.

Look it up. Volume 1, chapter 13, *The Purdue model as it is actually used* — `REQ-027` is the only requirement on the sheet that a plant network depends on, and it reads `Could`.

Step 12.10 — Age the assumptions and close them

Produce. `O08 Assumption` with an owner, a date and a closure route on every row.

You need. `O08`; `O62 Delivery Risk`; `O60`; `03 Change Requests`.

Where it lands.

> **`O08 Assumption` — add `Owner`, `Raised`, `Test`, `Due`, `Closed`**
>
> `O08` has three columns only: `Ref`, `Assumption`, `Status`. Six rows, four `Open` and two `At risk` — `ASM-03 Vendor selections complete before the dependent phase` and `ASM-04 Delivery capacity can be staffed to plan`. No owner, no date, no test. An assumption with no test is a hope.

How.

1. Add the five columns and fill `Owner` with a person. Match them to `O62` where the same thing is already a risk: `ASM-03` is `RSK-01`, `ASM-04` is `RSK-03`.
2. Write `Test` as the thing that would prove or disprove it. For `ASM-02 Data owners can be designated within two quarters`, the test is `DEC-04 Designate data owners`, which is `Outstanding`.
3. Set `Due` from the earliest `O30` start the assumption underpins.
4. Close an assumption by converting it to a fact with a reference, or to a risk with an owner. Never by deleting the row.
5. Review the six monthly. Six rows is five minutes, and `At risk` for a year is a decision nobody took.

Signed by. `GB-01 Architecture Board`.

Done when. Every row has an owner, a test and a due date, and no row has been `Open` for two reviews without a note.

Shows up in. Nothing reads `O08`. Put the two `At risk` rows in the monthly pack beside their risks in `O62`.

Goes wrong when. The assumptions are written at the start of the programme and never read again. Then `ASM-01 Executive sponsorship holds for the programme duration` fails quietly, and the first sign is a budget conversation.

Step 12.11 — Baseline the requirements and route a requirement change

Produce. A requirements baseline, and the route a change to a requirement takes afterwards.

You need. `04 Baselines` and `03 Change Requests` from chapter 11; `O06`; `R08`, `R09`, `R26`, `R27`.

Where it lands.

> **`04 Baselines` with `Scope` = the motivation domain**
>
> Cut a baseline covering `O01` to `O08`, `O63`, `R08`, `R09`, `R10`, `R11`, `R26` and `R27`. `Record count` is the counted total of those sheets, not the whole file. `Decision ref` points at the `DEC-` row that approved the catalogue — which does not exist yet, so raise it.

How.

1. Cut the baseline only once steps 12.2 to 12.9 are done. A baseline over a generated `Traced` column preserves the defect.
2. Raise the decision that approves the requirements catalogue. There is no such row in `O60`, and twenty requirements currently read `Approved` without one.
3. Set the route: a change to a requirement is a `03 Change Requests` row against `O06`, classified `Clarify`, `Extend` or `Re-architect`, and a `Re-architect` on a `Must` goes to the Investment Committee.
4. Make the rule that a requirement cannot change without its acceptance criterion changing with it, in the same change request.
5. Re-baseline at each `O66 Architecture Increment`, six of which are defined, landing between month 10 and month 56.

Signed by. `GB-01 Architecture Board` for the baseline; `GB-02 Investment Committee` where a `Must` moves.

Done when. The baseline exists with a counted total and a decision ref, and one requirement change has been through the route.

Shows up in. `RPT06`, which can now be read against a baseline rather than against yesterday.

Goes wrong when. The catalogue is baselined early to show progress. Every later correction then looks like scope creep, and the corrections stop.

Step 12.12 — Run the monthly requirements review

Produce. A one-page monthly review, computed, with five numbers and one list.

You need. `RPT06`; `O06`; `R27`; `R09`; `O04`; `O08`; `03 Change Requests`.

Where it lands.

> **`RPT06 Requirements Traceability, read with its defects known`**
>
> `RPT06` reprints `O06`'s `Ref`, `Name`, `Category`, `Priority` and `Driver`, adds `Reaches a block` copied from `R09`'s `Traced`, and footers one typed sentence. It computes nothing. Until `R09` is repaired, read its `Reaches a block` column as a four-row pattern and say so on the page.

The five numbers: requirements with no objective in `R27`; requirements with no building block in a repaired `R09`; requirements whose acceptance criterion is blank; measures with no target or no owner; assumptions `Open` for two reviews. The one list: every `Must` whose criterion has never appeared in `O59`.

How.

1. Compute all five. Each is a count over a column, so none of them needs judgement.
2. Publish them in the same order monthly, with last month's figure beside this month's.
3. Take the `Must` list to `GB-01 Architecture Board` by ref, not by count.
4. Re-run the spine check from step 12.1 quarterly. A new requirement added without an `R08` and an `R27` row breaks the chain silently.

Signed by. `GB-05 Architecture Function` publishes; `GB-01 Architecture Board` receives.

Done when. The five numbers have moved in the right direction for two consecutive months.

Shows up in. `RPT06` and `RPT09`, and in `O59` as the criteria start producing verdicts.

Goes wrong when. The review reports the requirement count. It goes up every month, which looks like progress and measures nothing.

What good looks like

A reader picks any requirement and walks it both ways without a gap: up through `R08` to a driver and through `R26` to a goal, down through `R27` to an objective and `R10` to a measure with a target, and sideways through `R09` to a building block whose acceptance criterion has a verdict in `O59`. That walk is currently impossible in five places. Closing them is the whole of Requirements Management.

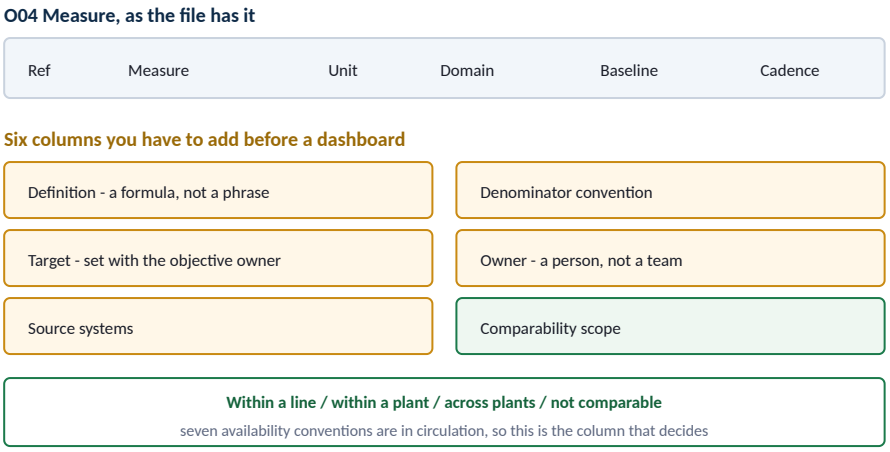


Figure 12.3 The measure definition register against `004 Measure`. The top row is what the sheet has. The six below are what it needs before any dashboard is built, and the green one is the column everyone forgets.

THE REVIEWER'S ASK — ANSWER THESE BEFORE YOU CALL REQUIREMENTS MANAGEMENT DONE

Pick a requirement at random. Walk it to a measure with a target. How many hops failed?

Which measure in `004` would two of your plants compute differently, and does the register say so?

Name the obligation on `063` that an inspector would actually arrive to inspect.

How many of the 31 requirements have an acceptance criterion a gate could refuse?

THE CHAPTER IN TEN LINES

- O01 holds 12 drivers, O02 6 goals, O03 14 objectives, O04 16 measures, O06 31 requirements, O07 8 constraints, O08 6 assumptions.
- R08 Requirement derives from Driver carries 31 instances and is the healthiest join in the file.
- Nothing joins a driver to a goal, and nothing joins a requirement to an objective. Create R26 Goal addresses Driver and R27 Requirement realises Object.
- R27's full name is cut at Excel's 31 characters, like twenty of the file's own tabs.
- R09 Requirement satisfied by Building Block is a positional walk down the 37 refs of O55, so every data requirement is off by four rows.
- R09's Traced column reads No on every fourth row starting at the first, eight of thirty-one. That is generated, not assessed.
- RPT06 Requirements Traceability copies that column row for row and computes nothing.
- R10 is a diagonal, which leaves MSR-15 and MSR-16 measuring nothing.
- O04 Measure has a Baseline and a Cadence but no target and no owner.
- There is no OEE standard, seven availability conventions are in circulation, and the micro-stop threshold moves loss between availability and performance without changing OEE.
- OEE is explicitly unsuitable for comparing machines, for benchmarking and for aggregating to plant level. So the measure definition register comes before the dashboard.
- ArchiMate 4 has no Constraint element, so every row of O07 has to be expressed as a Requirement.
- O63 Control Objective is written for a European manufacturer, and CE and UKCA marking have no application in South Africa. Rebuild it on the domestic instruments.
- Priority and Status on O06 both follow the Category column in contiguous blocks, which means neither was assessed.

SOURCES FOR THIS CHAPTER

- **1.** Manufacturing_TOGAF_Metamodel.xlsx, sheets O01 Driver to O08 Assumption
- **2.** Manufacturing_TOGAF_Metamodel.xlsx, sheets R08, R09, R10, R11 and RPT06 Requirements Traceability
- **3.** Manufacturing_TOGAF_Metamodel.xlsx, sheets O63 Control Objective, R16 and RPT09 Regulatory Coverage
- **4.** The Open Group, TOGAF Standard — ADM Requirements Management
  https://pubs.opengroup.org/architecture/togaf-standard/adm/chap13.html

APPENDIX A

The sheet-by-sheet reference

Every sheet, every column, every row count and every defect, read off the file.

This is the appendix you keep open beside the workbook. Every sheet in `Manufacturing_TOGAF_Metamodel.xlsx`, with its exact tab name, its reference prefix, its row count, its columns in sheet order, what it is for, which ADM phase fills it, and — where there is one — the defect in it [1]. Everything here was read off the file on 16 October 2026. Where this appendix and the file disagree, the file is right; re-read it.

How to read the tables

The file holds **105 sheets**: `00 Index`, **64 object sheets** numbered `O01` to `O66` with `O34` and `O51` absent, `EXT Industry Extension`, **25 relationship sheets** `R01` to `R25`, **13 report sheets** `RPT01` to `RPT13`, and `ZZ Checks`. A loop from `O01` to `O66` fails on the two holes.

Every sheet has the same layout: title in row 1, subtitle in row 2, **headers in row 4, data from row 5**. Row 3 is empty. The first column is always `Ref`, except on relationship sheets, where only `R06` has one. Read with `openpyxl` and set `min_row=5`. Sheets are grouped as `00 Index` groups them; its `Realization` domain appears below as *implementation*. Kestrel Industrial Group, whose data fills the file, is an invention, as is Karoo Industrial Group in volume 1.

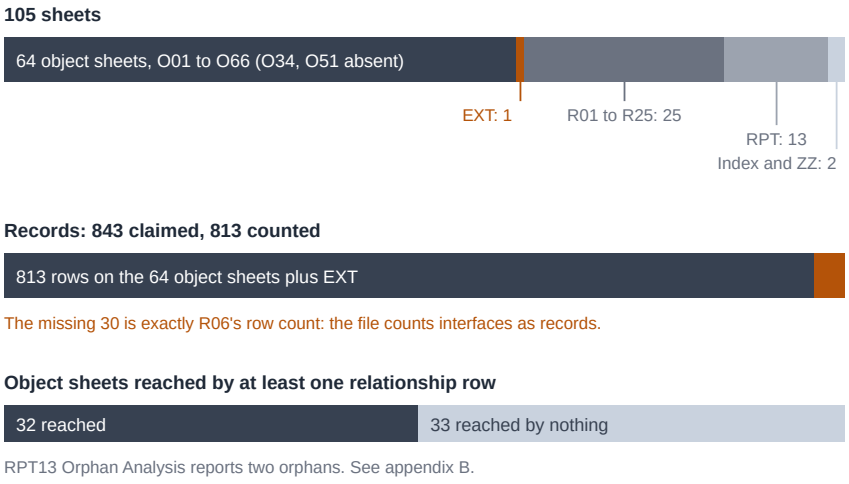


Figure 13.1 The shape of the repository, and the two places its own figures do not reconcile. **843 against 813** is exactly `R06`'s thirty rows; **32 against 33** is the reachability gap appendix B works out.

Motivation — `001` to `008`

Tab	Prefix	Rows	Columns, in sheet order	Phase
`001 Driver`	`DRV-`	12	Ref, Driver, Type, Description, Origin, Owner, Pressure	Prelim, A
`002 Goal`	`GOL-`	6	Ref, Goal, Description, Owner, Status	A
`003 Objective`	`OBJ-`	14	Ref, Objective, Goal, Horizon, Status	A
`004 Measure`	`MSR-`	16	Ref, Measure, Unit, Domain, Baseline, Cadence	A, RM
`005 Principle`	`PRN-`	24	Ref, Principle, Category, Statement, Owner, Status	Prelim
`006 Requirement`	`REQ-001`	31	Ref, Requirement, Statement, Category, Priority, Driver, Status	RM
`007 Constraint`	`CON-`	8	Ref, Constraint, Description, Type	A
`008 Assumption`	`ASM-`	6	Ref, Assumption, Status	A

Sheet	What it is for	The defect in it
`001`	The pressures behind the architecture	Six of twelve are `063` 's European obligations. `DRV-08` , `DRV-11` , `DRV-12` reach no requirement
`002`	Six enterprise goals	`GOL-03` 's owner, `Chief Commercial Officer` , is in neither `010` nor `011`
`003`	Statements under a goal	`Horizon` is a duration, not a date, so nothing can be late
`004`	What objectives are tested on	No definition, formula, denominator or owner column; no OEE, yield or on-time measure. `MSR-15` and `MSR-16` measure no objective
`005`	The 24 principles	`Owner` is `Architecture Board` on all 24; `Draft` on every fifth row; no `Tier` , `Rationale` , `Implications` , `Approved` or `Review date`
`006`	The requirements register	Statements generated from the driver name; three-digit reference
`007`	Eight limits	No owner, no expiry, so nothing can be retired
`008`	Six assumptions with a status	No owner, no test date

`005 Principle` cannot be tested at a gate: a principle with no tier, no implications and no approval date is a slogan. It also contradicts its own `PRN-13 Data Ownership is Individual` — *"An owner is a person, not a committee"* — on all 24 of its own rows.

BEFORE YOU TOUCH THE MOTIVATION SHEETS, ANSWER THESE

Which of `001`'s twelve drivers would survive a conversation with this plant's regulator, and which are imported?

`005` has no `Review date`. What date are you putting on its twenty-four rows, and who signs it?

Organisation and business — `009` to `019`, `035`

Tab	Prefix	Rows	Columns, in sheet order	Phase
`009 Organization Unit`	`ORG-`	12	Ref, Organisation unit, Type, Head, Headcount	B
`010 Actor`	`ACT-`	18	Ref, Actor, Type, Status	B
`011 Role`	`ROL-`	18	Ref, Role, Type, Performed by	B
`012 Location`	`LOC-`	8	Ref, Location, Type, Scale	B
`013 Business Capability`	`C01` dotted	100	Ref, Capability, Level, Parent, Description, Criticality, Current, Target, Disposition	B
`014 Value Stream`	`VS01`	5	Ref, Value stream, Scope, Stages, Capabilities	B
`015 Business Service`	`BSV-`	10	Ref, Business service, Capability, Description, Exposure, Criticality	B
`016 Business Process`	`BPR-`	20	Ref, Business process, Value stream, Description, Automation	B
`017 Business Function`	`BFN-`	10	Ref, Business function, Capability, Description	B
`018 Product`	`PRD-`	4	Ref, Product, Type, Description	B
`019 Contract`	`CTR-`	10	Ref, Contract, Counterparty, Type, Expiry	B
`035 Business Building Block`	`BBB-`	10	Ref, Business building block, Capability, What it must do, Acceptance criteria	E

Sheet	What it is for	The defect in it
`009`	The org chart	`ORG-11` and `ORG-12` own no capability. `Internal Audit` carries 212 people against `Operations` at 102
`010`	Who performs roles	The same eighteen names as `011`, all corporate, all `Filled`
`011`	The named roles	The same eighteen names as `010`. **No plant role at all**
`012`	Sites and plants	`LOC-08 Cloud Region` has 261 people. No `Line count`, `Shift pattern`, `Format` or `Ownership` column
`013`	A three-level capability map	A generated cross-product; `Disposition` is arithmetic; criticality does not inherit; the 90 sub-rows appear in no report
`014`	Five streams	`VS01` and `VS04` carry an identical capability list, as do `VS02` and `VS05`; `Stages` is a count
`015`	One per level-1 capability	Every description reads "Externally consumable" while `Exposure` is `Internal` on nine of ten
`016`	Four per value stream	Named "stage 1" to "stage 4", four per stream against `014`'s declared 4, 5, 6, 4, 5
`017`	One per capability	The same `Description` sentence on all ten rows
`018`	Four product lines	None found
`019`	Vendor agreements	`Counterparty` holds the application's own name; `A11` and `A12` have no contract
`035`	Business blocks	Acceptance criteria restate the block name

The organisation in this repository is a corporate information-technology org chart with no factory in it. `010` and `011` hold the same eighteen names — the chief executive, six other chiefs, the chief enterprise architect, four domain architects, three heads of function, a data steward, a security architect and an internal auditor — joined one-to-one by `R21`, every row `Filled`. There is no plant manager, production planner, scheduler, maintenance planner, reliability engineer, quality manager, process engineer, operator, setter or millwright anywhere in the file [2]. For a manufacturing repository that is the sharpest single defect. `012 Location` is the second: no column on it can say how many lines a plant runs or on what pattern.

Data — `020` to `022`, `036` to `040`, `042` to `044`

Tab	Prefix	Rows	Columns, in sheet order	Phase
`020 Data Entity`	`E01`	14	Ref, Data entity, Class, Definition, Classification, Owner, Master system, Lineage	C
`021 Logical Data Component`	`LDC-`	12	Ref, Logical data component, Entity, Description	C
`022 Physical Data Component`	`PDC-`	12	Ref, Physical data component, Application, Type	C
`036 Data Entity Relationship`	`REL-`	11	Ref, From entity, Relationship, To entity, Cardinality	C
`037 Data Attribute`	`ATR-001`	60	Ref, Entity, Attribute, Type, Key, Classification	C
`038 Data Set`	`DST-`	6	Ref, Data set, Description	C
`039 Data Migration`	`MIG-`	6	Ref, Entity, From system, To system, Status	F
`040 Source To Target Mapping`	`MAP-`	12	Ref, Entity, Source field, Target field, Rule	F
`042 Metadata Element`	`MTD-`	14	Ref, Metadata element, Category, What it records, Coverage	C
`043 Master Data Rule`	`MDR-`	10	Ref, Entity, Rule type, Description, Golden source	C
`044 Data Building Block`	`DBB-`	10	Ref, Data building block, Entity, What it must do, Acceptance criteria	E

Sheet	What it is for	The defect in it
`020`	Fourteen core entities, with owner, master and lineage	The master is wrong where it matters, three classifications are inverted, twelve of fourteen owners are titles nobody holds
`021`	A store per entity	Covers `E01` – `E12`; `E13` and `E14` have none, though both read `Full` lineage
`022`	A database per application	`PDC-01 ERP Core database` is typed `File`
`036`	Entity-to-entity links	`REL-01` reads *Product is derived from Bill of Material*, `1:1` — direction and cardinality both wrong; duplicated by `R19`
`037`	Five attributes per entity	Three-digit reference. `ATR-001 Product Identifier` is `Restricted` under an `Internal` entity; `ATR-002 Product Name` is `Public`
`038`	Six curated sets	No owner, no entity link, so none traces to a source
`039`	Six migrations	`MIG-01` migrates `Product` out of `PLM` into `SCADA / Historian`
`040`	Field-level mapping	Field names are placeholders — "PLM field" to "MES field"
`042`	The metadata layer	`MTD-02 Business owner` has `Coverage` = `Absent`, against `PRN-13`
`043`	A rule and golden source per entity	Disagrees with `020.Master system` on nine of ten shared entities; `Description` is a tautology on every row
`044`	Data blocks	Acceptance criteria restate the block name

Read the data group as one object. `020.Master system` names `SCADA / Historian` as the master of `E01 Product`; a historian is append-only time-series and cannot authorise the creation of an item. `APS`, a planning engine, masters `E06`, `E09` and `E11`, and `E13` is mastered in `Supplier Portal`. `043` disagrees with all of it.

Application — `023`, `024`, `045`

Tab	Prefix	Rows	Columns, in sheet order	Phase
`023 Application Component`	`A01`	12	Ref, Application, Type, Description, Disposition, Criticality, Health, In service, Users, Owner	C
`024 Information System Service`	`ISS-`	12	Ref, Information system service, Application, Description, Style, Published	C
`045 Application Building Block`	`ABB-`	10	Ref, Application building block, Application, What it must do, Acceptance criteria	E

Sheet	What it is for	The defect in it
`023`	The portfolio	`A04 SCADA / Historian` is the only `Critical` application and its `Disposition` is `Replace` with no successor. Owners include `Legal`, `Internal Audit`, `Risk & Compliance`. `A05 APS` has 266 users against the ERP's 196
`024`	One per application	`Style` cycles File, Batch, Event, API regardless of the application; seven of twelve are `Unpublished`, against `PRN-10`
`045`	A target per application	Covers `A01` – `A10`; `A11` and `A12` have none — the two `RPT13` flags as orphans

Note the column count on `023`: **ten headers, not nine.** The tenth is `Owner`, and four of twelve applications are owned by an assurance or legal function. A script that reads nine columns here drops the owner.

Technology — `025` to `028`, `046`, `048` to `050`

Tab	Prefix	Rows	Columns, in sheet order	Phase
`025 Technology Service`	`TSV-`	12	Ref, Technology service, Description, Category	D
`026 Logical Technology Componen`	`LTC-`	7	Ref, Logical technology component, Rationale, Status	D
`027 Physical Technology Compone`	`PTC-`	12	Ref, Physical technology component, Vendor, Class, Support end, Position, Location	D
`028 Platform Service`	`PSV-`	8	Ref, Platform service, Description	D
`046 Technology Building Block`	`TBB-` and `T01`	7	Ref, Technology building block, Ref, What it must do, Acceptance criteria	E
`048 Environment`	`ENV-`	4	Ref, Environment, Description, Personal data present	D
`049 Resilience Tier`	`RES-`	12	Ref, Application, Tier, Recovery objective, Tested	D
`050 Network Link`	`NWL-`	10	Ref, From, To, Bandwidth, Diverse path	D

Sheet	What it is for	The defect in it
025	Twelve services	TSV-02 Storage and TSV-09 Data Platform are categorised Security; TSV-11 Security Operations is Infrastructure
026	Seven plant-floor components	Tab cut at 31 characters; the same seven recur in 046 and as RME-01 – RME-07
027	The platform inventory	Vendor holds the literal string Vendor in **seven** of twelve cells; Position is typed, not computed; PTC-03 SCADA control servers sit in a data centre
028	Eight platform services	No status, no owner; PSV-02 API gateway is a service here while EP-03 says the product is not selected
046	Technology blocks	**Two columns headed Ref**; the second holds T01 – T07, a key space nothing else uses
048	Four environments	ENV-01 Production reads No for personal data; ENV-02 and ENV-04 read Yes
049	Tier and recovery objective	Tier and objective are uncorrelated; seven of twelve are Never tested
050	Ten links	NWL-01 and NWL-09 are one pair recorded twice with different diversity; NWL-02 and NWL-10 one pair at 10 Gbps and 100 Mbps

027 inverts physical reality four times: the historian cluster at Head Office, the SCADA control servers and the shop-floor terminals at Primary Data Centre, the plant switches at Central Distribution Centre. Control servers sit at the plant they control. Volume 1 chapter 13 carries the teaching, under **The Purdue model as it is actually used**.

Implementation — `029` to `033`, `052` to `056`, `062`, `066`

Tab	Prefix	Rows	Columns, in sheet order	Phase
`029 Gap`	`GAP-`	2	Ref, Gap, Domain, Building block, Severity, Status, Due	E
`030 Work Package`	`WP-`	12	Ref, Work package, Domains, Cost, Start, End, Approval, Delivery	E, F
`031 Deliverable`	`DLV-`	20	Ref, Deliverable, Phase, Status	all
`032 Standard`	`STD-`	16	Ref, Standard, Category, Status, Conformance	D, G
`033 Architecture Contract`	`ACO-`	8	Ref, Architecture contract, Work package, Compliance status	G
`052 Implementation Factor`	`IF-`	8	Ref, Implementation factor, Category, Deduction	E
`053 Transition Architecture`	`TA-1`	3	Ref, Transition architecture, Description, Work packages, Exit criteria	F
`054 Architecture Alternative`	`ALT-A`	5	Ref, Alternative, Description, Cost, Recommendation	A, E
`055 Building Block`	reused	37	Ref, Domain, Building block, What it must do, Acceptance criteria	E
`056 Project`	`PRJ-`	12	Ref, Project, Work package, Cost, Start, End, Lead	F
`062 Delivery Risk`	`RSK-`	8	Ref, Risk, Likelihood, Impact, Owner	F
`066 Architecture Increment`	`INC-1`	6	Ref, Increment, Lands, What is added	F

Sheet	What it is for	The defect in it
029	The gap list	Two rows, both High, both Business; no data, application or technology gap anywhere
030	Twelve packages with cost and dates	Sums to $65.0m against ALT-B at $46m. WP-01 is Delivered with an End of 2030-01; WP-07 is In flight and Delivered
031	ADM deliverables by phase	No owner, no date; Phase H has none
032	Sixteen standards	Not one is a manufacturing standard; STD-01 is Deprecated and Conformant; six read Breach
033	A contract per package	Covers WP-01 – WP-08; four packages worth $24.5m have none, which makes GT-0 unpassable for them
052	Delivery dependencies and constraints	IF-03 Peak trading freeze is retail vocabulary; IF-08 Site access windows is the right constraint and carries no dates
053	Three transition states	Exit criteria reads "5 exit criteria", "6 exit criteria", "7 exit criteria" — a count where a criterion belongs
054	Five costed alternatives	ALT-E at $31m is Partly adopted, which no 060 decision records; lettered reference
055	All 37 blocks in one sheet	Duplicates 035, 044, 045, 046 exactly. **No reference space of its own**
056	One project per package	The same names, costs and dates as 030, plus a Lead; PRJ-05 and PRJ-08 are led by the Internal Auditor
062	Eight delivery risks	Three are Certain, which makes them issues; RSK-06's owner, Head of Procurement, resolves to nothing
066	Six increments	Unpadded reference; Lands reads "Month 10" against no baseline date, so no increment has a date

Governance and security — `041`, `047`, `057` to `061`, `063` to `065`

Tab	Prefix	Rows	Columns, in sheet order	Phase
`041 Security Control`	`SEC-`	8	Ref, Security control, What it does, Position, Enforced at	D, G
`047 Reference Model Entry`	`RME-`	11	Ref, Reference model entry, Description, Status	Prelim
`057 Governance Body`	`GB-`	5	Ref, Governance body, Cadence, Membership, Mandate	Prelim, G
`058 Governance Gate`	`GT-0`	4	Ref, Gate, When, What it tests	G
`059 Compliance Assessment`	none	0	Ref, Building block, Block name, Acceptance criterion, Gate, Verdict, Evidence	G
`060 Governance Decision`	`DEC-`	7	Ref, Decision, Body, Status	G
`061 Governance Change Request`	`CR-G01`	5	Ref, Change request, Against, Type, Status, Expires	H
`063 Control Objective`	`CO-`	6	Ref, Obligation, Source, Control objective, Evidence	B, RM
`064 Security Policy`	`POL-`	4	Ref, Security policy, Scope, Position	D, G
`065 Enforcement Point`	`EP-`	5	Ref, Enforcement point, Acts on, Position	D

Sheet	What it is for	The defect in it
`041`	Eight controls	`SEC-07 Non-production masking` reads `In place` and is contradicted twice; `SEC-07` and `SEC-08` are reached by no relationship row
`047`	Eleven entries	All read `Adopted`, including entries whose components do not exist; the first seven duplicate `026` and `046`
`057`	Five bodies and mandates	The one sheet whose content carries no defect; it has no chair column
`058`	Four gates	Reference starts at zero; no holder column, so the holder is buried in `GB-01` and `GB-02`'s mandate text
`059`	Gate assessments	**Empty — header row only**; `R24` is empty in consequence, and there is no prefix
`060`	Seven decisions	No date column; `DEC-01` is `Outstanding` while `030` reports two packages `Delivered`
`061`	Five dispensations	`CR-G` prefix, alone in the file; one of five carries an `Expires` date
`063`	Six obligations and evidence	Written for a European manufacturer; `Control objective` is a tautology on every row
`064`	Four policies	`POL-03 Material activity is logged` reads `Met` while 21 of 30 interfaces carry no logging
`065`	Five enforcement points	`EP-05` is reached by nothing, and its `Position` reads *Most refreshes are unmasked*

`063` is the most useful single finding in the file for a South African reader. Its six obligations are `ISO 9001`, `ISO 14001 / 45001`, `Product safety and CE / UKCA marking`, `REACH / RoHS`, `Customs and origin rules` and `Extended producer responsibility`. **CE and UKCA marking have no application in South Africa** — the domestic equivalents are the compulsory specifications and letters of authority of the national regulator, and REACH and RoHS bind only on export [3]. The sheet omits every South African instrument that inspects a plant: the Occupational Health and Safety Act and its regulations, the atmospheric emission licence, the water use licence, legal metrology, the carbon tax and POPIA [3]. Rebuild it for the jurisdiction the plant is in, and keep the `Source` column — the only place a retrieval date can go.

The industry extension

`EXT Industry Extension` holds three columns — `Ref`, `Industry extension type`, `What it holds and why the core metamodel is not enough` — and six rows. It is a register of types, not a data sheet: **no instances anywhere.**

Ref	Type	Shadowed in 020 as	State
X01	Plant / Site	E06 Plant / Site	Shadowed
X02	**Production Line**	—	**Shadowed nowhere**
X03	Bill of Material	E02 Bill of Material	Shadowed
X04	Batch / Lot	E05 Batch / Lot	Shadowed
X05	Quality Specification	E11 Quality Specification	Shadowed
X06	Supplier	E09 Supplier	Shadowed

O12 Location holds sites, E07 Work Centre holds resources, and a production line is neither. That is the most consequential absence in the file: the line is the object effectiveness is measured on, a changeover belongs to and a capacity constraint lives on. Appendix C, pack 1, builds it.

The complete `Ref` prefix table

Derived from column A of every sheet. **Irregularities are in bold.**

Sheets	Prefix form	Example
`001` – `005`, `007` – `012`	three letters, hyphen, two digits	`DRV-01`, `PRN-24`, `LOC-08`
`006 Requirement`	**three letters, hyphen, three digits**	**`REQ-001` – `REQ-031`**
`013 Business Capability`	**dotted, no prefix, no hyphen**	**`C01`, `C01.01`, `C01.01.01`**
`014 Value Stream`	**two letters, two digits, no hyphen**	**`VS01` – `VS05`**
`015` – `019`, `021`, `022`	three letters, hyphen, two digits	`BSV-01`, `CTR-10`, `PDC-12`
`020 Data Entity`	**one letter, two digits, no hyphen**	**`E01` – `E14`**
`023 Application Component`	**one letter, two digits, no hyphen**	**`A01` – `A12`**
`024` – `033`, `035`, `036`	three letters, hyphen, two digits	`ISS-01`, `PTC-12`, `ACO-08`
`037 Data Attribute`	**three letters, hyphen, three digits**	**`ATR-001` – `ATR-060`**
`038` – `045`, `047` – `050`, `052`	three letters, hyphen, two digits	`MIG-01`, `RES-12`, `NWL-10`
`046 Technology Building Block`	**two `Ref` columns**	**`TBB-04` and `T04`**
`053 Transition Architecture`	**two letters, hyphen, unpadded digit**	**`TA-1` – `TA-3`**
`054 Architecture Alternative`	**three letters, hyphen, letter**	**`ALT-A` – `ALT-E`**
`055 Building Block`	**no space of its own — reuses `BBB-`, `DBB-`, `ABB-`, `TBB-`**	**10 + 10 + 10 + 7 = 37**
`056`, `057`, `060`, `062` – `065`	two or three letters, hyphen, two digits	`PRJ-12`, `GB-05`, `EP-05`

Then the tail of the space, where every row is irregular.

Sheets	Prefix form	Example
`058 Governance Gate`	two letters, hyphen, one digit, from zero	`GT-0` – `GT-3`
`059 Compliance Assessment`	none — the sheet is empty	—
`061 Governance Change Request`	two letters, hyphen, letter, two digits	`CR-G01` – `CR-G05`
`066 Architecture Increment`	three letters, hyphen, unpadded digit	`INC-1` – `INC-6`
`EXT Industry Extension`	one letter, two digits, no hyphen	`X01` – `X06`
`R06 Application interfaces with`	three letters, hyphen, three digits — the only reference space on a relationship sheet	`INT-001` – `INT-030`

Carry three things into any script. The space is not uniform: three sheets pad to three digits and the rest to two, four use no hyphen, two number from a letter and one from zero. `055` has no identity of its own, so a join on `Ref` across `035`, `044`, `045`, `046` and `055` returns every block twice. And `R06` holds the only reference space on a relationship sheet, which is why `ZZ Checks`'s record total reconciles only if interface rows count as records.

The tab-name truncation table

Excel caps a tab name at 31 characters. Every tab below is cut at that cap except the five marked **Intact**, which run 27 to 30 characters. **`R04 Business Service supported` ends in a space**, so `wb['R04 Business Service supported']` raises `KeyError` and only the trailing-space form works.

Tab as stored	Should read
`O26 Logical Technology Componen`	… `Component`
`O27 Physical Technology Compone`	… `Component`
`R02 Value stream uses Capabilit`	… `Capability`
`R03 Capability realised by Busi`	… `Business Service`
`R04 Business Service supported`	… `supported by Application`
`R05 Application manages Data En`	… `Data Entity`
`R06 Application interfaces with`	… `with Application`
`R07 Application deployed on Tec`	… `Technology`
`R08 Requirement derives from Dr`	… `Driver`
`R09 Requirement satisfied by Bu`	… `Building Block`
`R10 Objective measured by Measu`	… `Measure`

Tab as stored	Should read
`R13 Work Package delivers Build`	… `Building Block`
`R14 Project delivers Work Packa`	… `Work Package`
`R15 Transition contains Work Pa`	… `Work Package`
`R16 Control Objective met by Co`	… `Control`
`R17 Policy enforced at Enforcem`	… `Enforcement Point`
`R18 Standard applies to Applica`	… `Application`
`R19 Data Entity relates to Data`	… `Data Entity`
`R20 Application located at Loca`	… `Location`
`R23 Increment delivered by Proj`	… `Project`
`R24 Compliance Assessment tests`	… `tests Building Block`
`R25 Application supports Produc`	… `Product`
`R01 Capability decomposition`	**Intact**
`R11 Goal realised by Objective`	**Intact**
`R12 Gap closed by Work Package`	**Intact**
`R21 Role performed by Actor`	**Intact**
`R22 Org Unit owns Capability`	**Intact**

Three further tabs sit at or just under the cap and are complete as written — `O24 Information System Service` and `O45 Application Building Block` at 30 characters, and `RPT06 Requirements Traceability` at exactly 31. Do not "fix" them.

> **WRITE THE SHEET NAMES ONCE, FROM THE FILE**
>
> Never type a tab name from memory and never normalise whitespace. Read `wb.sheetnames`, keep the strings, and build a dictionary keyed on the first token of each name — `O27`, `R04`, `RPT10`. One line of defensive code saves an afternoon.

The relationship sheets

Twenty-five sheets, **430 instances**, which is the one figure in `ZZ Checks` that reconciles exactly. Only `R06` carries a `Ref` column.

Tab	Rows	Columns, in sheet order	Shape, and the defect
`R01 Capability decomposition`	90	Parent, Parent name, Child, Child name, Level	Genuine decomposition of a generated tree
`R02 Value stream uses Capabilit`	15	Value stream, Name, Capability, Capability name	A three-wide sliding window; agrees with `O14` on `VS01` – `VS03`, contradicts it on `VS04`, `VS05`, reaches `C01` – `C07` only
`R03 Capability realised by Busi`	10	Capability, Capability name, Business service, Service name	Positional diagonal, `C01` → `BSV-01`
`R04 Business Service supported`	10	Business service, Service name, Application, Application name	Positional diagonal; *Quality Management Service* is supported by `APS`
`R05 Application manages Data En`	28	Application, Application name, Data entity, Entity name, Role	**Two key schemes in one sheet**: `Master` rows hold a name, `Consumer` rows a ref, cycling `A01`, `A04`, `A07`, `A10`
`R06 Application interfaces with`	30	Ref, Source, Source name, Target, Target name, Type, Frequency, Managed, Logged	The only sheet with a `Ref`; `Source` and `Target` hold names
`R07 Application deployed on Tec`	12	Application, Application name, Platform component, Component name, Environment	Positional diagonal: `A03 MES` lands on `PTC-03 SCADA control servers`, `A04 SCADA / Historian` on `PTC-04 ERP application tier`
`R08 Requirement derives from Dr`	31	Requirement, Requirement name, Driver, Driver name	Six diagonal rows, then constants: 8 to `DRV-10`, 6 to `DRV-07`, 11 to `DRV-09`
`R09 Requirement satisfied by Bu`	31	Requirement, Requirement name, Building block, Block name, Traced	Positional walk through `BBB`, `DBB`, `ABB`, `TBB`. **`Traced` reads `No` on every fourth row, although all 31 rows name a block**
`R10 Objective measured by Measu`	14	Objective, Objective name, Measure, Measure name, Unit	Positional diagonal; `MSR-15` and `MSR-16` reach nothing
`R11 Goal realised by Objective`	14	Goal, Goal name, Objective, Objective name	The inverse of `O03.Goal`; correct
`R12 Gap closed by Work Package`	2	Gap, Gap name, Severity, Work package, Package name	Positional; two rows
`R13 Work Package delivers Build`	37	Work package, Package name, Building block, Domain, Block name	Positional, wrapping modulo twelve. **Every one of the 37 blocks is in a package**, so `CR-G03` has nothing to point at
`R14 Project delivers Work Packa`	12	Project, Project name, Work package, Package name, Cost	Positional; `O56` is `O30` with a `Lead`

`R15 Transition contains Work Pa`	7	Transition, Name, Work package, Package name	Hand-assigned; leaves `WP-06`, `WP-08`, `WP-09`, `WP-10`, `WP-12` — $26.1m — in no transition state
`R16 Control Objective met by Co`	6	Control objective, Obligation, Control, Control name, Position	Positional: **`ISO 9001` is evidenced by `SEC-01 Identity lifecycle`**; `Position` holds `O63`'s `Evidence`
`R17 Policy enforced at Enforcem`	4	Policy, Policy name, Enforcement point, Point name, Position	Positional; `EP-05` enforces nothing
`R18 Standard applies to Applica`	16	Standard, Standard name, Application, Application name, Conformance	Positional, step of two. **Only the six odd-numbered applications appear**
`R19 Data Entity relates to Data`	11	Ref, From, Relationship, To, Cardinality	`O36` with renamed headers
`R20 Application located at Loca`	12	Application, Application name, Location, Location name	Positional modulo eight; contradicts `O27` on where `A04` is
`R21 Role performed by Actor`	18	Role, Role name, Actor, Actor name, Status	Positional, all `Filled`, which is why `ZZ Checks` reports zero vacancies
`R22 Org Unit owns Capability`	10	Org unit, Unit name, Capability, Capability name	Positional; `Human Resources` owns `Manufacturing Operations`
`R23 Increment delivered by Proj`	6	Increment, Increment name, Lands, Project, Project name	Positional, step of two. **Only odd-numbered projects appear**
`R24 Compliance Assessment tests`	0	Assessment, Building block, Block name, Criterion, Verdict	Empty, because `O59` is
`R25 Application supports Produc`	4	Product, Product name, Application, Application name	Positional, step of three

Count the positional sheets: `R03`, `R04`, `R07`, `R09`, `R10`, `R12`, `R13`, `R14`, `R16`, `R17`, `R18`, `R20`, `R21`, `R22`, `R23`, `R25` — **sixteen of twenty-five**, with `R02`'s sliding window and `R05`'s `Consumer` half making eighteen generated by row position. Treat each as a placeholder with the right shape and the wrong content.

Where the repository contradicts its own documentation

Every item was checked against the file. This is the honest answer to the question a sponsor asks in week two: you can trust the structure and little of the content.

#	The claim	Where it is made	What the file shows
1	843 records	`00 Index` row 2, `ZZ Checks`, `RPT01`	813. The difference is **exactly 30 — `R06` 's row count**
2	65 types populated, every type carries records	`ZZ Checks`, `RPT01`	64 of 65. `O59` holds zero rows, and `00 Index` 's own row for it reads "0 records"
3	430 relationship instances	`00 Index`, `ZZ Checks`	**Correct.** The 25 sheets sum to exactly 430
4	Every relationship row names both ends by reference and by name	`00 Index`	False on two sheets. `R05` 's `Master` rows and `R06` 's `Source` and `Target` carry names only
5	Platform components past support, 4	`ZZ Checks`, `RPT01`, `RPT10`	2. `PTC-07` and `PTC-09` end 2026-05; `PTC-06` 2027-07 and `PTC-08` 2027-04 are future
6	Work packages past end date, 4	`ZZ Checks`, `RPT01`, `RPT08`	0. Every `End` falls between 2029-07 and 2031-10; the earliest `Start` is 2027-02
7	Vacant architecture roles, 0	`ZZ Checks`, `RPT01`	True by construction: all 18 rows of `R21` read `Filled`
8	Security controls absent, 1	`ZZ Checks`, `RPT11`	**Correct** — `SEC-02 Design review`
9	Control objectives without evidence, 5	`ZZ Checks`, `RPT09`	**Correct** — two *No mechanism*, three *In breach*, only `CO-05` *Evidenced*
10	Interfaces unmanaged 22, unlogged 21, direct reads 4	`ZZ Checks`, `RPT04`	**Correct** against `R06` 's 30 rows
11	`ZZ Checks` has seventeen lines	commonly stated	**Fifteen checks and a closing note — sixteen rows**

Now the contradictions between sheets, which are the ones that change a design.

1. **Two sheets claim the golden source and disagree on nine of the ten entities they share.** `O20.Master system` against `O43.Golden source`: `E02` `PLM` / `Track & Trace`; `E03` `Plant Analytics` / `Supplier Portal`; `E04` and `E05` `APS` and `WMS` / `Supplier Portal`; `E06` and `E09` `APS` / `QMS`; `E07` `QMS` / `WMS`; `E08` `SCADA / Historian` / `MES`; `E10` `ERP Core` / `QMS`. **The only row they agree on is `E01 Product`, and it is wrong in both** — each names `SCADA / Historian`, and `APS` masters three more entities besides.
2. **`O20.Classification` is inverted where it matters**: `E09 Supplier`, `E11 Quality Specification` and `E13 Inventory Position` read `Public` while `E04` and `E07` read `Restricted`. A quality specification is the testable definition of the product.
3. **`O20.Owner` names titles that resolve to nothing** — `Head of Finance` on `E01`, `Head of Quality Management` on `E03` and `E11`, `Head of Health, Safety` on `E08` and `E10`. Only `Data Steward` resolves, as `ROL-16`.
4. **`R05` uses application names on its `Master` rows and refs on its `Consumer` rows**, so a join on `Application` returns half the sheet — against `00 Index` 's own integrity promise.
5. **Sixteen of twenty-five relationship sheets are positional**, so `ISO 9001` is evidenced by `SEC-01 Identity lifecycle` and `Human Resources` owns `Manufacturing Operations`.

6. **`O05 Principle` carries `Owner = Architecture Board` on all 24 rows against its own `PRN-13`**, with `Draft` on every fifth row — `PRN-05`, `PRN-10`, `PRN-15`, `PRN-20` — which is a generation pattern, not an assessment.
7. **`O10` and `O11` hold the same eighteen names with no plant role among them.**
8. **`O27`'s past support is typed, not computed, so 4 is really 2**; `Vendor` reads literally `Vendor` in seven of twelve cells; and `PTC-03 SCADA control servers` sit in a data centre.
9. **`O48` records personal data in pre-production and development but not production**, against `SEC-07` reading `In place` and `EP-05` reading *Most refreshes are unmasked*. The data at issue is employee and biometric, which makes it a POPIA special-information question [3].
10. **`O49` tier and recovery objective are uncorrelated, with seven of twelve never tested**, and `MSR-13` carries a baseline of 21 against twelve rows.
11. **`O30` sums to $65.0m against `ALT-B` at $46m**, the file being in dollars, and **`O30` contradicts itself on approval and delivery**: `WP-07` is `In flight` and `Delivered`, `WP-05` `Not started` and `In flight`.
12. **`O53`'s `Exit criteria` is a count**, and **five work packages worth $26.1m sit in no transition state.**
13. **`O13` is a generated cross-product** with `Disposition` as arithmetic — `Build` if and only if `Target − Current ≥ 2` — and criticality that does not inherit: `C01.02.01` is `Critical` under a `Low` parent. **`O14` gives `VS01` and `VS04` one capability list.**
14. **`LOC-08 Cloud Region` has 261 people. `O63` is for a European manufacturer**, with CE and UKCA marking that have no application here and a tautological `Control objective` column. **`X02 Production Line` is declared in `EXT` and shadowed nowhere.**
15. **`R09.Traced` reads `No` on eight rows that each name a building block**, and **`R13` funds all 37 blocks**, which leaves `CR-G03` with nothing to point at.
16. **`O33` covers eight of twelve packages**, so `GT-0` cannot be passed for $24.5m of work; **`R18` leaves six applications under no standard** and **`R23` six projects in no increment.**
17. **There is no change control at all** — no version column, no change log, no baseline, no effective date, no date on `O60`.

The naming rules, and the change control to add

Three naming rules, which cost nothing at the start and cannot be retrofitted. **One reference space per sheet, declared in row 2** — three letters, a hyphen, three digits everywhere, because `O06` and `O37` already need three. **A reference is never reused and never renumbered**, which is what `O55` breaks. **A name is not a key** — every relationship column holds a reference, and the name beside it is regenerated, never edited.

Then the five columns the file lacks, on every object sheet.

Column	Holds	Why
`Version`	An integer, incremented on change	Without it no two readers can agree which repository they are reading
`Effective from`	A date	`003.Horizon` and `066.Lands` are durations because there is no baseline date to count from
`Changed by`	A role reference from `011`	A change with no author cannot be questioned
`Change request`	A `CR-` reference	`061` exists; give it something to point at
`Baseline`	A label	A gate assessment compares a design to a baseline, and there is nothing here to compare against

And one new sheet, `067 Change Log`, with `Ref`, `Date`, `Sheet`, `Row`, `From`, `To`, `Change request`, one row per change. It is the cheapest sheet in the repository and the one that makes Phase H possible at all.

What to fix first

1. **`011 Role` and `010 Actor`**, because everything downstream names an owner and none that matters resolves. Appendix C, pack 2.
2. **`X02 Production Line` as a new sheet, plus four columns on `012 Location`.** Appendix C, pack 1.
3. **`020.Master system` against `043.Golden source`** — one master per entity, or `CONTESTED` and a date. Appendix C, pack 3.
4. **`063 Control Objective`, rebuilt for the jurisdiction the plant is in.**
5. **`004 Measure`, extended into a measure definition register**, before any dashboard.
6. **The arithmetic** — `030` against `054`, `R15`'s five missing packages, `027`'s typed `Position` and `030`'s typed `Delivery`.
7. **Change control** — the five columns and `067 Change Log`.

> **THE ONE-LINE TEST**
>
> Pick any number on `RPT01 Dashboard` and try to recompute it from the object sheets. Fourteen of eighteen reconcile. The four that do not are the four `ZZ Checks` gets wrong, and the sheet's own closing line — that every figure is computed from the workbook and not asserted — is the nineteenth thing in the file that is not true. Appendix B works through all thirteen reports.

THE CHAPTER IN TEN LINES

- 105 sheets: 64 object, EXT, 25 relationship, 13 report, the index and ZZ Checks. O34 and O51 do not exist.
- Headers are in row 4 and data starts in row 5 on every sheet. Read with min_row=5.
- The Ref space is irregular: three sheets pad to three digits, four use no hyphen, two number from a letter and GT-0 starts at zero.
- O55 Building Block has no reference space of its own and duplicates 37 refs from four other sheets.
- O46 Technology Building Block has two columns headed Ref; the second holds T01 to T07.
- Twenty-two tab names are cut at 31 characters, and R04 Business Service supported ends in a space.
- 843 records is really 813, and the difference is exactly the 30 rows of R06.
- O59 Compliance Assessment is empty, so 65 populated object types is really 64.
- O20 Master system and O43 Golden source disagree on nine of the ten entities they share.
- There is no change control anywhere in the file: no version column, no change log, no baseline, no date on O60.

SOURCES FOR THIS CHAPTER

- **1.** Manufacturing_TOGAF_Metamodel.xlsx — all 105 sheets, read with openpyxl, 16 October 2026
- **2.** Manufacturing_TOGAF_Metamodel.xlsx, sheets O10 Actor, O11 Role and R21 Role performed by Actor
- **3.** Volume 1, chapter 16, Regulation, risk and what the inspector asks for — the South African instruments that inspect a plant, with their sources and retrieval dates

APPENDIX B

The report catalogue

What each of the thirteen reports actually computes, and which of their numbers you can quote.

Thirteen report sheets sit at the end of `Manufacturing_TOGAF_Metamodel.xlsx`, and `00 Index` describes them as *"Thirteen reports compute from the object sheets"* [1]. Read all thirteen before you put any of them in front of anybody, because most of them do not compute anything. This appendix says, for each one, what it actually computes, what it merely reprints and from where, whether its footer is arithmetic or a typed cell, and whether its number can be trusted [2].

Kestrel Industrial Group, whose data fills the file, is an invention.

How a report in this file is built

Every report follows the same layout as every other sheet: title in row 1, subtitle in row 2, headers in row 4, data from row 5. Several carry one extra row after the data holding a sentence — sometimes a number, sometimes a maxim. **Nothing in any of the thirteen is a formula.** Open the file with `data_only=False` and every cell comes back as a literal. That is the single most useful fact in this appendix: these are snapshots, and a snapshot of generated data taken at an unknown date is not a report.

On the shape, the sibling volumes in this series found that twelve of thirteen reports were column-subset reprints of one sheet and only the orphan report performed a join. **That is not quite the pattern here.** Counted against the file:

Kind	Count	Which
Straight column-subset reprint of one sheet	8	`RPT02`, `RPT04`, `RPT05`, `RPT08`, `RPT09`, `RPT10`, `RPT11`, `RPT12`
A typed list of measures with no source sheet	1	`RPT01`
Reprint, filtered, with one computed column	1	`RPT03`
Reprint joined to a relationship sheet	2	`RPT06`, `RPT07`
A genuine anti-join	1	`RPT13`

So three reports reach outside a single object sheet, not one. Six carry a footer with a number in it, all six typed by hand; four of those six are arithmetically right against their own sheet and two are wrong. One report, `RPT12`, carries no number of any kind.

The thirteen, one at a time

`RPT01 Dashboard`

Columns `Measure`, `Value`, `What it means`. Eighteen measure rows and a closing line.

What it computes: nothing. Every value is a typed integer. **What it reprints:** fifteen of the eighteen lines restate `ZZ Checks`; three are new — `Applications` 12, `Gaps overdue` 1 and `Level 1 capabilities` 10. **Footer:** a maxim, *"Every red figure is computed from this workbook, not asserted."* No figure on the sheet is computed.

Fourteen of the eighteen reconcile when you recount them from the object sheets. Four do not: `Object types populated` 65 against 64, `Records` 843 against 813, `Platform components past vendor support` 4 against 2, and `Work packages past end date` 4 against 0. `Gaps overdue` 1 reconciles against `O29`'s typed `Status` of `Overdue` on `GAP-02` but not against its `Due` date of 2028-05, which is in the future.

Trust: the four wrong lines are the four `ZZ Checks` gets wrong, and they are the four an executive reads first. Hand-correct them before this sheet leaves the room.

`RPT02 Application Portfolio`

Columns `Ref`, `Application`, `Type`, `Disposition`, `Criticality`, `Health`, `In service`, `Users`, `Owner`. Twelve rows, no footer.

Computes: nothing. **Reprints:** `O23 Application Component`, nine of its ten columns, dropping `Description` — the only column that says what each application does. **Trust:** faithful to its source, and its source is wrong in ways the report makes more visible rather than less. `A04 SCADA / Historian` is the only `Critical` application, its `Disposition` is `Replace` and nothing is named to replace it; it is owned by `Internal Audit`. `A01 ERP Core` is owned by `Legal`, `A05 APS` by `Risk & Compliance`. `A03 MES` has been in service since 2010 with `Health` = `Poor` and `Disposition` = `Retain`. `A05 APS` carries 266 users against the ERP's 196, which no plant resembles, so do not score business value off the `Users` column.

`RPT03 Capability Heatmap`

Columns `Ref`, `Capability`, `Criticality`, `Current`, `Target`, `Gap`, `Disposition`. Ten rows, no footer.

Computes: one column, and it is real. `Gap` = `Target` − `Current`, which appears nowhere in `O13 Business Capability`. **Reprints:** `O13`, filtered to `Level` = 1, dropping `Level`, `Parent` and `Description`. **Trust:** the subtraction is correct on all ten rows. Two cautions. `Disposition` is the same arithmetic restated — on `O13` it reads `Build` if and only if `Target − Current ≥ 2` — so the last two columns of this report carry one fact between them. And the reader loses the 90 sub-rows, which is where the contradiction lives: criticality does not inherit in `O13`, so `C01 Product Development` is `High` on this report while `C01.01` is `Low` and `C01.02.01` is `Critical`. A heatmap of ten rows over a hundred-row map is a cover, not a summary.

RPT04 Interface Catalogue

Columns Ref, Source, Target, Type, Frequency, Managed, Logged. Thirty rows and a footer.

Computes: nothing. **Reprints:** R06 Application interfaces with, dropping Source name and Target name — which costs the reader nothing, because R06's Source and Target columns already hold application names rather than refs. **Footer:** typed, *"22 of 30 interfaces are unmanaged and 21 carry no logging."* Recounted off the sheet: 22 Unmanaged and 21 No. Both right. The four Direct DB read rows — INT-017, INT-019, INT-025, INT-027 — match ZZ Checks too.

Trust: the most trustworthy report in the file. Every figure on it survives recomputation. It is also the report with the most immediate consequence, because INT-017 QMS reads the SCADA / Historian store directly, which is a read of a plant system by a corporate application across the boundary volume 1 chapter 13 is about.

RPT05 Data Ownership

Columns Ref, Entity, Class, Classification, Owner, Master system, Lineage. Fourteen rows and a footer.

Computes: nothing. **Reprints:** O20 Data Entity, dropping Definition. **Footer:** typed, *"6 of 14 entities have no lineage from report back to source."* Recounted: E03, E04, E05, E06, E10 and E14 read Absent. Six. Right.

Trust: the arithmetic is right and the content is not. The report puts in front of an executive that E01 Product is mastered in SCADA / Historian, that E09 Supplier and E11 Quality Specification are classified Public, and that E01 is owned by Head of Finance — a title that resolves to nothing in O11 Role. Dropping Definition also removes the reader's only means of noticing that the definitions are the sound part of the sheet.

RPT06 Requirements Traceability

Columns Requirement, Name, Category, Priority, Driver, Reaches a block. Thirty-one rows and a closing line with no number in it.

Computes: nothing, but it joins. Requirement, Name, Category, Priority and Driver come from O06 Requirement, dropping Statement and Status. Reaches a block is R09 Requirement satisfied by Bu's Traced column, renamed.

Trust: no. R09 is a positional walk — REQ-001 to BBB-01, REQ-011 to DBB-01, REQ-021 to ABB-01, REQ-031 to TBB-01 — so every one of its 31 rows names a building block. And yet Traced reads No on exactly every fourth row: REQ-001, REQ-005, REQ-009, REQ-013, REQ-017, REQ-021, REQ-025, REQ-029. **The report's own answer contradicts its own source**, and the pattern is a generation artefact, not an assessment. Recompute the column yourself from R09's Building block and it reads Yes thirty-one times, which is also useless, because the assignment behind it is positional.

RPT07 Gap Analysis

Columns Ref, Gap, Domain, Severity, Status, Due, Work package. Two rows, no footer.

Computes: nothing, but it joins. The first six columns come from `O29 Gap`, dropping `Building block`; `Work package` comes from `R12 Gap closed by Work Package`. This is the second report in the file that reaches outside one sheet, and almost nobody notices, because it has two rows.

Trust: the join is correct. The sheet has nothing to say. `O29` holds two gaps, both `High`, both `Business`, both capability uplifts — and no data, application or technology gap is recorded anywhere in the file, although `R13` funds all 37 building blocks and `ZZ Checks` reports six entities without lineage and 22 unmanaged interfaces. `GAP-02` reads `Overdue` against a `Due` of 2028-05, which has not happened. **This is the gap report. There is no "Gaps and Roadmap" sheet.**

`RPT08 Roadmap`

Columns `Ref`, `Work package`, `Cost`, `Start`, `End`, `Approval`, `Delivery`. Twelve rows and a footer.

Computes: nothing. **Reprints:** `O30 Work Package`, dropping `Domains`. **Footer:** typed, *"4 work packages are past their end date."*

Trust: no, and this is the most quotable error in the file. The footer counts the four rows whose `Delivery` column is typed `Past end date` — `WP-02`, `WP-03`, `WP-09`, `WP-12`. But the programme's earliest `Start` is **2027-02** (`WP-09`) and its latest `End` is **2031-10** (`WP-05`). Every `End` on the sheet falls between 2029-07 and 2031-10. In October 2026 nothing in this programme has started, so nothing can be late. The report also carries `WP-01` as `Delivered` with an `End` of 2030-01, `WP-07` as `In flight` and `Delivered` at once, and `WP-05` as `Not started` and `In flight` at $3.1m. And the twelve `Cost` cells sum to **$65.0m** against `ALT-B Foundations first` at **$46m** in `O54`, which is the `Recommended` alternative.

`RPT09 Regulatory Coverage`

Columns `Ref`, `Obligation`, `Source`, `Control objective`, `Evidence`. Six rows and a footer.

Computes: nothing. **Reprints:** `O63 Control Objective` exactly — all five columns, nothing dropped, nothing renamed. It is the only report that is a verbatim copy of its source. **Footer:** typed, *"5 obligations have no mechanism or a mechanism in breach."* Recounted: `CO-01` and `CO-03` *No mechanism*, `CO-02`, `CO-04` and `CO-06` *In breach*, `CO-05` *Evidenced*. Five. Right.

Trust: the count is right and the sheet is for the wrong country. The six obligations are `ISO 9001`, `ISO 14001 / 45001`, `Product safety and CE / UKCA marking`, `REACH / RoHS`, `Customs and origin rules` and `Extended producer responsibility`. CE and UKCA marking have no application in South Africa; the domestic equivalents are compulsory specifications and letters of authority, and REACH and RoHS bind only on export [3]. `Control objective` is a tautology on every row — *"Control that evidences ISO 9001."* Do not put this report in front of a South African executive committee with the obligations as written.

`RPT10 Technology Currency`

Columns `Ref`, `Component`, `Vendor`, `Class`, `Support end`, `Position`, `Location`. Twelve rows and a footer.

Computes: nothing. **Reprints:** `O27 Physical Technology Compone` in full, with `Physical technology component` renamed `Component`. **Footer:** typed, *"4 components are running past vendor support today."*

Trust: no, and the report disproves itself in the column beside the one it relies on. `Position` is a typed value on `O27`, not a comparison of `Support end` against today. Four rows read `Past support`: `PTC-06` with a support end of **2027-07**, `PTC-07` **2026-05**, `PTC-08` **2027-04** and `PTC-09` **2026-05**. Read on 16 October 2026, **only two of those four dates are in the past.** Two components typed as expired are supported for another six and nine months.

Three further cautions on this sheet. `Vendor` holds the literal string `Vendor` in seven of twelve cells, so it cannot be used to plan a vendor conversation. `Location` inverts physical reality: `PTC-03 SCADA control servers` at `Primary Data Centre`, `PTC-06 Shop-floor Windows terminals` at `Primary Data Centre`, `PTC-07 Plant network switches` at `Central Distribution Centre` and `PTC-01 Plant historian cluster` at `Head Office`. And a shop-floor terminal estate and a control server estate have different patch windows and different consequences of failure, which this report cannot show because it has no column for either.

`RPT11 Security Position`

Columns `Ref`, `Control`, `What it does`, `Position`, `Enforced at`. Eight rows and a footer.

Computes: nothing. **Reprints:** `O41 Security Control` in full, with `Security control` renamed `Control`. **Footer:** typed, *"1 controls are absent. A control that is absent is not a control."* One control is absent — `SEC-02 Design review` — so the count is right and the grammar is not.

Trust: the count is right; one row on it is contradicted twice elsewhere. `SEC-07 Non-production masking` reads `In place`, while `O48 Environment` records personal data in pre-production and development and `EP-05 Non-production provisioning` in `O65` reads *Most refreshes are unmasked*. Carry that contradiction to the meeting, not the report.

`RPT12 Standards Conformance`

Columns `Ref`, `Standard`, `Category`, `Status`, `Conformance`. Sixteen rows. **No footer at all.**

Computes: nothing. **Reprints:** `O32 Standard` exactly, all five columns. **Trust:** it adds nothing whatever to its source. Read `O32`. What the report does show, because the rows sit together, is that `STD-01 API design standard` is `Deprecated` and `Conformant` at the same time, that six of sixteen read `Breach`, and that not one of the sixteen standards is a manufacturing standard: no enterprise-to-control integration standard, no batch control standard, no measure definitions standard, no industrial security standard. For a manufacturer that absence is the finding, and it is invisible on a report that lists only what is there.

`RPT13 Orphan Analysis`

Columns `Ref`, `Object`, `Disposition`, `Finding`. Two rows and a closing line.

What it actually tests: which references in `O23 Application Component` do not appear in the `Application` column of `R04 Business Service supported`. That is a genuine anti-join, and it is

the only computation in the file that reaches across two sheets to find an absence rather than a value. `Disposition` is brought in from `O23`, and `Finding` is a fixed string, *"No business service maps to it."*

Recomputed: `R04` is a positional diagonal mapping `BSV-01` to `A01` through `BSV-10` to `A10`, and `O23` holds twelve applications, so `A11 Track & Trace` and `A12 Plant Analytics` fall out. The report is correct. `A11`'s `Disposition` is `Retire` and `A12`'s is `Replace`, which is the reading the subtitle invites: *"An orphan is not always wrong. It is always worth a question."*

Trust: correct, and far too narrow. It tests one object sheet against one relationship sheet.

The thirteen on one page

Row counts exclude the footer or closing line, which is counted separately in the last column.

Report	Source sheet	Rows	What it computes	Footer
`RPT01 Dashboard`	none — a typed list	18	nothing	maxim, no number
`RPT02 Application Portfolio`	`O23` less `Description`	12	nothing	none
`RPT03 Capability Heatmap`	`O13`, level 1 only	10	`Gap = Target - Current`	none
`RPT04 Interface Catalogue`	`R06` less two name columns	30	nothing	typed, correct
`RPT05 Data Ownership`	`O20` less `Definition`	14	nothing	typed, correct
`RPT06 Requirements Traceability`	`O06` joined to `R09.Traced`	31	nothing	note, no number
`RPT07 Gap Analysis`	`O29` joined to `R12`	2	nothing	none
`RPT08 Roadmap`	`O30` less `Domains`	12	nothing	**typed, wrong**
`RPT09 Regulatory Coverage`	`O63`, verbatim	6	nothing	typed, correct
`RPT10 Technology Currency`	`O27`, one header renamed	12	nothing	**typed, wrong**
`RPT11 Security Position`	`O41`, one header renamed	8	nothing	typed, correct
`RPT12 Standards Conformance`	`O32`, verbatim	16	nothing	**none at all**
`RPT13 Orphan Analysis`	`O23` anti-joined to `R04`	2	the orphan set	note, no number

And the reports a manufacturer needs that none of the thirteen is. There is no report on the production line, because `X02 Production Line` does not exist as a sheet. There is no report on plant

structure, because `O12 Location` has no line count and no shift pattern. There is no measure report, because `O04 Measure` has no definition, formula or denominator column. There is no report that joins a governance gate to a building block, because `O59 Compliance Assessment` is empty. There is no report on resilience testing, although `O49` records that seven of twelve applications have never been tested. And there is no change report, because the file holds no version column, no change log and no date on `O60 Governance Decision`.

TWO QUESTIONS FOR THE FIRST GOVERNANCE MEETING

Which report will you put in front of `GB-02 Investment Committee` first, and what is the one number on it you have recomputed yourself?

`RPT13` finds two orphans and the real figure is thirty-three sheets. Which four of the thirty-three will you raise, and to which body?

BEFORE YOU QUOTE A REPORT

Which three of the thirteen would you be willing to defend line by line, and what makes the other ten different?

`RPT08` says four packages are past their end date and its own `Start` column says the programme has not begun. Which number goes in the pack, and who signs it?

What `RPT13` misses

`RPT13` reports two orphans out of 813 rows. Run the same test properly — collect every string in every cell of all 25 relationship sheets, then ask of each object sheet whether any of its references appears in that pool — and the answer changes shape.

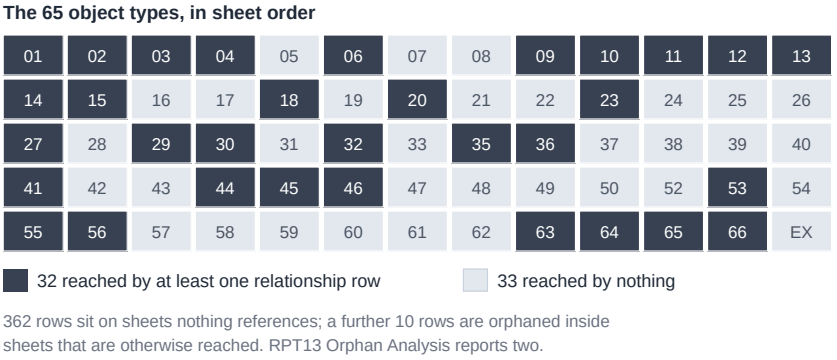


Figure 14.1 Reachability across the sixty-five object types. `EX` is `EXT Industry Extension`. **Thirty-three sheets are reached by no relationship row at all**, against the two orphans `RPT13` reports.

Thirty-two of the sixty-five object types are reached by at least one relationship row. Thirty-three are reached by nothing at all.

Reached by nothing	Rows
`005 Principle`, `007 Constraint`, `008 Assumption`	24, 8, 6
`016 Business Process`, `017 Business Function`, `019 Contract`	20, 10, 10
`021 Logical Data Component`, `022 Physical Data Component`	12, 12
`024 Information System Service`, `025 Technology Service`	12, 12
`026 Logical Technology Componen`, `028 Platform Service`	7, 8
`031 Deliverable`, `033 Architecture Contract`	20, 8
`037 Data Attribute`, `038 Data Set`, `039 Data Migration`	60, 6, 6

Reached by nothing	Rows
`O40 Source To Target Mapping`, `O42 Metadata Element`	12, 14
`O43 Master Data Rule`, `O47 Reference Model Entry`	10, 11
`O48 Environment`, `O49 Resilience Tier`, `O50 Network Link`	4, 12, 10
`O52 Implementation Factor`, `O54 Architecture Alternative`	8, 5
`O57 Governance Body`, `O58 Governance Gate`	5, 4
`O59 Compliance Assessment`, `O60 Governance Decision`	0, 7
`O61 Governance Change Request`, `O62 Delivery Risk`	5, 8
`EXT Industry Extension`	6

That is **362 rows on sheets nothing references**. Add the ten individual rows orphaned inside sheets that are otherwise reached — `DRV-08`, `DRV-11`, `DRV-12`, `MSR-15`, `MSR-16`, `ORG-11`, `ORG-12`, `SEC-07`, `SEC-08`, `EP-05` — and **372 of 813 rows, 46 per cent of the repository, are unreachable from any relationship sheet.** And `O55 Building Block` counts as reached only because its 37 references are copies of `O35`, `O44`, `O45` and `O46`; on its own identity nothing points at it, which takes the honest figures to 34 sheets and 399 rows, just under half.

Four of those absences are worth naming to a sponsor, because each one is a question the repository cannot answer.

- **`O43 Master Data Rule`** holds the golden source for ten entities and no relationship row connects it to `O20`, `O23` or anything else. That is precisely why the two sheets can disagree on nine entities without the file noticing.
- **`O49 Resilience Tier`** and **`O48 Environment`** name applications by name rather than by `A` reference and appear on no relationship sheet, so no report can ask whether the `Critical` application is in a tested tier. It is: `RES-04` is the only coherent row on `O49`.
- **`O57 Governance Body`**, **`O58 Governance Gate`** and **`O60 Governance Decision`** are joined to nothing. The gates hold no work packages, the decisions name their body in free text, and `O59 Compliance Assessment` — the sheet that would have joined a gate to a building block — is empty.
- **`EXT Industry Extension`** is reached by nothing because it holds no instances. The six industry types are a register, and `X02 Production Line` is not shadowed anywhere at all.

> **THE TEST TO RUN ON THE CLIENT'S OWN REPOSITORY**
>
> The orphan question is not "which applications have no business service". It is "which sheets does nothing point at". Eight lines of `openpyxl` answers it, and on this file the answer is half the repository. Run it in week three, before anybody asks you to build a report over the data.

The board set

Four reports are worth an executive committee's time, and one more with its arithmetic corrected. Each is mapped below to the body in `O57 Governance Body` whose mandate covers it and to the decision in `O60 Governance Decision` it should move.

Report	Body	Mandate it falls under	Decision it should move
`RPT04 Interface Catalogue`	`GB-01 Architecture Board`	Holds G1 and G2; approves dispensations with an expiry	`DEC-02 Select the integration product — Outstanding`
`RPT05 Data Ownership`	`GB-03 Data Governance Council`	Designates data owners and resolves disputes	`DEC-04 Designate data owners — Outstanding`
`RPT09 Regulatory Coverage` , rebuilt for the jurisdiction	`GB-02 Investment Committee`	Holds G0 funding and G3 benefit confirmation	`DEC-03 Fund compliance mechanisms — Outstanding`
`RPT11 Security Position`	`GB-01 Architecture Board`	Holds G1 and G2	`DEC-06 Adopt the four-gate regime — Approved` , while `SEC-02 Design review` is `Absent`
`RPT08 Roadmap` , with the dates and the $65.0m corrected	`GB-02 Investment Committee`	Holds G0 funding	`DEC-01 Adopt the recommended architecture alternative — Outstanding`

Three notes on that table. `RPT11` and `DEC-06` are the sharpest pairing in the file: the board has already approved a four-gate regime whose first gate is a design review, and the design review control is the one absent control on the estate. `RPT09` must be rebuilt before it goes anywhere, because an executive committee that approves funding against CE marking has funded the wrong thing. And `RPT08` goes to the Investment Committee with two corrections written on its face — that no package has started, and that the twelve packages sum to $19m more than the recommended alternative — or it does not go at all.

What stays out of the pack, and why. `RPT01 Dashboard` is four figures wrong of eighteen, and the four are the headline ones. `RPT02` and `RPT03` are faithful reprints of generated sheets, so they show the shape of the metamodel rather than the state of the estate. `RPT06` copies a generated traceability column. `RPT07` has two rows. `RPT10` contradicts itself in adjacent columns. `RPT12` adds nothing to `O32` . And `RPT13` , the one report that genuinely computes, understates its own answer by a factor of about twenty — which makes it the most useful of the thirteen to an architect and the least useful to a board.

> **ONE RULE FOR EVERY REPORT IN THIS FILE**
>
> Do not present a number off a report sheet until you have recomputed it from the object sheets. Six footers carry a number; four are right and two are wrong, and nothing on the face of the sheet distinguishes them. Put the recomputed figure in the pack and the sheet reference in the footnote.

THE CHAPTER IN TEN LINES

- Nothing in any of the thirteen reports is a formula. Every cell is a literal.
- Eight reports are column-subset reprints of one object sheet. Three reach outside one sheet, not one.
- RPT03 computes Gap as Target minus Current, the only computed column outside a footer.
- RPT06 joins O06 to R09, RPT07 joins O29 to R12, and RPT13 performs a genuine anti-join.
- Six footers carry a number and all six are typed. Four are right, RPT08 and RPT10 are wrong.
- RPT10 repeats O27's typed Position, so its four past-support components are really two.
- RPT08 reports four packages past their end date in a programme whose earliest start is 2027-02.
- RPT06 copies R09's generated Traced column, which reads No on every fourth row although all 31 rows name a block.
- RPT13 finds two orphan applications. Thirty-three of sixty-five object sheets are reached by nothing.
- The board set is RPT04, RPT05, RPT11, a rebuilt RPT09 and a corrected RPT08.

SOURCES FOR THIS CHAPTER

- **1.** Manufacturing_TOGAF_Metamodel.xlsx, sheet 00 Index
- **2.** Manufacturing_TOGAF_Metamodel.xlsx, report sheets RPT01 to RPT13 and the object and relationship sheets behind them, read with openpyxl on 16 October 2026
- **3.** Volume 1, chapter 16, Regulation, risk and what the inspector asks for — compulsory specifications, letters of authority and the instruments that inspect a South African plant

APPENDIX C

Workshop packs, checklists and the ninety-day calendar

Five packs to run, thirteen weeks to run them in, and the checks that change an engagement.

Five workshop packs, a thirteen-week calendar and three checklists. Run the packs in the order they appear: each one produces the input the next one needs, and the first two produce the objects the repository cannot name at all.

Every pack has the same shape — the output, the room, the ground rules, the rule or scale being applied, a worked example from the Kestrel repository, a running order with real timings, a capture grid, the three ways it derails and what gets signed. Kestrel Industrial Group is an invention, and so is Karoo Industrial Group in volume 1 [1].

Two standing rules before any of them. **Count, do not ask** — "yes, we have one item master" is an assertion; two extracts and a comparison is a count [2]. And **the output of a workshop is rows in a sheet, the same day.** A photograph of a flip chart is not a deliverable.

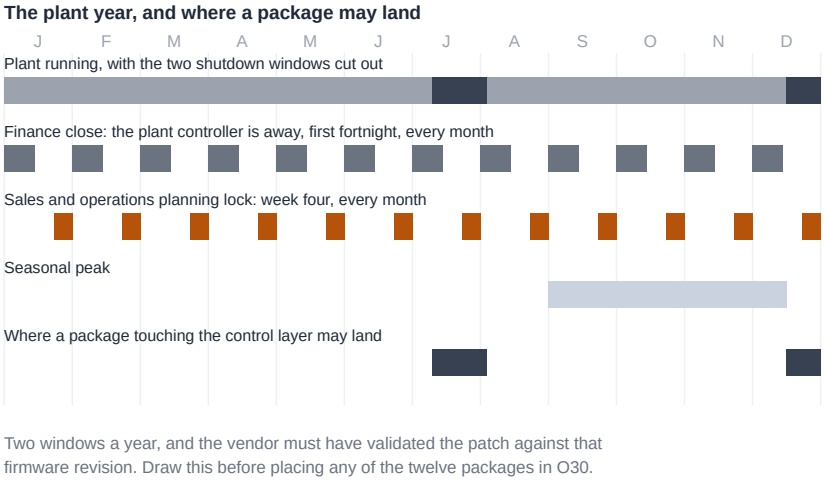


Figure 15.1 The plant year, drawn before any package is placed. **A package touching the control layer can land in two windows a year**, and the planning lock and the close remove most of what is left.

Pack 1 — The plant and line register

The output. Two things. A new sheet, `X02 Production Line`, because the type is declared in `EXT Industry Extension` and shadowed nowhere in the file. And four new columns on `O12 Location` — `Line count`, `Shift pattern`, `Format` and `Ownership` — because the sheet as shipped holds `Ref`, `Location`, `Type` and `Scale` and cannot say how many lines a plant runs or on what pattern it runs them [3].

The room. Per plant, and run it at the plant. The plant manager or works manager. The production manager. The maintenance planner. The industrial engineer. The detailed scheduler, with their spreadsheet open. One setter or line leader. Volume 1 chapter 5 has all six under **The plant**, **Engineering** and **Planning and supply chain**.

The ground rules. Two hours, one plant, no second plant in the room. The scheduler's file is evidence and the organisation chart is not. Nobody leaves without a number in every cell.

The rule being applied. *A production line is a sequence of work centres that can be started and stopped as a unit and that holds one changeover state at a time.* Rule it in the first ten minutes and apply it mechanically. Two lines that always run the same product at the same time are one line if they cannot be stopped separately, and two lines if they can.

The worked example. `O12 Location` holds eight rows: `LOC-01 Head Office` at 92 people, `LOC-02 Northern Plant` 198, `LOC-03 Coastal Plant` 259, `LOC-04 Components Plant` 111, `LOC-05 Central Distribution Centre` 85, `LOC-06 Regional Warehouse` 78, `LOC-07 Primary Data Centre` 181 and `LOC-08 Cloud Region` **261 people**. Three of the eight are `Production`, and not one column on the sheet says how many lines they run. Meanwhile `O27` places `PTC-02 MES application servers` at `Coastal Plant` and `PTC-03 SCADA control servers` at `Primary Data Centre`, while `R20` locates `A04 SCADA / Historian` at `Components Plant`. Three sheets, three answers, and no line anywhere.

Running order.

Minutes	What happens
0–10	State the rule for what counts as a line. Write it on the wall. No debate after minute ten
10–45	Count the lines, plant by plant, off the scheduler's file. One row per line
45–75	Shift pattern per line: crews, days, and whether the pattern is the same all year
75–100	Changeover families per line, and who sequences them
100–115	Name the constraint resource on each line, and say how you know
115–120	Read the rows back. Signatures

ASK THESE IN THE ROOM, BEFORE MINUTE TEN

What is the one sentence that decides whether two machines are one line or two lines here?

Which file is the real line list, and who maintains it?

Form C-1 — Production line register

Feeds a sheet the repository does not have. Create `X02 Production Line` with the columns below, in this order, and add `Line count`, `Shift pattern`, `Format` and `Ownership` to `O12 Location` the same day. A first pass needs one row per line: three to twelve rows per plant. Signed by the plant manager.

Ref	Production line	Location	Shift pattern	Changeover family	Constraint resource
LIN-001	EXAMPLE — delete this row	LOC-02	3 crew, 5 day	Coated / uncoated	

Form C-6 — Shift pattern and plant window grid

Feeds the four new columns on `O12 Location` and the `Shift pattern` column of `X02 Production Line`. One row per line, grouped by plant. A continuous line needs four or five crews, not three, once leave, training and absence cover are counted; record what is run, not what is established. Signed by the plant manager and the maintenance manager jointly.

Location	Line	Crews run	Days run	Shutdown window	Maintenance window
LOC-02	EXAMPLE — delete this row	3	Mon–Fri	Dec wk 3–4	Sun nights

Three ways it derails. The room spends an hour arguing about what a line is — which is why the rule goes on the wall at minute ten and is not reopened. Somebody offers the fixed-asset register as the line list, and it is not: it is the fourth tree over the same steel, kept for depreciation. And the plant manager says a number first and nobody contradicts it — so take the count from the scheduler's file, in front of everybody, and let the correction happen there.

What gets signed. The plant manager signs the line count and the shift pattern per plant. The `Domain Architect, Business` — `ROL-09` in `O11 Role` — countersigns that the rows are in the repository. Until all three plants are signed, no capacity, changeover or effectiveness work starts.

Pack 2 — The role register and the shift reality

The output. `O11 Role` and `O10 Actor`, rebuilt to hold the roles the plant actually performs, and the volume 1 interview order booked with names and dates against it.

The room. The plant manager. The production manager. **A shift supervisor from each shift, including night.** The maintenance manager and the maintenance planner. The quality manager. The industrial engineer. The master scheduler. Human resources, for the establishment. And the shop steward chair, in their own forum, before anything is written about measurement.

The ground rules. Three hours. A role is a job somebody is doing on a shift this week, **named as the plant names it** — scheduler, setter, millwright, not "the production scheduling resource". No role is recorded without a named person behind it or the word `Vacant`.

The rule being applied. `O11 Role` has four columns: `Ref`, `Role`, `Type`, `Performed by`. One row per role. `Performed by` is `Person` or `Vacant`, and `Type` is the function the role sits in — `Plant`, `Engineering`, `Maintenance`, `Quality`, `Planning`, `Data`, `Safety`. Those type values do not exist on the sheet today; add them.

The worked example. `O10 Actor` and `O11 Role` hold the **same eighteen names** — seven chiefs, the chief enterprise architect, four domain architects, three heads of function, a data steward, a security architect and an internal auditor. `R21 Role performed by Actor` joins them one-to-one and marks every row `Filled`, which is why `ZZ Checks` reports `Vacant architecture roles` as 0. There is no plant manager, production planner, scheduler, maintenance planner, reliability engineer, quality manager, process engineer, operator, setter or millwright anywhere in the file. The consequence is downstream: `O20.Owner` names `Head of Finance` as the owner of `E01 Product` and `Head of Quality Management` as the owner of `E03 Routing / Recipe`, and neither title resolves against `O11`. Only `Data Steward`, `ROL-16`, resolves at all.

Volume 1 chapter 5 carries the thirty-odd roles to put on the sheet: plant manager, production manager, shift superintendent, shift supervisor, line leader, operator, setter, materials handler, storeman; process engineer, industrial engineer, manufacturing engineer, design engineer, automation and controls engineer, project engineer; maintenance manager, maintenance planner, reliability engineer, millwright, electrician, instrument technician; quality manager and the laboratory; demand planner, supply planner, master scheduler, production planner, detailed scheduler, materials manager, buyer, expediter, logistics coordinator; master data steward, the item-master gatekeeper, the ERP functional lead, the execution-system administrator, the historian tag-list owner; the safety, health and environment officer, the environmental officer, the occupational health practitioner, the continuous improvement lead, the shop steward and the bargaining-council agent. Chapter 19's **The interview order** has the eleven to see and the order to see them in.

Running order.

Minutes	What happens
0–15	Read the eighteen rows of `O11` out loud. Let the room react. That reaction is the mandate
15–75	Build the role list by function, naming each role as the plant names it
75–110	Put a person or `Vacant` behind every row. Count the vacancies honestly
110–140	Identify which roles work shifts, and which shift. Night shift is a column, not a footnote
140–165	Book the eleven interviews with dates
165–180	Read back. Signatures. Separate note on anything touching individual measurement

Form C-2 — Role register

Feeds `O11 Role` directly — these are its four real columns, in sheet order, plus two working columns to transcribe into `O10 Actor` and the interview plan. A first pass runs to thirty to forty rows, so continue on a second copy of this form. Signed by the plant manager; accepted by the `Chief Enterprise Architect`, `ROL-08`.

Ref	Role	Type	Performed by	Shift	Interview date
ROL-19	EXAMPLE — delete this row	Plant	Person	Nights	

Three ways it derails. Human resources supplies the establishment list instead of the role list — the establishment is posts and grades, and a role is a job being done, and in a plant running structural overtime they are not the same set. The room gives you titles rather than the work behind them, which is answered by asking for the last three exceptions rather than for a description. And somebody proposes operator-level productivity measurement in the same session: **stop, park it, and take it to the steward forum**, because changes to work organisation are a consultation matter and a project that introduces individual measurement without consultation gets stopped late [1].

What gets signed. The plant manager signs the role list and the vacancy count. The shop steward chair acknowledges, in their own forum, the scope of what will and will not be measured. The `Chief Enterprise Architect` accepts the register into `O11` and `O10`, and `R21` is rebuilt from it rather than left as a diagonal.

Pack 3 — Master data and the golden source

The output. One master system per entity in `O20.Master system`, one `Golden source` per rule in `O43 Master Data Rule`, the two reconciled, and the word `CONTESTED` with a date wherever they are not. Plus a named individual in `O20.Owner` for every one of the fourteen entities.

The room. `GB-03 Data Governance Council` as `O57 Governance Body` defines it — the CDO, business data owners and stewards — plus five people who are not on it: the ERP functional lead, the product-lifecycle and engineering lead, the execution-system administrator, the historian tag-list owner and the item-master gatekeeper. The last of those is not a job title; find them by asking three people how to get an item code created and noticing whose name comes up twice.

The ground rules. Half a day. One entity at a time, in `O20` order. **An owner is a person, not a committee** — which is `PRN-13 Data Ownership is Individual`, and the sheet that states it carries `Architecture Board` as the owner of all 24 of its own rows.

The rule being applied. *A master system is the system in which a create is authorised. Every other system that holds the record is a consumer.* One master per entity. Two masters is a defect, not a design. Where the room cannot agree, the answer is `CONTESTED` plus a date and a named arbiter — which is a respectable outcome and far better than a cell that reads "both".

The worked example, and the opening exhibit. Put this on the screen first. `O20.Master system` says `E01 Product` is mastered in **`SCADA / Historian`**. `MDR-01` in `O43 Master Data Rule` agrees. A historian is an append-only time-series store; it cannot authorise the creation of an item, and `O39 Data Migration` then has `MIG-01` migrating `Product` *into* it. That is the only row on which the two sheets agree, and it is wrong on both.

Then the rest, which is the session's work:

Entity	O20.Master system	O43.Golden source	Verdict
`E01 Product`	SCADA / Historian	SCADA / Historian	Agreed and wrong
`E02 Bill of Material`	PLM	Track & Trace	
`E03 Routing / Recipe`	Plant Analytics	Supplier Portal	
`E04 Work Order`	APS	Supplier Portal	
`E05 Batch / Lot`	WMS	Supplier Portal	
`E06 Plant / Site`	APS	QMS	
`E07 Work Centre`	QMS	WMS	
`E08 Equipment / Asset`	SCADA / Historian	MES	
`E09 Supplier`	APS	QMS	
`E10 Purchase Order`	ERP Core	QMS	

Add two facts while the room is assembled. `APS`, a planning engine, is named as the master of `E06`, `E09` and `E11 Quality Specification`. And `E09 Supplier`, `E11` and `E13 Inventory Position` are all classified **`Public`**, while `E04 Work Order` and `E07 Work Centre` read `Restricted`. A quality specification is the testable definition of the product.

Running order.

Minutes	What happens
0–15	The `E01 Product` exhibit. Do not explain it; ask the room what it means
15–120	Fourteen entities, one at a time: master, owner, classification, lineage
120–150	Mark the contested rows, name an arbiter and a date for each
150–170	Walk `043`'s ten rules against the agreed masters and correct `Golden source`
170–180	Read back the contested list. Signatures

Form C-3 — Data entity ownership and mastering

Feeds `O20 Data Entity` — six of its eight real columns, in sheet order. `Class` and `Definition` are left to type straight into Excel, because `O20`'s definitions are the sound part of the sheet. Fourteen rows is the whole sheet. Signed by the chair of `GB-03 Data Governance Council`.

Ref	Data entity	Classification	Owner	Master system	Lineage
E01	EXAMPLE — delete this row	Confidential	Named person	ERP Core	Partial

ASK THE ROOM BEFORE THE ENTITY LIST STARTS

Who, by name, can create a product record today, and how many ways are there to do it?

For which entities is the honest answer `CONTESTED`, and who arbitrates each one?

Three ways it derails. The room settles on "both" — so write `CONTESTED`, a date and an arbiter, and move on; "both" is how the file reached nine disagreements without noticing. The room picks the system with the best reporting rather than the system that authorises the create, which is how a historian and an analytics platform ended up mastering an item and a routing. And the owner named

is a job title nobody holds: twelve of `O20`'s fourteen owners are titles that resolve to nothing in `O11`, so test each name against the register built in pack 2 before it is written down.

What gets signed. The `GB-03` chair signs the fourteen owners and the fourteen masters. `DEC-04 Designate data owners` in `O60 Governance Decision` moves off `Outstanding`, and the contested list goes to `GB-04 Executive Committee`, whose mandate in `O57` is the escalation point when bodies cannot agree.

Pack 4 — The measure definition register

The output. A measure definition register, and `004 Measure` extended to hold it. Volume 1 chapter 7 makes this the first deliverable before any dashboard, under **So the first deliverable is a measure definition register**, and chapter 19 repeats it in **What to count before you model anything**.

The room. The operations director or group manufacturing director. The plant controller. Each plant's production manager. The industrial engineer. The historian tag-list owner. The reporting lead. Nobody from the dashboard project.

The ground rules. Two sessions of two hours, one measure family each. **No measure leaves the room without six things:** an owner who is a person, a formula, a named denominator, the reason-code tree it depends on, the systems it is computed from, and a declared comparability scope — within a line, within a plant, across plants, or not at all.

The rule being applied, and the content that makes it necessary. `004 Measure` has `Ref`, `Measure`, `Unit`, `Domain`, `Baseline` and `Cadence`. It has **no definition, no formula, no denominator and no owner**, and not one of its sixteen rows is a manufacturing measure: no effectiveness, no yield, no on-time-in-full. `MSR-13 Recovery objectives tested` carries a baseline of 21 against twelve rows in `O49 Resilience Tier`.

Four facts to state out loud, because they are what the register is for [4][5].

1. **There is no standard for overall equipment effectiveness.** The publisher of the most-used definition says so itself.
2. **The numerator is uncontested and the denominator is not.** Volume 1's research found six conventions for what run time is divided by, and a seventh divergence in where the micro-stop threshold sits. Two physically identical plants can report availability thirty points apart, neither of them lying.
3. **The micro-stop threshold moves loss between availability and performance without changing effectiveness.** So two plants with identical behaviour and different thresholds publish different availability, different performance and the same headline number. If a group dashboard draws the three components side by side and the thresholds are not harmonised, those columns are not a comparison.
4. **Effectiveness is documented as unsuitable for comparing machines, for benchmarking and for aggregating to department or plant level.** A line is a serial system governed by its constraint, so a line's figure is not the mean of its stations'. Compute on the constraint, or on the line treated as one resource, and say which you did.

NEVER PUT THE 85 PER CENT FIGURE ON A SLIDE

The site that publishes the most-used definition calls it a convenient, compelling and completely artificial benchmark, and the general reference treatment says treating it as the mark of excellence is in many cases incorrect [4][5]. Put it in a deliverable and it will be quoted back as a target, and a plant will be measured against a number somebody made up. Set incremental targets from the plant's own baseline instead. One terminology correction while you are there: the relevant international standard's middle term is **effectiveness**, not performance, its published edition dates from 2014 and a second edition was still at draft stage as at October 2026 [6].

Running order, per session.

Minutes	What happens
0–15	Read `004` 's sixteen rows. Ask which of them a plant manager has ever used
15–30	State the six things a measure needs. Write them on the wall
30–90	One measure at a time. Stop at the denominator and do not move until it is named
90–105	Declare the comparability scope for each. "Not at all" is an answer
105–120	Read back. Signatures. Note the measures that could not be defined

Form C-4 — Measure definition register

Feeds `O04 Measure` for the first three columns and **a sheet the repository does not have** for the rest. Add `Definition`, `Formula`, `Denominator convention`, `Reason-code tree`, `Computed from` and `Comparability scope` to `O04`, or create `O04a Measure Definition` with `Ref`, `Measure`, `Owner`, `Formula`, `Denominator convention`, `Comparability scope`. A first pass covers eight to twelve measures. Signed by the operations director.

Ref	Measure	Unit	Owner	Denominator convention	Comparability scope
MSR-17	EXAMPLE — delete this row	%	Named person	Planned production time	Within a line

Three ways it derails. The room asks for the dashboard instead, twice — decline it twice, and the third request will not come. Two plants discover their availability denominators differ and the session becomes an argument about who was right: rule that neither is, that the convention is a choice, and that the choice is the deliverable. And somebody writes 85 per cent on the flip chart.

What gets signed. The operations director signs the register and the comparability scope per measure. `GB-03 Data Governance Council` records it. Nothing is built over an undefined measure, and the list of measures that could not be defined goes in the pack as a finding, not as an omission.

Pack 5 — Roadmap sequencing against the plant year

The output. `O30 Work Package`'s `Start` and `End` re-dated against the plant calendar. `R15 Transition contains Work Pa` completed, so no package sits outside a transition state. `O52 Implementation Factor` given dates. And the arithmetic reconciled or explicitly flagged.

The room. The operations director. Every plant manager. The maintenance manager, with the shutdown calendar. The master scheduler, with the planning fences. The project engineer, with the capital plan for the next twenty-four months. The head of delivery. The finance business partner.

The ground rules. Half a day, and **draw the plant year before you touch a package.** No package is placed on a calendar quarter. Two hard constraints: a package that touches the control layer is scheduled in a shutdown window and nowhere else, and a package that touches the planning engine is scheduled outside the sales and operations planning lock.

The rule being applied. The plant year has four features and they are not negotiable. The **shutdown windows**, which are when operational technology is patched — the vendor has to have validated the patch against that firmware revision, and the plant has to be down. The **maintenance windows** between shutdowns, which the maintenance manager and production fight over already. The **sales and operations planning lock**, a monthly cycle that occupies a week at executive level and the whole month at working level, with the plant controller unavailable for the first fortnight because of the close. And the **seasonal peak**, which is a physical constraint on a plant, not a trading freeze.

The worked example. `O30` holds twelve packages summing to **$65.0m**, against `ALT-B Foundations first` at **$46m** in `O54 Architecture Alternative`, which is the `Recommended` alternative — $19m unexplained, and `ALT-E Compliance led` at $31m marked `Partly adopted` with no decision in `O60` recording it. `R15` places seven packages in three transition states and leaves `WP-06`, `WP-08`, `WP-09`, `WP-10` and `WP-12` — **$26.1m** — in no state at all. `O53`'s `Exit criteria` column reads "5 exit criteria", "6 exit criteria", "7 exit criteria", which is a count where a criterion belongs. And `O52` names eight factors, of which `IF-08 Site access windows` is the one that matters — *"Physical work limited to shutdowns"* — with no dates on it, while `IF-03` is called `Peak trading freeze`, which is retail vocabulary and not a plant constraint.

Running order.

Minutes	What happens
0–30	Draw the plant year: shutdowns, maintenance windows, the planning lock, the peak
30–60	Classify the twelve packages: control layer, planning engine, corporate, or none of the three
60–120	Place them on the plant year. Record what will not fit and why
120–150	Assign every package to a transition state. Five currently have none
150–170	Date the dependencies in `O52`, and write the exit criteria `O53` is missing
170–180	Read the $65.0m against the $46m out loud. Signatures

Form C-5 — Work package sequencing

Feeds `O30 Work Package` for the first six columns, in sheet order, plus the transition state that `R15` holds. `Domains`, `Approval` and `Delivery` are typed straight into Excel. Twelve rows is the whole programme. Signed by `GB-02 Investment Committee`.

Ref	Work package	Cost	Start	End	Transition
WP-13	EXAMPLE — delete this row	$2.4m	2027-05	2028-03	TA-1

Three ways it derails. The roadmap is built on calendar quarters and then "validated" against the plant year, which produces a plan that moves twice — build it on the plant year from the first minute. The capital plan is not in the room, so a new line commissions in the same quarter as the rollout and one of the two slips. And the arithmetic is left to be fixed later: the $19m gap goes on the first slide or it goes to the committee unnoticed.

What gets signed. `GB-02 Investment Committee` signs the dates and the cost reconciliation. Its mandate in `O57` is G0 funding and G3 benefit confirmation, so this is the body with the authority. `DEC-01 Adopt the recommended architecture alternative` and `DEC-02 Select the integration product` both move off `Outstanding`, or the reason they cannot is minuted.

The ninety-day calendar

Thirteen weeks, built on volume 1 chapter 19's **The ninety-day shape** and the step order of this workbook.

Week	What is done	Workbook steps	Forms	Presented to
1	Plant walk, every plant, on shift, including a night shift. No modelling	1.1–1.3	none	nobody
2	Interviews 1–6 from the order. Photograph the paper	1.4, 2.1	none	verbal readout, five minutes, no slides
3	Pack 1 at each plant. The ten afternoon checks begin	2.2–2.5	C-1	plant managers
4	Pack 2. Interviews 7–11. Checks 1, 2, 3 closed	4.1–4.4	C-2	`GB-05 Architecture Function`
5	Checks 4–7 closed. `O12`, `X02`, `O10`, `O11` loaded	4.5–4.8	C-1, C-2	nobody
6	Checks 8–10 closed. The counted baseline written	5.1–5.3	none	**`GB-04 Executive Committee`: the counted baseline, one page per count**
7	Pack 3. `O20` and `O43` reconciled or marked `CONTESTED`	5.4–5.8	C-3	`GB-03 Data Governance Council`
8	Pack 4, session one. Application and interface baseline from `R06`	6.1–6.5	C-4	nobody
9	Pack 4, session two. Technology baseline, `O27` relocated and re-dated	7.1–7.5	C-4	`GB-01 Architecture Board`
10	Findings written: six to eight, each with evidence and a named quantifier	8.1–8.3	none	**`GB-04`: the findings, with exposures the business will quantify**
11	Target architecture to the depth the decisions need, no further	8.4–8.7	none	`GB-01`
12	Pack 5. Three transition states with written, verifiable exit criteria	9.1–9.5	C-5	`GB-02 Investment Committee`
13	The decision page. One page of what only the executive can decide	9.6, 10.1	none	**`GB-04`: target, transitions, roadmap, decisions, and what is not known**

Now the honest arithmetic, because thirteen weeks is not thirteen working weeks. Start in a month with a shutdown in it and the plant is unavailable for one to two weeks of it — not partly unavailable, unavailable, because during a shutdown every person on the list above is on the floor and will not sit in a workshop. Add the close: the plant controller is unavailable for the first fortnight of every month, which removes them from six of the thirteen weeks. Add the sales and operations planning cycle, which occupies the executive for one week in four. Add South African public holidays, which cluster

in March, April and late April. **The realistic figure is nine to ten weeks of genuine access out of thirteen**, and it is not evenly spread.

Three consequences. Protect weeks 1 to 3 absolutely: the plant walk and pack 1 cannot be rescheduled, because everything else depends on them. Put the three `GB-04` readouts in weeks 6, 10 and 13 and book them in week 1, before anybody's diary fills. And when the sponsor asks for the target architecture in week 4, the answer is that you have not yet seen the second plant.

The checklists

The ten afternoon checks

From volume 1 chapter 19, **What to count before you model anything**. Each takes an afternoon or less, and each either finds a defect that invalidates a design decision or removes a risk you would otherwise carry for a year [2]. The right-hand column says what the Kestrel repository already shows, which is where to start on a real one.

#	The check	What this repository already shows
1	Does the item master exist once, and do the instances agree?	`E01 Product` is mastered in `SCADA / Historian` on both `O20` and `O43`
2	Is there a measure definition register?	No. `O04` has no definition, formula, denominator or owner column
3	Is the time fence shorter than the cumulative material lead time?	Nothing in the file records a fence. Check it in the planning configuration
4	Does the system overwrite the promised date on reschedule?	Not recorded. `E14 Customer Order` has `Lineage` = `Absent`
5	Does the plant backflush?	Not recorded, and `E05 Batch / Lot` lineage is `Absent`, which is where genealogy would be
6	Is ownership an attribute of inventory?	No. `E13 Inventory Position` has no ownership column and reads `Public`
7	Are scrap factors applied as one over one minus s?	Not recorded. `E02 Bill of Material` has no scrap or yield attribute in `O37`
8	Do the equipment model and the plant hierarchy reconcile?	Cannot be asked. There is no line, and `O12` has eight rows with no structure
9	What are the historian's exception and compression settings?	Not recorded, and `A04 SCADA / Historian` is the only `Critical` application
10	What does the retention policy say, and which clocks does it breach?	No retention column anywhere. `O48` records personal data outside production

A pre-gate checklist, one per gate

`O58 Governance Gate` holds four gates and no holder column; the holder is in `O57`'s mandate text. Every line below is testable against a sheet.

Gate	Held by	Pass when
GT-0 G0 Charter	GB-02 for funding, GB-01 for scope	A row exists in O33 Architecture Contract against this package; acceptance criteria are written in O55; a transition state in R15 contains it; the cost in O30 matches O56
GT-1 G1 Design	GB-01 Architecture Board	Every O05 principle the design touches is Approved, not Draft; every O32 standard that applies is named with its conformance; a CR- dispensation exists for each departure, with an Expires date
GT-2 G2 Build complete	GB-01 Architecture Board	A row in O59 Compliance Assessment for every acceptance criterion, with a Verdict and Evidence; no Breach left unrecorded in O32; R06 shows no new unmanaged or unlogged interface
GT-3 G3 Benefit	GB-02 Investment Committee	The measure in O04 has a definition, an owner and a baseline; the benefit is stated against a date; the O49 tier has been tested; the decision is recorded in O60 with a date

Three of those lines cannot be met today. O59 is empty, so GT-2 has nothing to assess against. O05 has Draft on four rows and no Approved column, so GT-1 cannot be tested. And O60 has no date column, so GT-3 cannot record when the benefit was confirmed. Fix the three columns before the first gate, not after it.

The repository health checklist

`ZZ Checks` holds fifteen checks and a closing note. Four of the fifteen are wrong, and they are marked. Run this on the client's own file in week 5, and again before each `GB-04` readout.

#	Check	Stated	Recomputed
1	Object types populated	65	**64.** `059` is empty
2	Records	843	**813.** The difference is exactly `R06`'s 30 rows
3	Relationship types	25	25
4	Relationship instances	430	430, exactly
5	Reports	13	13
6	Interfaces unmanaged	22	22
7	Interfaces with no logging	21	21
8	Direct database reads	4	4
9	Platform components past support	4	**2.** Two of the four support ends are future
10	Security controls absent	1	1, `SEC-02`
11	Control objectives without evidence	5	5 of 6
12	Data entities without lineage	6	6 of 14
13	Vacant architecture roles	0	0, true only because `R21` is a diagonal of `Filled`
14	Work packages past end date	4	**0.** Nothing has started
15	Records validated against a real organisation	0	0

Six checks the sheet does not carry, and should. Object sheets reached by no relationship row — 33 of 65 on this file. Applications with no architecture contract — four packages, $24.5m. Resilience tiers never tested — seven of twelve. Entities whose owner resolves to a role in `011` — two of fourteen. Work packages in no transition state — five, $26.1m. And rows changed since the last baseline, which cannot be computed at all, because there is no version column and no change log.

Form C-7 — Repository health run sheet

Run the fifteen `ZZ Checks` lines plus the six missing ones, recompute each from the object sheets, and date the run. Twenty-one rows, so continue on a second copy of this form. Signed by the `Chief Enterprise Architect`, `ROL-08`, before each `GB-04` readout.

#	Check	Stated	Recomputed	Date run	By
1	EXAMPLE — delete this row	65	64	2026-11-06	

THE LAST PAGE, BEFORE YOU HAND ANYTHING OVER

Which three numbers in your pack have you recomputed yourself, and which three are you quoting?

Which of the five packs did you not run, and what are you building on instead of its output?

What is the one thing you do not know, written in a sentence you would read out to the executive committee?

THE CHAPTER IN TEN LINES

- Run the five packs in order: each produces the input the next one needs.
- Pack 1 builds X02 Production Line, which the repository declares and shadows nowhere, and four new columns on O12 Location.
- Pack 2 rebuilds O10 Actor and O11 Role, which today hold the same eighteen corporate titles and no plant role.
- Pack 3 reconciles O20 Master system against O43 Golden source, which disagree on nine of ten entities. CONTESTED plus a date is a respectable answer.
- Pack 4 produces the measure definition register, which comes before any dashboard. Never put the 85 per cent figure on a slide.
- Pack 5 sequences the roadmap on the plant year, reconciles 65.0 million against 46 million and places the five packages in no transition state.
- Every measure needs an owner, a formula, a named denominator, a reason-code tree, the systems it is computed from and a declared comparability scope.
- Thirteen weeks started in a month with a shutdown in it gives nine or ten weeks of real access. Protect weeks 1 to 3 and book the three readouts in week 1.
- Three gate lines cannot be met today: O59 is empty, O05 has no Approved column and O60 has no date column.
- Four of the fifteen lines in ZZ Checks are wrong, and six checks it should carry are absent.

SOURCES FOR THIS CHAPTER

- **1.** Volume 1, chapter 5, Who does what — the cast list, the interview order and the consultation obligations
- **2.** Volume 1, chapter 19, Your first ninety days — the ten afternoon checks and the ninety-day shape
- **3.** Manufacturing_TOGAF_Metamodel.xlsx, sheets O12 Location, O20 Data Entity, O27 Physical Technology Compone, O30 Work Package, O43 Master Data Rule, O49 Resilience Tier, O52 to O58, O60, R15 and ZZ Checks, read 16 October 2026
- **4.** Vorne Industries, OEE FAQ — the availability denominator conventions and the publisher's own description of the 85 per cent figure as artificial https://www.oee.com/faq/
- **5.** Wikipedia, Overall equipment effectiveness — the calculation conventions, the micro-stop threshold and the documented unsuitability for comparison, benchmarking and aggregation https://en.wikipedia.org/wiki/Overall_equipment_effectiveness
- **6.** ISO, ISO/DIS 22400-2 — second edition at draft stage as at 16 October 2026; ISO 22400-2:2014 remains the published part https://www.iso.org/standard/87563.html

APPENDIX D

The repository in ArchiMate 4

Sheet by sheet, what becomes an element, what becomes a relationship, and what the notation cannot hold at all.

This appendix is the authority on how the Kestrel Industrial Group repository maps onto ArchiMate 4. Where another chapter of this workbook says "draw it", this is the chapter that says what to draw. Kestrel Industrial Group is an invention, and so is Karoo Industrial Group in volume 1; their data must never be mixed.

Volume 1 chapter 17, *Modelling manufacturing in ArchiMate*, teaches the notation against the industry [4]. This appendix is narrower: it takes the 64 object sheets, the `EXT Industry Extension` register and the 25 relationship sheets you actually have [1], and says for each one what element or relationship it becomes, where the fit is approximate, and where the honest answer is that it does not map at all. Twenty-six of the 64 object sheets have no clean equivalent. That is the boundary between the repository and the notation, and knowing where it runs is what stops you drawing nonsense.

The ground rules, stated once

ArchiMate 4 gives you **42 elements across seven domains** — generic, strategy, business, application, technology, motivation, and implementation and migration — plus **11 relationships** and junctions [2] [3]. Nothing else. Four rules follow.

The element is `Role`, with no layer word in front of it. ArchiMate 4 has no business-layer variant of the role element, and a tool that offers you one is not ArchiMate 4. `Role`, `Collaboration`, `Process`, `Function`, `Service`, `Event`, `Location`, `Path` and `Grouping` are generic: they are not tied to a layer, and what layer they read as comes from what they connect to. That collapses four sheets — `O15`, `O24`, `O25` and `O28` — onto one element.

There is no Constraint element. Express a constraint as a `Requirement`. This matters because `O07 Constraint` is a whole sheet, eight rows of it.

Relationships carry no properties. You cannot put a quantity, a date, a cardinality or a scrap factor on an edge. Every time the repository states a fact about a pairing — `R09`'s `Traced`, `R18`'s `Conformance`, `R06`'s `Managed` and `Logged` — that fact must be held as a property of one end, or outside the model.

An element is a type, not an instance. ArchiMate models the kinds of thing a plant has, not the individual things. This is what makes `X04 Batch / Lot` undrawable, and it is the one rule architects break most often on a plant floor.

The object mapping

Six tables, one per domain group. The verdict column reads `Clean` where the element means the same thing the sheet means, `Approx.` where a different concept is pressed into service and something stated on the sheet is lost, and `Absent` where the notation holds nothing at all.

`O34` and `O51` do not exist. A loop from `O01` to `O66` fails on both holes. There are 64 object sheets, not 66. The two missing numbers sit between `O33 Architecture Contract` and `O35 Business Building Block`, and between `O50 Network Link` and `O52 Implementation Factor` — in both cases between sheets that belong to different domains. That pattern says export artefact, not real gap: two types were cut from the generator's list and the numbering was never closed up. Nothing in `00 Index`, `ZZ Checks` or any `R` sheet refers to either number. Treat them as absent by design and do not go looking for the rows.

Motivation and governance

Sheet	What it holds	ArchiMate 4	Verdict and note
`001 Driver`	12 pressures, with `Origin` and `Pressure`	`Driver`	Clean
`002 Goal`	6 corporate goals	`Goal`	Clean
`003 Objective`	14 dated objectives	`Outcome`	Clean. `Horizon` is a property
`004 Measure`	16 metrics with `Unit`, `Baseline`, `Cadence`	—	**Absent.** See below
`005 Principle`	24 principles with `Statement`	`Principle`	Clean. The sheet has no `Tier`, `Rationale` or `Review date`, so it cannot be tested at a gate
`006 Requirement`	31 requirements	`Requirement`	Clean
`007 Constraint`	8 constraints with `Type`	`Requirement`	Approx. No Constraint element; `Type` becomes a property
`008 Assumption`	6 assumptions with `Status`	`Assessment`	Approx. That it is unverified is lost
`032 Standard`	16 standards with `Conformance`	`Requirement`	Approx. The verdict is lost
`041 Security Control`	8 controls with `Position`	`Requirement` plus realising element	Approx. See the `Absent` pattern below
`052 Implementation Factor`	8 factors across five categories	`Assessment` or `Requirement`	Approx. The `Dependency` rows have no element at all
`062 Delivery Risk`	8 risks, likelihood by impact	`Assessment`	Approx. The grading is a property
`063 Control Objective`	6 obligations and their objectives	`Driver` plus `Requirement`	Approx. One row becomes two elements
`064 Security Policy`	4 policies with `Position`	`Principle`	Approx. `Position` is lost
`065 Enforcement Point`	5 points	`Node`, `Process` or `Grouping` by row	Approx. The sheet mixes kinds

`041`'s `Position` column is the sheet's whole point and the notation has no cell for it — but one value draws beautifully. **An `Absent` control is a `Requirement` with no incoming `Realization`.** Nothing realises it, and on a requirements realization view the hole is visible without a report. `SEC-02 Design review` is the one row that reads `Absent`, and `ZZ Checks` counts it correctly as 1.

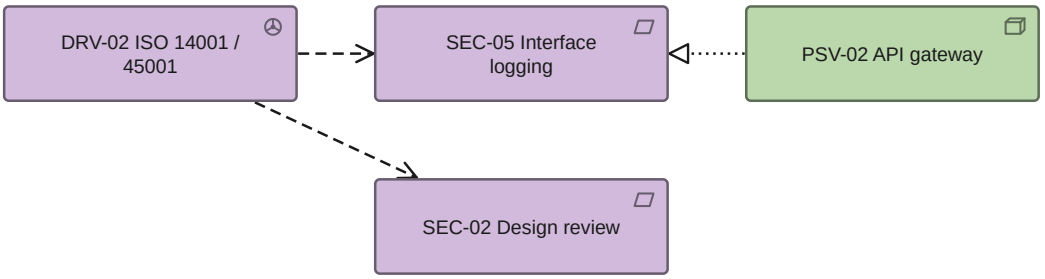


Figure 16.1 An `Absent` control is a `Requirement` with no incoming `Realization`. `SEC-05 Interface logging` is realised by the gateway that enforces it. `SEC-02 Design review` is realised by nothing, and the hole is visible on the view without a report. This is the only value in `O41 Security Control`'s `Position` column that the notation can express at all.

Strategy

Sheet	What it holds	ArchiMate 4	Verdict and note
`O13 Business Capability`	100 rows over three levels	`Capability`	Clean. `Current`, `Target`, `Criticality` and `Disposition` are properties
`O14 Value Stream`	5 streams	`Value Stream`	Clean. `Stages` is a count, not stage names, so the stages themselves cannot be drawn from this sheet
`O54 Architecture Alternative`	5 alternatives with `Cost`	`Course of Action`	Approx. Cost and the recommendation are properties

Business

Sheet	What it holds	ArchiMate 4	Verdict and note
`009 Organization Unit`	12 units with `Head`, `Headcount`	`Business Actor`	Clean
`010 Actor`	18 named positions	`Business Actor`	Clean, and duplicates `011` name for name
`011 Role`	the same 18 names	`Role`	Clean. The element is `Role`
`012 Location`	8 locations with `Scale`	`Location`, and `Facility` for the four plants	Clean. No `Line count`, `Shift pattern` or `Ownership` column exists
`015 Business Service`	10 services	`Service`	Clean
`016 Business Process`	20 processes with `Automation`	`Process`	Clean
`017 Business Function`	10 functions	`Function`	Clean, and one-for-one with `013` level 1
`018 Product`	4 products	`Product`	Clean
`019 Contract`	10 vendor agreements with `Expiry`	`Business Object`	Approx. **No Contract element.** Counterparty and expiry become properties; enforceability is lost
`035 Business Building Block`	10 blocks with acceptance criteria	`Grouping`	Clean. The criteria become `Requirement` elements the grouping aggregates

Application and data

Sheet	What it holds	ArchiMate 4	Verdict and note
`020 Data Entity`	14 conceptual entities	`Business Object`	Clean. Conceptual, so business layer, not `Data Object`
`021 Logical Data Component`	12 logical stores	`Data Object`	Clean
`022 Physical Data Component`	12 databases with `Type`	`Artifact`	Clean. See the historian rule below
`023 Application Component`	12 applications	`Application Component`	Clean
`024 Information System Service`	12 services with `Style`	`Service`, or `Application Interface` where `Style` is `API`	Clean
`036 Data Entity Relationship`	11 rows with `Cardinality`	—	**Absent.** It is a relationship, not an object
`037 Data Attribute`	60 attributes with `Key`, `Classification`	—	**Absent.** Attributes are properties
`038 Data Set`	6 curated sets	`Grouping` over `Business Object`s	Clean
`039 Data Migration`	6 movements with `Status`	—	**Absent.** A movement is a `Flow` between two components
`040 Source To Target Mapping`	12 field rules	—	**Absent.** A transformation rule is not a construct
`042 Metadata Element`	14 metadata kinds with `Coverage`	—	**Absent.** Metadata about the model is not in the model
`043 Master Data Rule`	10 rules with `Golden source`	`Requirement` on the mastering component	Approx. Survivorship is policy; the rule logic does not survive
`044 Data Building Block`	10 blocks	`Grouping` over `Business Object`s	Clean
`045 Application Building Block`	10 blocks	`Grouping` over `Application Component`s	Clean

Technology

Sheet	What it holds	ArchiMate 4	Verdict and note
`025 Technology Service`	12 services with `Category`	`Service`	Clean. `Category` is wrong on at least four rows and is a property either way
`026 Logical Technology Componen`	7 logical components	`Node`, `System Software` or `Communication Network` by row	Clean per row. See the control layer below
`027 Physical Technology Compone`	12 components with `Class`	`Node`, `Device`, `System Software` or `Communication Network` by `Class`	Clean per row
`028 Platform Service`	8 platform services	`Service`	Clean
`046 Technology Building Block`	7 blocks, and a second `Ref` column	`Grouping` over `Node`s	Clean. The second column, `Ref (T)`, holds `T01` – `T07` and resolves to nothing
`047 Reference Model Entry`	11 entries with `Status`	`Grouping`	Clean. Adoption status is a property
`048 Environment`	4 environments	`Grouping` of `Node`s	Clean. `Personal data present` is a property, and it says production holds none while development does
`049 Resilience Tier`	12 rows: `Tier`, `Recovery objective`, `Tested`	—	**Absent.** All three are properties of the `Application Component`, so the sheet leaves the diagram entirely
`050 Network Link`	10 links with `Bandwidth`	`Path`, or `Communication Network` where it is a physical segment	Clean. `Diverse path` is a property

`049` deserves a sentence of its own, because an architect will try to draw it. Resilience is not a thing in a plant; it is a statement about a thing. Tier, recovery objective and test date belong in the `Application Component`'s property set, and the view that communicates them is a heatmap coloured by tier, not an ArchiMate view.

Implementation, migration and the governance record

Sheet	What it holds	ArchiMate 4	Verdict and note
`029 Gap`	2 gaps with `Severity`, `Due`	`Requirement` with no `Realization`	Approx. The implementation aspect holds `Work Package`, `Deliverable` and `Plateau` only — there is no Gap element
`030 Work Package`	12 packages with `Cost`, `Approval`, `Delivery`	`Work Package`	Clean
`031 Deliverable`	20 deliverables with `Phase`	`Deliverable`	Clean
`033 Architecture Contract`	8 contracts with `Compliance status`	`Deliverable` plus `Requirement`s	Approx. **No Contract element.** Signature and status are lost
`053 Transition Architecture`	3 transition states	`Plateau`	Clean. `Exit criteria` states a count where a criterion belongs
`055 Building Block`	37 rows over four domains	`Grouping`	Clean, and a duplicate. See below
`056 Project`	12 projects with `Lead`	`Work Package`	Approx. A project and a work package collapse to one element
`057 Governance Body`	5 bodies with `Membership`, `Mandate`	`Collaboration`	Clean
`058 Governance Gate`	4 gates `GT-0` – `GT-3`	`Process` plus `Assessment`	Approx. `When` and `What it tests` are properties; there is no holder column
`059 Compliance Assessment`	header row only, zero rows	`Assessment`	Clean, and empty
`060 Governance Decision`	7 decisions with `Body`, `Status`	—	**Absent.** The thing decided is already an element elsewhere; the decision is a property of it
`061 Governance Change Request`	5 requests with `Against`, `Expires`	—	**Absent.** A change request is a workflow record about the model
`066 Architecture Increment`	6 increments with `Lands`	—	**Absent.** No Increment element, and `Plateau` is taken

The count, and what to do with each group

Thirty-eight sheets map clean, sixteen need a substitution, and ten have nothing. **Twenty-six of 64 have no clean equivalent** — higher than the sibling volumes in this series, and the extra load falls in

three places: the measure register, the governance workflow sheets, and the control layer. Split the twenty-six two ways.

Those that can carry the fact another way. The element is named above, and something stated on the sheet is lost. `O07` becomes a `Requirement` and loses the constraint-versus-requirement distinction, which costs you at a gate: a constraint is not negotiable and a requirement is. `O19` and `O33` become a `Business Object` and a `Deliverable` and lose counterparty, expiry and signature — keep the contract register in Excel and let the model carry the obligation only. `O29` becomes a `Requirement` with no `Realization`, the best substitution in this list: severity and due date go, but the gap becomes visible on a view. `O41` becomes a `Requirement` plus whatever realises it. `O43` becomes a `Requirement` on the mastering component and loses the rule. `O52` splits across `Assessment` and `Requirement`, and its two `Dependency` rows have nothing. `O54` becomes a `Course of Action`. `O56` collapses into `Work Package`, which is honest: `R14` maps `PRJ-01` → `WP-01` one for one, so the two sheets hold one fact twice. `O58`, `O62`, `O63`, `O64` and `O65` land on `Process`, `Assessment`, `Driver`, `Principle` and `Node`, each losing a verdict column.

Those that stay in the repository because the notation has nothing for them. `O04 Measure`, `O36 Data Entity Relationship`, `O37 Data Attribute`, `O39 Data Migration`, `O40 Source To Target Mapping`, `O42 Metadata Element`, `O49 Resilience Tier`, `O60 Governance Decision`, `O61 Governance Change Request` and `O66 Architecture Increment`. Three of those are not objects at all — `O36` is a relationship, `O37` is a property set, `O39` is a `Flow` — and four are records *about* the model rather than *in* it. Do not try. Point the view at the sheet.

> `O04 Measure` is the one that matters in manufacturing, and it is worth being blunt. **There is no metric element in ArchiMate 4, and `Value` is not a number** — `Value` expresses worth to a stakeholder, not a measured quantity. So OEE, first-pass yield and on-time-in-full have no home in the notation at all. Not an approximate home: none. You cannot draw the baseline, the unit, the cadence or the calculation. `MSR-07 Components within vendor support` at a baseline of 42 and `MSR-11 Lineage coverage of regulatory reports` at 57 are the two numbers the whole programme is judged on, and neither can appear on any view except as text in a note.
>
> This is exactly why the measure definition register is a separate artefact, and why every measure needs a written definition that names the numerator, the denominator, the exclusions and the source sheet. Attach the measure to an `Outcome` as a property and keep the arithmetic where it can be reviewed.

`O55 Building Block` is a composed filter over `O35`, `O44`, `O45` and `O46`: 10 plus 10 plus 10 plus 7 is 37, and all 37 refs are duplicates of refs already on those four sheets. It has no reference space of its own. **Draw `O55` and every building block appears twice** — once from its domain sheet and once from the composed one — which is how a 37-element view becomes a 74-element view and a reader stops trusting it. Pick one source. If you are drawing by domain, use `O35`, `O44`, `O45` and `O46`. If you are drawing the whole programme, use `O55` and nothing else. `R13 Work Package delivers Build` carries 37 rows and resolves against both, so a careless join doubles that too.

The historian, and the rule about tags

`O22 Physical Data Component` maps cleanly to `Artifact`, and `PDC-04 SCADA / Historian database` is one of its twelve rows. The trap is not the mapping. It is what you do next.

A historian's content is a tag namespace: tens of thousands of named points, each a sensor, a setpoint, a calculated value or a status flag, each with its own type, scan rate and compression settings. An architect who has just learned that a database is an `Artifact` will reach for `Data Object` and start modelling tags. Do not. **A tag is not a data entity**, and a view full of tags is a view nobody can read — a single bottling line carries hundreds of them, and none of them is a business concept.

The rule: model the namespace, not the tags. One `Artifact` for the historian archive. One `Data Object` for the namespace as a governed structure — the naming convention, the asset-to-tag hierarchy, the units register. One `Service` for the query interface the analytics estate consumes. Then stop. The tag list is reference data and belongs in the historian's own asset framework, not in the architecture repository and not on a view.

Two things here make the point. `O37 Data Attribute` holds 60 attributes across 10 entities, six apiece from one list — `Identifier`, `Name`, `Status`, `Effective date`, `Owner`, `Classification`. That is the scale at which attribute modelling is affordable; a historian has three orders of magnitude more points. And `O22` types `PDC-04 SCADA / Historian database` as `Relational` while typing `PDC-05 APS database` as `Time series` — the two are swapped. A historian is an append-only time-series store; a planning engine is not. If you were modelling tags off that sheet you would be modelling them off the wrong storage shape.

The control layer

`O26 Logical Technology Componen` and `O27 Physical Technology Compone` are where a manufacturing repository earns its keep, and where a view most easily degrades into a network diagram. You have `Node`, `Device`, `System Software`, `Communication Network`, `Path`, `Service` and `Technology Interface`. That is enough, used with discipline.

`O27.Class` takes eight values. Map each:

Class	Rows	ArchiMate 4	Note
Historian	PTC-01	Node for the cluster, System Software for the product, Artifact for the archive	Three elements, one row
Application	PTC-02, PTC-04, PTC-05	Node	The hosting tier, not the application
Control	PTC-03	Node for the server, System Software for the SCADA product	The controllers below it are Device
End user	PTC-06	Device	Shop-floor terminals are physical and sited
Network	PTC-07	Device for the switch, Communication Network for the segment it forms	Two elements, two purposes
Database	PTC-08	Node plus System Software	
Storage	PTC-09, PTC-10	Node	Device where it is a fixed appliance
Compute	PTC-11, PTC-12	Node; Device for edge gateways fixed at the line	PTC-12 Edge gateways is the row where this choice matters

Now the five things a reader will actually be asked to draw.

A PLC is a `Device`. It is physical hardware that runs control logic on a fixed scan. Its program is an `Artifact` deployed on it; the scanning behaviour, if you need it, is a `Function` assigned to the device. Do not model a PLC as `System Software` and do not model one per rack slot.

A DCS is a `Node`. A distributed control system is not one box: it is controllers, operator stations and engineering stations that behave as one system. Draw the `Node`, aggregate the `Device`s that matter, and leave the rest out.

A SCADA server is a `Node` carrying `System Software`. The `Node` is the server or cluster; the `System Software` is the SCADA product. `PTC-03 SCADA control servers` is that pair. Note that `027` sites it at `Primary Data Centre`, which inverts physical reality — control servers sit at the plant they control — so correct the location before you draw it.

An OPC UA server is `System Software` exposing a `Technology Interface`. This is the one element in the control layer that earns an interface, because OPC UA is the published contract across the boundary. Everything else talks over a `Communication Network`; the OPC UA endpoint is the thing another team integrates against, so give it an interface and name it.

A historian is a `Node`, `System Software`, `Artifact` and `Service` — as above. The `Service` is what you put on a layered view; the rest stays on the technology view.

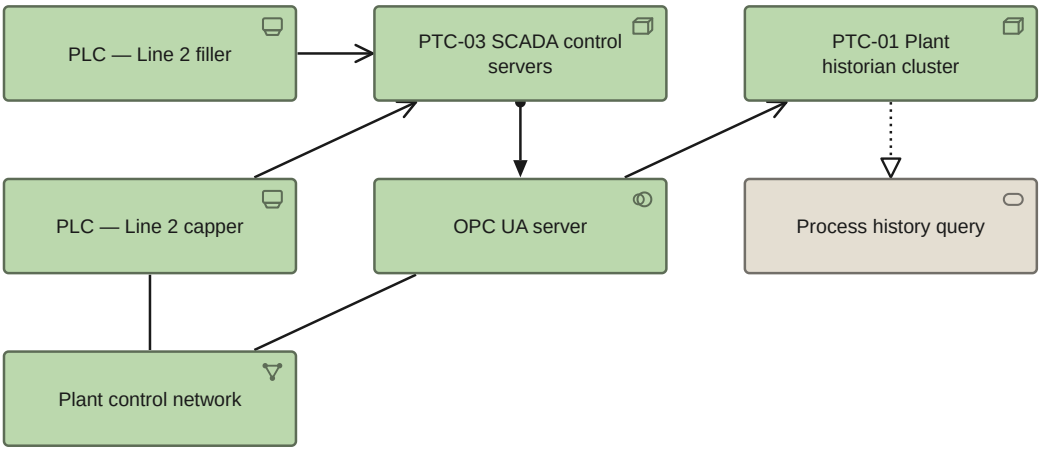


Figure 16.2 The control layer drawn with the elements the notation has. A PLC is a `Device`; the SCADA server is a `Node` carrying `System Software`; the OPC UA server is the one thing here that earns an interface, because it is the published contract; the historian is a `Node` whose archive is an `Artifact` and whose query interface is a `Service`. **Draw the service row, not the device row** — one network per zone, and no cables.

The rule that keeps it readable: draw the service row, not the device row. One `Communication Network` per zone, one `Path` per conduit, no cables, no port counts, no IP ranges. If a reader can count switches on your view, you have drawn the wrong thing. Volume 1 chapter 13 is the teaching half of all of this [5].

Zones and conduits

`TBB-01 OT/IT segmentation` exists as a building block on `O46` and `O55`, and `LTC-01 OT/IT segmentation` and `RME-01 OT/IT segmentation` repeat it on `O26` and `O47`. **There is no zone register anywhere in the file.** Four sheets say segmentation is a thing to build, and nothing says between what and what.

Draw it like this. **A zone is a `Grouping`.** Its border renders dashed, which reads correctly as a boundary rather than a container. **A conduit is a `Path`.** The elements that enforce the boundary — a broker in the demilitarised zone, a firewall pair, a data diode — are `Node`s or `Device`s that sit on the `Path` and realise the `Requirement` the zone exists for.

And then the honest note: **the notation was not designed for this.** A `Grouping` has no semantics. It constrains nothing, enforces nothing and is not checked by any tool. A zone drawn as a `Grouping` is a picture of an intention. The enforcement lives in a firewall rule base, in `O65 Enforcement Point`, and in a conduit register that this repository does not have. Draw the `Grouping` so the reader can see the boundary; keep the rule base, the allowed protocols and the direction of each flow in a document that an engineer will maintain.

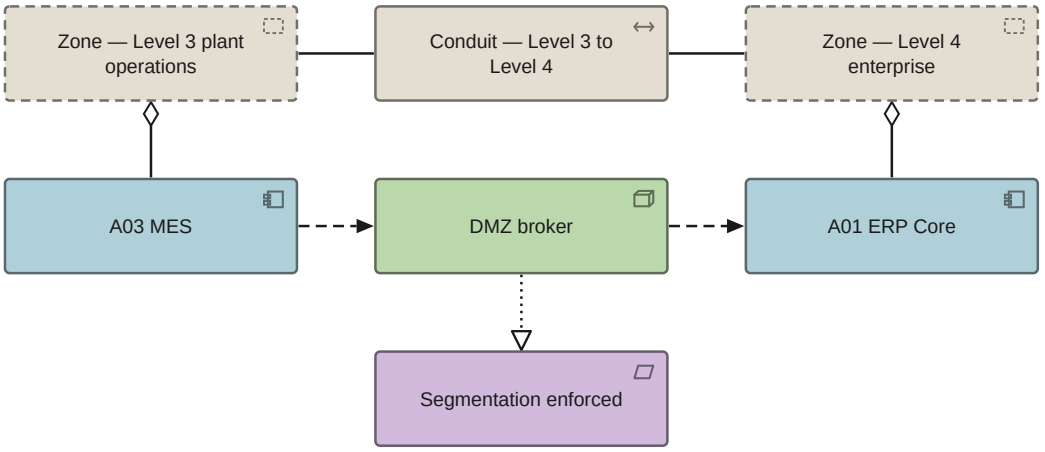


Figure 16.3 A zone is a `Grouping`, a conduit is a `Path`, and the element that enforces the boundary realises the `Requirement` the zone exists for. The honest note goes on the view's face: **a `Grouping` has no semantics** — it constrains nothing and no tool checks it. `TBB-01 OT/IT segmentation` exists on four sheets of this repository and there is no zone register behind any of them.

The relationship mapping

Twenty-five sheets. For each: the ArchiMate relationship, and whether the sheet's stated source-to-target direction reverses against the notation's. The reversals are not errors in the repository — they are the natural reading of a passive title such as "supported by" or "realised by". They become errors when you draw the arrow the way the columns run. Check this table before every view. It is split in two so the header repeats.

R01 to R13

Sheet	ArchiMate relationship	Reverses?	Note
R01 Capability decomposition	Composition	No	Parent to child; 90 rows, all generated
R02 Value stream uses Capabilit	Serving	**Yes**	Notation runs Capability to Value Stream. Aggregation the other way is defensible; pick one and keep it
R03 Capability realised by Busi	Realization	**Yes**	Notation runs Service to Capability. Positional diagonal, C01 → BSV-01 onward
R04 Business Service supported	Serving	**Yes**	Notation runs Application Component to Service. Tab name ends in a space
R05 Application manages Data En	Access	No	Write access for Master, read for Consumer. Two key schemes in one sheet
R06 Application interfaces with	Flow	No	30 rows, and the only Ref space on a relationship sheet
R07 Application deployed on Tec	Assignment	**Yes**	Notation runs Node to Application Component
R08 Requirement derives from Dr	Influence	**Yes**	Notation runs Driver to Requirement
R09 Requirement satisfied by Bu	Realization	**Yes**	Notation runs block to Requirement. The Traced column has nowhere to go
R10 Objective measured by Measu	—	**Cannot be expressed**	Measure is not an element
R11 Goal realised by Objective	Realization	**Yes**	Notation runs Outcome to Goal
R12 Gap closed by Work Package	Realization	**Yes**	Two rows only, against 37 building blocks
R13 Work Package delivers Build	Realization	No	37 rows; resolves against 055 and the four domain sheets both

R14 to R25

Sheet	ArchiMate relationship	Reverses?	Note
R14 Project delivers Work Packa	Composition	No	Parent Work Package to child. One for one down the sheet
R15 Transition contains Work Pa	Realization	**Yes**	A Plateau is a state; work realises it, it does not contain work
R16 Control Objective met by Co	Realization	**Yes**	Notation runs control to Requirement. Positional diagonal
R17 Policy enforced at Enforcem	Realization	**Yes**	Notation runs point to Principle
R18 Standard applies to Applica	Realization	**Yes**	Conformance runs from the conformant thing to the Requirement
R19 Data Entity relates to Data	Composition, Aggregation or Association by verb	**Mixed**	Reverses on the three is derived from rows; holds on has and is composed of
R20 Application located at Loca	Assignment	**Yes**	Notation runs Location to the element sited there
R21 Role performed by Actor	Assignment	**Yes**	Notation runs Business Actor to Role. 18 rows, all Filled
R22 Org Unit owns Capability	Assignment	No	Positional diagonal, so the owner is whoever shares a row number
R23 Increment delivered by Proj	—	**Cannot be expressed**	No Increment element
R24 Compliance Assessment tests	Association	Neutral	Direction-neutral, and the sheet is **empty**
R25 Application supports Produc	Serving	**Yes**	And the sheet's own title and column order disagree: the title says application to product, the columns run product to application

The count: fifteen of the twenty-five reverse outright, sixteen counting R19, seven hold, one is direction-neutral, and two cannot be expressed at all. Two further points of hygiene. R24 is **empty**, header row only, because 059 Compliance Assessment holds zero rows — so 00 Index's claim that every object type carries records is false, and it is 64 of 65. R12 is **nearly empty**, two rows against 37 building blocks.

The edges every architect needs, that exist on no sheet

This is the appendix's most actionable finding, and the Requirements Management chapter depends on it. Four edges that any requirements traceability view needs are not in the file — not as a relationship sheet, and in three of four cases not as an inline column either.

- **`Driver` to `Goal`.** `R08` runs requirement to driver and `R11` runs goal to objective, so the chain *driver → goal → objective → requirement* is broken at its first link. You cannot answer "which goal does `DRV-03 Product safety and CE / UKCA marking` serve?" from this repository.
- **`Requirement` to `Objective`.** No sheet. Traceability runs requirement to driver and requirement to building block, never requirement to objective — so nothing shows that a requirement moves a measure. Combined with `O04` having no element, the line from money to metric is cut twice.
- **`Role` to `Process`.** No sheet. `O11 Role` joins only to `O10 Actor`, through `R21`. Nothing says who performs `BPR-01` to `BPR-20`. In the notation that edge is `Assignment` from `Role` to `Process`, which is the most-used edge in business architecture, and it is absent. `O16 Business Process` appears in no relationship sheet at all: twenty processes with no edges.
- **`Application Component` to `Service`.** No sheet. `O24 Information System Service` carries an inline `Application` column, so the fact exists as a column but not as a row, which means no report can join on it. The notation edge is `Realization`.

The pattern is broader than those four. **Thirty-two of the 64 object sheets have no reference appearing in any relationship sheet.** Half the repository is unconnected: `O05 Principle`, `O07 Constraint`, `O16 Business Process`, `O17 Business Function`, `O19 Contract`, `O21`, `O22`, `O24`, `O25`, `O26`, `O28`, `O31 Deliverable`, `O33`, `O37` through `O43`, `O47` through `O50`, `O52`, `O54`, `O57`, `O58`, `O60`, `O61`, `O62`. Some are joined by an inline column, readable by eye and unjoinable by report. The rest are islands, and the missing edges are your Phase B and Phase C backlog.

The six `EXT` types

`EXT Industry Extension` has three columns, six rows and **no instances**. It is a register of types, not a data sheet: it declares that six manufacturing concepts need a home and does nothing about it. Five of the six are shadowed in `O20 Data Entity` — `X01` as `E06`, `X03` as `E02`, `X04` as `E05`, `X05` as `E11`, `X06` as `E09`. One is shadowed nowhere.

`X01 Plant / Site`

Elements. `Location` for the site as a place, `Facility` for the buildings and the plant itself. `O12 Location` already holds `LOC-02 Northern Plant`, `LOC-03 Coastal Plant` and `LOC-04 Components Plant`, so the register's first type is modellable without building anything.

What the notation cannot hold. Capacity. Shift pattern. Line count. The plant calendar. `O12` has no column for any of them either, which is the fact everything else in manufacturing turns on. `LOC-08 Cloud Region` is given 261 people, which tells you the `Scale` column is not a headcount in any useful sense.

Draw it? Yes. `Location` on an application-placement view, `Facility` on a plant-floor view. Never both on one view.

`X02 Production Line`

Elements. `Equipment` for the line, composing the `Equipment` that are its work centres, inside the plant's `Facility`, with a `Function` for what it does and a `Service` for the capacity it offers.

`X02` has no row anywhere in the repository. Not in `O12 Location`, which holds sites. Not in `O20 Data Entity`, where `E07 Work Centre` holds resources. A production line is neither a place nor a resource, and nothing in 813 records is one. **This is the most consequential absence in the file for a manufacturer**, and it is worth being exact about why: the line is the object OEE is measured on, the object a changeover belongs to, the object a capacity constraint lives on, and the object a maintenance window is negotiated against. Four of the questions a plant manager asks you in week one are about a line, and the repository cannot name one.

What the notation cannot hold. Rated rate. Changeover matrix. The product families it is qualified to run. Takt time.

Draw it? Yes — and build the sheet first. Any step that tells a reader to use the extension must tell them to create `X02`'s object sheet with its own `Ref` prefix before anything else.

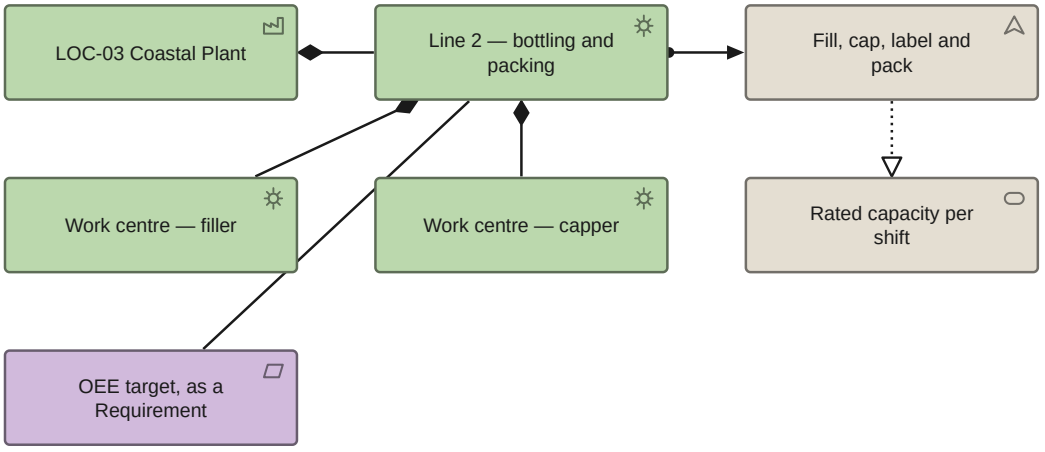


Figure 16.4 `X02 Production Line`, the type with no row anywhere in the repository, drawn as `Equipment` composing its work centres inside the plant's `Facility`, with a `Function` for what it does and a `Service` for the capacity it offers. **The OEE target can only appear as a `Requirement`**, because there is no metric element and `Value` is not a number — which is why the measure definition register is a separate artefact.

`X03 Bill of Material`

Elements. A structure over `Business Object`s: one `Business Object` per item, `Composition` from the parent item to each component. Shadowed as `E02`, mastered — correctly, for once — in `PLM`.

What the notation cannot hold. Four things, and all four are load-bearing. **Effectivity** — a component valid from one date to another — has nowhere to live, because relationships carry no properties. **Revision** — the structure at revision C against revision D — has nowhere to live for the same reason, and naming the element "Bracket rev C" turns one item into many. **A scrap factor on a line** — quantity-per plus yield loss — is a number on an edge, and there are no numbers on edges. **A**

phantom — a structural level that exists for the engineer and is never stocked — has no way to be marked as one.

Best approximation. Model the item types and the composition shape, put nothing numeric on the view, and state on the view's face that quantity-per, effectivity, revision and scrap live in the PLM and the ERP item master. The repository's own hint is `O37 : ATR-010 Bill of Material · Effective date` exists as an attribute, which is the right place for it and the wrong place to draw it from.

Draw it? A three-level example, yes, to show the shape. A real structure, never. Volume 1 chapter 9 carries the full treatment [6].

`X04 Batch / Lot`

Elements. `Business Object` for the type. That is all.

A genealogy is a graph of instances, not of types. Batch 24-A-0117 consumed lots of three raw materials, was split across two lines, and ended in eleven customer shipments. Those are individual things with individual identities, and **ArchiMate models types, so a genealogy is not drawable in ArchiMate at all.** This is not a judgement call or a question of taste: an instance graph has no representation in the language.

What to do instead. The view names the type — one `Business Object`, *Batch / Lot* — and points at the data model. `O36 Data Entity Relationship` and `R19` hold the same eleven rows and carry the shape, although `REL-05 Batch / Lot has Plant / Site 1:1` has the cardinality backwards. The recall simulation, which is the real test of a genealogy, runs against the Track & Trace database, not against a view.

Draw it? The type, once, on an information structure view. The genealogy, never.

`X05 Quality Specification`

Elements. `Business Object` for the specification, and a `Requirement` per characteristic where you need to show that a product conforms. Shadowed as `E11`.

What the notation cannot hold. The limits. A specification is a table of characteristics, methods, units, targets and tolerances, and none of that is modellable — it is content, held in the QMS.

And `O20` classifies it `Public`, which is wrong twice over. Commercially: it is the testable definition of the product, the most sensitive document a manufacturer owns, and it usually encodes a customer's requirements too. And as a compliance record: in a regulated plant the specification and the results against it are what an inspector asks for, and an inspection record is not public. `E11`'s `Owner` reads `Head of Quality Management`, which resolves against nothing in `O11 Role`, so there is also nobody to ask.

Draw it? Yes, as a `Business Object` on the information structure view, with its classification corrected first.

`X06 Supplier`

Elements. Two, and the sheet conflates them. `Business Actor` for the supplier as an organisation you contract with, and `Business Object` for the supplier master record you hold about it. The qualification is an `Assessment`; the qualification status is a property. Shadowed as `E09`, also classified `Public`, also mastered in `APS` — a planning engine, which cannot master a counterparty.

What the notation cannot hold. Approved-manufacturer status, the source list, the call-off structure, the audit date.

Draw it? Yes, both elements, and label them so a reader can see which is the company and which is the record.

> Three things an extension type needs before it is real:
>
> 1. **The `EXT` row** — the declaration that the concept exists and why the core metamodel is not enough.
> 2. **An `O` sheet with its own `Ref` prefix** — somewhere for the instances to live, with columns that hold the facts the concept turns on.
> 3. **At least one `R` sheet** — because an object with no relationships is an island, and 32 of the 64 object sheets already are.
>
> An `EXT` row alone changes nothing. All six rows in this file are declarations with no sheet and no relationship behind them, which is why five of the six quietly turned into `O20` rows and the sixth, `X02 Production Line`, turned into nothing.

THE CHAPTER IN TEN LINES

- ArchiMate 4 gives you 42 elements, 11 relationships and junctions. The element is Role; there is no Business Role and no Constraint element.
- Relationships carry no properties, and an element is a type, not an instance. Those two rules decide most of this appendix.
- O34 and O51 do not exist. There are 64 object sheets plus the EXT register, not 66.
- Thirty-eight object sheets map clean, sixteen need a substitution, ten have nothing. Twenty-six of 64 have no clean equivalent.
- O04 Measure has no element at all. There is no metric element and Value is not a number, so OEE, yield and on-time-in-full cannot be drawn.
- An Absent security control is a Requirement with no incoming Realization, and the hole shows on the view.
- Model the historian's namespace, not its tags. A tag is not a data entity and a view of tags is unreadable.
- A PLC is a Device, a DCS is a Node, a SCADA server is a Node carrying System Software, and only the OPC UA server earns an interface.
- A zone is a Grouping and a conduit is a Path, but a Grouping has no semantics, so keep the rule base outside the model.
- Fifteen of the 25 relationship sheets reverse direction against the notation, sixteen counting R19. Two cannot be expressed: R10 and R23.
- Driver to goal, requirement to objective, role to process and application to service exist on no sheet. Thirty-two of 64 object sheets are unconnected.
- X02 Production Line has no row anywhere. It is the object OEE, changeover and capacity all belong to, and it is the most consequential absence in the file.
- A batch genealogy is a graph of instances, so it is not drawable in ArchiMate. Name the type and point at the data model.
- An extension type needs three things: the EXT row, an O sheet with its own Ref prefix, and at least one R sheet.

SOURCES FOR THIS CHAPTER

- **1.** Manufacturing_TOGAF_Metamodel.xlsx — the 64 object sheets O01–O66 (O34 and O51 absent), EXT Industry Extension, and the 25 relationship sheets R01–R25; read with openpyxl, 16 October 2026
- **2.** The Open Group, ArchiMate 4 Specification (Open Group publication C260) — element, relationship and layer inventory; described, not reproduced https://publications.opengroup.org/c260
- **3.** The Open Group blog, "Discussing the Release of the ArchiMate 4 Specification", 20 May 2026 (retrieved 16 October 2026) https://blog.opengroup.org/2026/05/20/discussing-the-release-of-the-archimate-4-specification/
- **4.** Manufacturing — a starter pack for enterprise architects (volume 1), chapter 17, "Modelling manufacturing in ArchiMate"
- **5.** Manufacturing — a starter pack for enterprise architects (volume 1), chapter 13, "The plant floor — OT, ISA-95 and the IT/OT boundary"
- **6.** Manufacturing — a starter pack for enterprise architects (volume 1), chapter 9, "The conceptual data model"

APPENDIX E

Where to look in volume 1

Every step in this workbook, and the section of the learner guide that teaches it — checked line by line against the real headings.

Every numbered step with teaching behind it carries a `**Look it up.**` line naming a chapter of volume 1 and the exact section inside it. This appendix collects all of them, so you can go the other way: from the step you are on, straight to the page that explains why.

Two tables. The **step index** runs forward — one row per numbered step across chapters 2 to 12, what it produces, and where volume 1 teaches it. The **reverse map** runs backward, one row per volume 1 item, and is computed from the forward one rather than asserted.

Volume 1 is *Manufacturing — a starter pack for enterprise architects*; its worked example is Karoo Industrial Group and this workbook's is Kestrel Industrial Group, both inventions whose data must not be mixed.

How this index was built, and what the check found

Every `**Look it up.**` line in chapters 2 to 12 was read out of the chapter files and each heading matched against the real `##` and `###` headings of volume 1. A dangling cross-reference is the defect a reader notices first, so the check was run as code.

140 numbered steps, carrying 220 citations between them, and all 220 resolve. Every section heading cited exists in volume 1, and exists in the chapter the step names. Nothing in the two tables below is asserted from memory.

> **The one thing that makes this check harder than it looks**, and the reason to write it carefully if you run it yourself. A heading binds to the nearest preceding chapter mention, and that mention is not always in the same sentence, nor in the same case. Step 7.7 names three headings across two chapters: two of them follow the words *volume 1 chapter 13*, and the third follows *Chapter 17's* — sentence-initial, capitalised and possessive. Five more citations in chapters 6 and 7 take the same shape. A checker that matches only a lowercase chapter mention, or that refuses to carry a binding across a full stop, binds those to the wrong chapter and reports them as misfiled when they are correct. Match the chapter mention without regard to case, let the binding carry until the next one, and admit a sentence that names two chapters at once. Get that wrong and the check invents defects, which is worse than not running it.

Twenty-five of the 140 steps carry no `**Look it up.**` line at all, and not evenly: chapter 11 has eight of thirteen without one; chapter 7 five of thirteen; chapter 12 five of twelve; chapter 6 four of thirteen; chapter 10 three of eleven. That is defensible where a step is pure repository mechanics:

volume 1 does not teach governance workflow, so there is nothing to look up for "grant, date and expire a dispensation". Those rows read `—` in the index.

The step index

One row per numbered step. The **Volume 1** column gives the chapter number and section heading as `9, *Effectivity*`, and shows both where a step cites two.

Chapter 2 — Standing the practice up

Step	What it produces	Volume 1
2.1 Fix the boundary of the enterprise and the plants inside it	One-page scope statement	7, *Sites, plants, lines, cells and stations*; 2, *What to do with this in week one*
2.2 Count the repository instead of believing it	Completed Form 1-1	19, *What to count before you model anything*
2.3 Stand up the measure definition register before anything else	Measure definition register	7, *So the first deliverable is a measure definition register*; 10, *The measure definition register*
2.4 Settle the plant hierarchy against the ERP hierarchy	Signed control-to-ERP hierarchy mapping	9, *The plant hierarchy, and the reconciliation nobody budgets for*; 20, *Map 2 — The ISA-95 equipment hierarchy to the ERP plant hierarchy*
2.5 Build the `X02 Production Line` sheet	New sheet, `X02 Production Line`	7, *The four capacities*; 3, *OEE, and the three traps inside it*
2.6 Give `O05 Principle` the columns a gate can test	`O05 Principle` with five testable columns	14, *The constructive answer: build the house reference model*; *Then declare it*
2.7 Give every principle a person, and re-assess the Draft rows	`O05 Principle` with named individual owners	5, *Why you need the cast list before you need the model*

Chapter 2, continued

Step	What it produces	Volume 1
2.8 Stand up the governance bodies and put a plant voice in them	`057` with a plant voice on it	5, *Above the plant*; 7, *Centralised, or plant-autonomous*
2.9 Give the four gates a holder and an entry condition	`058 Governance Gate` made schedulable	14, *Then declare it*
2.10 Date and own the deliverables register	`031 Deliverable` with owner and date	19, *The ninety-day shape*
2.11 Choose the tool, and declare what you have extended	Tool choice and extension statement	15, *Enterprise architecture tooling, and the honest treatment of ArchiMate 4*; 17, *The ground rules, stated once*
2.12 Write down the reference conventions, including the bad ones	One-page reference convention note	3, *How to use a list like this*
2.13 Put change control in place before you change anything	Four change-control sheets, five columns	9, *Revision, and engineering change as a data event*

Chapter 3 — Architecture Vision

Step	What it produces	Volume 1
3.1 Secure the mandate and name a sponsor who exists	Request for architecture work, one page	19, *Week one: see the plant*
3.2 Build the stakeholder sheet, because there is not one	New sheet, `067 Stakeholder`	5, *Who to see first*
3.3 Put the plant on the role sheets	`010 Actor` and `011 Role`	5, *The plant*; 19, *The interview order*
3.4 Repair the organisation units and make their heads resolve	`009` with heads that resolve	5, *Where these people live*
3.5 Set the scope: plants, lines, products and period	Scope table, with what is out	2, *The full matrix*; 4, *The customer writes your architecture*
3.6 Test the drivers against the jurisdiction the plant is in	Corrected driver list	16, *Compulsory specifications, letters of authority, and section 20*; *Who inspects, and what they ask for*

Chapter 3, continued

Step	What it produces	Volume 1
3.7 Put dates on the objectives and attach them to measures	`003 Objective` with dates and a measure	20, *Map 5 — Measure to definition to source*
3.8 Take the value streams as the spine, and repair them	`014` repaired against `R02`	17, *Value stream*; 8, *The twelve processes a manufacturer runs*
3.9 Sketch the baseline on one page	One-page baseline of the estate	12, *The shape of a South African estate*
3.10 State the target in capability terms, and reject the arithmetic	Target capability state	17, *Capability map*
3.11 Write the Architecture Vision and take it through G0	Architecture Vision document	18, *The decisions that must be taken before anything can start*
3.12 Write the Statement of Architecture Work and sign it	Signed statement of architecture work	19, *The ninety-day checklist*

Chapter 4 — Business Architecture

Step	What it produces	Volume 1
4.1 Settle the archetype, order type by order type	One-page archetype statement	2, *The archetype is a property of the order, not only of the plant*; *The full matrix*; *Where the axes meet: the mixed-mode plant*
4.2 Build the plant register the repository does not have	Populated `X01 Plant / Site` sheet	7, *Sites, plants, lines, cells and stations*; 9, *The plant hierarchy, and the reconciliation nobody budgets for*; 17, *The plant hierarchy*
4.3 Add the columns `O12 Location` is missing, and take the headcount off the cloud	Corrected `O12 Location` sheet	7, *Sites, plants, lines, cells and stations*; 13, *The air gap, and where the real path is*
4.4 Record the shift pattern and the crew count	Shift-pattern statement per line	7, *Shifts, crews, and what a pattern costs*; *The four capacities*; *Why OEE is not comparable, stated hard*
4.5 Rebuild the role register against the plant	`O11 Role` holding the plant roles	5, *The plant*; *Engineering*; *Maintenance*; *Quality*; *Planning and supply chain*; *Who to see first*; 19, *The interview order*
4.6 Separate the actor from the role, and name people	`O10 Actor`, `O11 Role` and `R21` separated	5, *Where these people live*; 18, *Finding 8 — Biometric clocking, with consent taken in the employment contract*
4.7 Put the plant into `O09 Organization Unit`	Organisation sheet containing the factories	7, *Centralised, or plant-autonomous*; *Multi-plant, multi-company and the transfer price*

Chapter 4, continued

Step	What it produces	Volume 1
4.8 Rebuild the level-2 decomposition against real manufacturing capabilities	Level-2 decomposition a plant manager knows	17, *Capability map*; 11, *From business capability to application capability*
4.9 Re-rate criticality, and stop deriving disposition from arithmetic	Criticality that inherits, maturity with a basis	6, *Gross margin is not a property of the company*; 7, *So the first deliverable is a measure definition register*
4.10 Reassign capability ownership in `R22`	Capability-ownership map that resolves	7, *Centralised, or plant-autonomous*
4.11 Rebuild the value streams and name their stages	Five value streams with named stages	8, *The twelve processes a manufacturer runs*; *Plan to produce*; 17, *Value stream*
4.12 Give `016 Business Process` a level, a trigger, an outcome, an owner and a volume	Process sheet holding two levels	8, *The four levels, and who reads each*; *The conventions this book uses at level 2*; *Testing a level-2 model for completeness*; *The notation exception*; 17, *The BPMN exception, stated once*
4.13 Re-point the business services and functions at something real	Business-service catalogue, assessed	11, *The twelve*; *Three things that are not application capabilities*; 4, *What is actually sold*; *Product, conversion, or capacity*
4.14 Fix the product, the contract and the counterparty	Product sheet separating made from sold	6, *Tolling, where you never own the material*; 3, *Ownership: tolling and contract manufacturing*; 4, *Ownership is an attribute of inventory, and this one you cannot fix later*

Chapter 5 — Data Architecture

Step	What it produces	Volume 1
5.1 Decide which sheet is the authority on the golden source	Single written golden-source decision	11, *One: what is the system of record for the item, the bill of material and the stock position?*; 20, *Map 4 — System class to what it masters*
5.2 Correct the master system on the entities that cannot be mastered where it says	Corrected `Master system` per entity	13, *Historians, and the number that changes itself*; 10, *Why a historian's data model is not relational, and should not be forced to be*; 9, *The item master, and why it is the hardest master data object in any industry*
5.3 Fix the key scheme in `R05`	Relationship sheet keyed by reference	12, *The three things it is always asked to do and always does badly*
5.4 Name a person for every entity owner	`Owner` value on every `O20` row	10, *Master data governance, and the item-creation workflow*; 9, *One row, six owners who disagree*
5.5 Re-classify the entities, starting with the quality specification	Defensible classification on every entity	10, *Classification, and why an item has several*; 16, *Compulsory specifications, letters of authority, and section 20*; *POPIA, where it actually bites a manufacturer*
5.6 Make lineage exist for the six entities that have none	Lineage statement per entity	10, *The reporting layer, and the reconciliation that is always wrong*; *The plant-to-finance reconciliation*; 16, *The retention clocks, and why a single corporate period is always wrong*
5.7 Split the bill of material into its four structures	`X03 Bill of Material` sheet	9, *Several structures over the same parts*; *Header and line*; *Effectivity*; *Phantom assemblies*; *Scrap, yield, and the arithmetic done properly*; *Ten questions that settle a bill-of-material design*; 20, *Map 3 — The four bills of material, side by side*

Chapter 5, continued

Step	What it produces	Volume 1
5.8 Build the batch and genealogy sheet	`X04 Batch / Lot` sheet	9, *Batch, lot, serial, and genealogy as a graph*; *Genealogy is a graph*; *Recall simulation is the real test*; 12, *Traceability and serialisation*; 18, *Finding 4 — The batch record is in a spreadsheet estate, against a traceability obligation*
5.9 Build the quality specification sheet	`X05 Quality Specification` sheet	9, *Quality*; 16, *Compulsory specifications, letters of authority, and section 20*; 3, *Quality words, where the distinction has legal force*
5.10 Make ownership an attribute of inventory	Ownership attribute on the inventory entity	4, *Ownership is an attribute of inventory, and this one you cannot fix later*; 9, *Inventory, and ownership as a first-class attribute*; 6, *Tolling, where you never own the material*
5.11 Rebuild the entity relationship model	Entity relationship model	9, *The model on one page*; 17, *Information structure*
5.12 Give every entity its attributes, with a unit of measure	Attribute list a developer can build from	9, *Units of measure, and the error that never announces itself*; *Standard times, and why they are always wrong*
5.13 Join the logical components to the physical ones	Entity to logical to physical to application	10, *Moving data between systems*; 12, *The shape of a South African estate*
5.14 Declare the tag namespace	Written tag-naming convention and a register	13, *Why a tag is not a data entity*; *Exception and compression, and how they silently change a reported number*; 10, *Time-series and tag data, which no other industry in this series has*

Chapter 6 — Application Architecture

Step	What it produces	Volume 1
6.1 Give every application a named individual owner	`023` with an `Owner` value	5, *Who to see first*
6.2 Rebuild Disposition against Health and In service, and name the successor	`Disposition` per application, with a successor	12, *Six questions, asked of every class*
6.3 Replace the Users column with a business-value score	Scored portfolio position	7, *So the first deliverable is a measure definition register*
6.4 Split A04 SCADA / Historian into two rows	Two `023` rows	13, *DCS, SCADA, and the difference that survives*; *Historians, and the number that changes itself*
6.5 Write the ERP/MES boundary paper and get it signed	One-page signed statement	11, *The two questions that decide every manufacturing application architecture*; 12, *The boundary war, and how it actually gets settled*; 17, *The two you have to invent: the ERP/MES boundary view*
6.6 Log the application classes the estate has no row for	Undeclared-systems log	11, *The twelve*; 12, *Shop-floor data collection*; *LIMS and QMS*; *The shape of a South African estate*
6.7 Rebuild R04 and R07 from evidence	`R04` and `R07` rebuilt from evidence	17, *Application co-operation*

Chapter 6, continued

Step	What it produces	Volume 1
6.8 Rebuild R20 and R25, and let an application sit at more than one site	`R20` and `R25` rebuilt	—
6.9 Repair R05's two key schemes	`R05` with one key scheme	9, *The item master, and why it is the hardest master data object in any industry*
6.10 Triage the thirty interfaces, and fix the rows where Type fights Frequency	`R06` carriers and cadences consistent	13, *Integration, as it really is*; *Polling, events and the queue that must not fill*
6.11 Publish the information system services	`024` describing real interfaces	—
6.12 Complete O45 and make the acceptance criteria testable	`045` covering every application	—
6.13 Read all thirteen reports and write down what each one computes	One-line statement per report	—

Chapter 7 — Technology Architecture

Step	What it produces	Volume 1
7.1 Recompute Position from Support end, and write the currency paper	`027` with a `Position` value	—
7.2 Put real vendors into the Vendor column	`027.Vendor` with real vendors	—
7.3 Relocate the control servers, and record why	`027.Location` corrected	13, *The Purdue model as it is actually used*; *Where the diagram and the plant diverge*; *The test*
7.4 Rebuild O25's Category column, and add the services the list omits	`025` with a `Category` value drawn	—
7.5 Reconcile O26, O47 and O46, and resolve the second Ref column	One register of technology patterns	14, *The constructive answer: build the house reference model*; *Then declare it*
7.6 Add the ISA-95 equipment-level column	`027` with an ISA-95 level column	13, *The equipment hierarchy, and the correction that matters most*; 9, *The plant hierarchy, and the reconciliation nobody budgets for*; 14, *ISA-95 and IEC 62264 — the one everybody has heard of*
7.7 Build the zone and conduit register	Zone and conduit register on Form 7-B	13, *The standards, with their uneven dates*; *The air gap, and where the real path is*; 17, *Zones and conduits*; *And the IT/OT zone view*

Chapter 7, continued

Step	What it produces	Volume 1
7.8 Record the historian's exception, compression and namespace settings	Tag-class statement of historian settings	13, *Exception and compression, and how they silently change a reported number*; *Why a tag is not a data entity*; 10, *Time-series and tag data, which no other industry in this series has*
7.9 Rebuild O49 against what stops production, and record dip tolerance per line	`049` tiers tied to stopping production	13, *Voltage quality is the design input, and it is written down*; *What breaks, in order*; *Load-shedding ended; the discipline did not*
7.10 Settle O48's personal-data answer	`048` with a defensible answer	16, *POPIA, where it actually bites a manufacturer*
7.11 Close the design gate	`059` carrying its first verdicts	—
7.12 Rebuild O50 Network Link	`050` with one row per segment	—
7.13 Fix the Position columns in R16 and R17	`R16` and `R17` positions fixed	—

Chapter 8 — Opportunities and Solutions

Step	What it produces	Volume 1
8.1 Read both rows of O29 Gap and establish which direction the register was written in	One-page note on the gap register's direction	18, *The findings*
8.2 Rebuild the register from the four domain assessments you already hold	Gap register traceable to a cell	19, *What to count before you model anything*
8.3 Reconcile the package total against the recommended alternative	Reconciliation of the $19m	4, *Maintenance capital versus growth capital*
8.4 Get the alternative decided, and record what Partly adopted means	Dated decision in `O60` adopting one alternative	18, *The decisions that must be taken before anything can start*
8.5 Test all four acceptance-criteria families and mark which a gate could use	Verdict per family	7, *So the first deliverable is a measure definition register*
8.6 Collapse the duplication, and decide what O46's second Ref column is for	Authoritative-sheet statement and the `T01` fix	12, *IIoT platforms, unified namespace, edge and the digital twin*
8.7 Write the exit criteria, because the column currently holds a count	Named, testable exit criteria for `TA-1`	10, *The measure definition register*

Chapter 8, continued

Step	What it produces	Volume 1
8.8 Put the five orphaned work packages into a transition state, or out of the plan	`R15` with twelve assignments	18, *The target and the transition*
8.9 Break the two positional diagonals before Phase F inherits them	`R13` and `R09` rebuilt off the diagonal	17, *Requirements realization*
8.10 Turn the eight implementation factors into dated, owned constraints	Eight factors each carrying an owner	7, *Time fences, and why nobody respects them*
8.11 State the comparison-tool contract before anyone extracts anything	One-page contract note, agreed	10, *Moving data between systems*
8.12 Size the extract from the counted rows, not from ZZ Checks	Stated extract size	10, *Duplicate parts as a measurable cost*
8.13 Take Phase E through G0 and hand the sequencing to Phase F	Pack the Investment Committee can vote	14, *Maturity models, and what they are actually for*

Chapter 9 — Migration Planning

Step	What it produces	Volume 1
9.1 Reconcile Approval against Delivery, and stop the package nobody approved	Twelve-row status reconciliation, four contradictions named	18, *What it costs to be wrong*
9.2 Replace the typed Past end date with a comparison the sheet performs	Computed overdue flag	7, *On-time-in-full, and its four date bases*
9.3 Decide whether O56 Project is a sheet or a transcription, and fix the leads	A project sheet, or a deletion	5, *Who to see first*
9.4 Build the window and freeze calendar before you date anything	Dated change calendar per plant	18, *The roadmap and what the year allows*
9.5 Test every package's start and end against the calendar	Twelve-row calendar test per package	7, *Sales and operations planning*
9.6 Cost a cutover per plant, in lost production and in the shortest safe window	Per-site cutover cost	2, *The non-stoppable asset, and why power interruption hit manufacturing hardest*

Chapter 9, continued

Step	What it produces	Volume 1
9.7 Design the parallel run per application, and name who is on shift	Parallel-run design per application	7, *Shifts, crews, and what a pattern costs*
9.8 Record the change-control obligation where a system carries a quality record	Per-application statement of whether a change	16, *Pharmaceutical*
9.9 Rebuild O39 Data Migration so a migration can be planned from it	Migration register with reconciliation tests	9, *Batch, lot, serial, and genealogy as a graph*
9.10 Assign the five orphaned packages and re-date the six increments	Sequence tying packages to plateaux	7, *The planning cadence*
9.11 Re-own the eight delivery risks, starting with the owner who does not exist	`062` in which every owner resolves	5, *Maintenance*
9.12 Set O31 Deliverable honestly and hand Phase G something it can govern	Phase F pack Phase G can govern	13, *What you are actually expected to produce*

Chapter 10 — Implementation Governance

Step	What it produces	Volume 1
10.1 Write the first compliance assessment, and say out loud that it is the first	First rows of `059 Compliance Assessment`	16, *What a standard is, and what it is not*
10.2 Give each gate a holder, an evidence demand and testable criteria	Two columns on `058`, filled for four gates	14, *The constructive answer: build the house reference model*
10.3 Date every decision and every dispensation, and clear the four that block the roadmap	`Decided` column on `060 Governance Decision`	—
10.4 Re-assess every contract in `033`, because nothing was assessed	`033` compliance status corrected	—
10.5 Complete `057`, because there is no plant in it	`057` completed with plant membership	5, *Safety, improvement, and the industrial-relations layer*; *The plant*
10.6 Record the plant's management-of-change regime and where it meets your gate	One-page seam document	13, *What you are actually expected to produce*; *Why patching a controller is not like patching a server*

Chapter 10, continued

Step	What it produces	Volume 1
10.7 Record the engineering-change path, with the effectivity decision named	Engineering-change path across `023`	8, *Engineering change*; 9, *Revision, and engineering change as a data event*; *Effectivity*; 12, *Revision, effectivity and change*
10.8 Build the inspector-evidence register	Inspector-to-record register	16, *Who inspects, and what they ask for*; *The retention clocks, and why a single corporate period is always wrong*; 20, *Map 7 — Regulator to evidence*; *Map 8 — Retention clock by record class*
10.9 Stop `063`'s tautology at the gate	`063` control objectives made testable	16, *Compulsory specifications, letters of authority, and section 20*; *What a standard is, and what it is not*
10.10 Put the plant's calendar on the gate	Column on `058 Governance Gate`	7, *Shifts, crews, and what a pattern costs*; *Sales and operations planning*; *Time fences, and why nobody respects them*
10.11 Publish the monthly governance pack	Four-page pack, same shape every month	—

Chapter 11 — Architecture Change Management

Step	What it produces	Volume 1
11.1 Count the records before you version anything	Counted record inventory, sheet by sheet	19, *What to count before you model anything*
11.2 Create `01 Version Control` and publish version 1.0	First sheet of the repository	—
11.3 Create `02 Change Log`, one row per field	Sheet that makes every edit	—
11.4 Create `03 Change Requests`	Register that decides what gets changed	—
11.5 Create `04 Baselines`	Sheet that lets somebody say what	—
11.6 Append five columns to all 64 object sheets and back-fill `Proposed`	Every record in the repository carrying	—
11.7 Cut baseline `BL-00` and publish it	Frozen, dated baseline of the repository	—

Chapter 11, continued

Step	What it produces	Volume 1
11.8 Write the trigger list	Published list of change triggers	8, *Engineering change*; 16, *Who inspects, and what they ask for*
11.9 Classify a change and route it	Change-classification rule and routing	—
11.10 Grant, date and expire a dispensation	A dispensation that ends	—
11.11 Route a plant-floor change into the right register	Routing for the five plant changes	9, *Effectivity*; *Revision, and engineering change as a data event*; 13, *What you are actually expected to produce*
11.12 Govern a change to a computerised system holding a quality record	Change route for systems	14, *What each scheme compels, which is what makes it architecture*; 16, *POPIA, where it actually bites a manufacturer*
11.13 Measure whether change management is working	Six numbers, published monthly	7, *So the first deliverable is a measure definition register*

Chapter 12 — Requirements Management

Step	What it produces	Volume 1
12.1 Walk the spine and record where it breaks	One-page spine diagnostic	17, *The practical discipline*
12.2 Create the two relationship sheets the spine needs	Two new relationship sheets	—
12.3 Repair `R09`, and stop `RPT06` reporting a pattern as an assessment	Corrected `R09 Requirement satisfied by Bu`	20, *Map 4 — System class to what it masters*
12.4 Give every requirement an acceptance criterion and a verification method	Two new columns on `006 Requirement`	17, *Requirements realization*
12.5 Build the measure definition register against `004`	Measure definition register	7, *Why OEE is not comparable, stated hard*; *So the first deliverable is a measure definition register*; 10, *The measure definition register*; 3, *OEE, and the three traps inside it*
12.6 Re-express the constraints as requirements	Eight constraints restated as requirements	—

Chapter 12, continued

Step	What it produces	Volume 1
12.7 Rebuild `063 Control Objective` for the jurisdiction the plant is in	Replacement `063` for this jurisdiction	16, *Compulsory specifications, letters of authority, and section 20*; *The Act, and why the regulations are the architecture*; 22, *South Africa*
12.8 Work one obligation end to end	One obligation traced end to end	16, *Air quality — and the deadline that is not final*; 10, *Time-series and tag data, which no other industry in this series has*
12.9 Set priority and status so they mean something	`Priority` and `Status` on `006`, defensible	13, *The Purdue model as it is actually used*
12.10 Age the assumptions and close them	`008 Assumption` with owner and closure	—
12.11 Baseline the requirements and route a requirement change	Requirements baseline and the change route	—
12.12 Run the monthly requirements review	One-page monthly review, computed	—

The reverse map

Computed from the 220 citations, not asserted. **Citations** counts individual section references, so a chapter cited twice by one workbook chapter counts twice.

Volume 1	Workbook chapters that cite it	Citations
1 The industry in one chapter	**none**	0
2 The archetypes	2, 3, 4, 9	6
3 The vocabulary	2, 4, 5, 12	5
4 How the business model works	3, 4, 5, 8	6
5 Who does what	2, 3, 4, 6, 9, 10	17
6 How manufacturers make money	4, 5	3
7 The operating model and the planning cadence	2, 4, 6, 8, 9, 10, 11, 12	26
8 The process model	3, 4, 10, 11	9
9 The conceptual data model	2, 4, 5, 6, 7, 9, 10, 11	26
10 The information model	2, 5, 7, 8, 12	14
11 The conceptual application model	4, 5, 6	6
12 The application landscape	3, 5, 6, 8, 10	11
13 The plant floor — OT, ISA-95 and the IT/OT boundary	4, 5, 6, 7, 9, 10, 11, 12	24
14 The reference architecture	2, 7, 8, 10, 11	9
15 Technology providers	2	1
16 Regulation, risk and what the inspector asks for	3, 5, 7, 9, 10, 11, 12	18
17 Modelling manufacturing in ArchiMate	2, 3, 4, 5, 6, 7, 8, 12	15
18 Karoo Industrial Group, end to end	3, 4, 5, 8, 9	8
19 Your first ninety days	2, 3, 4, 8, 11	8
20 Appendix A, The cross-maps	2, 3, 5, 10, 12	7
21 Appendix B, Glossary	**none**	0
22 Appendix C, Standards, bodies and where to find them	12	1
23 Appendix D, References	**none**	0

What the reverse map shows

Three volume 1 items are cited by no step: chapter 1, *The industry in one chapter*; Appendix B, the glossary; and Appendix D, the references. Only one is a problem. Chapter 1 is orientation, written before any modelling vocabulary exists, so no step's teaching sits there; and nothing cites a reference list.

Appendix B, the glossary, is the one that matters. It holds 154 terms, and this workbook turns on vocabulary: re-classifying `E11 Quality Specification`, splitting `A04 SCADA / Historian` into two rows, recording a scrap factor — each is a step about getting a word right. That no step points at the glossary is a gap in the workbook, not in volume 1, and the cheapest cross-reference you can add.

Two items are cited once. Chapter 15, *Technology providers*, only by workbook chapter 2, at the tooling step — right, because a workbook about a repository you already have has one tooling decision in it. Appendix C, *Standards, bodies and where to find them*, only by chapter 12. That one is thin: Phase D and Phase G both touch standards conformance and neither points at it.

Three items carry a third of the whole index between them. Volume 1 chapter 7, the operating model and the planning cadence, and chapter 9, the conceptual data model, lead on 26 citations each; **volume 1 chapter 13, the plant floor chapter, is third at 24, and all three reach eight of the eleven workbook chapters.** Chapter 13 has no equivalent in the sibling volumes of this series, so the question worth asking was whether the workbook uses it or writes around it. Eleven of its 24 citations land in Phase D, and the remaining thirteen are spread across seven other chapters — it is both concentrated where the IT/OT boundary has to be carried and drawn on everywhere the plant floor touches a step. The workbook is using what differentiates the set.

The rest is unremarkable, which is the point. Chapter 16 at 18 and chapter 17 at 15 are used broadly rather than in one place — what you expect of material that constrains every phase. Chapter 5, *Who does what*, at 17, concentrates in the early chapters, because the first thing this repository needs is people in it.

> **Rebuilding this index.** When a step is renumbered, added or re-pointed, this appendix goes stale silently. The check is mechanical: read the `**Look it up.**` lines out of chapters 2 to 12, pull the chapter number and italicised heading from each, and match against volume 1's own `##` and `###` headings. Run it before every reissue, and read the binding rule in the note above before you trust the result.

THE CHAPTER IN TEN LINES

- 140 numbered steps across chapters 2 to 12, carrying 220 citations into volume 1.
- All 220 resolve: every heading cited exists, and exists in the chapter the step names.
- A heading binds to the nearest chapter mention, which may be capitalised, possessive and in the next sentence.
- A checker that matches only a lowercase chapter mention invents misfiled citations that are not there.
- 25 steps carry no Look it up line, 8 of them in chapter 11 and 5 in chapter 12.
- Three volume 1 items are cited by no step: chapter 1, Appendix B the glossary, and Appendix D the references.
- The glossary going uncited is a gap in the workbook, not in volume 1, and it is the cheapest cross-reference to add.
- Chapter 15 and Appendix C are each cited once, which is thin for a workbook that touches standards in two phases.
- Volume 1 chapters 7 and 9 lead on 26 citations each, and chapter 13, the plant floor chapter, is third at 24.
- All three reach eight of the eleven workbook chapters.
- 11 of chapter 13's 24 citations land in Phase D, and the rest are spread across seven other chapters.
- Chapter 5, Who does what, at 17 citations, concentrates in the early chapters, because the first thing this repository needs is people in it.
- Rebuild this index by script before every reissue, and get the binding rule right before you trust it.

SOURCES FOR THIS CHAPTER

- **1.** Manufacturing — the architecture workbook (volume 2), chapters 2 to 12; the Look it up lines read from the chapter files, 16 October 2026
- **2.** Manufacturing — a starter pack for enterprise architects (volume 1), 23 items; section headings read from the chapter files, 16 October 2026

APPENDIX F

The forms

Thirty printable pages, each one feeding a named sheet, and six of them feeding a sheet the repository does not have.

Thirty forms. Each one is a page. Print the page, take it to the room or the plant, fill it in by hand, and transcribe it into the sheet named on its face the same day. That last clause is the whole discipline: a form that sits in a folder for a fortnight has already stopped being evidence and started being a memory.

Every form here was built backwards from a real sheet in the repository you were handed [1]. The columns are that sheet's columns, in that sheet's order, spelled the way the sheet spells them. If a form's column reads `Master system`, it is because `O20 Data Entity` reads `Master system` and not "the master". Transcription has to be mechanical or it is not transcription.

Six of the thirty feed a sheet the repository does not have. They say so on their face, and they give the columns of the sheet you are to create. That is not a defect in the forms. It is a defect in the repository, and printing it on a form the reader fills in by hand is the most honest way to carry it.

Kestrel Industrial Group is the invented manufacturer in the repository. Karoo Industrial Group is volume 1's invented group. Both are inventions, and their data never mixes.

How to use these forms

The forms are numbered by the phase of the Architecture Development Method they serve [2], but the numbering is not a running order. Work in this order, every time.

1. **Read the face.** Each form names the sheet it feeds, how many rows a first pass usually needs, and who signs it. If you cannot name the signatory before the room starts, you have the wrong room.
2. **Print one copy per block.** A block is one sitting of one group of people. Two plants is two copies. Do not run two plants through one sheet of paper — you will lose which answer came from where, and that is the one fact the form exists to preserve.
3. **Delete the example row.** Every form carries exactly one, marked `EXAMPLE — delete`. It shows the grain, not the answer. Readers who leave it in transcribe a fiction into the repository and then defend it.
4. **Write refs as the sheet writes them.** The reference prefixes in this file are irregular and you must derive them from the sheet, not from a convention. `O06 Requirement` runs `REQ-001` with three digits. `O13 Business Capability` runs `C01`, `C01.01`, `C01.01.01`, dotted and with no hyphen. `O20 Data Entity` runs `E01`. `O53 Transition Architecture` runs `TA-1`. `EXT Industry Extension` runs `X01` to `X06`. Guessing a prefix is how a repository acquires two key spaces for one object.
5. **Transcribe the same day, and initial the paper when you do.** The paper is the audit trail for the sheet. Keep it.

6. **Where the form says continue on a second copy, continue on a second copy.** Eight to twelve ruled rows is one page. A thirty-row first pass is three copies, not one page of eight-point handwriting.

> Do not fill a form in a meeting where nobody present can be the signatory. The signature is not ceremony — it is the test that the room had the authority to answer the question. A capability map signed by nobody is a drawing.

The index

Thirty forms, numbered by the phase they serve. The numbering is not a sequence you work through: you open the block you need.

Form	What it captures	Feeds	Rows	Signed by
P-1	Architecture practice roles and the people in them	`011`, `010`, `R21`	18–24	Chief Enterprise Architect
P-2	Governance bodies and the gate each one holds	`057`, `058`	5–8	Chief Enterprise Architect
P-3	Measure definition register	New sheet	20–40	Chief Data Officer
P-4	Deliverable commitment, with an owner and a date	`031`	20	Chief Enterprise Architect
A-1	Stakeholder register	New sheet	25–45	Chief Enterprise Architect
A-2	Driver, obligation and the instrument that creates it	`001`, `063`	12–20	Chief Compliance Officer
B-1	Plant and site, with format and shift pattern	New sheet `X01`	4–12	Chief Operating Officer
B-2	Production line	New sheet `X02`	10–40	Plant manager
B-3	Plant role register	`011` extended	15–30	Plant manager
B-4	Organisation unit and where its people are	`009`, `012`	12–20	Chief Operating Officer
B-5	Capability, identify	`013`	10–30	Domain Architect, Business
B-6	Capability, assess	`013`	10–30	Domain Architect, Business
B-7	Value stream and its named stages	`014`, `016`	20–30	Domain Architect, Business
B-8	Business service and the function that performs it	`015`, `017`	10–20	Domain Architect, Business
C-1	Data entity, identify	`020`	14–25	Chief Data Officer
C-2	Data entity owner, master and lineage	`020`, `043`	14–25	Data Governance Council
C-3	Bill of material, batch and quality specification	New sheets `X03` – `X05`	6–15	Chief Data Officer
CA-1	Application component, identify	`023`	12–25	Domain Architect, Application
CA-2	Application component, assess	`023`	12–25	Domain Architect, Application
CA-3	Interface	`R06`	30–80	Domain Architect, Application
D-1	Physical technology component and its support date	`027`	12–30	Domain Architect, Technology

D-2	Network link and the zone it crosses	`050` extended	10–25	Domain Architect, Technology
D-3	Resilience tier and the test that proves it	`049`	12	Chief Risk Officer
E-1	Gap	`029`	15–40	Chief Enterprise Architect
E-2	Work package and its transition state	`030`, `053`, `R15`	12	Investment Committee
F-1	Project and its named lead	`056`	12	Head of Delivery
G-1	Compliance assessment at a gate	`059`	37	Architecture Board
H-1	Change request against the architecture	`061`, new sheet	5–15	Architecture Board
H-2	Change log entry	New sheet	Continuous	Chief Enterprise Architect
R-1	Requirement	`006`	31–60	Chief Enterprise Architect

Six columns, and what that costs you

A printed form that has to be written in by hand fits six columns across a 170mm text column and no more. Seven columns means either handwriting too small to read back or a column so narrow the answer gets abbreviated into uselessness. The repository has eleven sheets wider than six columns: `O01` at seven, `O06` at seven, `O13` at nine, `O20` at eight, `O23` at ten, `O27` at seven, `O30` at eight, `O56` at seven, `O59` at seven, and `R06` at nine.

Three things happen to those sheets here.

Split. `O13 Business Capability` becomes B-5 and B-6: identify, then assess. `O23 Application Component` becomes CA-1 and CA-2 on the same pattern. The split is not arbitrary — identify is answered by whoever knows the estate, assess is answered by whoever carries the consequence, and those are rarely the same people in the same room. Splitting the form splits the meeting, which is an improvement.

Type straight into Excel. Some columns are lookups or derivations, not answers. `R06`'s `Source name` and `Target name` repeat the application names that the refs already resolve. `O27`'s `Position` is a function of `Support end` against today's date — and the fact that it is typed by hand in this repository is why `ZZ Checks` reports four components past support when two of the four support dates, `PTC-06` at 2027-07 and `PTC-08` at 2027-04, are still in the future. Never put a computed column on a form. A hand-written derived value is a hand-written error.

Drop a column and add a better one. `O49 Resilience Tier` has `Tested` as a two-value column, and seven of its twelve rows read `Never tested`. D-3 keeps `Tested` and adds `Test date`, because "Tested" without a date is a claim and "Tested 2026-03-11" is a record.

The reference prefixes, derived from the file

A form's `Ref` column is the one place a reader is most likely to invent something, because the prefixes in this repository follow no single rule. Do not guess. These are the irregular ones, read off the sheets.

Sheet	Prefix	What is irregular about it
`006 Requirement`	`REQ-001`	Padded to three digits
`013 Business Capability`	`C01`, `C01.01`, `C01.01.01`	Dotted, no hyphen, no padding, three levels
`014 Value Stream`	`VS01`	No hyphen
`020 Data Entity`	`E01`	No hyphen, one letter
`023 Application Component`	`A01`	No hyphen, one letter
`030 Work Package`	`WP-01`	Two letters where most sheets use three
`037 Data Attribute`	`ATR-001`	Padded to three digits
`046 Technology Building Block`	`TBB-01` plus `T01` – `T07`	Two columns headed `Ref`; the second is a key space nothing else uses
`052 Implementation Factor`	`IF-01`	Two letters
`053 Transition Architecture`	`TA-1`	Single digit, no padding
`054 Architecture Alternative`	`ALT-A` to `ALT-E`	A letter, not a number
`055 Building Block`	`BBB-`, `DBB-`, `ABB-`, `TBB-`	No space of its own; duplicates 37 refs from four other sheets
`057 Governance Body`	`GB-01`	Two letters
`058 Governance Gate`	`GT-0`	Starts at zero
`061 Governance Change Request`	`CR-G01`	A letter inside the number part
`066 Architecture Increment`	`INC-1`	Single digit, no padding
`EXT Industry Extension`	`X01`	No hyphen, one letter
`R06 Application interfaces with`	`INT-001`	The only reference space on a relationship sheet
`059 Compliance Assessment`	None issued	Zero rows, so no prefix exists; you choose it

Two further traps worth a sentence. Several sheet tab names are cut at Excel's thirty-one-character limit, and **`R04 Business Service supported` ends in a space** — which means a formula that references it by a retyped name will fail silently. `026 Logical Technology Componen` and `027 Physical Technology Compone` are missing their final letters for the same reason. Only `R01`, `R11`, `R12`, `R21` and `R22` survive intact among the relationship sheets. Copy a tab name; never retype it.

And **`034` and `051` do not exist.** A loop from `001` to `066` fails on the holes. There are 64 object sheets, plus `EXT Industry Extension`, which `00 Index` counts as the sixty-fifth object type.

BEFORE YOU PRINT ANYTHING

Which sheet does this form feed, and who in the room can sign for it?

What will you do with the paper after you transcribe it, and for how long?

Which column on this form is computed rather than answered, and have you removed it?

Preliminary — standing the practice up

Four forms. These are the ones you fill in before you have any right to ask anybody else a question, because they establish who decides, who signs, what gets counted and what gets delivered.

`057 Governance Body` ships five rows and they are the best-constructed sheet in the file: the Architecture Board holds `G1` and `G2`, the Investment Committee holds `G0` funding and `G3` benefit confirmation, the Data Governance Council designates data owners. What `058 Governance Gate` lacks is a holder column, so the gate rows `GT-0` to `GT-3` float free of the bodies in `057`. P-2 adds it.

`031 Deliverable` ships twenty rows across phases A to G plus `RM`, with nine `Approved` and eleven `Draft`, and no owner and no date on any of them. Three of the `Approved` rows are interesting: `DLV-03 Stakeholder Map` is approved against no sheet at all, `DLV-10 Interface Catalogue` is approved against a catalogue where twenty-two of thirty rows are unmanaged, and `DLV-12 Technology Standards` is approved against a standards sheet with no manufacturing standard on it. Form P-4 adds the owner and the date, which is what turns a status into a commitment.

`004 Measure` ships sixteen rows with a `Baseline` and a `Cadence`, and no target and no owner. Sixteen measures none of which is a manufacturing measure: the `Domain` values run Compliance, Finance, Operations, Data, Commercial, Technology, Delivery, Resilience and Governance, and there is no effectiveness measure, no first-pass yield, no schedule adherence and no mean time between failures anywhere on the sheet. P-3 is the form that fixes this, and volume 1 makes the register the first deliverable before any dashboard exists.

Form P-1 — Architecture practice roles and actors

Feeds. `O11 Role` (`Ref`, `Role`, `Type`, `Performed by`), `O10 Actor` (`Actor`) and `R21 Role performed by Actor` (`Status`). One form, three sheets: the role ref goes to `O11`, the actor name to `O10`, and the pair plus `Status` to `R21`.

First pass. 18 to 24 rows.

Signed by. Chief Enterprise Architect (`ROL-08`), countersigned by the CIO for any row reading `Vacant`.

`O10` and `O11` currently hold the same eighteen names joined one-to-one, and all eighteen `R21` rows read `Filled` [3] — which is why `ZZ Checks` reports zero vacant architecture roles. That figure is true by construction and worth nothing. Write `Vacant` where it is vacant.

Ref	Role	Type	Performed by	Actor	Status
ROL-19	EXAMPLE — delete	Architecture	Person		Vacant

Form P-2 — Governance bodies and the gate each holds

Feeds. `O57 Governance Body`, all five columns, plus one column `O58 Governance Gate` does not have. Add `Gate held` to `O57`; it carries the gate refs `GT-0` to `GT-3`, comma separated.

First pass. 5 to 8 rows. More than eight bodies is not governance, it is a diary.

Signed by. Chief Enterprise Architect (`ROL-08`), tabled at the Executive Committee (`GB-04`).

A gate with no holder is not a gate. `O58` names four gates and what each tests, and nothing in the file says who sits at them — `GB-01` and `GB-02` carry the answer inside their `Mandate` prose, where no query can reach it. This form moves it into a column.

Ref	Governance body	Cadence	Membership	Mandate	Gate held
GB-06	EXAMPLE — delete	Monthly			GT-1

Form P-3 — Measure definition register

Feeds. A sheet the repository does not have. `O04 Measure` carries `Ref`, `Measure`, `Unit`, `Domain`, `Baseline` and `Cadence` — no target and no owner, so a measure cannot be missed and nobody can be asked why. Add `Definition`, `Target` and `Owner` to `O04`, or build the register as a sheet of its own and key it to `O04.Ref`. `Domain`, `Baseline` and `Cadence` are typed straight into Excel from the existing rows.

First pass. 20 to 40 rows. Continue on a second copy of this form.

Signed by. Chief Data Officer (`ROL-05`) for the definition; the named owner for the target.

`Definition` is the column that earns the register. Write the arithmetic, the denominator and the exclusion rule in one sentence — "scheduled hours less planned downtime, by line, excluding changeover under eight minutes" — and never write a measure name without one.

Ref	Measure	Unit	Definition	Target	Owner
MSR-17	EXAMPLE — delete	%			

Form P-4 — Deliverable commitment

Feeds. `O31 Deliverable`, all four columns, plus `Owner` and `Due`, which the sheet does not have. Twenty rows ship with `Phase` and `Status` and nothing else — nine `Approved`, eleven `Draft` — so no deliverable has a date or a name against it.

First pass. 20 rows, one per shipped deliverable. Add rows for anything your engagement commits to that is not already there.

Signed by. Chief Enterprise Architect (`ROL-08`).

`DLV-19 Compliance Assessment` reads `Draft` and `O59 Compliance Assessment` holds zero rows. Those two facts belong in one sentence at the next board, and this form is where the date gets written.

Ref	Deliverable	Phase	Status	Owner	Due
DLV-21	EXAMPLE — delete	B	Draft		2027-03

Phase A — vision

Two forms, and the first of them feeds a sheet that does not exist anywhere in the repository.

There is no stakeholder sheet among the 64 object sheets. `O09 Organization Unit` holds units and headcounts, `O10 Actor` holds eighteen corporate titles and `O11 Role` holds the same eighteen again — and not one of the three carries an interest column or an influence column. `DLV-03 Stakeholder Map` reads `Approved` on `O31 Deliverable`, which means a deliverable is recorded as approved with no sheet behind it. Form A-1 is the sheet.

The second form is the one that saves a South African reader the most time. `O01 Driver` ships twelve rows of which six are regulatory, and those six are `ISO 9001`, `ISO 14001 / 45001`, `Product safety and CE / UKCA marking`, `REACH / RoHS`, `Customs and origin rules` and `Extended producer responsibility`. `O63 Control Objective` repeats them, `O06 Requirement` repeats them again as `REQ-001` to `REQ-006`, and `RPT09 Regulatory Coverage` reprints `O63` with a footer. One European obligation set propagates through four sheets.

CE and UKCA marking have no application in South Africa. REACH and RoHS bind only on export to the relevant markets, and then on the exporter. The instruments that actually inspect a South African plant are the Occupational Health and Safety Act and its regulations, the atmospheric emission licence under the Air Quality Act, the water use licence under the National Water Act, NRCS compulsory specifications and letters of authority, the Legal Metrology Act, the carbon tax and POPIA. Form A-2 is where the substitution gets written down, obligation by obligation, with the instrument named rather than the standard — because the instrument, not the standard, is the citation.

Form A-1 — Stakeholder register

Feeds. A sheet the repository does not have. Create `STK Stakeholder` with these columns and the repository's convention — title row 1, subtitle row 2, headers row 4, data from row 5, `Ref` first. `Position` carries the role ref from `011 Role` or the body ref from `057 Governance Body` where one exists, and free text where it does not.

First pass. 25 to 45 rows. Continue on a second copy of this form.

Signed by. Chief Enterprise Architect (`ROL-08`).

`Interest` and `Influence` take High, Medium or Low and nothing else. `Engagement` takes Inform, Consult, Involve or Decide. Resist adding a fifth value to either: the point of the register is that it sorts, and a five-value scale does not sort.

Ref	Stakeholder	Position	Interest	Influence	Engagement
STK-01	EXAMPLE — delete	ROL-14	High	Medium	Consult

Form A-2 — Driver, obligation and the instrument that creates it

Feeds. `001 Driver` (`Ref`, `Driver`, `Type`, `Owner`) and `063 Control Objective` (`Obligation`, `Source`). `Instrument` is a column neither sheet has and both need. `Description`, `Origin` and `Pressure` on `001`, and `Control objective` and `Evidence` on `063`, are typed straight into Excel.

First pass. 12 to 20 rows.

Signed by. Chief Compliance Officer (`ROL-07`), countersigned by the General Counsel.

Write the instrument, not the standard. "SANS 10142-1" is a standard and is voluntary; "NRCS compulsory specification VC 8055, letter of authority" is an instrument and is not. Where the obligation is a certification scheme rather than law, say so in `Type`: a customer contract requiring IATF 16949 is commercial, not regulatory, and it binds just as hard.

Ref	Driver or obligation	Instrument	Type	Owner	Pressure
DRV-13	EXAMPLE — delete	OHSA s 24	Regulatory		High

Phase B — business architecture

Eight forms, and this is where the repository's sharpest defect gets repaired.

The organisation in this repository is a corporate IT org chart with no factory in it. `O10 Actor` and `O11 Role` hold the same eighteen names — Chief Executive, CFO, COO, CIO, CDO, CRO, Chief Compliance Officer, Chief Enterprise Architect, four domain architects, Head of Delivery, Head of Operations, Head of Service Management, Data Steward, Security Architect and Internal Auditor. There is no plant manager, no production planner, no scheduler, no maintenance planner, no reliability engineer, no quality manager, no process engineer, no operator, no setter and no millwright anywhere in the 105 sheets. Form B-3 is the repair, and it is not optional: every downstream owner column in this file — `O20.Owner`, `O23.Owner`, `O62.Owner` — resolves against a role list that does not contain anyone who works in a plant.

The second repair is physical. `O12 Location` has four columns — `Ref`, `Location`, `Type`, `Scale` — and `Scale` holds a headcount, which is how `LOC-08 Cloud Region` comes to have 261 people in it. The sheet cannot say what a plant makes, who owns it, how many lines it runs or on what shift pattern, and those are the facts everything else in manufacturing turns on. `EXT Industry Extension` declares `X01 Plant / Site` and `X02 Production Line` as types the core metamodel cannot hold, and then neither type gets a sheet. Five of the six extension types are at least shadowed in `O20 Data Entity` as `E06`, `E02`, `E05`, `E11` and `E09`. `X02 Production Line` is shadowed nowhere. `O12` holds sites, `E07 Work Centre` holds resources, and a production line is neither. For a manufacturer that is the most consequential absence in the file: the line is the object effectiveness is measured on, the object a changeover belongs to, and the object a capacity constraint lives on. Forms B-1 and B-2 build both sheets.

The headcounts do not reconcile either. `O09 Organization Unit` sums to 1,896 people across twelve units; `O12 Location` sums to 1,265 across eight locations. The three plants — `LOC-02 Northern Plant`, `LOC-03 Coastal Plant`, `LOC-04 Components Plant` — hold 568 people between them, while `ORG-07 Operations` is recorded at 102. Form B-4 makes the two sheets agree or records why they do not.

Form B-1 — Plant and site

Feeds. A sheet the repository does not have. `EXT Industry Extension` declares `X01 Plant / Site` and no instance of it exists. Create `X01 Plant / Site` with these columns. `O12 Location` gives you `Ref`, `Location`, `Type` and `Scale` and nothing else — it cannot say how many lines a plant runs or on what pattern, and it gives `LOC-08 Cloud Region` 261 people.

First pass. One row per site, including the ones that are not plants. 4 to 12 rows.

Signed by. Chief Operating Officer (`ROL-03`), with each plant manager signing their own row.

`Format` is the archetype — discrete, batch, process or mixed. `Ownership` is owned, leased, tolled or contract. `Shift pattern` is written as the pattern actually run, not the pattern in the handbook: "3 x 8, 5 days" and "3 x 8, 7 days" are different plants with the same equipment.

Ref	Plant / Site	Format	Ownership	Line count	Shift pattern
X01-01	EXAMPLE — delete	Discrete	Owned	4	3 x 8, 5 days

Form B-2 — Production line

Feeds. A sheet the repository does not have, and the most consequential absence in the file. `X02 Production Line` is declared in `EXT Industry Extension` and shadowed nowhere — not in `O12 Location`, which holds sites, and not in `E07 Work Centre`, which holds resources. Create `X02 Production Line` with these columns and key `Plant` to `X01 Plant / Site`.

First pass. 10 to 40 rows. Continue on a second copy of this form.

Signed by. The plant manager for the plant named in the row. There is no such role in `O11 Role` yet — create it on Form B-3 first.

`Design rate` is the nameplate rate with its unit — "1,200 units per hour", "14 tonnes per hour". `Changeover class` groups lines by how expensive a product change is on them: minor, major or wash-out. Until this sheet exists, no effectiveness number in this repository has an object to attach to and no capacity constraint has a home.

Ref	Production line	Plant	Product family	Design rate	Changeover class
X02-01	EXAMPLE — delete	X01-01		1200 units/h	Major

Form B-3 — Plant role register

Feeds. `O11 Role` (`Ref`, `Role`, `Type`, `Performed by`) plus `Plant` and `Shift`, two columns the sheet does not have and a multi-site manufacturer cannot do without. Each new role also needs a matching row in `O10 Actor` and a join in `R21 Role performed by Actor`.

First pass. 15 to 30 rows. Continue on a second copy of this form.

Signed by. The plant manager, countersigned by the Chief Enterprise Architect (`ROL-08`) when the ref is issued.

Start with these and do not stop there: plant manager, production manager, production planner, scheduler, shift supervisor, team leader, setter, operator, maintenance planner, reliability engineer, millwright, instrument technician, quality manager, quality inspector, laboratory analyst, process engineer, industrial engineer, warehouse supervisor, storeman, SHE officer. `Type` takes Operations, Engineering, Quality, Maintenance or Planning — five values the shipped sheet has none of.

Ref	Role	Type	Performed by	Plant	Shift
ROL-19	EXAMPLE — delete	Operations	Person	X01-01	A

Form B-4 — Organisation unit and where its people are

Feeds. `009 Organization Unit`, all five columns, plus `At location`, which keys to `012 Location` and reconciles two headcounts that currently do not agree. One row per unit per location: a unit split across three plants is three rows.

First pass. 12 to 20 rows.

Signed by. Chief Operating Officer (`ROL-03`), with HR confirming the headcounts.

`009` sums to 1,896 and `012` sums to 1,265. Until the difference of 631 has a reason written beside it, no capability heatmap weighted by headcount means anything, and `R22 Org Unit owns Capability` — which currently hands capability ownership to whichever unit shares a row number — has nothing to be rebuilt from.

Ref	Organisation unit	Type	Head	Headcount	At location
ORG-13	EXAMPLE — delete	Line		84	LOC-02

Form B-5 — Capability, identify

Feeds. `013 Business Capability`, the first five of its nine columns. Form B-6 carries the other four. Split because identify is answered by whoever knows the business and assess is answered by whoever carries the consequence, and those are different rooms.

First pass. 10 rows at level 1, then one copy per level-1 capability for its children. 10 to 30 rows per copy.

Signed by. Domain Architect, Business (`ROL-09`), with the owning executive signing level 1.

`013` ships 100 rows that are a generated cross-product: ten level-1 capabilities, three level-2 children each drawn from a word list — Administration, Optimisation, Onboarding, Reporting, Assurance — and two level-3 children each, `– Define` and `– Operate`. The descriptions restate the names. The ninety sub-rows appear in no report. Do not edit them; replace them.

Ref	Capability	Level	Parent	Description
C11	EXAMPLE — delete	1		

Form B-6 — Capability, assess

Feeds. `013 Business Capability`, columns six to nine, plus `Evidence`, which the sheet does not have. `Disposition` on the shipped sheet is pure arithmetic — `Build` exactly when `Target` less `Current` is two or more, and only two values appear across 100 rows — so do not write it by hand; compute it after the fact.

First pass. Same rows as the copy of B-5 you are working from.

Signed by. Domain Architect, Business (`ROL-09`), countersigned by the capability owner from `R22`.

Criticality does not inherit on the shipped sheet: `C01 Product Development` is `High` while `C01.01` is `Low` and `C01.01.01` is `Medium`, and `C01.02.01` is `Critical` under a `Low` parent. Decide your inheritance rule before the room starts and write it at the top of the page. `Evidence` is one line naming what you saw that justifies `Current` — a report, a count, an observation on the floor.

Ref	Criticality	Current	Target	Evidence	Disposition
C11	High	2	4	EXAMPLE — delete	

Form B-7 — Value stream and its named stages

Feeds. `O14 Value Stream` (`Ref`, `Value stream`, `Capabilities`) and `O16 Business Process` (`Ref`, `Business process`, `Value stream`, `Automation`). One row per stage, not per stream.

First pass. 20 to 30 rows across five streams. Continue on a second copy of this form.

Signed by. Domain Architect, Business (`ROL-09`).

`O14.Stages` holds a count — "4 stages", "5 stages", "6 stages" — not stage names, and the counts sum to 24 against `O16`'s 20 process rows. `VS01 Concept to Product` and `VS04 Order to Cash` carry the identical capability list `C01, C02, C03`; `VS02` and `VS05` both carry `C02, C03, C04`. `R02 Value stream uses Capabilit` is a three-wide sliding window reaching only `C01` to `C07`, so three level-1 capabilities are used by no value stream at all. This form replaces the count with names and the window with answers.

Ref	Value stream	Stage no	Stage name	Capability	Automation
BPR-21	EXAMPLE — delete	1		C04	Manual

Form B-8 — Business service and the function that performs it

Feeds. `O15 Business Service` (`Ref`, `Business service`, `Capability`, `Exposure`, `Criticality`) and `O17 Business Function` (`Business function`). `Description` on both sheets is typed straight into Excel; on the shipped rows it restates the name and is worth nothing.

First pass. 10 to 20 rows.

Signed by. Domain Architect, Business (`ROL-09`).

Check `Exposure` against reality before you copy it. The shipped sheet marks `BSV-01 Product Development Service` as `External` with `Criticality Low`, and `BSV-04 Manufacturing Operations Service` as `Internal` with `Criticality Low` — in a manufacturer. `R04 Business Service supported` is a positional diagonal mapping `BSV-01` to `A01` through `BSV-10` to `A10`, so a service called *Quality Management Service* is supported by whichever application sits in the matching row, and `A11` and `A12` support nothing.

Ref	Business service	Capability	Business function	Exposure	Criticality
BSV-11	EXAMPLE — delete	C04		Internal	High

Phase C — data architecture

Three forms. Two of them split `O20 Data Entity` and the third builds three sheets the repository declares and does not hold.

`O20` is where this repository does the most damage [4]. `E01 Product` — the item master — is recorded as mastered in `SCADA / Historian`. A historian is an append-only time-series store; it cannot master anything. `E06 Plant / Site`, `E09 Supplier` and `E11 Quality Specification` are all mastered in `APS`, a planning engine. `E13 Inventory Position` is mastered in `Supplier Portal`. The `Owner` column names job titles that resolve against nothing in `O11 Role` except `Data Steward` at `ROL-16`, and names the wrong ones besides: `E01 Product` is owned by `Head of Finance`, `E02 Bill of Material` by `Head of Sales`, `E08 Equipment / Asset` and `E10 Purchase Order` both by `Head of Health, Safety`. `Classification` is inverted exactly where it matters — `E09 Supplier`, `E11 Quality Specification` and `E13 Inventory Position` all read `Public`, while `E04 Work Order` and `E07 Work Centre` read `Restricted`. A quality specification is the testable definition of the product; it is the most commercially sensitive document a manufacturer owns, and in a regulated plant it is also a compliance record. Six of fourteen `Lineage` cells read `Absent`, and they include `E03 Routing / Recipe`, `E04 Work Order` and `E05 Batch / Lot` — the three things a traceability obligation turns on.

`O37 Data Attribute` compounds it. Sixty rows, six attributes each — `Identifier`, `Name`, `Status`, `Effective date`, `Owner`, `Classification` — across ten of the fourteen entities, so `E11` to `E14` have no attributes at all. Both `Identifier` and `Name` are marked `Key` = `Yes` on every entity, which gives each one two primary keys. And the `Classification` values on the attributes contradict the classification on the entity above them: `Product` is `Internal` on `O20` while its `Name` attribute is `Public` and its `Effective date` is `Confidential`. There is no attribute form in this appendix because attribute analysis is desk work against a physical schema, not a room exercise — but when you do it, start by deciding which single column is the key.

And `O43 Master Data Rule` disagrees with `O20` on nine of the ten entities they share. The one row they agree on is `E01 Product`, where both say the item master lives in the historian. Form C-2 is the form that settles it, and it has an `Agreed` column for exactly that reason.

Form C-1 — Data entity, identify

Feeds. `O20 Data Entity`, the first five of its eight columns. Form C-2 carries `Owner`, `Master system` and `Lineage`.

First pass. 14 to 25 rows.

Signed by. Chief Data Officer (`ROL-05`).

`Class` takes Master, Transactional or Reference. `Definition` is one sentence that a person in the plant would recognise, not a restatement of the name. Add the entities this repository is missing before you start: there is no work centre capacity, no changeover matrix, no downtime reason, no tag and no measure result among the fourteen.

Ref	Data entity	Class	Definition	Classification
E15	EXAMPLE — delete	Master		Confidential

Form C-2 — Data entity owner, master and lineage

Feeds. `O20 Data Entity` (`Owner`, `Master system`, `Lineage`) and `O43 Master Data Rule` (`Golden source`). `Agreed` is a column neither sheet has. The two sheets disagree on nine of their ten shared entities and nothing in the repository records which one is right, so the first job of this form is to make somebody say.

First pass. 14 to 25 rows.

Signed by. Data Governance Council (`GB-03`), with the named owner signing their own row.

`Owner` is a person, not a committee and not a job family — `PRN-13 Data Ownership is Individual` says so, in the repository's own words. `Master system` takes an application ref from `O23`, never an application name: `R05 Application manages Data En` already puts names in its `Master` rows and refs in its `Consumer` rows, so a join on `Application` returns half the sheet.

Ref	Owner	Master system	Golden source	Lineage	Agreed
E01	EXAMPLE — delete	A02	A02	Partial	Yes

Form C-3 — Bill of material, batch and quality specification

Feeds. Three sheets the repository does not have. `EXT Industry Extension` declares `X03 Bill of Material`, `X04 Batch / Lot` and `X05 Quality Specification` and holds no instance of any of them. They are shadowed as data entities `E02`, `E05` and `E11`, which gives you a definition and nothing structural. Create all three with these columns; put the object type in `Object` and file each row to the sheet its type names.

First pass. 6 to 15 rows. One row per structure, not per record.

Signed by. Chief Data Officer (`ROL-05`), countersigned by engineering for `X03` and by the quality manager for `X05`.

`Structure` is what the object is shaped like — single-level, multi-level, phantom-bearing, co-product-bearing. `Effectivity` is date, serial, revision or lot. `Retention clock` is the obligation that sets how long the record is kept, named as an instrument. Get these three wrong and no recall simulation will complete.

Ref	Object	Structure	Identifier rule	Effectivity	Retention clock
X03-01	Bill of Material	EXAMPLE — delete		Date	

Phase C — application architecture

Three forms. `O23 Application Component` has ten columns and splits; `R06 Application interfaces with` has nine and loses three to Excel.

`O23` ships twelve applications and retires and replaces the wrong things [5]. `A04 SCADA / Historian` is the only `Critical` application on the sheet and its `Disposition` is `Replace`, with nothing named to replace it — and it is owned by `Internal Audit`. `A01 ERP Core` is owned by `Legal`, `A05 APS` by `Risk & Compliance`, `A07 QMS` by `Internal Audit`. Those are organisation unit names from `O09`, not roles, and an assurance function cannot own a production system. `A03 MES` has been in service since 2010 with `Health` `Poor` and `Disposition` `Retain`. `A02 PLM` is `Poor` and `Consolidate` with nothing to consolidate into. Four of twelve read `Replace`. And the `Users` counts are useless for business-value scoring: `A05 APS` carries 266 users against the ERP's 196, which no plant resembles — a finite-capacity scheduler is used by a handful of planners.

`R06` is the only relationship sheet in the file with a reference space of its own, `INT-001` to `INT-030`. Twenty-two of its thirty rows are `Unmanaged`, twenty-one carry no logging, and four are `Direct DB read` — one application reading another's store. Those three counts are the only numbers in `ZZ Checks` that are both correct and worth acting on.

Form CA-1 — Application component, identify

Feeds. `023 Application Component`, columns one to four plus `In service`. Form CA-2 carries `Disposition`, `Criticality`, `Health`, `Users` and `Owner`.

First pass. 12 to 25 rows. Continue on a second copy of this form.

Signed by. Domain Architect, Application (`ROL-11`).

`Type` takes Package, Platform, Custom or Service. `In service` is the year the current instance went live, not the year the vendor was first engaged. Add the plant systems this sheet is missing: there is no shop-floor data collection, no laboratory information management, no calibration register and no historian client layer among the twelve.

Ref	Application	Type	Description	In service
A13	EXAMPLE — delete	Package		2019

Form CA-2 — Application component, assess

Feeds. `023 Application Component`, columns five to ten. `Owner` takes a role ref from `011 Role`, not an organisation unit name — the shipped sheet uses unit names and puts `Internal Audit` on two production systems.

First pass. Same rows as the copy of CA-1 you are working from.

Signed by. Domain Architect, Application (`ROL-11`), countersigned by the named owner.

`Disposition` must name its destination. `Replace` with nothing named is not a disposition, it is a mood — `A04 SCADA / Historian`, the only `Critical` row on the sheet, reads `Replace` today with no successor anywhere in the file. Write the successor ref in `Health` notes or raise it as a gap on Form E-1 the same day.

Ref	Disposition	Criticality	Health	Users	Owner
A13	EXAMPLE — delete	High	Adequate	40	ROL-14

Form CA-3 — Interface

Feeds. `R06 Application interfaces with`, six of its nine columns. `Source name` and `Target name` are lookups from `023` and are typed straight into Excel; `Frequency` is typed with them. `Ref` continues the `INT-` space, which is the only reference space in the file living on a relationship sheet.

First pass. 30 to 80 rows. Continue on a second copy of this form. The thirty shipped rows are a sample, not a catalogue.

Signed by. Domain Architect, Application (`ROL-11`).

`Type` takes Batch, Event, API, File or Direct DB read. Record `Direct DB read` honestly — every one of them is a gap, and four are already on the sheet. `Managed` and `Logged` are observations, not intentions: walk the integration platform and the log store before you tick either.

Ref	Source	Target	Type	Managed	Logged
INT-031	A03	A04	EXAMPLE — delete	Unmanaged	No

Phase D — technology architecture

Three forms, and this is the phase where the repository's physical model is furthest from any real plant.

`O27 Physical Technology Compone` places `PTC-03 SCADA control servers` at `Primary Data Centre` [6] and `PTC-07 Plant network switches` at `Central Distribution Centre`. Control servers sit at the plant they control — that is not a preference, it is what keeps the line running when the corporate link drops. `PTC-01 Plant historian cluster` is placed at `Head Office`. The `Vendor` column holds the literal string `Vendor` in eight of twelve cells. And `Position` is typed rather than computed, which is why `ZZ Checks` and `RPT10 Technology Currency` both report four components past support when `PTC-06` runs out in 2027-07 and `PTC-08` in 2027-04 — both still ahead of today.

`O50 Network Link` carries ten rows describing a ring between the eight locations, and it contradicts itself twice: `NWL-01` and `NWL-09` both run Northern Plant to Coastal Plant at 10 Gbps, one with `Diverse path` `No` and one with `Yes`; `NWL-02` and `NWL-10` both run Coastal Plant to Components Plant, one at 10 Gbps diverse and one at 100 Mbps not. Nothing in the sheet describes a plant's internal network, and nothing describes a zone boundary. Form D-2 adds the column that matters.

`O49 Resilience Tier` has tier and recovery objective uncorrelated and seven of twelve rows never tested. `RES-01 ERP Core` is `Tier 3` at 24 hours; `RES-02 PLM` is `Tier 1` at 24 hours; `RES-06 WMS` and `RES-12 Plant Analytics` are both `Tier 2` at 15 minutes. The only internally coherent row is `RES-04 SCADA / Historian` — `Tier 1`, one hour, `Tested`.

Form D-1 — Physical technology component

Feeds. `027 Physical Technology Compone`, six of its seven columns. `Position` is a function of `Support end` against today's date and is computed in Excel, never written by hand.

First pass. 12 to 30 rows. Continue on a second copy of this form.

Signed by. Domain Architect, Technology (`ROL-12`).

`Vendor` takes the vendor's name. Eight of the twelve shipped cells hold the word `Vendor`, which tells you the sheet was generated and never read. `Location` takes a ref from `012 Location`, and if your answer puts control servers anywhere but the plant they control, check it twice before you write it.

Ref	Physical technology component	Vendor	Class	Support end	Location
PTC-13	EXAMPLE — delete		Control	2029-03	LOC-02

Form D-2 — Network link and the zone it crosses

Feeds. `O50 Network Link`, all five columns, plus `Zone crossing`, which the sheet does not have. Write the zone pair the link joins — enterprise to plant DMZ, plant DMZ to supervisory, supervisory to control — using the zone names your own segmentation design uses.

First pass. 10 to 25 rows. One row per physical path, and two rows where two paths exist.

Signed by. Domain Architect, Technology (`ROL-12`), countersigned by the Security Architect (`ROL-17`) on every row whose `Zone crossing` is not blank.

A link with no zone crossing recorded is a link nobody has assessed. `LTC-01 OT/IT segmentation` reads `Emerging` on `O26 Logical Technology Componen` while `RME-01` reads `Adopted` on `O47 Reference Model Entry` — the same seven technologies appear on both sheets with different statuses. One of those two sheets is wrong, and this form is where you find out which.

Ref	From	To	Bandwidth	Diverse path	Zone crossing
NWL-11	LOC-02	LOC-07	EXAMPLE — delete	Yes	Plant DMZ to enterprise

Form D-3 — Resilience tier and the test that proves it

Feeds. `049 Resilience Tier`, all five columns, plus `Test date`, which the sheet does not have. Twelve rows ship and seven read `Never tested`; "Tested" with no date is a claim and "Tested 2026-03-11" is a record.

First pass. One row per application in `023`. Twelve rows today.

Signed by. Chief Risk Officer (`ROL-06`), countersigned by the application owner.

Set the tier from the consequence of the outage, then set the recovery objective from the tier, then test it. The shipped sheet does none of those in order, which is how `Tier 1` ends up at 24 hours and `Tier 2` at 15 minutes on the same page. For a plant system, state the recovery objective against the shift, not the working day: four hours into a 3 x 8 pattern is half a shift of lost output and a batch that may no longer be dispositionable.

Ref	Application	Tier	Recovery objective	Tested	Test date
RES-13	A13	Tier 2	4 hours	EXAMPLE — delete	

Phase E — opportunities and solutions

Two forms. `O29 Gap` holds two rows, both `High`, both business-domain capability uplifts, and both of them are restatements of `O35 Business Building Block` rather than findings. Everything genuinely wrong with this estate — the golden-source disagreement, the unmanaged interfaces, the absent compliance records, the organisation with no factory in it — is a gap that was never raised. Form E-1 is the most useful page in this appendix.

`O30 Work Package` sums to $65.0m across twelve packages [7] against `ALT-B Foundations first` at $46m, recorded as `Recommended` on `O54 Architecture Alternative`. Nineteen million dollars is unexplained, and `O54` also carries `ALT-E Compliance led` at $31m as `Partly adopted`, which no decision in `O60 Governance Decision` records. The whole file is in dollars, which for a South African manufacturer means read the magnitudes and not the currency. `R15 Transition contains Work Pa` carries seven rows covering `WP-01`, `WP-02`, `WP-04` and `WP-05` under `TA-1`, `WP-03` under `TA-2`, and `WP-07` and `WP-11` under `TA-3` — leaving `WP-06`, `WP-08`, `WP-09`, `WP-10` and `WP-12`, $26.1m, in no transition state at all. Form E-2 has a `Transition` column so that cannot happen again.

Form E-1 — Gap

Feeds. `029 Gap`, six of its seven columns, and `R12 Gap closed by Work Package` once a package exists. `Status` is typed straight into Excel and takes Open, Overdue or Closed.

First pass. 15 to 40 rows. Continue on a second copy of this form. Two rows is not a gap analysis.

Signed by. Chief Enterprise Architect (`ROL-08`).

A gap is the distance between the baseline and the target, stated so that a reader can tell whether it has closed. "Product Development capability uplift not in place" — both shipped rows read like this — is not a gap, it is a building-block name with a negation attached. Write what is true now, what must be true, and what the difference costs.

Ref	Gap	Domain	Building block	Severity	Due
GAP-03	EXAMPLE — delete	Data	DBB-01	High	2028-06

Form E-2 — Work package and its transition state

Feeds. `O30 Work Package` (`Ref`, `Work package`, `Domains`, `Cost`, `End`), `O53 Transition Architecture` and `R15 Transition contains Work Pa` (`Transition`). `Start`, `Approval` and `Delivery` are typed straight into Excel.

First pass. 12 rows, one per shipped package. Add rows for anything you raise.

Signed by. Investment Committee (`GB-02`).

Every package gets a transition state or it is not in the plan. Five of the twelve shipped packages have none, and they carry $26.1m between them. Check `Approval` against `Delivery` when you transcribe: `WP-07` is `In flight` for approval and `Delivered` for delivery; `WP-05` is `Not started` for approval and `In flight` for delivery, at $3.1m; `WP-08` is `Approved` and `Not started`. Two of those three are governance failures and one is a typing error, and only the room can tell you which.

Ref	Work package	Domains	Cost	End	Transition
WP-13	EXAMPLE — delete	Data	$2.4m	2029-11	TA-1

Phase F — migration planning

One form. `O56 Project` repeats `O30 Work Package` row for row — same names, same costs, same dates — and adds a `Lead`. The leads are drawn from `O10 Actor` without regard to what the project does: `PRJ-01 Governance foundations` is led by the Chief Executive, `PRJ-05 Access and security` and `PRJ-08 Process automation` by `Internal Auditor`, `PRJ-09 Resilience uplift` by `Domain Architect, Data`, `PRJ-11 Customer experience` by the Chief Compliance Officer. An internal auditor cannot lead a delivery project and remain an internal auditor.

Form F-1 — Project and its named lead

Feeds. `056 Project`, six of its seven columns, and `R14 Project delivers Work Packa`. `Start` is typed straight into Excel.

First pass. 12 rows, one per work package. More than one project per package means the package is too big.

Signed by. Head of Delivery (`ROL-13`), countersigned by the named lead.

`Lead` takes a role ref from `011 Role`, and the role must be one that can hold a delivery budget. Check every row against the independence rule before you transcribe: assurance roles — `ROL-18 Internal Auditor` and anyone in `ORG-11 Internal Audit` — cannot lead what they will later assess.

Ref	Project	Work package	Cost	End	Lead
PRJ-13	EXAMPLE — delete	WP-13	$2.4m	2029-11	ROL-14

Phase G — implementation governance

One form, and it feeds the sheet that proves this practice has never governed anything.

`O59 Compliance Assessment` is empty [8]. Header row only, zero data rows, seven columns waiting. `R24 Compliance Assessment tests` is header-only in consequence. `SEC-02 Design review` reads `Absent` on `O41 Security Control` and `RPT11 Security Position`, while `POL-04 Security is designed in, not added afterwards` reads `Met` on `O64 Security Policy` and `EP-01 Design review gate` reads `Newly established` on `O65 Enforcement Point`. Four sheets, three positions, one truth: the design gate has produced no records.

Meanwhile `O55 Building Block` carries 37 acceptance criteria across four domains — ten business, ten data, ten application and seven technology — every one of them a testable sentence, and not one has ever been tested. Thirty-seven criteria and zero assessments is the whole finding.

Form G-1 — Compliance assessment at a gate

Feeds. `059 Compliance Assessment`, six of its seven columns, and `R24 Compliance Assessment tests`. `Block name` is a lookup from `055 Building Block` and is typed straight into Excel.

First pass. 37 rows — one per acceptance criterion in `055`. Continue on a second copy of this form.

Signed by. Architecture Board (`GB-01`), which holds `GT-1 G1 Design` and `GT-2 G2 Build complete`.

`Verdict` takes Compliant, Non-compliant, Dispensation or Not yet assessable, and `Evidence` names the artefact, the date and where it is held. A verdict with no evidence reference is a verdict the board cannot defend, and the first thing an auditor asks for is the row that was refused.

Ref	Building block	Acceptance criterion	Gate	Verdict	Evidence
CA-01	DBB-01	EXAMPLE — delete	GT-1	Non-compliant	

Phase H — architecture change management

Two forms, and between them they add the thing this repository has none of.

There is no change control anywhere in the file [9]. No version columns on any of the 64 object sheets, no change log, no change requests against records, no baselines. `O61 Governance Change Request` holds five rows and they are change requests against work packages and against the repository as a whole — `CR-G03 Raise unfunded building blocks as gaps`, `CR-G04 Record benefit realisation dates per package` — not against records. Two of the five have no expiry date and one, `CR-G05 Refuse or re-baseline the integration package`, reads `Board decision required` with nothing in `O60 Governance Decision` to match it. `O60` itself has no date column at all.

Add two sheets and five columns. The sheets are `02 Change Log` and `03 Change Requests`, numbered to sit beside `00 Index` at the front of the workbook. The five columns go on all 64 object sheets, in this order, after the last existing column: `Version`, `Record Status`, `Changed By`, `Change Ref`, `Approved By`. `Version` is an integer that increments. `Record Status` takes Draft, Current or Superseded. `Change Ref` points at a row in `03 Change Requests`. Without those five columns, every number in `ZZ Checks` is a snapshot of an unknown date.

Form H-1 — Change request against the architecture

Feeds. `061 Governance Change Request`, all six columns, and a new sheet `03 Change Requests` with the same shape for requests against individual records rather than against packages.

First pass. 5 to 15 rows.

Signed by. Architecture Board (`GB-01`). A dispensation with no expiry is not a dispensation.

`Against` takes a work package ref, a record ref or the literal `Repository`. `Type` takes Dispensation, Scope, Correction or Compliance. `Expires` is mandatory on every row of type Dispensation and must be a date, not a phase — `CR-G01` expires 2029-06, which is the only row on the shipped sheet that does this correctly.

Ref	Change request	Against	Type	Status	Expires
CR-G06	EXAMPLE — delete	E01	Correction	Raised	2027-06

Form H-2 — Change log entry

Feeds. A sheet the repository does not have. Create `02 Change Log` with these columns. Every row records one change to one record on one sheet, and `Version` must match the `Version` column you have added to that object sheet.

First pass. Continuous. One row per change, written when the change is made. Continue on a second copy of this form as often as it fills.

Signed by. Chief Enterprise Architect (`ROL-08`) weekly, not per row.

`Sheet` takes the sheet name exactly as the tab spells it — and several tabs are cut at Excel's 31 characters, so `R04 Business Service supported ` ends in a space and `O26 Logical Technology Componen` and `O27 Physical Technology Compone` have no final letter. Copy the tab, do not retype it.

Ref	Sheet	Record	Version	Changed By	Approved By
CL-0001	O20 Data Entity	E01	2	EXAMPLE — delete	

Requirements management — continuous

One form. `006 Requirement` holds 31 rows and they are generated from three sources: six from the European obligation set in `001 Driver`, eight from the data entities, six from the applications, six from the level-1 capabilities and five from the reference model entries. Every statement is a template with a name substituted into it. `R08 Requirement derives from Dr` and `R09 Requirement satisfied by Bu` are both positional, and `RPT06 Requirements Traceability` reports nine of 31 requirements reaching no building block — which is computed correctly from a sheet that was generated positionally, so the nine are arbitrary.

Form R-1 — Requirement

Feeds. `O06 Requirement`, six of its seven columns, and `R08 Requirement derives from Dr`. `Status` is typed straight into Excel and takes Proposed, Approved or Withdrawn.

First pass. 31 rows to re-state what is there, then as many as the engagement raises. Continue on a second copy of this form.

Signed by. Chief Enterprise Architect (`ROL-08`), with the requesting stakeholder from Form A-1 countersigning.

`Ref` runs `REQ-001` with three digits — the only sheet in the file that pads to three. `Statement` is testable or it is not a requirement: "Product is mastered in one system and consumed elsewhere through a published interface" passes; "Product Development exposes measures that the executive can act on" does not, and six of the shipped rows read like the second.

Ref	Requirement	Statement	Category	Priority	Driver
REQ-032	EXAMPLE — delete		Data	Must	DRV-13

The sheets that get no form, and why

Thirty forms feed 27 of the 64 object sheets, seven of the twenty-five relationship sheets — `R06`, `R08`, `R12`, `R14`, `R15`, `R21` and `R24` — and eight sheets that do not exist yet. Thirty-seven object sheets have no form in this appendix. Two of them, `O05 Principle` and `O32 Standard`, have forms in the next appendix instead; `O58 Governance Gate` is reached by the `Gate held` column on Form P-2. That leaves **thirty-four object sheets, all thirteen `RPT` sheets, `ZZ Checks`, `EXT Industry Extension` and eighteen of the twenty-five relationship sheets** with no printable form anywhere in this volume.

That is not an omission to apologise for. A form exists to get an answer out of a room onto paper. Where the answer is not in a room, a form is the wrong instrument. Four reasons cover all of them.

Derived from a judgement another form already records. These sheets are written once a row exists elsewhere, and writing them by hand is transcription with extra steps.

Sheet	Rows	What it is derived from
`O02 Goal`, `O03 Objective`	6, 14	The vision document, signed before any sheet
`O21 Logical Data Component`	12	One per entity on Form C-1
`O22 Physical Data Component`	12	One per application on Form CA-1
`O24 Information System Service`	12	One per application on Form CA-1
`O33 Architecture Contract`	8	One per work package on Form E-2
`O35`, `O44`, `O45`, `O46`	10, 10, 10, 7	One per capability, entity, application, technology
`O55 Building Block`	37	The same 37 refs as `O35`, `O44`, `O45`, `O46`
`O66 Architecture Increment`	6	`O53 Transition Architecture` and `O30`

Produced by named analysis, not by a room. Somebody sits down with a tool and works these out. A workshop produces worse answers more slowly.

Sheet	Rows	What produces it
`036 Data Entity Relationship`	11	Logical data modelling in the modelling tool
`037 Data Attribute`	60	Attribute analysis against the physical schemas
`038 Data Set`	6	Report-to-source analysis
`039 Data Migration`	6	Migration design, after Form C-2 settles the master
`040 Source To Target Mapping`	12	Field-level mapping in the migration tool
`042 Metadata Element`	14	A catalogue capability assessment
`052 Implementation Factor`	8	Planning analysis against `007` and `008`
`054 Architecture Alternative`	5	An options paper with costed alternatives
`047 Reference Model Entry`	11	The reference-architecture decision, taken once
`025` , `026` , `028`	12, 7, 8	The same decision, expressed three ways

Owned by another function with its own register. Copying somebody else's register into yours creates two registers that drift.

Sheet	Rows	Whose register it is
`041` , `064` , `065`	8, 4, 5	Security's control register
`062 Delivery Risk`	8	The programme risk register
`019 Contract`	10	Procurement's contract register
`048 Environment`	4	The platform team's environment register
`060 Governance Decision`	7	The minute book, which is also where its missing date column lives
`018 Product`	4	The commercial product master
`007` , `008`	8, 6	Three and four columns; the vision document carries them

Computed, never typed. All thirteen `RPT` sheets and `ZZ Checks` . A report that can be filled in by hand is not a report. `EXT Industry Extension` joins them for the opposite reason: three columns, six rows, a register of types rather than data, and the six types it declares are built by Forms B-1, B-2 and C-3.

And eighteen of the twenty-five relationship sheets get no form because they are resolved from the object sheets by query. Thirteen of them are positional diagonals — `R03` maps `C01` to `BSV-01` through `C10` to `BSV-10` ; `R07` maps `A01` to `PTC-01` onward; `R16` maps `CO-01` to `SEC-01` onward, so `ISO 9001` is evidenced by `SEC-01 Identity lifecycle` ; `R22` hands capability ownership to whichever organisation unit shares a row number. Those thirteen must be deleted and rebuilt by query against the repaired object sheets, not corrected by hand. Hand-correcting a diagonal gives you a diagonal with exceptions.

WHEN YOU HAVE WORKED THROUGH A BLOCK OF FORMS

Which form in this block produced an answer that contradicts a sheet you have already transcribed, and which of the two is now wrong?

Which sheet did you have to create because the repository does not have it, and who has agreed to maintain it?

What did the room refuse to answer, and whose signature is now missing as a result?

What a form cannot do

Three things, and knowing them keeps you from using paper where paper is the wrong tool.

A form cannot settle a disagreement between two sheets. It can record which one the room chose, and that is different. Form C-2 has an `Agreed` column precisely because the disagreement between `O20.Master system` and `O43.Golden source` on nine of ten entities is not a data-entry error that a careful transcription would catch. Somebody with authority has to decide, and the form records the decision and the date, not the reasoning. Put the reasoning in the minutes.

A form cannot make a generated row true. `O13 Business Capability`'s ninety sub-rows, `O16 Business Process`'s twenty stage placeholders, `O35`'s ten templated acceptance criteria and `O55`'s thirty-seven duplicated refs are all generated artefacts. Editing them row by row on paper produces a generated structure with manual corrections in it, which is harder to reason about than either a clean generation or a clean rebuild. Delete and replace. Forms B-5 and B-6 are written for a rebuild, not a review.

A form cannot tell you what it was not asked. Every one of these thirty pages has a fixed column set, which is its strength when you want comparable answers across three plants and its weakness when the third plant does something the first two do not. Leave a margin, and when a plant volunteers a fact the form has no column for — a tolling arrangement, a shared line, a product made under somebody else's licence — write it in the margin and raise it as a gap on Form E-1 rather than squeezing it into a column where it will be averaged away.

Then there is the thing a form does better than anything else, which is why this appendix exists. A printed page with eight blank rows and a signature line is the only instrument that reliably converts a conversation into a record on the same day. Software does not do this on a plant floor. A notebook does not do it either, because a notebook has no columns and therefore no fixed questions. Thirty pages of columns, carried in order, is a method.

One last thing about paper

The forms in this appendix are deliberately ugly. They have no branding, no instructions beyond what fits on the page, and no space for commentary. That is because a form with room for commentary gets commentary instead of answers, and because every one of these pages is going to be carried into a plant where somebody will put a coffee cup on it.

Print them single-sided. Write in pen. Date the top right corner. Initial the bottom when you transcribe, and keep the paper for as long as the longest retention clock in `X05 Quality Specification` says to keep a quality record — which in a South African plant is set by an instrument and not by a corporate policy, and is almost never the single period the policy states.

The repository you were given holds 813 rows across its object sheets and claims 843 [10]. The difference is exactly 30, the row count of `R06 Application interfaces with`, which is the only relationship sheet with a `Ref` column. The total reconciles only if an interface counts as a record, which concedes that an interface is an object with no object sheet. Thirty forms will not fix that. Twenty-seven of them will at least make the rows they touch true.

THE CHAPTER IN TEN LINES

- Thirty forms, numbered by phase, each one a printed page that feeds a named sheet.
- A form's columns are the real columns of the real sheet, in sheet order, spelled exactly.
- Six columns is the limit. O13 and O23 split into identify and assess; computed columns never go on paper.
- Six forms feed a sheet the repository does not have, and say so on their face.
- X02 Production Line is declared in EXT and shadowed nowhere. Form B-2 builds it.
- O12 Location cannot say a plant's format, ownership, line count or shift pattern. Form B-1 adds them.
- O10 Actor and O11 Role hold eighteen corporate titles and nobody who works in a plant. Form B-3 repairs that.
- O04 Measure has a baseline and a cadence and no target and no owner. Form P-3 is the register.
- There is no change control anywhere in the file. Forms H-1 and H-2 add two sheets and five columns.
- Thirty-four object sheets get no form, because they are derived, analysed, owned elsewhere or computed.

SOURCES FOR THIS CHAPTER

- **1.** Manufacturing_TOGAF_Metamodel.xlsx — sheet names, column headings and shipped values as read from the workbook
- **2.** The Open Group, the TOGAF Standard — the Architecture Development Method https://www.opengroup.org/togaf
- **3.** Manufacturing_TOGAF_Metamodel.xlsx, sheets O10 Actor, O11 Role and R21 Role performed by Actor
- **4.** Manufacturing_TOGAF_Metamodel.xlsx, sheets O20 Data Entity, O43 Master Data Rule, O37 Data Attribute and R05 Application manages Data En
- **5.** Manufacturing_TOGAF_Metamodel.xlsx, sheets O23 Application Component and RPT02 Application Portfolio
- **6.** Manufacturing_TOGAF_Metamodel.xlsx, sheets O27 Physical Technology Compone, O50 Network Link, O49 Resilience Tier and RPT10 Technology Currency
- **7.** Manufacturing_TOGAF_Metamodel.xlsx, sheets O30 Work Package, O53 Transition Architecture, O54 Architecture Alternative and R15 Transition contains Work Pa
- **8.** Manufacturing_TOGAF_Metamodel.xlsx, sheets O59 Compliance Assessment, O55 Building Block, O41 Security Control, O64 Security Policy, O65 Enforcement Point and RPT11 Security Position
- **9.** Manufacturing_TOGAF_Metamodel.xlsx, sheets O61 Governance Change Request and O60 Governance Decision
- **10.** Manufacturing_TOGAF_Metamodel.xlsx, sheets 00 Index, ZZ Checks and R06 Application interfaces with

APPENDIX G

The principles and standards starter set

Forty-eight principles and thirty-one standards, each marked with where it came from, and every version fact dated.

The repository ships 24 principles on `005 Principle` and 16 standards on `032 Standard` [1]. Not one of the 40 mentions a plant, a line, a batch, a recipe, a tag or a unit of measure. This appendix gives you about 48 live principles and 31 standards, each marked with where it came from, so that the first argument you have about either of them is about substance rather than about provenance.

Kestrel Industrial Group is the invented manufacturer in the repository. Both it and volume 1's Karoo Industrial Group are inventions, and their data never mixes.

What `005 Principle` actually ships, and four things wrong with it

Twenty-four rows, six columns: `Ref`, `Principle`, `Category`, `Statement`, `Owner`, `Status`. The statements are short and several are good — `PRN-13 Data Ownership is Individual` reads "An owner is a person, not a committee", which is as clean a statement as you will find in a generated repository. The `Category` values are `Business` (four rows), `Application` (four), `Data` (six), `Technology` (five) and `Security` (five). There is no `Governance` value. `Status` has two values, `Approved` and `Draft`.

One. All 24 rows carry `Owner = Architecture Board`, which is a committee, against the sheet's own `PRN-13`. The principle that says an owner is a person is owned by a committee, and so is every other principle on the sheet. This is not a subtle inconsistency. It is the sheet failing its own test on every row, and it is the single fastest way to open a conversation about what a principle register is for. `057 Governance Body` records the Architecture Board as `GB-01`, a monthly forum of the Chief Enterprise Architect, the domain architects and CIO and CISO delegates. A forum cannot be woken at two in the morning and asked to decide. A person can.

Two. `Status = Draft` falls on every fifth row. `PRN-05`, `PRN-10`, `PRN-15` and `PRN-20` are `Draft` and the other twenty are `Approved`. That is a generation pattern, not an assessment: nobody looked at twenty-four principles and concluded that these four were unready. It matters more than it looks, because two of the four are the ones the rest of the repository leans on. `PRN-10 Data is Shared` — "Data is shared through published interfaces, not copies" — is the principle that twenty-two unmanaged interfaces in `R06 Application interfaces with` breach, and it is `Draft`. `PRN-15 Requirements-Based Change` is the principle the whole of `006 Requirement` and `RPT06 Requirements Traceability` assume, and it is `Draft`. Fix the status before you cite either.

Three. There is no `Tier`, no `Rationale`, no `Implications`, no `Approved` and no `Review date` column, so a principle cannot be tested at a gate. `058 Governance Gate` says `GT-1 G1 Design` tests whether "design meets principles, standards and reference architecture". With these six columns, that test cannot be run. There is nothing to say which principle outranks which when two conflict, nothing

to say what the principle costs anybody, nothing to say when the board last looked at it, and no date on which it became binding. `O59 Compliance Assessment` is empty and `R24 Compliance Assessment tests` is empty with it, which is the observable consequence. Your starter set ships all five columns, and they are additions to the sheet, not re-uses of existing ones.

Four. `Status` has no `Retired` value. A register with no way to retire a principle grows and never shrinks, and after three years nobody can tell which of the forty-odd rows the board would still defend. Add `Retired` to the permitted values of `Status` and `Governance` to the permitted values of `Category` before you load anything.

What a principle has to carry to be testable

Five columns, all of them additions to `O05 Principle`. Each one exists because without it a specific question cannot be answered at a gate.

`Tier`. Answers "who may let this one go, and for how long". Without it every dispensation request arrives at the Architecture Board, including the ones a domain architect could settle in a sentence, and the board spends its monthly hour on naming conventions.

`Rationale`. Answers "why is this here at all". One or two sentences, and the test of a good one is that it names a consequence somebody has actually experienced. For `PRN-27 One Item Master` the rationale is that this estate records `E01 Product` as mastered in `SCADA / Historian` — an append-only time-series store cannot master anything, and until that is fixed no item count in the business is defensible. For `PRN-40 Clocks Agree` the rationale is that genealogy is reconstructed from timestamps and timestamps from unsynchronised systems disagree, so the recall simulation stalls at the first join.

`Implications`. Answers "what does this cost, and who pays". This is the column that makes a principle honest, and it is the one most often left blank. `PRN-25 The Plant Keeps Running` implies local buffering at the boundary, a store-and-forward design on every plant-to-enterprise path, and a standing refusal of any requirement that makes a line wait on a corporate service. That is a real cost on a real budget and the register should say so. `PRN-51 Interfaces Are Managed and Logged` implies that the twenty-two unmanaged interfaces in `R06 Application interfaces with` are either remediated or dispensed, and that four `Direct DB read` rows are rewritten. Write the number into the implication.

`Approved`. A date. Not a status — a date. `Status` = `Approved` with no date cannot answer "was this binding when that design went through the gate", which is the only question that matters in an argument about a past decision.

`Review date`. A date within twelve months. `O58 Governance Gate` has four gates and `O60 Governance Decision` has no date column, so the repository currently has no way at all to say when anything was last looked at. A register with no review date becomes a wall decoration in about eighteen months.

Two further columns are worth adding once the register is live: `Supersedes`, which carries the ref of the retired row a new principle replaces, and `Breach count`, which is computed and not typed. Nothing else. A principle register with fifteen columns is a register nobody fills in.

The tier ladder

Four tiers. A principle's tier says who may grant an exception to it and for how long, and nothing else. Tier is not importance, and it is not a priority order for work.

Tier	What sits here	Exception granted by	Maximum term
1	Statutory obligation, safety, and the plant's ability to produce	Nobody	No exception
2	Evidence, record and traceability obligations	Architecture Board, in writing	12 months
3	Estate integrity — mastering, interfaces, currency	Architecture Board	24 months
4	Preference, consistency, convenience	Domain architect	36 months

Tier 1 is deliberately small and deliberately absolute. If a tier-1 principle can be dispensed, it was not tier 1. In a manufacturer the tier-1 set is short: the plant keeps running, the safety system is untouched, genealogy resolves, units travel with quantities, the obligation names its instrument, the gate produces a record, retention is set by the instrument, and personal data in a plant is mostly employee data. Everything else negotiates.

The tie-break, when two principles in the same tier conflict. Apply in order, and stop at the first that decides.

1. The breach that is irreversible wins. A batch shipped without resolvable genealogy cannot be un-shipped; a naming convention breach can be renamed.
2. If both are reversible, the one with the shorter remediation window wins.
3. If both windows are the same, the one whose breach is visible to a regulator or a customer wins.
4. If it is still tied, it escalates to the Architecture Board (`GB-01`), and if the board cannot agree, to the Executive Committee (`GB-04`), which `O57 Governance Body` already records as the escalation point. Write the decision into `O60 Governance Decision` — and note that `O60` has no date column, so add one.

> **The arithmetic, honestly.** Forty-eight live principles make 1,128 unordered pairs. The tier distribution below is 12 in tier 1, 21 in tier 2, 11 in tier 3 and 4 in tier 4, which puts 66 plus 210 plus 55 plus 6, or **337 pairs, inside a single tier** — just under thirty per cent. For those 337 the ladder says nothing at all, and the tie-break rule above is doing the whole job. Worse, 210 of the 337 are in tier 2 alone, because tier 2 holds 21 of the 48 principles. So the ladder settles a large share of pairs that were never going to meet, and is silent on a large share of the pairs that reach a board. Do not sell it as a conflict-resolution mechanism. Sell it as a dispensation-authority mechanism, which is what it is.

The 48, by tier

`Source` reads `Repository` where the row is unchanged from `O05`, `Edited` where the statement or the category has changed, and `New` where it is not in the repository at all. The three retired rows are listed after.

Tier 1 — no exception

Ref	Principle	Category	Statement	Source
PRN-04	Business Continuity	Business	The plant produces when the enterprise network, the ERP or the corporate site is unavailable.	Edited
PRN-19	Security is Built In	Security	Security is a design input, not a review gate.	Repository
PRN-22	Defence in Depth	Security	Protection is layered as zones and conduits, and no single control is relied on alone.	Edited
PRN-25	The Plant Keeps Running	Manufacturing	No plant function depends on a synchronous call across the IT and OT boundary.	New
PRN-26	The Safety System Is Out of Scope	Manufacturing	No architecture decision changes a safety instrumented function.	New
PRN-27	One Item Master	Manufacturing	The item is mastered once, in a transactional system of record, and never in a historian.	New
PRN-28	Genealogy Resolves Both Ways	Manufacturing	Every lot resolves to its inputs and its outputs inside the recall window without reconstruction.	New
PRN-29	Units Travel With Quantities	Manufacturing	Every quantity carries its unit of measure; conversions are held once and never inferred.	New
PRN-30	Every Obligation Names Its Instrument	Governance	An obligation is cited by the instrument that creates it, in this jurisdiction.	New
PRN-31	The Gate Produces a Record	Governance	A gate that produced no assessment record was not held.	New
PRN-32	Retention Is Set by the Instrument	Data	Each record class keeps its own clock, from the instrument that compels it.	New
PRN-33	Plant Personal Data Is Employee Data	Data	Biometric and employee records are treated as special information wherever they sit.	New

Tier 2 — exception by the board, twelve months

Ref	Principle	Category	Statement	Source
PRN-03	Compliance is Designed, Not Added	Business	Obligations are architected in, not bolted on.	Repository
PRN-09	Data is an Asset	Data	Data has an owner, a definition and a lifecycle.	Repository
PRN-12	Common Vocabulary	Data	One definition per concept, and the definition names its unit of measure.	Edited
PRN-13	Data Ownership is Individual	Data	An owner is a person, not a committee.	Repository
PRN-14	Classification Travels with Data	Data	Classification is a property of the data, not its location.	Repository
PRN-20	Least Privilege	Security	Access is the minimum the role requires, and never blocks a safety action.	Edited
PRN-21	Attributable Access	Security	Every access to sensitive data is attributable to a person.	Repository
PRN-34	The Line Is a First-Class Object	Manufacturing	Rate, changeover and loss attach to a line, which is a record and not a label.	New
PRN-35	Definition Before Dashboard	Manufacturing	No measure is published until its register row names arithmetic, denominator, exclusions and owner.	New
PRN-36	Loss Metrics Are Not Comparable	Manufacturing	A loss metric is reported against its own line's definition and is not aggregated upward.	New
PRN-37	The Boundary Is Declared	Manufacturing	The split between transactional record-keeping and production execution is written down object by object.	New
PRN-38	The Record Is Made Where the Event Happens	Manufacturing	Production, quality and movement records are captured at the point of the event.	New
PRN-39	Engineering Change Is a Data Event	Manufacturing	A revision change carries an effectivity and a disposition for stock in flight.	New
PRN-40	Clocks Agree	Manufacturing	Every system that timestamps a production record synchronises to one plant time source.	New
PRN-41	A Dispensation Has an Expiry	Governance	No exception is granted without a date and a named approver.	New
PRN-42	Every Record Has a Version	Governance	Every repository record carries version, status, who changed it and who approved it.	New
PRN-43	Evidence Beats Assertion	Governance	A position is recorded only where a dated, held artefact supports it.	New

PRN-44	Lineage Reaches the Source	Data	Every regulatory or financial figure traces to the system that captured it.	New
PRN-45	Classification Follows Consequence	Data	Sensitivity follows what disclosure would cost, not where the data sits.	New
PRN-46	Resilience Is Tested or Not Claimed	Technology	A recovery objective is recorded only with a dated test behind it.	New
PRN-47	Power Quality Is a Design Input	Technology	Dip immunity, restart order and ride-through are designed, not assumed.	New

Tier 3 — exception by the board, twenty-four months

Ref	Principle	Category	Statement	Source
PRN-05	Common Use Applications	Application	Prefer a shared capability to a duplicated one.	Repository
PRN-07	Ease of Use	Application	An interface used on a shop floor is used in gloves, in poor light, under time pressure.	Edited
PRN-10	Data is Shared	Data	Data is shared through published interfaces; a store-and-forward buffer at the boundary is not a copy.	Edited
PRN-15	Requirements-Based Change	Governance	Change is driven by a stated, testable requirement.	Edited
PRN-17	Control Technical Diversity	Technology	One approved technology per capability, except at the control layer, where the installed base decides.	Edited
PRN-18	Interoperability	Technology	Standards are chosen for interoperability, and every standard is named with its edition.	Edited
PRN-23	Models are Governed	Governance	Analytical and machine-learning models are governed like any other asset.	Edited
PRN-48	A Tag Is Not a Data Entity	Manufacturing	Tag data enters the information model through a defined aggregation, never as an entity.	New
PRN-49	Standard Times Are Owned and Dated	Manufacturing	Every standard time carries an owner and the date it was last measured.	New
PRN-50	Reports Are Computed	Governance	No reported figure is typed; a typed figure is an opinion in a numeric font.	New
PRN-51	Interfaces Are Managed and Logged	Application	No interface enters production unmanaged or unlogged, and no application reads another's store.	New

Tier 4 — exception by the domain architect

Ref	Principle	Category	Statement	Source
PRN-06	Technology Independence	Application	Applications are independent of the platform beneath them.	Repository
PRN-08	Service Orientation	Application	Capability is exposed as a service with a contract.	Repository
PRN-11	Data is Accessible	Data	Those entitled to data can reach it without a project.	Repository
PRN-24	Sustainable by Design	Technology	Energy and resource cost per unit produced is a design consideration.	Edited

Retired

Ref	Principle	Why it goes
PRN-01	Primacy of Principles	"Applies to every part of the enterprise without exception" is a scope statement, not a principle. It cannot be breached, so it cannot be tested.
PRN-02	Maximise Benefit to the Enterprise	"Enterprise benefit outranks local optimisation" has no counterfactual. Every decision can be argued to satisfy it, which makes it useless at a gate.
PRN-16	Responsive Change Management	"The estate changes at the pace the business needs" is unfalsifiable and duplicates `PRN-15`.

The provenance split. Of the 24 shipped rows, 10 are carried unchanged, 11 are edited, and 3 are retired. The 48 live principles are therefore 10 `Repository`, 11 `Edited` and 27 `New`. Twenty of the 27 new rows sit in the two categories the shipped sheet does not have: `Manufacturing` holds 14 rows, all of them new, and `Governance` holds 8, of which 6 are new and 2 — `PRN-15` from `Technology` and `PRN-23` from `Security` — are recategorised. That is the honest measure of how far a generated register sits from a manufacturing one: fourteen of the forty-eight principles a manufacturer needs have no counterpart at all in what shipped.

Where the fourteen manufacturing principles come from

None of them is invented here. Each one compresses a chapter of volume 1 into a sentence a board can adopt, and the cross-reference is what you cite when somebody asks why the principle exists.

Principle	Volume 1 chapter	Section
PRN-25 The Plant Keeps Running	13 The plant floor	The air gap, and where the real path is
PRN-26 The Safety System Is Out of Scope	13 The plant floor	Why patching a controller is not like patching a server
PRN-27 One Item Master	9 The conceptual data model	The item master, and why it is the hardest master data object in any industry
PRN-28 Genealogy Resolves Both Ways	9 The conceptual data model	Recall simulation is the real test
PRN-29 Units Travel With Quantities	9 The conceptual data model	Units of measure, and the error that never announces itself
PRN-34 The Line Is a First-Class Object	7 The operating model	Sites, plants, lines, cells and stations
PRN-35 Definition Before Dashboard	7 The operating model	So the first deliverable is a measure definition register
PRN-36 Loss Metrics Are Not Comparable	7 The operating model	Why OEE is not comparable, stated hard
PRN-37 The Boundary Is Declared	11 The conceptual application model	Two: where is the boundary between transactional record-keeping and production execution?
PRN-38 Record Made Where the Event Happens	12 The application landscape	Shop-floor data collection
PRN-39 Engineering Change Is a Data Event	9 The conceptual data model	Revision, and engineering change as a data event
PRN-40 Clocks Agree	13 The plant floor	Why timestamps disagree
PRN-48 A Tag Is Not a Data Entity	13 The plant floor	Why a tag is not a data entity
PRN-49 Standard Times Are Owned and Dated	9 The conceptual data model	Standard times, and why they are always wrong

Two of the fourteen will be argued about in the room, so prepare for both.

`PRN-36 Loss Metrics Are Not Comparable` will be resisted by whoever has already built a plant-comparison dashboard, and the resistance will take the form of "surely we can at least rank the lines". You cannot, and volume 1 states it hard: the micro-stop threshold alone moves loss between availability and effectiveness without changing the headline figure, and the availability denominator has six or seven incompatible conventions in circulation. A ranking built on inconsistent denominators ranks the definitions, not the plants.

`PRN-25 The Plant Keeps Running` will be resisted by whoever owns the integration budget, because local buffering and store-and-forward cost money on every path. The counter-argument is `049 Resilience Tier`, where seven of twelve applications have never been tested and `RES-04 SCADA / Historian` is the only internally coherent row on the sheet. Ask what the group's lost-production

figure is for one shift on one line, and then ask what the buffering costs. The conversation ends quickly.

Form PS-1 — Principle register

Feeds. `005 Principle` (`Ref`, `Principle`, `Category`, `Owner`) plus `Tier` and `Review date`, two of the five columns the sheet does not have. `Statement`, `Rationale`, `Implications` and `Approved` are typed straight into Excel from the adoption workshop minutes.

First pass. 48 rows. Continue on a second copy of this form, and on a third.

Signed by. Architecture Board (`GB-01`) for the tier; the named individual owner for the row.

`Owner` is a person's name, not `Architecture Board` and not a job family. Every one of the 24 shipped rows fails this. `Review date` is a date within twelve months, no exceptions — a register nobody is obliged to re-read is a register nobody re-reads.

Ref	Principle	Tier	Category	Owner	Review date
PRN-52	EXAMPLE — delete	2	Manufacturing		2027-06

What `O32 Standard` ships

Sixteen rows, five columns: `Ref`, `Standard`, `Category`, `Status`, `Conformance`. `Status` runs `Current` (twelve rows), `Deprecated` (two) and `Emerging` (two). `Conformance` runs `Breach` (seven), `Not assessed` (five), `Conformant` (three) and `Partial` (one). The categories are `Integration`, `Data`, `Security`, `Platform`, `Delivery` and `Governance`.

Three things to notice before you touch it.

Not one of the sixteen is specific to manufacturing. This is a generic corporate standards list: API design, event schema, data classification, retention, identity, encryption, logging, cloud landing zone, container image, database, network segmentation, coding and review, environment, naming, records management, supplier security. A manufacturer's standards list is half plant and half corporate, and the plant half is entirely absent.

Two rows are `Deprecated` with nothing named to succeed them. `STD-01 API design standard` is `Deprecated` and `Conformant`, which is a sentence with no meaning — conformant to a standard that has been withdrawn. `STD-07 Logging and monitoring standard` is `Deprecated` and `Not assessed`. A deprecated standard with no successor is not a standard; it is a historical note sitting in a live register.

`R18 Standard applies to Applica` assesses each standard against exactly one application, and cycles through six of twelve. The sheet runs `A01`, `A03`, `A05`, `A07`, `A09`, `A11` and repeats — three rows each for `A01`, `A03`, `A05` and `A07`, two each for `A09` and `A11`. Six applications appear and six never do, and the six that never do include `A04 SCADA / Historian`, the only `Critical` application in `O23 Application Component`. The `Conformance` column in `R18` reprints `O32`'s `Conformance` value row for row, so the sheet records no per-application assessment at all. `RPT12 Standards Conformance` reprints `O32` with no footer.

Ref	Standard	Status	Conformance	Verdict
STD-01	API design standard	Deprecated	Conformant	Retire; content moves into `STD-02`
STD-02	Event schema standard	Current	Not assessed	Edit — becomes the interface standard, absorbing `STD-01` and `STD-07`
STD-03	Data classification standard	Current	Breach	Keep
STD-04	Data retention standard	Current	Conformant	Edit — clock per record class, set by instrument
STD-05	Identity and access standard	Current	Breach	Keep
STD-06	Encryption standard	Current	Not assessed	Keep
STD-07	Logging and monitoring standard	Deprecated	Not assessed	Retire; content moves into `STD-02`
STD-08	Cloud landing zone standard	Current	Breach	Keep
STD-09	Container image standard	Current	Conformant	Keep
STD-10	Database standard	Current	Breach	Edit — carve out the time-series store
STD-11	Network segmentation standard	Emerging	Breach	Edit — enterprise only; OT goes to `STD-24`
STD-12	Coding and review standard	Current	Breach	Edit — carve out control code to `STD-25`
STD-13	Environment standard	Current	Not assessed	Keep
STD-14	Naming convention	Current	Breach	Edit — carve out the tag namespace to `STD-23`
STD-15	Records management standard	Current	Partial	Keep
STD-16	Supplier security standard	Emerging	Not assessed	Keep

Eight keep, six edit, two retire. Fourteen carried plus the seventeen new manufacturing standards below gives **thirty-one**, of which **seventeen are manufacturing-specific** — which is the proportion a manufacturer's register should have and the shipped sheet has none of.

The edit to `STD-02` is the largest of the six. It becomes the interface standard: one document covering the API design rules that `STD-01` held, the event schema rules it already held, and the interface logging rules that `STD-07` held. Three deprecated or overlapping documents become one live one, and `R18 Standard applies to Applica` gets rebuilt against every application rather than against six of twelve.

The version facts, and the rule about them

> **Never state a standard's version from memory, and never state one at all where you have not verified it.** Every fact in this section has a dated source behind it in volume 1's reference list. Several of them contradict material that is still circulating in vendor decks, and two of them are explicitly marked unverified here because they could not be confirmed against a primary document. Print the unverified ones as unverified. A wrong edition in a standards register is worse than a blank cell, because a blank cell gets filled in.

ISA-95. The current American National Standard for part 1 is **ANSI/ISA-95.00.01-2025** [2][3]. The standard has **eight parts plus the technical report ISA-TR95.01-2018** — and **the designations and publication dates of parts 2 through 8 could not be verified**, so no part table appears here as fact. Circulating summaries, including a widely-cited community encyclopedia entry, state five parts and contradict themselves in their own reference lists [4]. If your register needs the part list, licence the documents and read the covers.

There is no ISA-95 conformance scheme. "ISA-95 compliant" is therefore an uncertifiable claim, and you should strike it from any vendor response that contains it. Contrast IEC 62443, which does have a conformance scheme: ISASecure, run by the ISA Security Compliance Institute, certifies under SDLA, CSA and SSA [5]. That difference is the whole reason 62443 appears in contracts and ISA-95 does not.

B2MML belongs to MESA International, not to ISA. It is an XML implementation of the ISA-95 and ISA-88 models, published royalty-free with attribution [6]. Its current release — reported as version 7 in a trade-press announcement dated 24 November 2020 [7] — **predates ANSI/ISA-95.00.01-2025 by several years**, and because the version number and date rest on trade press rather than on a primary release record, treat both as provisional and check the repository's release page before you cite them. The architectural consequence stands regardless: an interface specified "to B2MML" is specified against an older model than the standard it claims to implement.

ISA-88. `ANSI/ISA-88.00.01-2010` carries conformance criteria in its own clause 9 [8], which makes it a rarity in this family: a batch-control standard you can actually be held to. Where a plant is organised around recipes, the recipe model reorganises the data model under it, and that is a design input and not a preference.

ISO 22400. The current published edition of part 2 is **ISO 22400-2:2014** [9]. A second edition is in development; as at October 2026 it sits at an ISO draft stage and is not published [10]. The three terms are availability, **effectiveness** and quality — **effectiveness, not performance** — and secondary summaries routinely write "performance rate" instead. **The exact wording must be confirmed against the licensed document before it enters a measure definition register**, because a register row that uses the wrong term for the middle factor will not reconcile against anything the standard says.

IEC 62443 has uneven editions, and you must not describe the family as one thing. `IEC 62443-2-1` is **Edition 2.0, dated 2024-08**, and it added a maturity model [11]. `IEC 62443-3-3` — the one that actually carries the system security requirements and the security levels — is **still Edition 1.0, dated 2013-08** [12]. A conformance claim against "62443" without a part and an edition is not a claim.

Alongside it, **NIST SP 800-82 Revision 3, September 2023, is the current final publication** [13]; Revision 4 exists only as an Initial Public Draft released 21 September 2026 [14].

IEC 61131-3. `IEC 61131-3:2013` Edition 3.0 was **withdrawn on 22 May 2025** and superseded by **`IEC 61131-3:2025`** [15][16]. Any control-code standard in your register that cites the 2013 edition is citing a withdrawn document, which an auditor will find in about four minutes.

Sparkplug is Eclipse Foundation work, not ISA's and not OPC's. Sparkplug 3.0 is published as ISO/IEC 20237:2023 [17][18]. MQTT carries it; the two are not the same thing and a standards register should hold both.

OPC UA's maturity is uneven across the family. The core specification parts sit at **1.05.x** while the UAFX field-exchange parts sit at **1.00.x** [19]. The OPC Foundation's release of 20 April 2026 states **"over 430" companion specifications** [20] — not the "about 60" that older material cites, and not a number you should repeat without the release date beside it. A companion specification is where the actual semantics live, so "we support OPC UA" says nothing until the companion specifications are named.

MTConnect: print no current version. The ANSI designation in circulation is a major version behind the documentation site's own part numbering, and **which version holds ANSI approval could not be verified** [21][22]. Record the standard, record that the version is unverified, and name the specific document your integrator will build against.

GS1. Identification keys are useful above the factory floor, and the Core Business Vocabulary gives you event types and business steps for traceability [23]. **Do not write that GS1 requires a two-dimensional carrier by 2027.** The GS1 US "Sunrise 2027" material is a voluntary industry capability ambition with no day and no month, and its own page says it is not necessary to implement all two-dimensional capabilities by that date [24].

The quality schemes moved, and two of them moved recently. `ISO 9001:2015` is **withdrawn**; **`ISO 9001:2026`, the sixth edition, published 16 September 2026** [25]. **`ISO 14001:2026` published 15 April 2026** [26]. **`ISO 45001:2018` is still current** — a revision is in draft [27]. **`IATF 16949` is still the 2016 first edition**, with a second edition planned for mid-2027 [28][29]. **`AS9100` is becoming `IA 9100`**, with the series expected early 2027 [30]. `O63 Control Objective` names `ISO 9001` and `ISO 14001 / 45001` with no edition at all, which is how a register goes stale without anybody noticing.

South Africa: the instrument is the citation. SABS publishes SANS standards; the NRCS regulates compulsory specifications and issues letters of authority [31][32]. A SANS standard is **voluntary unless an instrument makes it compulsory**, so cite the compulsory specification or the letter of authority, never the SANS number alone. On power quality, **`NRS 048-2:2025` Edition 5, approved by NERSA on 6 May 2025**, is the document, and it **assigns window-Y dip immunity to the customer** [33] — which makes ride-through a design obligation on your plant and not on the utility.

And there is no South African FSMA-204. The domestic food traceability obligation is **`R638`** [34], and it has **no Food Traceability List, no prescribed critical tracking events and no prescribed key data elements**. Anybody selling you a United States traceability data model as a South African compliance requirement is selling you a United States traceability data model.

`O63 Control Objective` is written for a European manufacturer. Its six obligations are `ISO 9001`, `ISO 14001 / 45001`, `Product safety and CE / UKCA marking`, `REACH / RoHS`, `Customs and origin rules` and `Extended producer responsibility`. **CE and UKCA marking have no application in South Africa.** REACH and RoHS bind on export to the relevant markets, and then on the exporter. The domestic instruments are the Occupational Health and Safety Act and its regulations [35], the atmospheric emission licence under the Air Quality Act [36], the water use licence under the National Water Act [37], NRCS compulsory specifications and letters of authority [31][32], the Legal Metrology Act [38], the carbon tax [39] and POPIA — whose section 26 on special personal information is the one that bites a plant with biometric clocking [40]. Rebuild the sheet against that list.

The seventeen manufacturing standards

Refs continue the `STD-` space. `Conformance basis` says how you could ever claim to conform — which for several of these is "by declaration only", and saying so is the point.

Ref	Standard	What it fixes	Conformance basis
STD-17	ISA-95 object naming at the ERP and MES boundary	One name per object either side of the boundary; the hierarchy levels named once	Declaration only — no conformance scheme exists
STD-18	ISA-88 recipe structure	Recipe, procedure, unit procedure and operation, and what a phase may do	Clause 9 of the 2010 part 1 carries conformance criteria
STD-19	ISO 22400 measure definitions	Availability, effectiveness and quality defined once, from the published 2014 part 2	Internal audit against the register row
STD-20	OPC UA and its named companion specifications	Which parts and which companion specifications an integration may use	Vendor certification against named parts
STD-21	MTConnect for machine-tool data	The agent, the device model and the data items a machine must expose	Declaration; version unverified, name the document
STD-22	MQTT and Sparkplug for the unified namespace	Topic structure, birth and death certificates, payload encoding	Sparkplug 3.0 is ISO/IEC 20237:2023
STD-23	Historian tag naming, exception and compression	Tag namespace, and the exception and compression settings per tag class	Internal audit against the historian configuration
STD-24	IEC 62443 zones and conduits	Zone and conduit definition, security levels, and the part and edition cited	ISASecure certification exists for products and processes
STD-25	IEC 61131-3 for control code	Language subset, naming, structuring and review of control code	Declaration against the 2025 edition
STD-26	Engineering change and effectivity	Effectivity basis, disposition of stock in flight, and who may release	Internal audit against the change records
STD-27	Bill-of-material structure and revision	Levels, phantoms, alternates, scrap and yield, and revision rules	Internal audit against the item master
STD-28	Batch and lot numbering and genealogy	Numbering rule, genealogy graph and the recall simulation that tests it	Recall simulation, timed, against the obligation
STD-29	GS1 keys above the factory floor	Which keys are used, where, and the event vocabulary for traceability	GS1 member compliance; keys are licensed, not certified
STD-30	The EDI message set with customers	Which messages, which version, which partner, and the acknowledgement rule	Partner conformance testing
STD-31	Unit of measure and conversion	The unit list, the conversion table, and where it is held once	Internal audit against the item master
STD-32	The applicable quality scheme's record obligations	Which records the scheme compels, for how long, and who holds them	Third-party certification audit
STD-33	Plant time synchronisation	One time source per plant, the hierarchy beneath it and the tolerance	Internal audit against the configured sources

Six of the seventeen deserve a sentence more.

STD-17 is the one that stops the ERP and MES argument from being re-fought every six months. The argument is settled object by object — who masters the item, who masters the order, who masters the

stock position, who masters the operation confirmation — and written down. Volume 1 chapter 11 makes it the second of its two questions that decide every manufacturing application architecture.

STD-19 is the standard that makes a dashboard mean anything, and it is also the one your register will most often cite incorrectly. Confirm the wording of the middle factor against the licensed 2014 document before the register row is approved.

STD-23 is the standard nobody writes and everybody needs. Exception and compression settings silently change a reported number: a historian configured to discard values inside a deadband will reproduce a trend faithfully and a total incorrectly, and no report will tell you which. Record the settings per tag class in the standard, and audit them.

STD-24 is the only item in this list that routinely becomes contractually binding, because ISASecure gives a customer something to demand. Cite the part and the edition every time: `62443-2-1` Edition 2.0 of 2024-08 and `62443-3-3` Edition 1.0 of 2013-08 are eleven years apart and are not interchangeable.

STD-31 sounds trivial and is not. Volume 1 chapter 9 calls units of measure the error that never announces itself, and in a repository where `O37 Data Attribute` gives every entity an `Identifier` and a `Name` both marked `Key` = `Yes`, the chance that unit handling has been thought about is low.

STD-18 matters where the plant is organised around recipes, which is every batch and process plant. A recipe standard that does not say what a phase may do will be interpreted differently by the control engineer and the MES vendor, and the difference surfaces as a batch that cannot be released.

STD-20 is the one vendors answer with a tick and no content. The semantics live in the companion specifications, of which the OPC Foundation's April 2026 release counts over 430, so the standard in your register must name the specific ones an integration may use. Core parts and field-exchange parts are at different maturities and cannot be treated as one family.

STD-26 and **STD-27** are a pair, and `O20 Data Entity` shows why: `E02 Bill of Material` is recorded as `Confidential` and owned by `Head of Sales`, and `E03 Routing / Recipe` has `Lineage` = `Absent`. An engineering change with no effectivity basis and no disposition rule for stock in flight produces scrap nobody can explain and a cost variance nobody can attribute.

STD-30 is the standard that stops the customer integration from being rebuilt per customer. Name the message set, the version, the partner and the acknowledgement rule, and refuse any partner-specific variant that is not recorded against a row.

STD-32 is where the register meets the auditor. The scheme compels records; the records are the architecture. Name the edition — `ISO 9001:2026`, `ISO 14001:2026`, `ISO 45001:2018`, `IATF 16949:2016` — because an uncited edition is how a register goes stale.

STD-33 is what makes `STD-28` possible. Genealogy is reconstructed from timestamps, and timestamps from unsynchronised systems disagree. One plant time source, a declared hierarchy and a stated tolerance, or the recall simulation will not complete.

Form PS-2 — Standards register

Feeds. `032 Standard`, all five columns, plus `Owner`, which the sheet does not have. `Edition` and `Scope` are typed straight into Excel; `Edition` is the column that keeps the register honest and it must hold a date or the word Unverified.

First pass. 31 rows. Continue on a second copy of this form.

Signed by. Architecture Board (`GB-01`); the named owner signs the conformance position.

`Status` takes Current, Emerging, Deprecated or Retired — add `Retired`, which the shipped sheet does not have. `Conformance` takes Conformant, Partial, Breach or Not assessed, and `Not assessed` is the honest answer far more often than the shipped sheet admits.

Ref	Standard	Category	Status	Conformance	Owner
STD-34	EXAMPLE — delete	Manufacturing	Current	Not assessed	

Form PS-3 — Dispensation request

Feeds. `061 Governance Change Request`, all six columns, with `Type` = `Dispensation`. One row per request, against one principle or one standard.

First pass. As many as are requested. Five rows ship on `061` and two of them have no expiry date.

Signed by. Architecture Board (`GB-01`), or the domain architect where the principle is tier 4. A tier-1 principle has no dispensation and this form does not apply to it.

`Expires` is a date, never a phase and never a milestone. `Remediation` is what will be true when the dispensation ends, written so that somebody can test it — and if you cannot write it, the request is a request to breach the principle permanently and should be refused.

Ref	Against	Tier	Reason	Remediation	Expires
CR-G06	PRN-51	3	EXAMPLE — delete		2028-03

Testing a principle at a gate

A principle that cannot fail a design is not a control. The mechanism already exists in this repository and has never been used: `O55 Building Block` carries 37 acceptance criteria across four domains, `O58 Governance Gate` names four gates, `O59 Compliance Assessment` has seven columns and zero rows, and `R24 Compliance Assessment tests` is header-only in consequence.

Wire it up in three moves.

1. **Map each principle to the acceptance criteria that would fail if it were breached.** Most tier-1 and tier-2 principles map to between one and four of the 37. `PRN-27 One Item Master` maps to `DBB-01 Product mastering`, whose criterion already reads "No product record without a named owner" — which `O20 Data Entity` breaches, because `E01 Product` is owned by `Head of Finance`, a title that resolves against nothing in `O11 Role`.
2. **Write the assessment row even when the verdict is bad.** `O59` takes `Ref`, `Building block`, `Block name`, `Acceptance criterion`, `Gate`, `Verdict` and `Evidence`. A `Non-compliant` verdict with evidence is worth more to the board than an empty sheet, because it starts a clock.
3. **Report the breach count, computed.** `RPT12 Standards Conformance` reprints `O32` and computes nothing. Replace it with a count over `O59` by principle and by tier. If tier 1 shows anything other than zero, that is the only line of the report anybody needs to read.

One warning about the mapping. Do not map every principle to a criterion. Tier-4 principles — `PRN-06 Technology Independence`, `PRN-08 Service Orientation`, `PRN-11 Data is Accessible`, `PRN-24 Sustainable by Design` — are preferences, and forcing a preference through a gate turns the gate into a design review with a budget. Four of the forty-eight are deliberately untested, and saying so protects the other forty-four.

The adoption workshop

One day, one room, and the output is a signed register rather than a discussion. Run it after Form A-1 is filled in and before anything is modelled, because a principle adopted after the design is a rationalisation.

Time	What happens	Output
08:30	The four defects on `005` tabled, with the sheet on screen	Agreement that the register is being rebuilt, not reviewed
09:00	The tier ladder and the tie-break rule adopted or amended	A ladder, written on the wall
09:45	Tier 1 read aloud, one row at a time, challenged	Twelve rows, or fewer, with no dispensation route
11:00	Tier 2 and tier 3 taken in blocks by category	Statements agreed; wording parked where contested
13:30	Owners named — a person per row, no committees	Form PS-1 filled in, `Owner` column complete
14:45	The three retirements argued and recorded	`Status` = `Retired` with a reason in the minutes
15:30	Standards: the seventeen manufacturing rows read against the estate	Form PS-2 started, `Conformance` left honest
16:30	Review dates set, all within twelve months	Form PS-1 `Review date` column complete
17:00	Decision recorded	A row in `060 Governance Decision`, with the date column you added

Three rules for the room.

1. **A principle nobody will defend out loud is not adopted.** Read each tier-1 row aloud and ask who will defend it at the Executive Committee. Silence retires the row.
2. **No wording is parked twice.** Contested wording goes to the owner to redraft and comes back by a date. A second parking is a retirement.
3. **The owner is in the room or the row is not adopted.** Twenty-four rows owned by a committee is what you are there to fix.

BEFORE THE REGISTER IS LOADED

Which tier-1 principle does this estate breach today, and who has been told?

For each of the three retirements, who would object, and have they been asked?

Which standard in your register holds an edition you have not seen with your own eyes?

Which measure in `004 Measure` will be published before its definition row exists, and who is stopping it?

What to do on the Monday

Load the 48 and the 31 into `005 Principle` and `032 Standard` with their new columns, and set every `Conformance` value you have not personally verified to `Not assessed`. That will look worse than the shipped sheet, which records seven breaches and three conformances against sixteen standards assessed against six applications by a positional rule. Looking worse is the correct outcome: the shipped position is a generated artefact and yours is a measurement.

Then run Form G-1 from the previous appendix against the thirty-seven acceptance criteria in `055 Building Block`, because `059 Compliance Assessment` is empty and a principle register with no compliance assessments behind it is a document, not a control. The first ten rows you write into `059` will be the first evidence this practice has ever produced at a gate.

THE CHAPTER IN TEN LINES

- O05 Principle ships 24 rows and every one is owned by a committee, against its own PRN-13.
- Status equals Draft falls on every fifth row, which is a generation pattern and not an assessment.
- The sheet has no Tier, Rationale, Implications, Approved or Review date, so no principle can be tested at a gate.
- The starter set is 48 live principles: 10 carried unchanged, 11 edited, 27 new, and 3 retired.
- Four tiers set who may grant an exception and for how long. Tier is not importance.
- Of 1,128 principle pairs, 337 share a tier and the ladder decides nothing for them.
- O32 Standard ships 16 rows and not one is specific to manufacturing.
- The set is 31 standards: 14 carried and 17 new, all 17 manufacturing-specific.
- Never state a version from memory. ISA-95 part 1 is the 2025 edition and parts 2 to 8 are unverified here.
- CE and UKCA marking have no application in South Africa. Cite the instrument, not the standard.

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APPENDIX H

Running the maturity assessment

Two assessments, twenty-six dimensions, two weight columns that each sum to 100, and no averaging between them.

Two assessments, run in the same fortnight, scored on separate sheets, reported on separate pages. One asks how good the architecture practice is. The other asks how sound the manufacturing estate is. They answer different questions, they are evidenced from different places, and the single most common mistake in this exercise is to average them into one number.

Keep them apart because the two scores move independently and because confusing them costs you the conversation. A practice can score well on repository hygiene while the plant cannot resolve a lot genealogy; a plant can run beautifully while the architecture function has never produced a compliance record. Kestrel Industrial Group, the invented manufacturer in the repository you were handed, manages both at once: a tidy 105-sheet repository with 843 claimed records, and no plant manager anywhere in it [1]. Karoo Industrial Group is volume 1's invented group, and the two never share data.

How to score

Both assessments use the same mechanics.

- **Five levels, integers only.** No halves. A half-point is a disagreement nobody has had.
- **Level 1 and level 5 are written out.** Levels 2, 3 and 4 are interpolations and are deliberately not defined, because defining them produces an argument about the rubric instead of about the estate.
- **A score needs evidence you have seen.** Not evidence that exists. Not evidence somebody says exists. A named artefact, a date, and where it is held.
- **Weights sum to 100 within each assessment.** They do not sum across the two.
- **Score 1 where there is nothing.** The temptation is to score 2 out of politeness. Resist it: a 1 with a reason is actionable and a 2 out of politeness is noise.
- **Every dimension scored below 3 becomes a candidate row on `029 Gap`.** That conversion is the point of the whole exercise and it is worked through later in this appendix.

Both assessments belong to the Preliminary Phase the first time you run them and to Phase H every time after that [4]. Run the first one before you have promised anybody a gate, and the second one before you have promised anybody a roadmap, because a roadmap built on an unscored estate is a list of projects with dates on it.

The weighted score is the sum of weight times score, divided by five. It lands between 20 and 100. Report the number once and the dimension scores every time — a single headline figure invites a target, and a target invites the scoring to drift.

Assessment one — EA practice maturity

Eleven dimensions. Each is evidenced from a named report sheet or from `ZZ Checks`, because a practice that cannot evidence its own maturity from its own repository has answered the question already.

#	Dimension	Weight	Evidence source	Volume 1
1	Repository completeness and integrity	8	`ZZ Checks`, `RPT13 Orphan Analysis`	19 · What to count before you model anything
2	Reference-data and key discipline	8	`R05`, `046`, `055`	9 · The plant hierarchy, and the reconciliation nobody budgets for
3	Requirements traceability	10	`RPT06 Requirements Traceability`	16 · Who inspects, and what they ask for
4	Measure definition and the dashboard	9	`004 Measure`, `RPT01 Dashboard`	7 · So the first deliverable is a measure definition register
5	Data ownership	10	`RPT05 Data Ownership`	10 · Master data governance, and the item-creation workflow
6	Application portfolio truth	9	`RPT02 Application Portfolio`, `RPT13`	12 · Six questions, asked of every class
7	Interface management	9	`RPT04 Interface Catalogue`	13 · Integration, as it really is
8	Technology currency	8	`RPT10 Technology Currency`	15 · What goes stale first, in order
9	Governance gate evidence	12	`059`, `R24`, `RPT11 Security Position`	16 · Who inspects, and what they ask for
10	Roadmap and transition integrity	9	`RPT07 Gap Analysis`, `RPT08 Roadmap`	18 · The target and the transition
11	Change control over the repository	8	No sheet exists; the absence is the evidence	19 · What not to do

Weights sum to 100. Three of the weights are worth defending out loud, because they are the ones a board will try to move.

Governance gate evidence carries 12, the heaviest weight in the assessment, because it is the only dimension that measures whether the practice has ever changed an outcome. A repository with perfect hygiene and no assessment records is a filing system. `055 Building Block` carries thirty-seven testable acceptance criteria and `059 Compliance Assessment` carries none, and that ratio is the whole argument.

Requirements traceability and data ownership carry 10 each, because they are the two dimensions that every other dimension depends on. A requirement that reaches no building block will not be delivered, and `RPT06 Requirements Traceability` already reports nine of thirty-one in that state —

computed correctly from a sheet generated positionally, so the nine are arbitrary and the real figure is unknown. An entity with no named owner has nobody to ask when the number is wrong.

Change control carries 8 despite having no evidence source at all, because weighting it at zero would be the same mistake the repository made. A dimension with no sheet behind it is still a dimension, and giving it a weight is how it gets a sheet.

#	Level 1	Level 5
1	Counts are typed; several are wrong and nobody has noticed	Every count is computed on open and reconciles to the rows
2	One key space has two schemes; refs resolve to nothing	One key space per object type, every ref resolving
3	Requirements are generated from names and reach nothing	Every requirement traces to a block, a package and a test
4	Measures have a baseline and no target, definition or owner	Every published figure has a register row behind it
5	Owners are job titles that resolve to nothing	Every entity has a named person and a tested lineage path
6	Dispositions name no destination; owners are assurance functions	Every disposition names its successor and its date
7	Most interfaces are unmanaged and unlogged	Every interface managed, published, logged and owned
8	Support position is typed, not computed, and is wrong	Position computed daily from support dates
9	The gate has produced no assessment records	Every criterion assessed at a gate, with dated evidence
10	Gaps are restated building blocks; packages outside transitions	Every gap costed, packaged and in a transition state
11	No version columns, no change log, no baselines	Every record versioned, with an approved change reference

What this repository cannot score above 1 as it ships

Four of the eleven are capped at 1 by the file itself, and two more at 2. Say so in the report and name the sheet, because a reader who thinks the score is a judgement will argue with you, and a reader who sees the empty sheet will not.

Dimension 9, governance gate evidence, scores 1 and cannot score higher. `059 Compliance Assessment` holds seven columns and zero data rows. `R24 Compliance Assessment tests` is header-only in consequence. The design gate has therefore produced no records at all, while `SEC-02 Design review` reads `Absent` on `041 Security Control` and on `RPT11 Security Position`. Three more sheets disagree with that: `POL-04 Security is designed in, not added afterwards` reads `Met` on `064 Security Policy`, `EP-01 Design review gate` reads `Newly established` on `065 Enforcement Point`, and `DLV-19 Compliance Assessment` reads `Draft` on `031 Deliverable`. One control, four positions, zero records. At weight 12 this is the heaviest dimension in the assessment and it is the one with nothing behind it.

Dimension 11, change control, scores 1 and cannot score higher. There is no change control anywhere in the 105 sheets: no version column on any of the 64 object sheets, no change log, no change requests against records, no baselines. `O61 Governance Change Request` holds five rows, but they are requests against work packages and against the repository as a whole, not against records, and two of the five have no expiry. `O60 Governance Decision` has no date column at all. The consequence is that every figure in `ZZ Checks` is a snapshot of an unknown date.

Dimension 4, measure definition, scores 1. `O04 Measure` has `Baseline` and `Cadence` and no target and no owner, so no measure can be missed and nobody can be asked why. Sixteen measures ship and not one of them is a manufacturing measure: the `Domain` values are Compliance, Finance, Operations, Data, Commercial, Technology, Delivery, Resilience and Governance, and there is no effectiveness figure, no first-pass yield, no schedule adherence and no mean time between failures.

Dimension 5, data ownership, scores 1. `RPT05 Data Ownership` reprints `O20 Data Entity` with a computed footer. Of fourteen `Owner` values only `Data Steward` resolves against `O11 Role`, at `ROL-16`. `E01 Product` is owned by `Head of Finance`, `E02 Bill of Material` by `Head of Sales`, `E08 Equipment / Asset` and `E10 Purchase Order` both by `Head of Health, Safety`. Six of fourteen `Lineage` cells read `Absent`, and they include the routing, the work order and the batch.

Dimension 1 is capped at 2. `ZZ Checks` carries fifteen checks and four of them are wrong. Records are reported as 843 against a counted 813 across the 64 object sheets plus `EXT Industry Extension`; the difference is exactly 30, which is the row count of `R06 Application interfaces with`, the only relationship sheet with a `Ref` column. "Object types populated 65, every type carries records" is false, because `O59` holds zero rows — it is 64 of 65. "Platform components past support 4" is really 2, because `PTC-06` runs to 2027-07 and `PTC-08` to 2027-04, both ahead of today, and `Position` in `O27` is typed rather than computed. "Work packages past end date 4" is impossible, because every `End` in `O30 Work Package` falls between 2029-07 and 2031-10. The counts that do hold — security controls absent at 1, control objectives without evidence at 5 of 6, the interface figures against `R06`'s thirty rows — hold because somebody counted them.

Dimension 8 is capped at 2 for the same reason: `RPT10 Technology Currency` repeats `O27`'s typed `Position` column and so repeats its error.

> Scoring your own practice at 1 on the heaviest dimension is uncomfortable and correct. The alternative — scoring 2 because a gate exists on paper — produces a report nobody can act on, because the remedy for "we scored 2" is unclear and the remedy for "the gate has produced zero records against thirty-seven acceptance criteria" is a date in a diary.

Assessment two — manufacturing health

Fifteen dimensions. This is the assessment that gets left out, and it is the one that tells the executive whether the estate can make and ship product with records behind it. Evidence comes from the plant as often as from the repository, and where the repository has nothing to say, the table says so.

#	Dimension	Weight	Evidence source	Volume 1
1	Item and bill-of-material mastering	9	`O20`, `O43`, `R05`	9 · The item master, and why it is the hardest master data object in any industry
2	The engineering and manufacturing BOM boundary	7	`O20 E02`, the PLM and ERP extracts	12 · The handover that nobody automates
3	Routing and standard times	6	`O20 E03`, the routing master	9 · Standard times, and why they are always wrong
4	Batch genealogy and traceability	9	`O20 E05`, a timed recall simulation	9 · Recall simulation is the real test
5	Measure definition register and loss comparability	8	`O04`, the register if one exists	7 · So the first deliverable is a measure definition register
6	The ERP and MES boundary	8	`R05`, `R06`, `O23`	12 · The boundary war, and how it actually gets settled
7	The IT and OT boundary and zoning	8	`O26`, `O47`, `O50`	13 · The Purdue model as it is actually used
8	Historian fidelity and tag governance	7	The historian configuration, per tag class	10 · Exception and compression silently change a reported number
9	Plant resilience, and whether it has been tested	7	`O49 Resilience Tier`, test records	13 · What breaks, in order
10	Maintenance and asset data	6	`O20 E08`, the asset register	9 · Equipment, functional location, and the two trees over the same steel
11	Quality records and the gate	7	`O20 E11`, `O20 E12`, the hold log	9 · Quality
12	Supplier and inbound data	5	`O20 E09`, `O19 Contract`	9 · Supplier, approved manufacturer, source list, call-off
13	The plant calendar and the delivery year	4	No sheet exists; the shift roster	7 · Shifts, crews, and what a pattern costs
14	Power quality and restart design	5	No sheet exists; the site electrical design	13 · Voltage quality is the design input, and it is written down
15	Change control over product and process	4	No sheet exists; the change log if one exists	8 · Engineering change

Weights sum to 100.

#	Level 1	Level 5
1	The item is mastered in a system that cannot master	One system of record, agreed, with the others consuming
2	The two structures are reconciled by hand, if at all	A governed transformation with a named owner at each end
3	Standard times are undated and nobody owns them	Every time owned, dated and re-measured on a cadence
4	Genealogy is reconstructed from spreadsheets after the event	Recall resolves both directions inside the obligation window
5	Figures are published with no definition behind them	Every figure has a register row naming its denominator
6	The split is re-argued per project	The split is written object by object and enforced at a gate
7	One flat network; the plant stops when the office does	Zones and conduits, with the line producing through an outage
8	Exception and compression are unknown per tag	Settings recorded per tag class and audited
9	Tiers are asserted and have never been tested	Every tier tested, dated, against the shift not the working day
10	One tree over the steel, or two that disagree	Functional location and equipment both held, both reconciled
11	The specification is uncontrolled and holds have no log	Specification versioned; every hold and disposition recorded
12	Suppliers duplicated; qualification status not held	One supplier record with qualification and approved sources
13	The calendar is a spreadsheet; the delivery year is a guess	One calendar, patterns recorded, capacity derived from it
14	Dip immunity unknown; restart is manual and undocumented	Immunity designed, restart ordered and tested
15	Product and process change by email	Every change has effectivity, disposition and a record

What the repository can and cannot tell you

Dimensions 13, 14 and 15 have no evidence source in the repository at all. There is no plant calendar sheet, no shift pattern column anywhere — `O12 Location` has four columns and `Scale` holds a headcount, which is how `LOC-08 Cloud Region` comes to have 261 people. There is no electrical design, no dip-immunity record and no restart sequence. And there is no change control, as dimension 11 of the first assessment already established. Score those three from the plant or do not score them, and say which you did.

Dimension 7 has two sheets that contradict each other. `LTC-01 OT/IT segmentation` reads `Emerging` on `O26 Logical Technology Componen` while `RME-01 OT/IT segmentation` reads `Adopted` on `O47 Reference Model Entry` — the same seven technologies appear on both sheets with different statuses, which means one of the two sheets is a wish. `O50 Network Link` describes a ring between eight locations and contradicts itself twice: `NWL-01` and `NWL-09` both run Northern Plant to Coastal

Plant at 10 Gbps with opposite `Diverse path` values, and `NWL-02` and `NWL-10` both run Coastal Plant to Components Plant at different bandwidths with opposite diversity. Nothing in the sheet describes a plant's internal network and nothing describes a zone boundary.

Dimension 9 scores 1 on the evidence as it stands. `049 Resilience Tier` has twelve rows, seven of them `Never tested`, and tier and recovery objective are uncorrelated: `RES-01 ERP Core` is `Tier 3` at 24 hours, `RES-02 PLM` is `Tier 1` at 24 hours, `RES-06 WMS` and `RES-12 Plant Analytics` are both `Tier 2` at 15 minutes. The only internally coherent row is `RES-04 SCADA / Historian` — `Tier 1`, one hour, `Tested`.

A gap, worked end to end

Dimension 1 of the manufacturing assessment, item and bill-of-material mastering, weight 9. Here is the whole path from evidence to a row somebody is accountable for.

Step one — read both sheets. `O20 Data Entity` has a `Master system` column. `O43 Master Data Rule` has a `Golden source` column. They cover the same ten entities, `E01` to `E10`, by name.

Entity	O20.Master system	O43.Golden source	Agreement
Product	SCADA / Historian	SCADA / Historian	Agree
Bill of Material	PLM	Track & Trace	Disagree
Routing / Recipe	Plant Analytics	Supplier Portal	Disagree
Work Order	APS	Supplier Portal	Disagree
Batch / Lot	WMS	Supplier Portal	Disagree
Plant / Site	APS	QMS	Disagree
Work Centre	QMS	WMS	Disagree
Equipment / Asset	SCADA / Historian	MES	Disagree
Supplier	APS	QMS	Disagree
Purchase Order	ERP Core	QMS	Disagree

Step two — state the finding in one sentence. Nine of the ten entities disagree, and the one that agrees agrees on the wrong answer: both sheets record the item master, `E01 Product`, as mastered in `SCADA / Historian`. A historian is an append-only time-series store. It cannot master anything, and the two sheets agreeing on that is worse than their disagreeing about the other nine, because an agreement looks like a decision.

Step three — check whether anything downstream depends on it. It does. `R05 Application manages Data En` has fourteen `Master` rows that put the application *name* in the `Application` column — `SCADA / Historian`, `PLM`, `Plant Analytics`, `APS` — where its fourteen `Consumer` rows correctly put a ref. A join on `Application` therefore returns half the sheet, which breaks the referential-integrity promise `00 Index` makes in its own text. `O39 Data Migration` then has `MIG-01` migrating `Product` *from* `PLM` *to* `SCADA / Historian`, which plans the error rather than fixing it. And `DBB-01 Product mastering` on `O44 Data Building Block` carries the acceptance criterion "No product record without a named owner", against an `Owner` of `Head of Finance` that resolves against nothing in `O11 Role`.

Step four — score the dimension. Level 1: "The item is mastered in a system that cannot master." That is literally the case. Score 1, weight 9.

Step five — write the gap. `029 Gap` has seven columns and two rows, both `High`, both restatements of a building-block name with a negation attached. Add this:

Column	Value
`Ref`	GAP-03
`Gap`	The item master is recorded in a time-series store, and `020` and `043` disagree on the golden source for nine of ten entities
`Domain`	Data
`Building block`	DBB-01
`Severity`	High
`Status`	Open
`Due`	2028-03

Step six — connect it. Write a row in `R12 Gap closed by Work Package` joining `GAP-03` to `WP-02 Data ownership and mastering`, which is already `Approved` at $8.8m and already `Past end date`. `RPT07 Gap Analysis` then picks the gap up automatically, because `RPT07` reprints `029` with the work package joined from `R12`. That is the one report in this set that will show your work without being rebuilt.

Step seven — say what it costs. Not in dollars, because the whole file is in dollars and this is a South African group; state magnitudes and consequences. The consequence of an item master nobody owns is duplicate part numbers, which means duplicate stock, which means a stock position that cannot be trusted at month end, which means the plant-to-finance reconciliation that volume 1 chapter 10 says is always wrong is wrong for a reason nobody can name. Write that sentence into the gap description if there is room. A gap that reads like arithmetic gets funded; a gap that reads like a sheet reference does not.

BEFORE THE GAP GOES TO THE BOARD

Which of the two sheets is right about the golden source for each disagreeing entity, and who said so?

Which named person will own `E01 Product` from the date the board sits?

What is the duplicate-part count today, and who counted it?

Two warnings

Never use the 85% figure as a benchmark. The number circulates as a best-practice target for overall equipment effectiveness and should not appear in your report at any level. The site that publishes it describes it as convenient, compelling and completely artificial [2]. Three further facts make it worse than merely unsourced.

There is no standard for the measure. ISO 22400-2's published 2014 edition defines the component terms, and the middle one is **effectiveness**, not performance — a wording that secondary summaries routinely get wrong and that must be confirmed against the licensed document before it enters a register [5]. But there is no conformance scheme and no single arithmetic, and **six or seven incompatible availability conventions** are in circulation: calendar time, scheduled time, loading time, planned production time, manned time, and variants that subtract changeover or do not.

The micro-stop threshold moves loss between availability and effectiveness without changing the headline figure. Set it at thirty seconds and a stoppage becomes a speed loss; set it at five minutes and the same event becomes downtime. The composite number is identical and the diagnosis is opposite.

And the measure is explicitly unsuitable for comparing machines, and for aggregating to department or plant level. A line with a constraint machine and a line without are not comparable; a plant average weights the slowest asset against the fastest and means nothing. Report it per line, against that line's own register definition, and refuse every request to rank or to roll up.

There is no published South African figure for most manufacturing benchmarks worth quoting here. If a dimension needs a comparator, say there is none. Do not construct one from a global average, a vendor survey or a conference slide, and do not infer one from the three national series that do exist — Statistics South Africa's production and sales index [7] measures volume, not effectiveness, and Eskom's availability figures [6] describe a utility's plant, not yours.

Two comparators do exist and are worth using because they are instruments rather than surveys. `NRS 048-2:2025` Edition 5, approved by NERSA on 6 May 2025, is the power-quality document, and it assigns window-Y dip immunity to the customer [3] — so for dimension 14 the comparator is the standard's own window, and the question is whether your plant meets it. For dimension 4, the comparator is the obligation that compels the record, not an industry figure.

Where you have no comparator, score against the level-1 and level-5 poles and nothing else. A score with no benchmark is honest. A score with an invented benchmark is a liability, and it is the first thing a sceptical executive will test.

Form MA-1 — EA practice maturity score sheet

Feeds. No sheet in the repository holds a maturity score. Create `04 Maturity` with these columns and key `Dimension` to the numbers in this appendix. `Weighted` is computed in Excel, never written by hand.

First pass. 11 rows, one per dimension.

Signed by. Chief Enterprise Architect (`ROL-08`), tabled at the Architecture Board (`GB-01`).

`Evidence seen` names the artefact, the date and where it is held. Leave it blank and the score is an opinion. Dimensions 4, 5, 9 and 11 cannot score above 1 against this repository as it ships, and 1 and 8 cannot score above 2 — check your working if they do.

#	Dimension	Weight	Score	Evidence seen	Weighted
1	EXAMPLE — delete	8	2		

Form MA-2 — Manufacturing health score sheet

Feeds. The same new sheet `04 Maturity`, with `Assessment` = `Manufacturing`. Keep the two assessments on one sheet and separated by that column; never total across them.

First pass. 15 rows, one per dimension. Continue on a second copy of this form.

Signed by. Chief Operating Officer (`ROL-03`) for the score, with each plant manager signing the rows that cover their plant. There is no plant manager role in `011 Role` — create it on Form B-3 before you run this.

Dimensions 13, 14 and 15 have no evidence source in the repository. Score them from the plant and write the plant artefact in `Evidence seen`, or leave them unscored and say so in the report. An unscored dimension with a reason is better than a guess.

#	Dimension	Weight	Score	Evidence seen	Weighted
1	EXAMPLE — delete	9	1		

Form MA-3 — Dimension score to gap row

Feeds. `029 Gap`, six of its seven columns, and `R12 Gap closed by Work Package`. `Status` is typed straight into Excel. One row per dimension scored below 3, from either assessment.

First pass. As many rows as you scored below 3. On this repository as it ships that is most of them, so continue on a second copy of this form.

Signed by. Chief Enterprise Architect (`ROL-08`), countersigned by the dimension's evidence owner.

`Gap` is a sentence a board member can act on, not a dimension name with a negation attached — both shipped rows on `029` read like the second and neither has ever moved. Name the dimension and its score in the sentence so the gap can be traced back to the assessment that produced it.

Ref	Gap	Domain	Building block	Severity	Due
GAP-04	EXAMPLE — delete	Data	DBB-01	High	2028-03

Running both assessments in one fortnight

Two weeks, and the output is two score sheets and a set of gap rows rather than a presentation.

Days	What happens	Output
1–2	Read all thirteen `RPT` sheets and `ZZ Checks` against the object sheets	The four wrong checks confirmed by your own count
3–4	Score assessment one from the reports; cap dimensions 4, 5, 9 and 11 at 1	Form MA-1, complete, with evidence references
5	First plant visit: lines, shift board, historian screens, hold cage	Dimensions 7, 8, 9, 13, 14 observed, not asked about
6–7	Second and third plants, same walk, same order	Three sets of observations, kept separate by plant
8	Recall simulation, timed, from a lot number somebody picks	Dimension 4 scored against the obligation window
9	Item master and BOM extracts counted for duplicates	Dimensions 1, 2, 3 scored with a number behind them
10	Score assessment two; leave 13, 14 and 15 unscored if the plant cannot evidence them	Form MA-2, complete
11–12	Convert every dimension below 3 into a gap row	Form MA-3, and rows in `029` and `R12`
13	Table both score sheets at the Architecture Board, separately	Two numbers, never averaged
14	Join the gaps to work packages and check the money	`RPT07 Gap Analysis` showing your rows

Three rules for the fortnight.

1. **Walk the plant before you score anything about the plant.** Dimensions 7, 8, 13 and 14 cannot be scored from a spreadsheet, and the answers you get by asking are the answers in the handbook rather than the answers on the floor.
2. **Time the recall simulation with a watch.** Dimension 4 has one honest test: pick a lot, resolve it to its inputs and its outputs in both directions, and record how long it took and how many systems were touched. If the obligation window is four hours and the simulation takes two days, the score is 1 and the argument is over.
3. **Never average the two assessments.** If somebody asks for one number, give them two numbers and the sentence that explains why. A combined figure hides the case where the practice is tidy and the plant cannot trace a lot, which is the most common shape in this industry and the most expensive.

What the two scores are for

The first assessment tells the Architecture Board whether this practice can be trusted with a gate. On this repository the answer is no, and it is evidenced by two empty sheets rather than by an opinion.

The second tells the executive whether the group can make product, ship it, and produce the records that prove it. That question is not answered anywhere in the 105 sheets, because the repository has no plant in it: no plant manager, no production planner, no scheduler, no maintenance planner, no quality manager, no process engineer, no operator, no setter and no millwright appears in `O10 Actor` or `O11 Role`, and `X02 Production Line` is declared in `EXT Industry Extension` and shadowed nowhere. Until that is repaired, the second assessment is scored from the floor and not from the file — and the fact that it has to be is itself the finding you lead with.

THE CHAPTER IN TEN LINES

- Two assessments, scored on separate sheets and never averaged into one number.
- EA practice maturity has eleven dimensions, weighted to sum to 100, each evidenced from a named report sheet.
- Manufacturing health has fifteen dimensions, weighted to sum to 100, and three of them have no source in the repository.
- Five levels, integers only. Level 1 and level 5 are written out; the middle three are interpolations.
- Governance gate evidence carries the heaviest weight and cannot score above 1, because O59 and R24 are empty.
- Change control also scores 1: no version columns, no change log, no baselines anywhere in 105 sheets.
- O20 and O43 disagree on the golden source for nine of ten entities, and agree on the wrong answer for the tenth.
- The item master is recorded in a time-series store, which cannot master anything.
- Never use the 85% figure as a benchmark. There is no OEE standard and the availability conventions are incompatible.
- Where there is no published South African comparator, say so rather than inventing one.

SOURCES FOR THIS CHAPTER

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